



Yalong YL-335B Automatic Production Line Training & Testing Equipment

Training Guide Book (Mitsubishi version)

Zhejiang Yalong Educational Equipment Joint-stock Co.,Ltd

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Item1 Understand YL-335B Automatic Production Line Trainer

1.1 YL-335B basic composition

YL-335B Automatic production line trainer is installed on an AL alloy rail, compose of feeding unit,process unit,assembly unit,transmission unit and sorting unit.outsie view as shown in diagram 1-1



Diagram1-1

On the equipment, each unit is an independent system, as well as an electromechanical system. Each unit is mainly a pneumatic execution unit, but the transmission unit mechanical arm movement is driven by servo motor,with precise position control, the drive system has a long travel, with one-dimension position control system. The sorting unit conveyor drive adopts general inverter drive 3-phase asynchronous motor AC transmission.The position control and inverter technology is the most widely used electrical control technology in modern industry.

On YL-335B, there are many types of sensors, used to detect materials movement position, status, colors and natures. The sensor technology is a key technology in electromechanical technology for modern industrial automation.

In the control, YL-335B adopts RS485 PLC network control mode, that is to say, each unit has a PLC to control task, each PLC communication with each other by RS485 serial ports for distribution control. The users can select various PLCs and the related RS485 communication modes according to actual requirements, to compose a small PLC network. The small PLC network has simple structure and low cost, and is widely used in the small automatic production lines, occupies a large share in modern industrial network

communication. Furthermore, with RS485 serial communication PLC network technology, the students can learn fieldbus technology, industrial Ethernet technology in the future.

1.2 YL-335B basic functions

All units of YL-335B trainer are shown in diagram 1-2(vertical view)

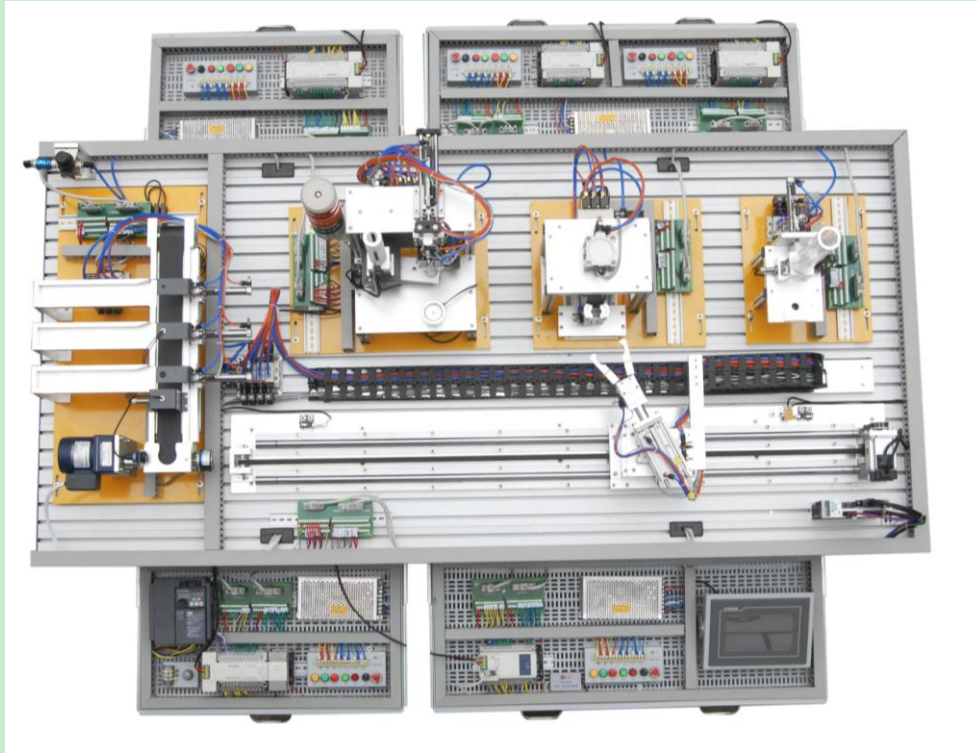


Diagram 1-2

Each unit basic functions:

1. The feeding unit

The feeding unit basic functions: the feeding unit is the initial unit of YL-335B, in the whole system , it provides workpieces to the other units. Specific function: according to requirements, push the workpiece to the material console, for mechanical arm to take.

The panorama of feeding unit as shown in diagram 1-3.

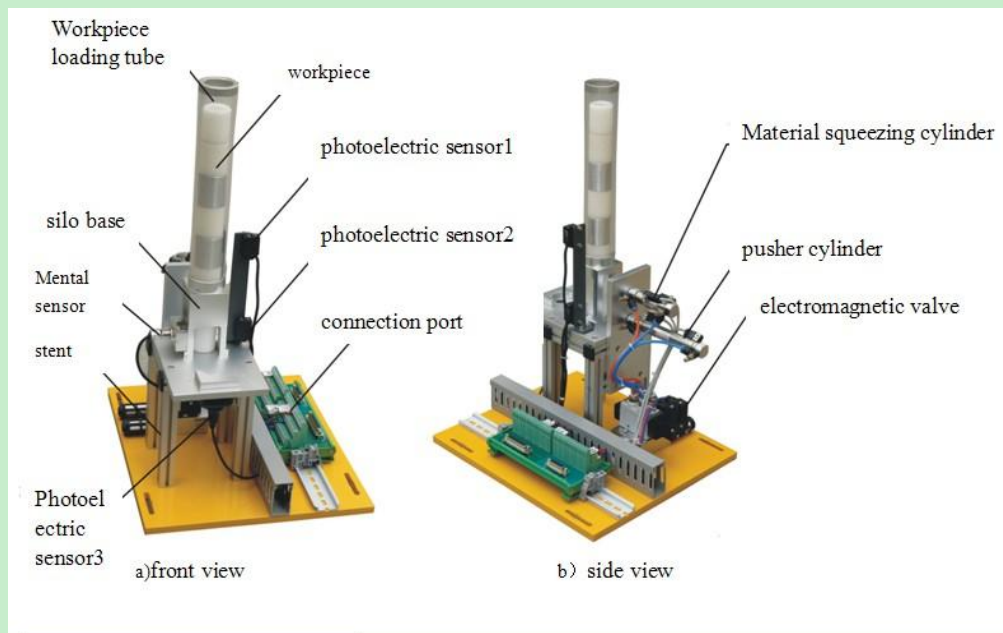


Diagram 1-3

2. The process unit

The process unit basic functions: processing the workpiece under the punching cylinder .After punching and processing,then return to wait for taking. The panorama of process unit as shown in diagram 1-4

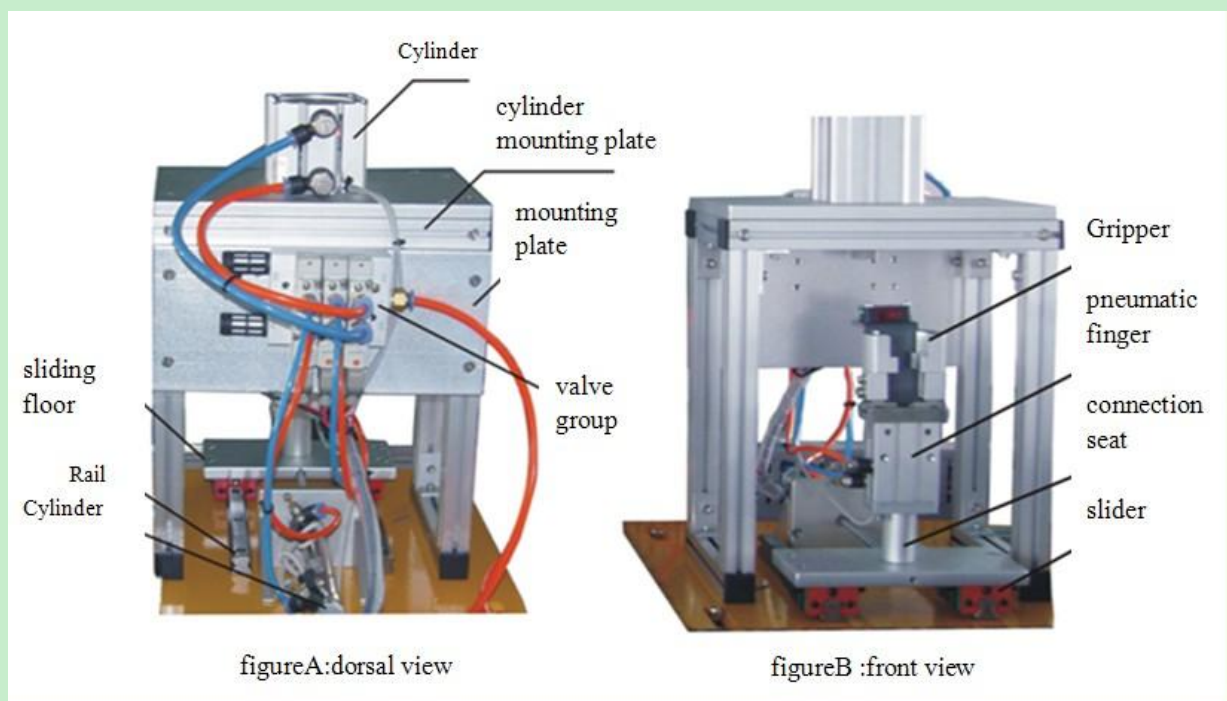


Diagram 1-4

3. Assembly unit functions: in the assembly process, the black or white small cylindrical workpiece are embedded into the workpiece from process unit. The panorama of assembly unit as shown in diagram 1-5

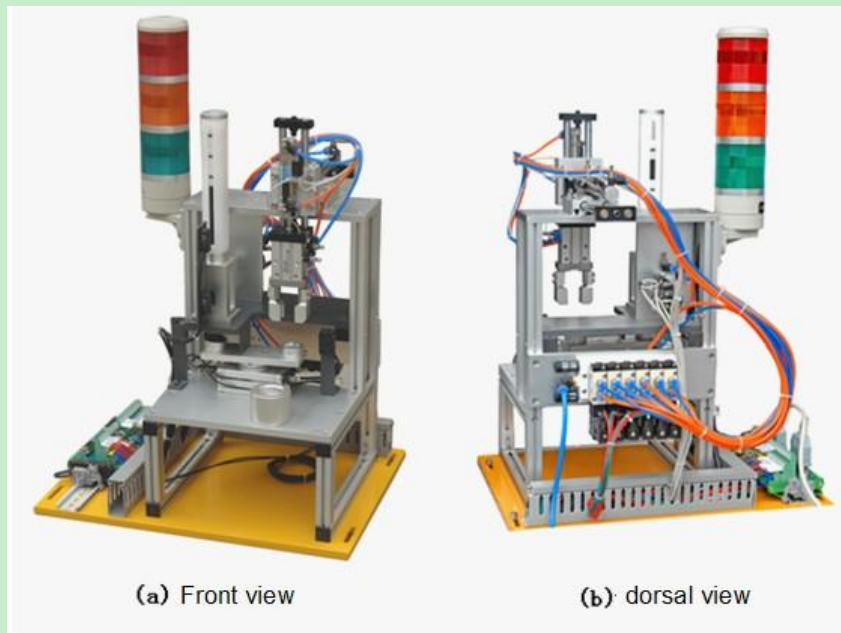


Diagram 1-5

4. The sorting unit basic functions:

with the sensor technology, sorting the workpieces from the assembly unit or process unit. The panorama of sorting unit as shown in diagram 1-6 .

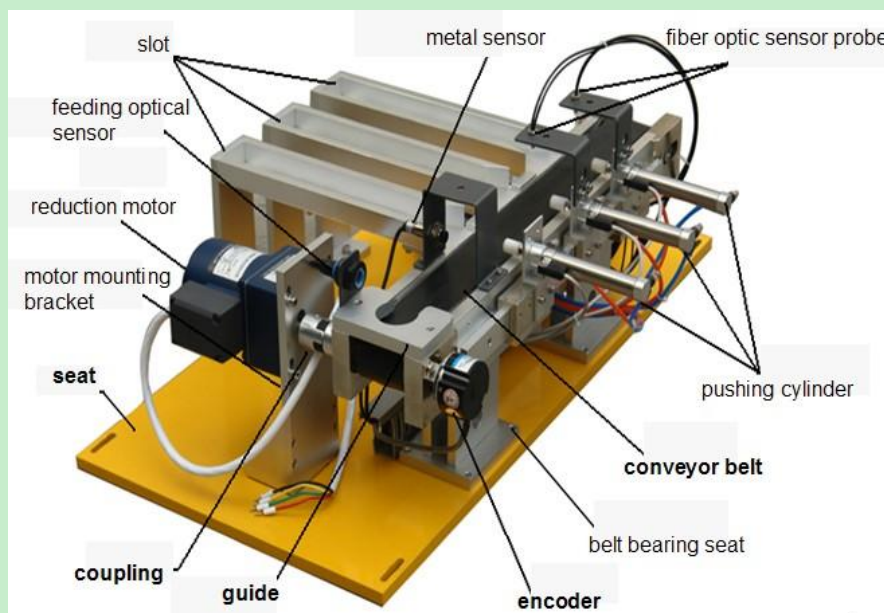


Diagram 1-6

5. The transmission unit basic functions:

When the workpiece is positioned, and caught, it will be transferred to the specific position and dropped. Achieve the transfer function of the workpiece. The panorama of transfer unit as shown in diagram 1-7 .

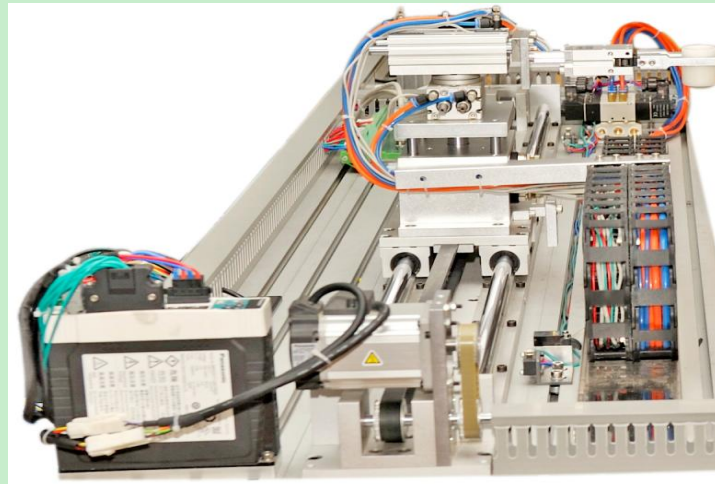


Diagram 1-7

According to the requirement, we adopt servo motor or stepper motor to control YL-335B .The standard configuration for YL-335B is servo motor .

1.3 YL-335B Electrical control

1.3.1 YL-335B Structural characteristics of working unit

Characteristics of unit YL-335B device structure is mechanical device and electrical control relative separation. Each working unit mechanical device are installed on the floor, on the whole, control of the production process PLC device unit is mounted in the table drawer on both sides of the board. Therefore, the exchange of information between the PLC devices and the working unit mechanical devices is a key issue. YL-335B 's solution is: the solenoid valve and sensor wiring are installation on the terminal in the side of the connection port. PLC's I / O pinout are connected to the connection port. Connection between the two ports interconnect through a multi-core signal cable. As shown in diagram 1-8 and 1-9 .



Diagram 1-8



Diagram 1-9

The connection port terminals by the side of device use three-tier terminal structure, the upper terminal connect the DC 24V power supply ,match with the +24V side.the bottom terminal connect the DC 24V power supply,match with 0V side .the middle tier terminal connect the separate signal lines.

The connection port terminals by the side of PLC use two-terminal structure, the upper terminal connect separate signal lines, its terminal ID machth with the terminal by the side of device. The bottom terminal connect +24 V terminal and the 0V terminal, supply DC 24V power.

Device side of the wiring terminal by the side of Device and terminal by the side of PLC connect with special cables.25-pin connector cable connect PLC input signals, 15-pin connector cable connect PLC output signal.

1.3.2 YL-335B Control System

Each unit is an independent system,as well as an electromechanical system.

1.When the unit as an independent system, the signal of equipment and the status of process show signal from the button indicator light module unit. Button indicator light module as shown in Diagram 1-10. Module on the lights and buttons all diverted to the terminal-side foot rafts.

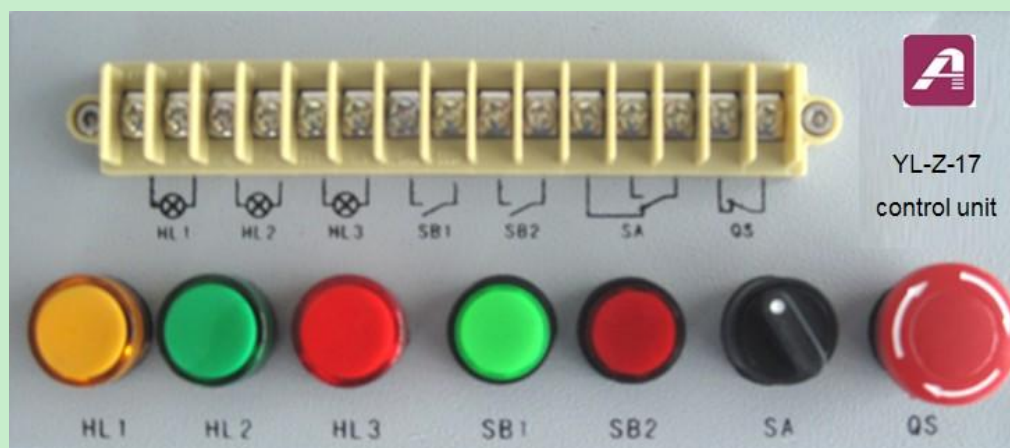


Diagram 1-10

Module box include :

- (1)Indicator light (DC24V) :yellow (HL1)、 green (HL2)、 red (HL3)。
- (2)Master device: a normally green open button SB1
a normally red open buttonSB2
selector switch SA (a pair of converted contacts)
emergency stop button QS (one normally closed contact)

2. When each station is connected into a control system network, for Mitsubishi FX series PLC, YL-335B standard serial RS485communication N: N is used. The initial control plan is shown in diagram 1-11.

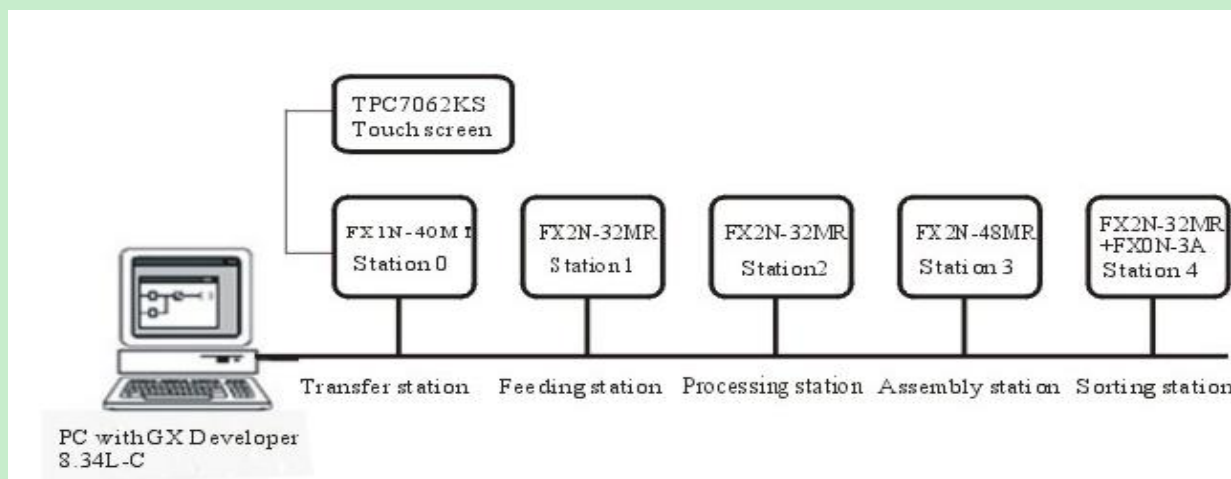


Diagram 1-11

Each station **PLC** configuration:

- (1) transmission station: **FX3U-48MT** master station, totally 24 input ,24 transistor output.
- (2) Feeding station: **FX3U-32MR** master station, totally 16 input ,16 relay output.
- (3) Processing station: **FX3U-32MR** master station, totally 16 input, 16 relay output.
- (4) Assembly station: **FX3U-48MR** master station, totally 24 input, 24 relay output.
- (5) Sorting station: **FX3U-32MR** master station, totally 16 input, 16 relay output.

3. Human-machine interface

The main system signals (reset, start, stop, etc.) is given through the touch-screen. Meanwhile, the human-machine interface display various status information. Human-machine interface is the bridge between the personnel and equipment communication. The use of man-machine interface can clear instructions and tell the operator the current status of work. make the operation simple, intuitive, image, vivid, and reduce operational errors, even the novice can easily operate. The use of human-machine interface also enables the machine wiring standardization, simplification, but also can reduce the required I / O points in PLC control , lower production costs, due to the control panel miniaturization and high performance, added the value of a complete set .

YL-335B adopt (MCGS) TPC7062KS touch screen as its human-machine interface. TPC7062KS is a low-power CPU with an embedded core (frequency 400MHz) the integration of high-performance embedded industrial computer. The product is designed with a 7-inch high-brightness TFT LCD screen (resolution 800 × 480), four-wire resistive touch screen (resolution 4096 × 4096), but also pre-installed Microsoft operating system embedded real-time multi-task WinCE.NET (Chinese version) and MCGS embedded configuration software (running version). TPC7062KS touch screen use, human-machine interface configuration methods will be introduced in the item five (sorting unit).

1.4 Power supply

An external power supply is three-phase five-wire AC 380V/220V, as shown in diagram 1-12 power supply module primary circuit diagram. In the diagram, the total power switch use DZ47LE-32/C32 three-phase four-wire leakage switch. The main load in the system by automatically switching separately. Among, inverter power supply through DZ47C16/3P three-phase breaker (auto switch) power supply. The station PLC use DZ47C5/2P single-phase breaker circuit (auto switch) power supply. In addition, the system configuration 4 sets DC 24V 6A were used for the feeding, processing and sorting unit, and transmission unit's DC power.

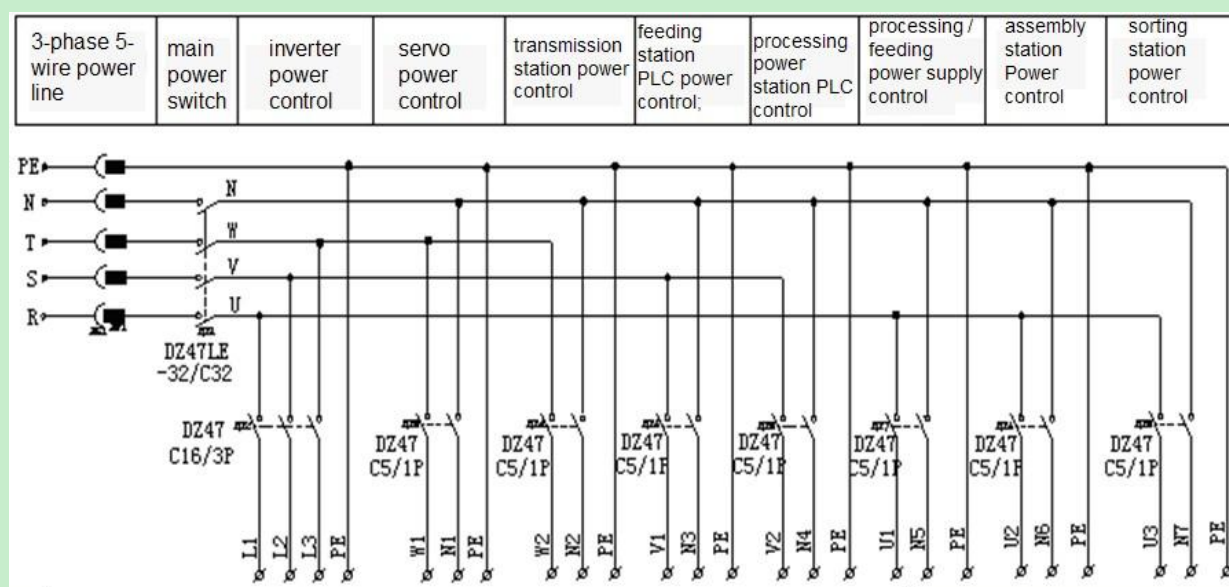


Diagram 1-12

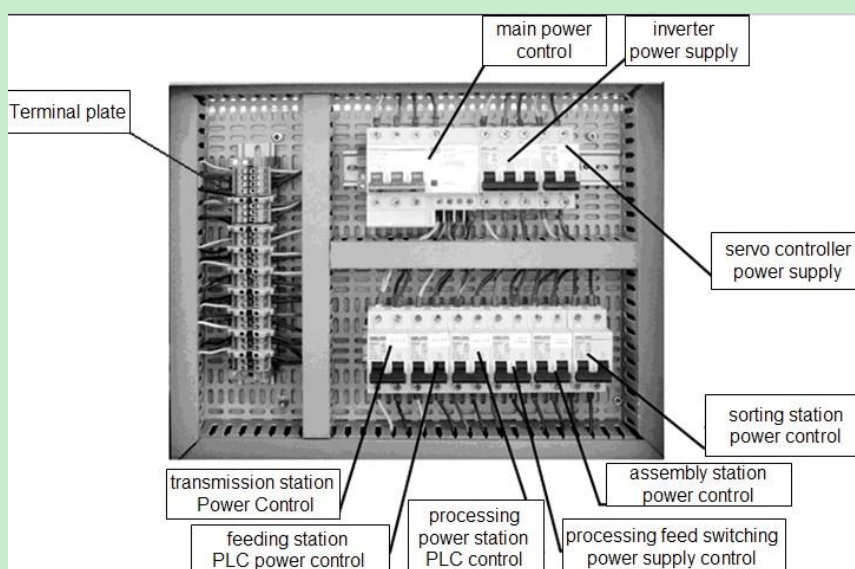


Diagram 1-13 Distribution box installation

1.5 Air source processing unit

YL-335B Air source processing components and circuit schematic are shown in diagram 1-14. Air source processing components is the basic components of the pneumatic control system, which is to remove the the impurities and condensate contained in compressed air, adjust and maintain a constant working pressure. In using, it should be noted that the water level of the condensate in the filter should always be checked and must be discharged before exceeding the maximum mark to avoid re-inhalation.

Gas inlet of air source processing components install a quick air switch for the start or close air source, when the gas switch in the left , the air connected to the gas source and the gas switch in the right ,which is closed.

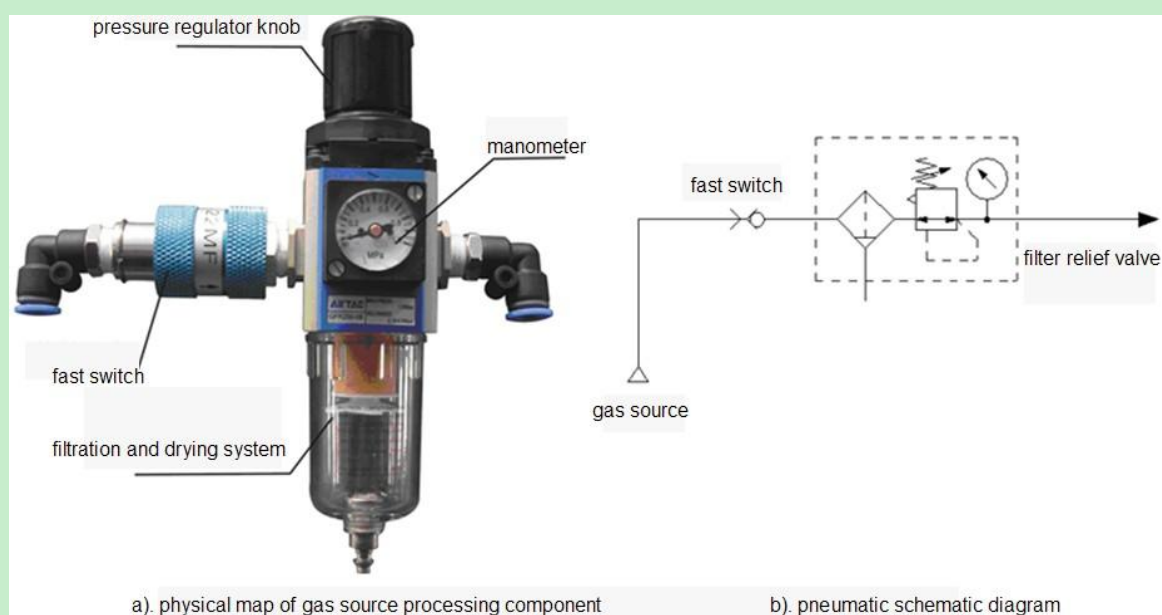


Diagram 1-14

The input air resource is from air compressor, the pressure shall be 0.6~1.0MPa, the output pressure 0~0.8MPa adjustable . The output air transfer to each working unit through quick tee connector and air tube.

Item 2 Control system of feeding unit

2.1 The structure and working process of the feeding unit

The main structure of the feeding unit is composed of: workpiece feeding pipe, workpiece pushing device, support frame, valve group, terminal block components, PLC, emergency stop button and start / stop button and etc. Among them, the mechanical part of the structure shown in diagram 2-1.

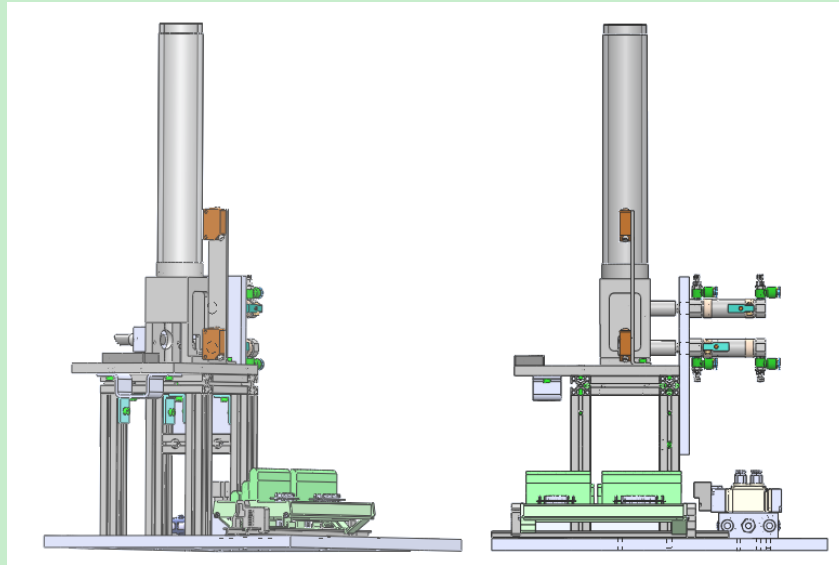


Diagram 2-1

Workpiece support and pushing device used to store workpiece, when necessary, it will push the workpiece from the bottom of the warehouse to the feeding unit. It mainly consists of workpiece tube, push cylinder, top cylinder, magnetic proximity switches, and diffuse optical sensors.

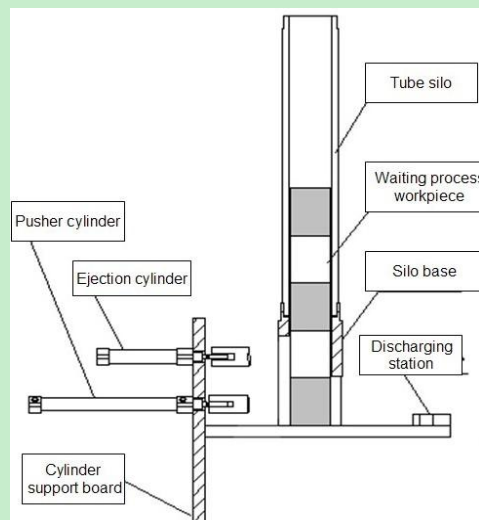


Diagram 2-2

The fundamental:workpiece in the warehouse. push cylinder on the bottom. the piston shaft returns.it is on the same position as the lowest workpiece,then the tighten cylinder is on a same level as the workpiece.when needed,push workpiece onto feeding unit,you shall push out tighten cylinder piston shaft onto the workpiece onto feeding unit.when the cylinder returns from the warehouse.you can tighten cylinder again and return.loosen the workpiece on the bottom,thus the workpiece in the warehouse will move automatically for next operation.

An diffuser optical switch is installed on the fourth layer workpiece,used to reflecte the workpiece enough or not.if there is no workpiece .the bottom and the fourth switches will be normal.if there are only three workpieces on the bottom,the bottom switch acts.the fourth switch is in nomal

Push cylinder release the workpiece to the discharge stage.The discharg table has a small hole on the table-board ,under the table sets a diffuser optical switch, through the small hole ,issued light up at work to test whether the workpiece is in there or not. So discharge the workpiece signal to the unit. in the transportation unit control,use this signal to determine whether need the mechanical arm to take the workpiece.

2.2 Knowledge Point

2.2.1 The pneumatic components of feeding unit

1.Standard double-acting linear cylinder

The standard cylinder is that the features and specifications are widely used, the structure easy to manufacture, as a generic cylinder in the market.

Reciprocating piston of the double-acting cylinder is driven by compressed air. Diagram 2-3 is a a half straight-line profiles of the standard double acting cylinder. As show,intake and exhaust openings are both on the two end caps, inlet gas from the rodless side cover, then push the piston forward movement; On the contrary, from the rod side cover, then push the piston backward movement.

Double-acting cylinder,with a simple structure and a stable output, and stroke can be selected, but it use compressed air acting on the piston to achieve flexible movement, when retraction,an effective role in an area of compressed air is small, so the produced force is less than the thrust generated when stretched out.

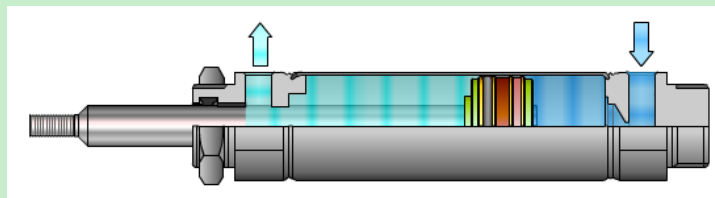


Diagram 2-3

In order to make the cylinder move smoothly and reliable, control the velocity of the cylinder, commonly used method is using one-way valve to realize.

One-way throttle valve is flow control valve paralleled by one-way valve and throttle valve, normally used to control cylinder movement speed, so called as the speed control valve.

As shown in diagram 2-4 shows connection diagram of the double-acting cylinder fitted with two one-way throttle valve, which called exhaust throttle mode. That is, when compressed air from the A-side air intake, exhaust from the B-side, the one-way throttle valve A open, take the air fast to the inflatable rodless chamber in cylinder; due to one-way throttle valve B closed, the gas with rod chamber only through the throttle valve exhaust, adjusting the opening degree of throttle valve B can change the extended cylinder when the velocity. On the contrary, adjust the opening degree of throttle A change in the cylinder can be retracted when the velocity. This control method, piston stable operation, is the most common way.

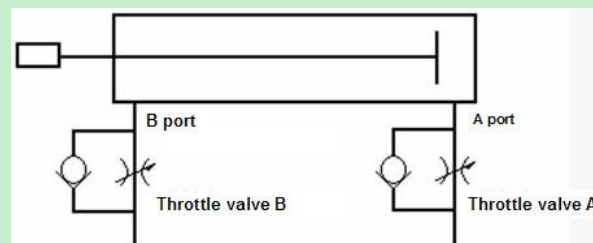


Diagram 2-4

There is a fast connection of air pipe on the throttle valve. Diagram2-5 shows the cylinder appearance of the cylinder throttle valve.

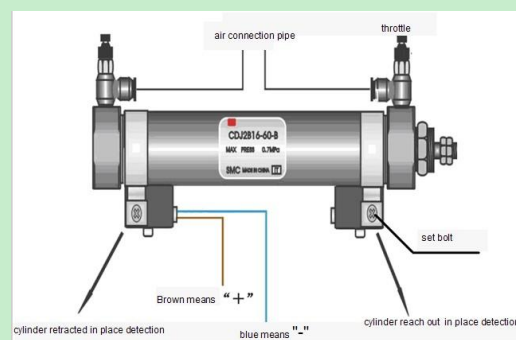


Diagram2-5

2. Single electric control solenoid valve, electro magnetic valve

As mentioned, the retract or push cylinder, the piston movement is rely on cylinder intake and exhaust from the other end, then, in turn, from the other side of the inlet, one end of the exhaust. Change in the direction of gas flow from gas flow change direction or control of on-off valves that control the directional control valves. In automatic control, the directional control valve solenoid control method often used to achieve directional control, name as the electromagnetic valve.

As shown in diagram 2-6, the working principle of single electronically controlled two position three way electromagnetic valve.

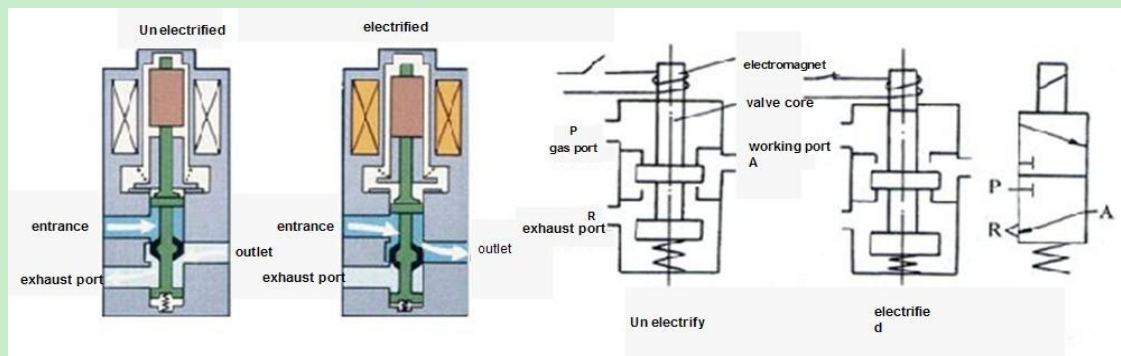


Diagram 2-6

The "position" refers to change the direction of gas , the spool has a different working position relative to the valve body. "Way" refers to the openings between valve connected with the system. Diagram 3-5, only two working positions with the gas supply port P, the work port A and exhaust port R, so named two position three way valve.

as shown in diagram2-7 , the graphic symbols of two position three way valve, two position four way valve, and two position five way single-solenoid control valve, how many grids there are, just called the position. "┐" and "└" symbol indicates the interface not to pass each other.

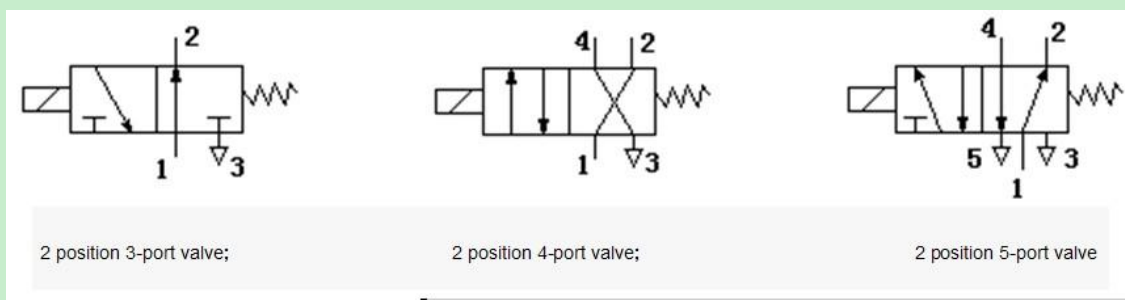


Diagram2-7

YL-335B the executive cylinder is double acting cylinder, so the solenoid valve shall with two working ports and two exhaust ports, as well as a supply port , the used solenoid valves are two position five way electromagnetic valve.

The feeding unit adopts two two-position five-way single controlled solenoid valve. The two solenoid valves have manual reversing and locking button, with lock (LOCK) and open (PUSH) 2 positions. With a small screwdriver to lock the button to the LOCK position, the manual switch down into the concave, can not be manual operation. Only in the PUSH position, the tool can be pressed down, the signal is "1", equivalent to the side of the electromagnetic signal is "1"; normal, the manual switch signal is "0". In the equipment debugging, you can use the manual switch to control the valve, in order to achieve the control of the corresponding gas to change the push cylinder and other actuator control, to achieve the purpose of debugging.

Two solenoid valves are centrally mounted on a busbar. The muffler is connected to the ends of the two exhaust ports in the busbar. The muffler serves to reduce the noise emitted by the compressed air to the atmosphere. The integration of a plurality of valves and a set of control valves, such as mufflers, busbars, is called a valve set, and the functions of each valve are independent of each other. The structure of the valve group is shown in diagram 2-8.

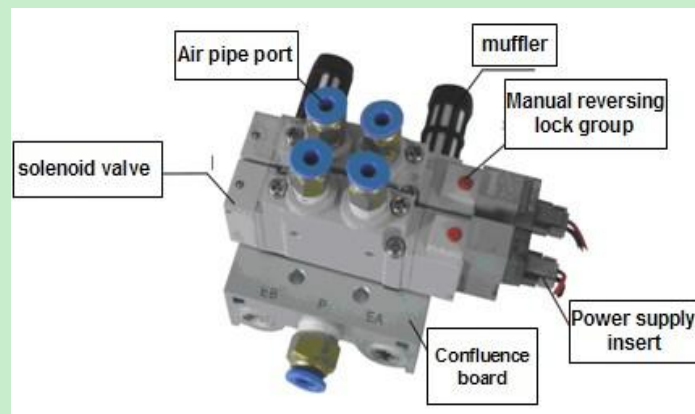


Diagram 2-8

3. Pneumatic control circuit

Pneumatic control circuit is the executive mechanism of this unit , and the logic control function of the executive mechanism is implemented by the PLC. The working principle of pneumatic control circuit as shown in diagram 2-9. 1A ,2A which is respectively two cylinder. 1B1 and 1B2 are magnetic induction proximity switches mounted at two extreme operating positions of the push cylinder, and 2B1 and 2B2 are magnetic proximity switches mounted at the two extreme operating positions of the top cylinder. 1Y1 and 2Y1 are the electromagnetic control terminals for controlling the solenoid valve of the push cylinder and the top cylinder respectively. Normally, the initial positions of the two cylinders are set in the retracted state.

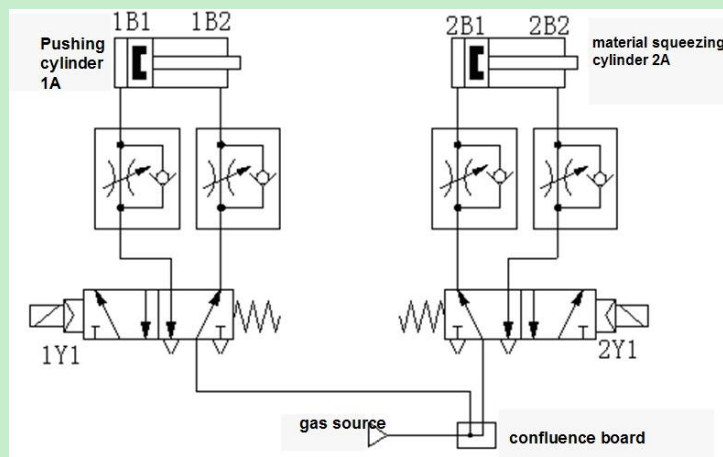


Diagram 2-9

2.2.2 Knowledge about sensor (Proximity Switches)

YL-335B all used sensors in this working units are proximity sensors, identify the object by approaching ,the sensitive character of the object,and output a corresponding switch signal, therefore, these sensors also call as proximity switch.

Proximity sensors have various ways to detect, including the use of electromagnetic induction caused by the eddy currents in detection of metallic objects , the capacity changes of electrical signals caused by capture test body approach, the use of magnets and guide switch ,the use of photoelectric effect and the Media Converter as a testing component parts and so on. YL-335B use proximity switches magnetic induction (or magnetic switch), inductive proximity switches, photoelectric switches and fiber optic diffuse reflection type photoelectric sensors. Here only introduce magnetic switches, inductive proximity switches and photoelectric switches diffuse reflection,The optical fiber type photoelectric sensor remain describ in the assembly unit practical training.

1. Magnetic switch

YL-335B the used cylinders are with magnetic switch,using permeability weak, across strong magnetic materials, such as a hard aluminum, stainless steel and etc. In the non-magnetic body of the piston install a permanent magnet magnetic ring, thus providing a magnetic field to reflect the location of the cylinder piston. Installed in the cylinder and the outer magnetic switch is used to detect the cylinder piston position, which detect movement of the piston stroke.

A contact-type magnetic switches with reed switch as magnetic field detection devices. Reed switches synthetic resin molding on the block, and usually has a movement indicator,over-voltage protection circuit is also included. Figure 2-10 is the schematic of the cylinder with a magnetic switch. When the magnetic ring near the switches, Reed switches's 2 reed is magnetized to attract each other, contact closure; when the magnetic ring remove the switch, the reed excitation loss, contact disconnected. When the contact is closed or disconnected send electrical signal, in the PLC automatic control, you can use this signal to determine the movement status or position of the top materials and material cylinder,and determine whether the workpiece has been launched or return.

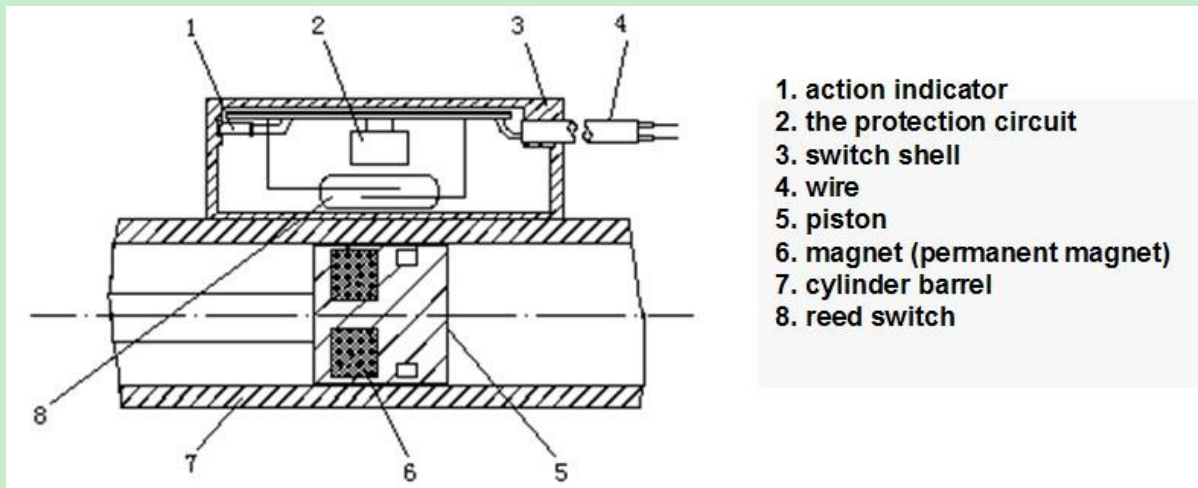


Diagram 2-10

LED display setting in the magnetic switch used to display the signal status, for debugging. Magnetic switch moves, the output signal "1", LED light; magnetic switch does not move, the output signal "0", LED does not indicate.

The installation position of magnetic switch can be adjusted, the method is to loose its positioning bolts, let the magnetic switch slide along the cylinder, arrived at the designated location, then tighten the fixed bolts.

Magnetic switch has a blue and brown two-outlet line, the blue pin-out should be connected to the PLC input public end, brown pin-out should be connected to the PLC input end.

Magnetic switch's internal circuitry show in diagram 2-11.

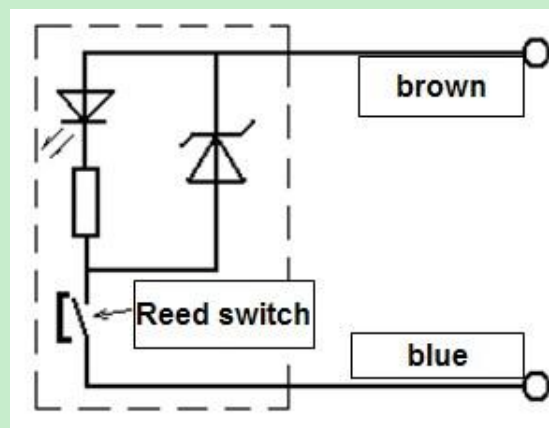


Diagram 2-11

2. Inductive proximity switches

Inductive proximity switches are sensors that use eddy current effects. The eddy current effect is that metal objects in an alternating magnetic field will produce internally alternating eddy currents. The eddy current will generate its magnetic field, which reacts on such a physical effect. If this alternating magnetic field is generated by an inductor coil, then the inductor current in the coil will change for the balance of eddy current magnetic field.

Using this principle, an induction coil in a high-frequency oscillator (LC oscillator) as a testing component, when the measured metal objects near the inductor coil produce the eddy current effect, causing changes in oscillator amplitude or frequency, the sensor signal conditioning circuit (including the detection, amplification, shaping and output circuit) changes into the switch output to achieve the detection purpose. Inductive proximity sensor working principle diagram shown in diagram 2-12. Feeding unit, in order to test whether the workpiece is a metal workpiece, in the bottom side of feeding tube installed a inductive sensor, as shown in Diagram 2-13

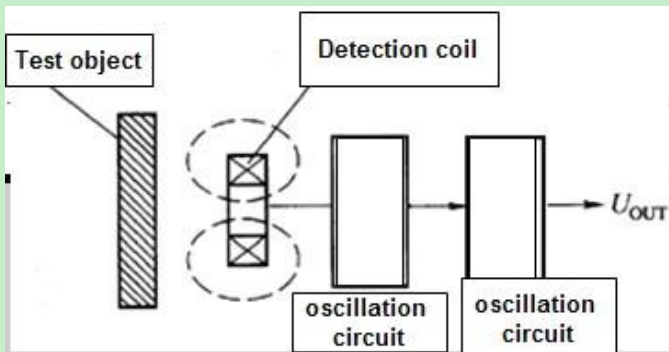


Diagram 2-12

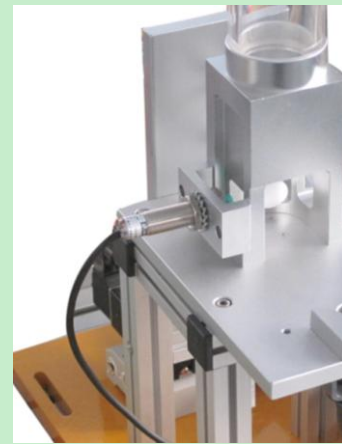


Diagram 2-13

In the selection and installation of proximity switches, we must seriously consider the detection distance, set distance to ensure a reliable production line of sensors moves. Installation instructions as shown in diagram 2-14.

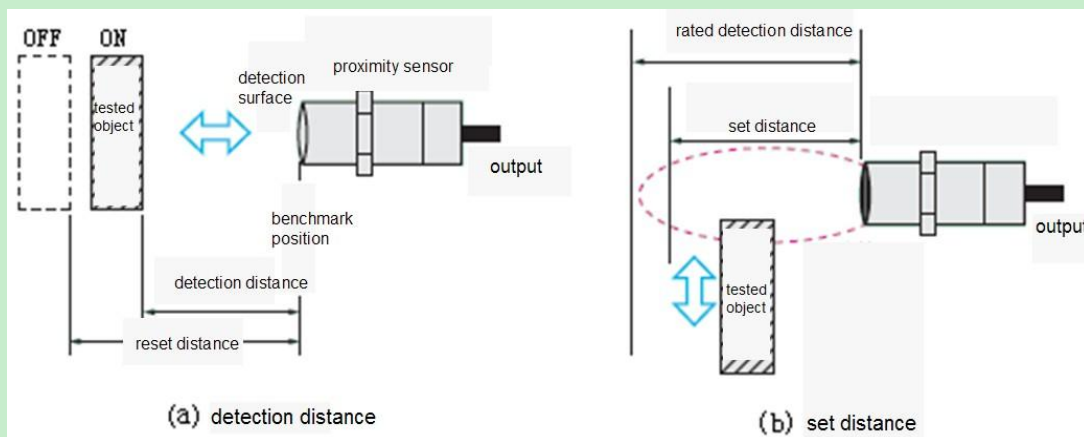


Diagram 2-14

3. Diffuse photoelectric proximity switch

(1) Photoelectric proximity switches

"Optical sensor" use various properties of light to detect whether have object and changes in the surface.Amount,output form sensor switch is Photoelectric proximity switches.

Photoelectric proximity switches mainly compose of optical transmitter and optical receiver. If the optical transmitter emitted light due to detection different objects, hidden, or reflection,the optical receivers will change. Sensitive optical receiver components will detect this change, and converted to electrical signals, electrical signals sent to the PLC. Most use visible light near the switch (mainly red, green, blue to determine the color)and infrared light.

According to the different ways of receive light, photoelectric proximity switches can be divided into pairs of radio type, reflective type and diffuse-type,3 species, as shown in diagram 2-15.

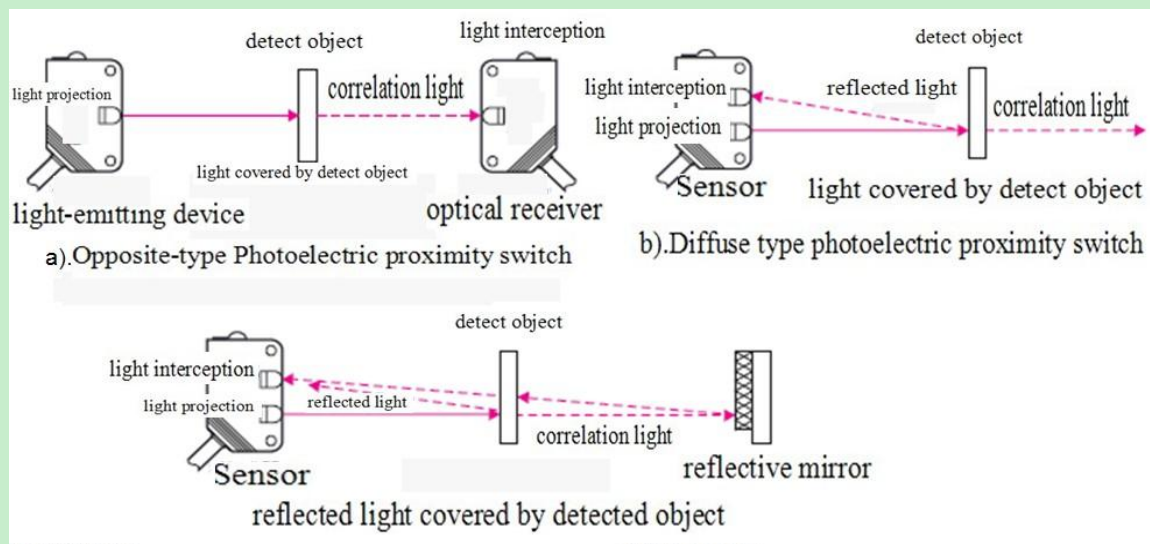


Diagram 2-15

(2) Diffuse-type photoelectric switch

Diffuse photoelectric switch use light shines on the measured object reflected back to work, because of the reflected light from an object is diffuse light, it is known as the diffuse photoelectric proximity switch. Its optical transmitter and optical receiver in the same side position, and is a integrated structure. At work, the light emitter emission testing is always light switch near the front if there is no object within a certain distance, there is no light is reflected into the receiver, proximity switches are the norm rather than action; the other hand, if the proximity switches occur within a certain distance in front of objects, as long as the reflected light intensity sufficient,Then the receiver receives enough diffuse light would make moves closer to switch to change the output state. Diagram 2-15 (b) is the diffuse type photoelectric proximity switch works diagram.

Feeding unit, the diffuse photoelectric proximity switch we choose sunx (OMRON) the company's CX-441 (E3Z-L61) amplifier built-in photoelectric switch (small beam-type, NPN type transistor open-collector output). The shape and the top surface of the adjustment knob and display of lights as shown in diagram 2-16.

The function selector switch choose moves by the light (Light), or shading action (Drag) model. That is, when this switch full clockwise rotation (L side), then enter the test-ON mode; when this switch fully counter-clockwise rotation (D side), then enter the test-OFF mode.

Setting knob is five weeks turning regulator, adjust distance should progressive slight rotation, if the regulator fully rotated,the regulator will idling. Adjustment method is: first,clockwise rotation to a minimum detection range (E3Z-L61 is about 20mm),then upon to distance place test objects, and gradually rotate clockwise from the regulator to find sensor into the test conditions point; pull test object distance,clockwise rotation until the sensor back to the point of non-detection state.The midpoint between two points, a stable detection of objects in the best position.

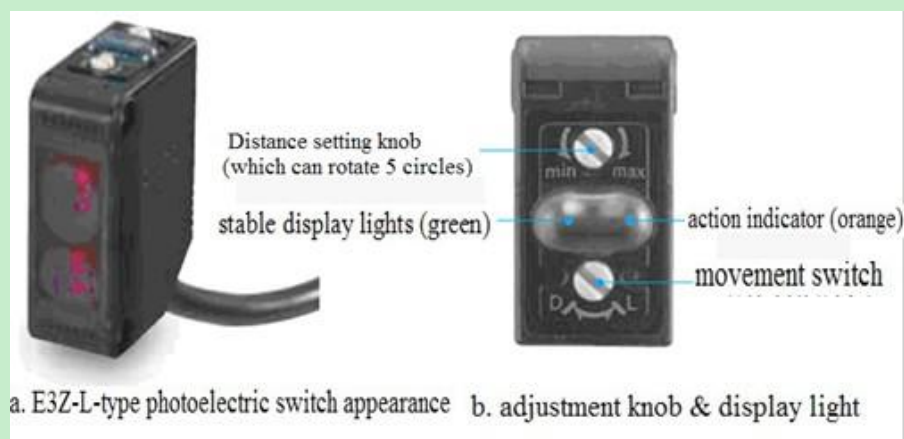
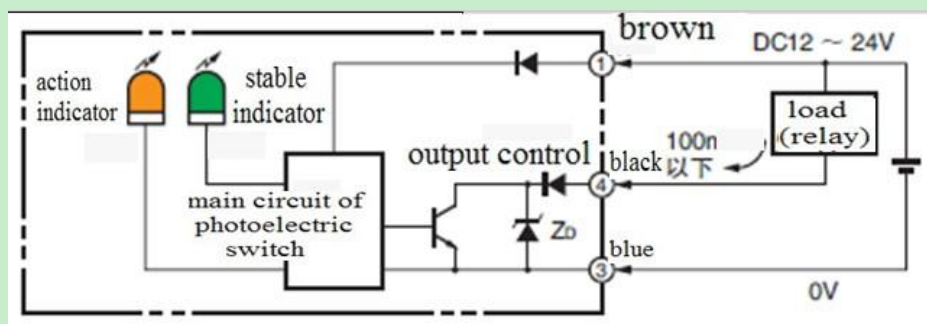


Diagram 2-16

Diagram 2-17 the internal circuitry of the photoelectric switch



The diffuse photoelectric switch detect whether the materials on the platform or not is a cylindrical photoelectric proximity switches, sent light to test whether the piece in or not through the hole exists. The photoelectric switch use SICK products MHT15-N2317 type, its form as shown in Diagram 2-18.



Diagram 2-18

4. Proximity switch graphic symbols

The graphic symbols of part proximity switch shown in Diagram 2-19. The diagram (a) (b) (c) the circumstances all use NPN-type transistor open collector output. If PNP-type is used , positive and negative polarity should in turn.

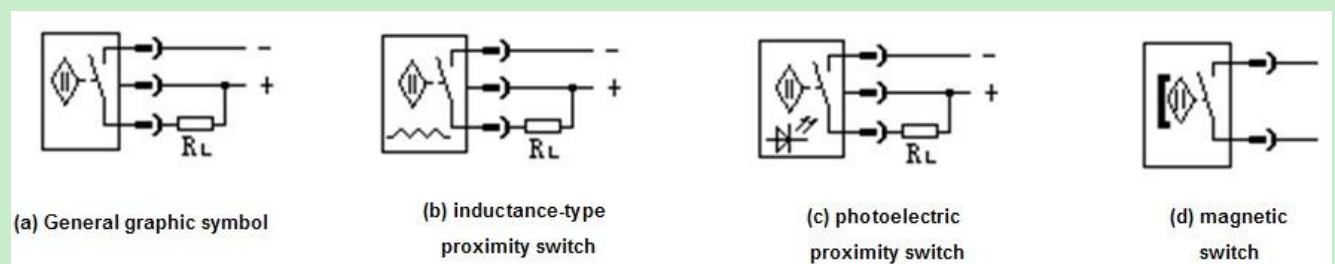


Diagram 2-19

2.3 Feeding unit install skill training

1. Training targets

Let the feed unit into components and spare parts, then assembled into intact, the installation includes the assembly of mechanical parts, gas connections and electrical wiring and adjustment.

2. Installation procedures and methods

(1) Mechanical parts installation

Firstly, install components, then assemble components. The combined components include:

- ① aluminum alloy components supporting frames;
 - ② discharge stations and silo base components;
 - ③ pusher body components.
- As shown in Diagram 2-20.

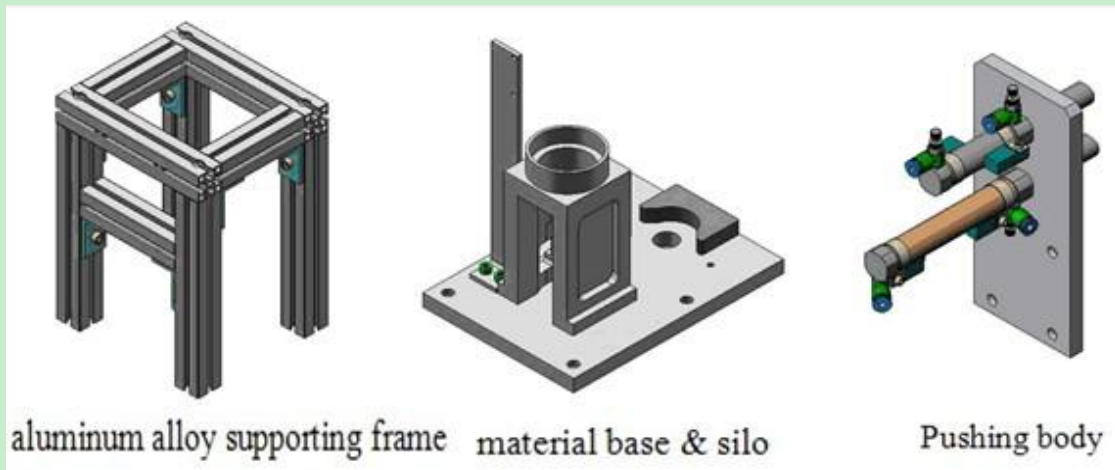


Diagram 2-20

After the components assembled, with bolts to connect them as a whole, and then loading tube rubber hammer to knock silo base. Then will be connected to a good feeding station and the electromagnetic valve, PLC, and fixed to the bottom row of terminals, the final completion of the fixed bottom feeding station installation.

During the installation process should pay attention to:

- ① assembly aluminum alloy Backstop, pay attention on readjust the edges of the parallel and vertical degrees, locking bolts.
- ②connected with install panels and aluminum alloy cylinder supporting frames, depending on specific location in advance of the aluminum "T"-type bath with matching nut placed on reserve, so that part of the aluminum profiles to connect, must be placed in the corresponding position of the corresponding nut. If there is no place to place nuts or not enough nuts, it will cause can not be installed or the installation is not reliable.
- ③ Fixed to the floor on the machine body when the floor needs to be moved to the console the edge of the opposite bolt from the screw into the bottom, the floor and mechanical support of part of the profile link.

(2) Gas line connect and debug

Connection steps: From the busbar started, as shown in diagram 2-9 Schematic diagram of pneumatic control circuit to connect solenoid valves, cylinder.attention should be sequentially-arranged into the air pipe, even beautiful, can not cross, discount; air pipe to be inserted in the fast joints tight, without any leakage phenomenon

Gas Path debugging include: ① Electromagnetic valve on the lock button to manually change to verify that the top feed cylinder and pusher cylinder location wether correct. ② Adjust the throttle valve to control the piston rod cylinder reciprocating motion speed, the speed up to not pushed down the workpiece.

(3)Electrical wiring

Electrical wiring included ,complete various sensors, solenoid valve, power leads to the device terminals and other wiring;the PLC connect power, I / O points wiring and so on. Feeding unit device connection solenoid valve and sensor wire as shown in Table 2-1.

Table 2-1

Middle layer input port			Middle layer output port		
Terminal No.	Equipment Symbols	Signal	Terminal No.	Equipment Symbols	Signal
2	1B1	Top material in place	2	1Y	Top feed solenoid valve
3	1B2	Top Material Reset	3	2Y	Push solenoid valve
4	2B1	Push material in place			
5	2B2	Push material Reset			
6	SC1	Discharge materials testing			
7	SC2	Lackof materials testing			
8	SC3	Whether have material testing			
9	SC4	Metal Material Testing			
10#~17# Terminal not connected			4#~14# Terminal not connected		

Pay attention on installation of wiring,the input signal terminal of the upper terminal (+24 V) can only serve as sensors for the positive supply-side, do not use electromagnetic valve actuators load. Electromagnetic valve actuators of the positive supply-side and 0V terminal should be connected to the output signal terminal on the lower terminal of the corresponding terminal. Device-side wiring is complete, use cable ties banding, and strive to tidy.

PLC side wiring, including power wiring, PLC's I / O point and the PLC side of the connection between the wiring port, PLC's I / O point and the button indicator of the connection between the module terminals. The specific wiring requirements is according to the work assignment.

Electrical wiring process should be consistent with the provisions of the national occupational standards, for example, wires connect to the terminal, the use of pressed terminal crimping method; connecting cable must have the required label; each terminal connected to wire not more than two cables .

2.4 Feeding unit PLC control system

2.4.1 Work task

The project only considers feeding unit as an independent device in the situation of running, the main signal and work status display from the button signal / light modules next to PLC。 Also, the button / indicator module work on the selector switch SA should be placed in the "single-station approach" position. The specific control requirements as follows:

- ① Connected with gas source, if the two cylinders are in a retracted position, and the silo have sufficient parts to be processed, then the "normal work" light HL1 light, indicates that the device ready. Otherwise, the indicator light flashes to 1Hz frequency.

- ② If the device is ready, press the start button, the working unit start, "equipment running" indicator HL2 light. Once started, if there are no expected stage parts, it should be pushed out workpiece material to stage. After the workpiece has been artificially removed, if it did not stop the signal, then the introduction of the workpiece for the next operation.
- ③ Press button while in operation, then completion of mission work cycle, each work unit stop working, HL2 light extinguished.
- ④ If there is not enough workpiece, then the unit continues work, but "normal" light HL1 frequency flashing 1Hz, "equipment running" light HL2 maintain light. If the silo has no parts, then the HL1 and HL2 indicator lights are flashing 2Hz frequency. Workstation after the completion of the task in this cycle stop. Added to the silo unless adequate workpiece, the workstation can no longer boot.

Complete the following tasks as required

1. Planning I / O allocation and distribution terminals of PLC.
2. System installation wiring.
3. According to the control requirement, PLC programming.
4. Debugging and running.

2.4.2 I / O wiring of PLC

The each station I/O signals (table 2-1) and tasks, supply station **PLC** is FX2N-32MR master station, total 16 point input and 16 point relay output. **I/O** signals of PLC are shown in table 2-2, wiring schematic is shown in diagram 2-22.

Table 2-2 I / O signal table of feeding unit PLC

Input signal				Output signal			
No.	PLC Input point	Signal Name	Signal Source	No.	PLC output point	Signal Name	Signal Source
1	X0	Top feed cylinder out in place	Device Side	1	Y0	Top feed solenoid valve	Device Side
2	X1	push material cylinder retract in place		2	Y1	Pusher solenoid valve	
3	X2	Push cylinder out in place		3	Y2		
4	X3	Push cylinder retract in place		4	Y3		
5	X4	Discharging materials testing		5	Y4		
6	X5	Lack of materials		6	Y5		

		testing				
7	X6	Out of materials testing		7	Y6	
8	X7	Metal parts testing		8		
9	X10	metal parts testing		9	Y7	Work instructions
10	X11			10	Y10	Running instructions
11	X12	Stop button	Button / LED Module	11	Y11	unused
12	X13	Open button				
13	X14					
14	X15	working mode choice				

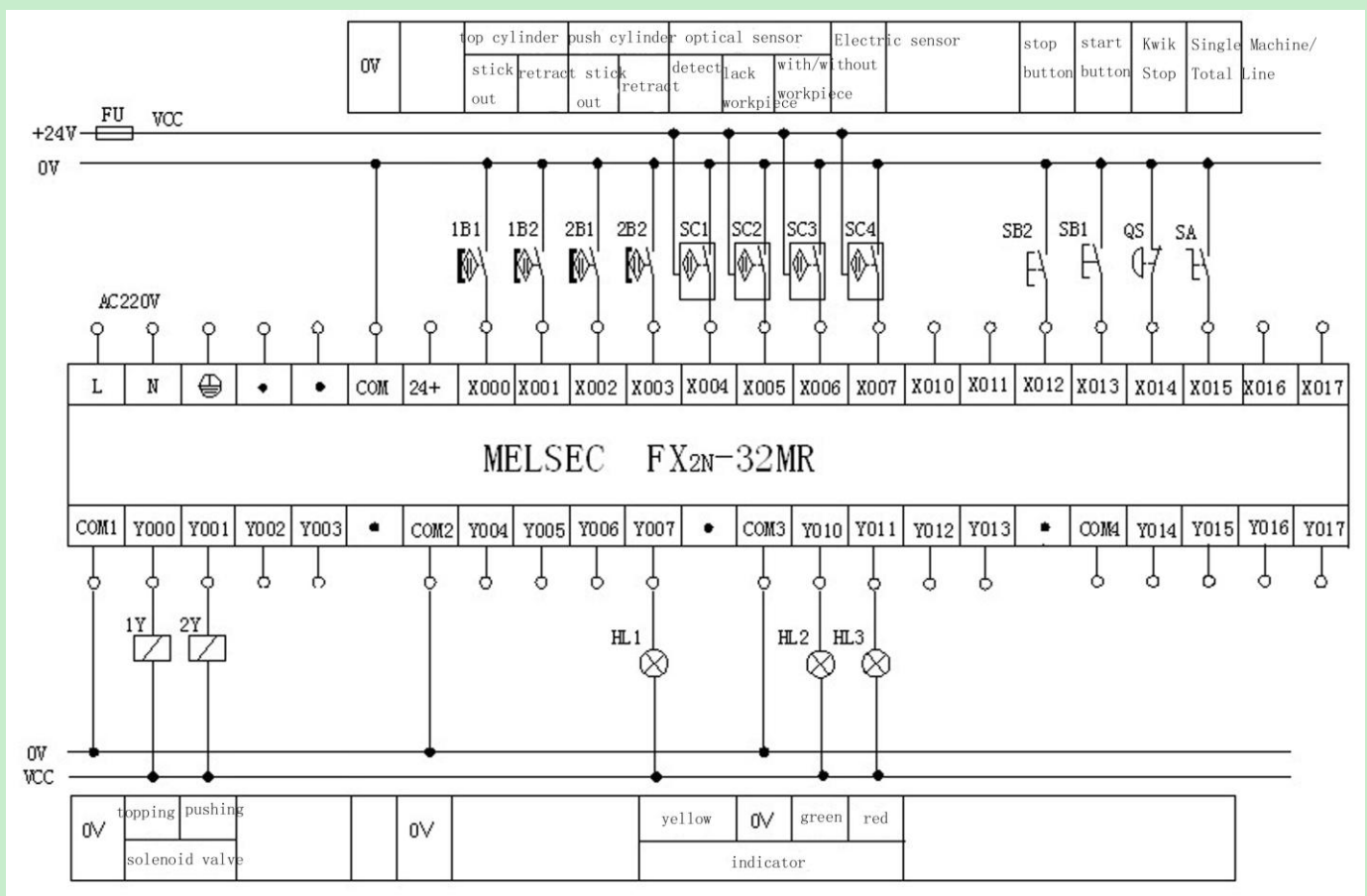
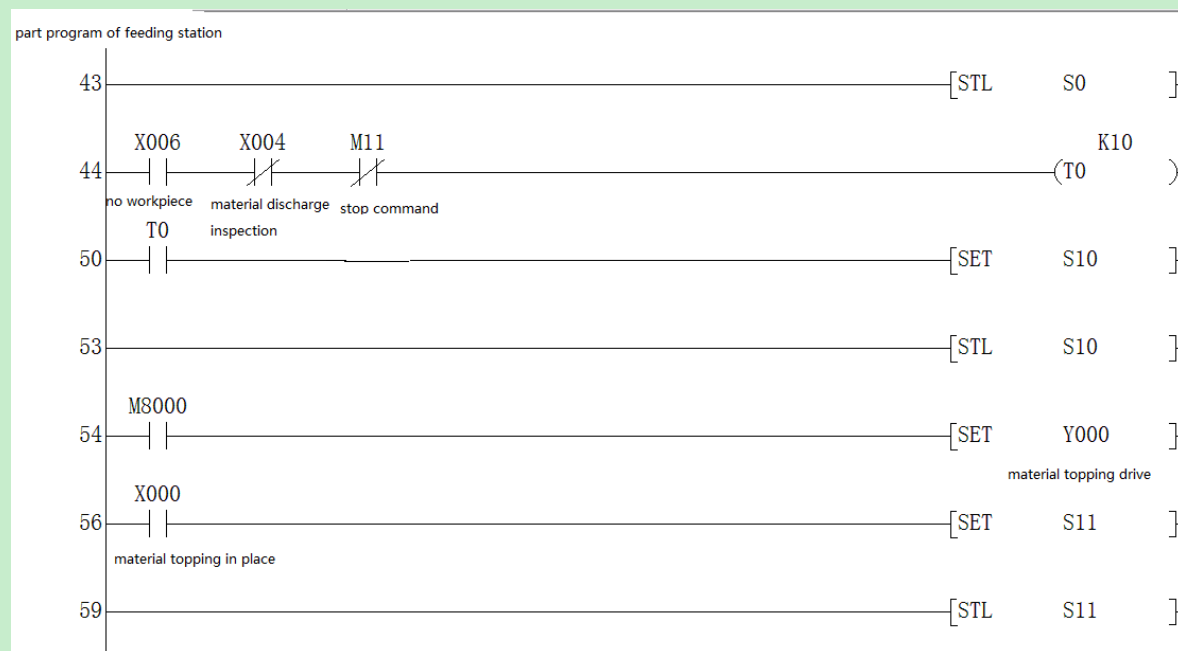


Diagram 2-21 I / O wiring diagram of feeding unit PLC

2.4.3 Feeding station control programming

1. Program structure: There are two subroutines, one is the system status display, and the other is material control. The main program in each scan cycle the system status display subroutine is called only when it has been established in the running for the feed control may call the subroutine.
2. PLC after power should first enter the initial state inspection stage, confirm the system is ready, then will be allowed into operation, so that problems can be found in timely, avoiding accident. For example, if the two cylinders in the source and access power not in the initial position, this because of the gas line connection error, apparently in this case does not allow the system operational. The PLC control system is usually tend to have such a routine requirement.
3. Feeding station work process: supply control, a sequence control process.
4. If there is no stop request, the control process will continue to cycle to cycle. Common sequence control system stops the normal requirement is received instruction to stop, the system completed the task of this working cycle, return to the initial step.
5. When the last workpiece is pushed out, workpiece loss alarm will come. Push cylinder reset, it returns to initial step, and stop. And the workpiece loss alarm starts, you shall add enough workpiece, then push start button to restart.

According to above analysis, system feeding control program as diagram 2-22



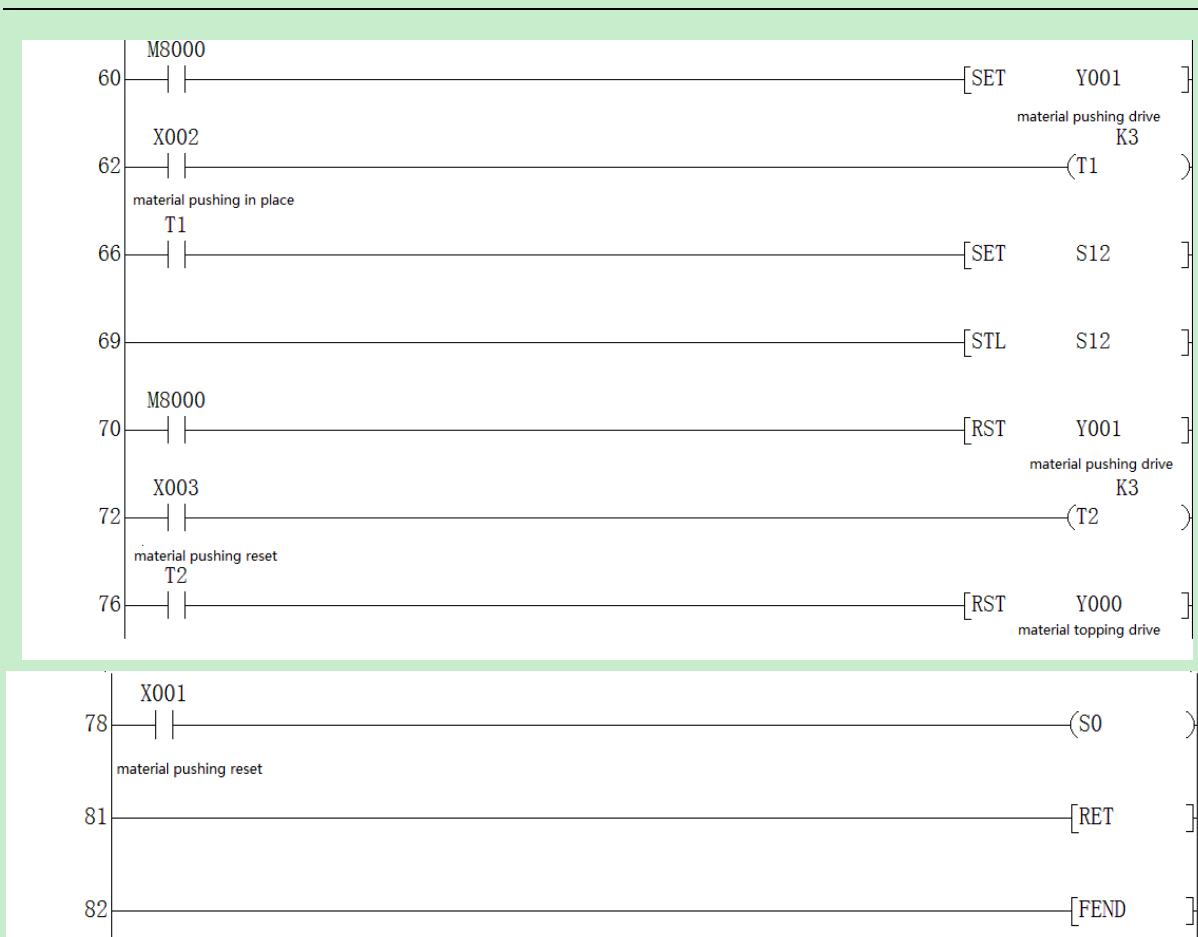


Diagram 2-22

Stepper sequence flow chart of feeding control program is shown in diagram 2-23. in the diagram, initial step S0 in the master program, when system is ready, it will be set.

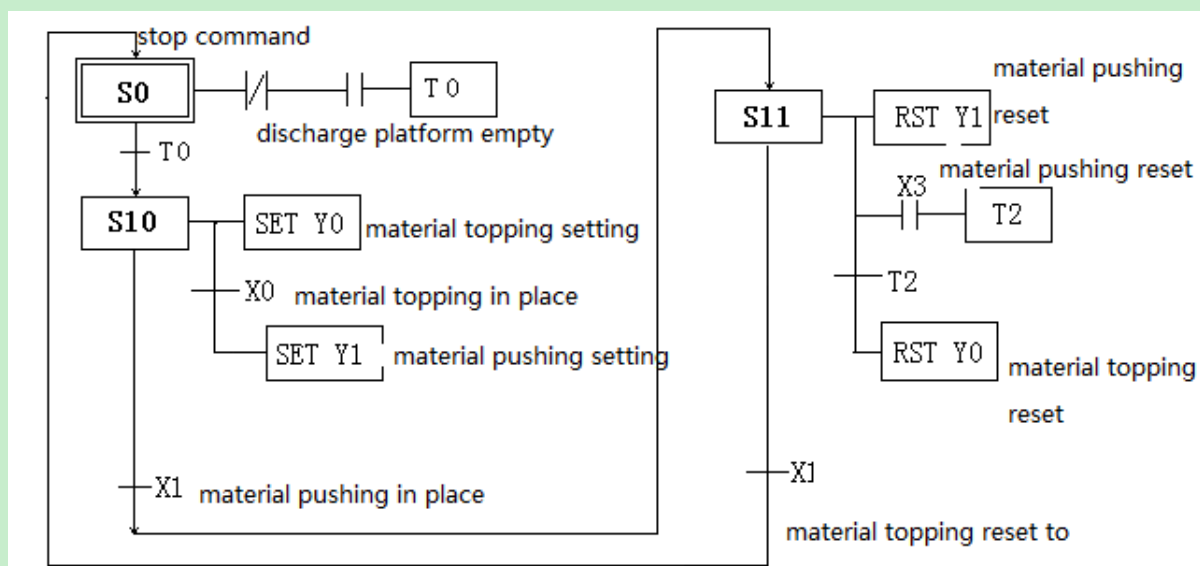


Diagram 2-23 Fedding control subprogram flow chart

2.4.4 Debugging and runing

- (1) Adjust pneumatic part, check the gas line wether correct, pressure wether reasonable, appropriate action cylinder speed wether appropriate.
- (2) Check the magnetic switch installation location wether in place, magnetic switches wether working correctly.
- (3) Check the I / O wiring wether correct.
- (4) Check photoelectric sensor install whether reasonable, sensitivity whether appropriate, to ensure the reliability of detection.
- (5) Put into workpiece, run the program, observe the processing unit action whether meets the mission requirements.
- (6) To debug a variety of possible scenarios, such as in any case are likely to join the workpiece, the system must be able to work reliably.
- (7) Optimization process.

3. Problems and thinking

- (1) Summary and learn to check the pneumatic connection, sensor wiring, I / O testing and troubleshooting methods.
- (2) How to deal with unexpected situations in the process.
- (3) Thinking if we adopt the network control,how to achieve?
- (4) Thinking variety of problems in the processing unit.

Item 3 Processing Unit Control System Training

3.1 Structure and working process of the Processing unit

The function of processing unit is to complete the processing of the transfer workpiece to the below regional press cylinder; complete the processing of stamping parts and then processed workpiece material back to re-process.

The main structure of processing unit consists of: processing platform and the sliding mechanism, processing (stamping) mechanism, electromagnetic valve group, wiring terminals, baseplate and etc. Among them, the module mechanical structure assembly as shown in diagram 3-1.

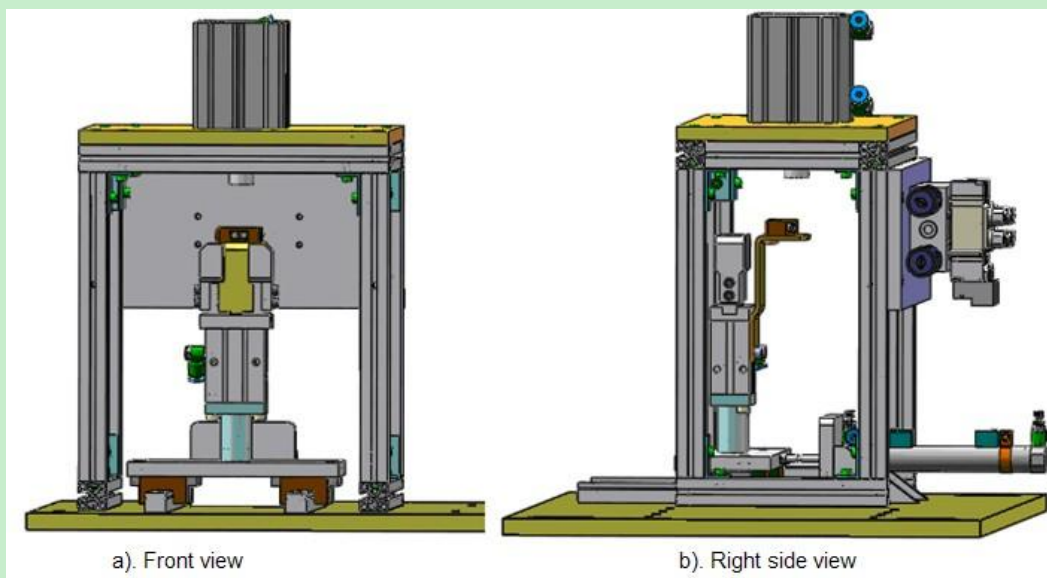


Diagram 3-1

1. The material platform and sliding mechanism

Processing platform and the sliding mechanism as shown in diagram 3-2. Processing platform are used to fix processed parts, and move workpiece under the stamping process to process. It is mainly composed of the pneumatic gripper fingers, processing, telescopic cylinders, linear guide rail and slider, magnetic proximity switches, diffuse photoelectric sensor.

The principle of slide processing platform works: slide processing platform working in the system after the initial state for the telescopic cylinders, this is the processing pneumatic fingers open state. After sending materials to the feed stage, the material detection sensor detects the workpiece, PLC control program-driven pneumatic workpiece clamping fingers → processing units will be returned to processing region → stamping cylinder press cylinder piston rod down to the bottom of stamping out parts. After the completion of stamping action → → processing units retracted up out → in place to re-release of the

order of the completion of pneumatic finger workpiece machining processes, and complete the signal processing system sounds. prepare for the next arrival process.

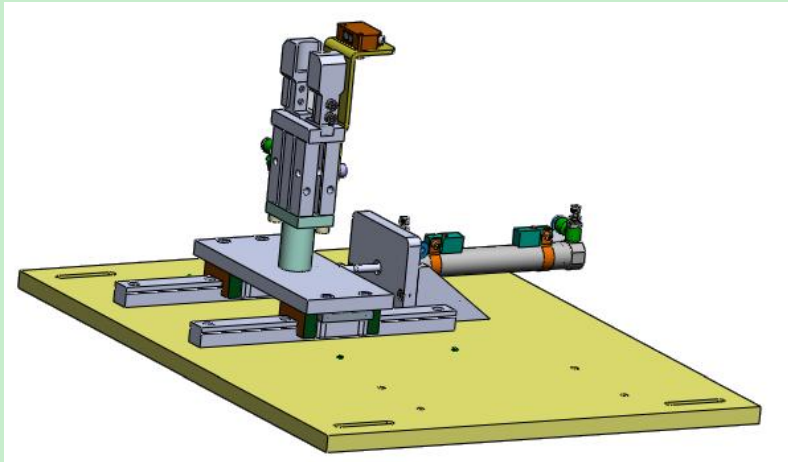


Diagram 3-2

In the mobile platform is installed of a diffuse-type photoelectric switch. If there is no workpiece on processing platform, then the diffuse photoelectric switches are in the normal state; if there are parts on processing platform, then the photoelectric proximity switch moves, indicating that the stage has processed workpiece. The photoelectric sensor's output signal to the PLC input processing unit determining whether there are workpiece need to process; after the machining process complete, processing units return to its original position. At the same time, PLC through the communication network, make the processing signal back to the system to coordinate control.

The diffuse photoelectric switch in the mobile platform still select E3Z-L61 amplifier built-in type photoelectric switch (small beam type), the principles ,structure and debugging methods of the photoelectric switch have been introduced in the previous before.

Mobile material platform stretch and return in place by adjusting the telescopic cylinders on the two magnetic switch location positioning. retracted position beneath in the processing of punch; location holding out should be grasping with the transport unit with the robot device action, to ensure the transport unit grasping robot can be smoothly put into the workpiece material on platform.

2. Processing (stamping) mechanism

Processing (stamping) mechanism as shown in diagram 3-3. Processing mechanism for the workpiece to press. It is mainly composed of press cylinder, stamping head, install panels and etc .

Working principle of stamping platform: when the workpiece reaches punching position, that is, retracted telescopic cylinder piston rod in place, pressing cylinder out of the workpiece processing, after the completion of processing actions stamping cylinder retracted in preparation for the next punch.

Punch according to the requirements of the workpiece on the workpiece stamping, Punch stamping cylinder mounted on the head. Stamping cylinder mounting plate for mounting on

the press cylinder to fixed.

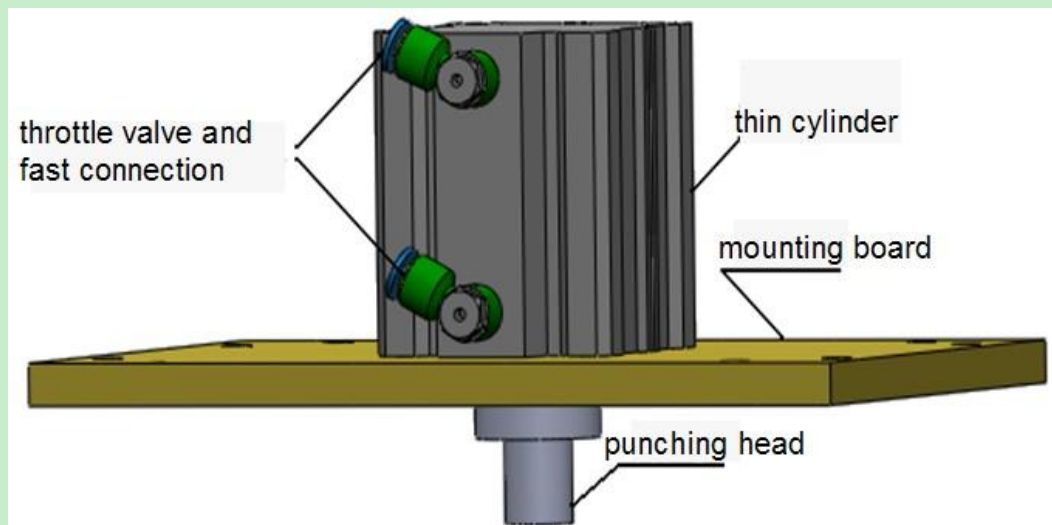


Diagram 3-3

3.2 Related knowledge points

3.2.1 Understanding of Linear Guide

Linear guide is a rolling guide, which consists of steel ball between the slide and the slide to make an infinite rolling loop, allowing the load platform along a rail with high precision for linear movement, its coefficient of friction can be reduced to the traditional sliding guide cited in $1/50$, so that it can achieve a high positioning accuracy. In the straight-line transmission areas, Linear Guideway has been a key product, has become a variety of machine tools, CNC machining centers, precision electronic machinery in the important and indispensable features.

Linear Guideway usually based on the rail and the slider ball contact between the tooth-type classification, mainly 2 type and four type two kinds. YL-335A use an ordinary level on both the accuracy of two linear guideway, their contact angle can remain constant in motion, rigidity is also relatively stable. Diagram 3-4 (a) give schematic cross-section in guideways, Diagram (b) assembled linear guideway.

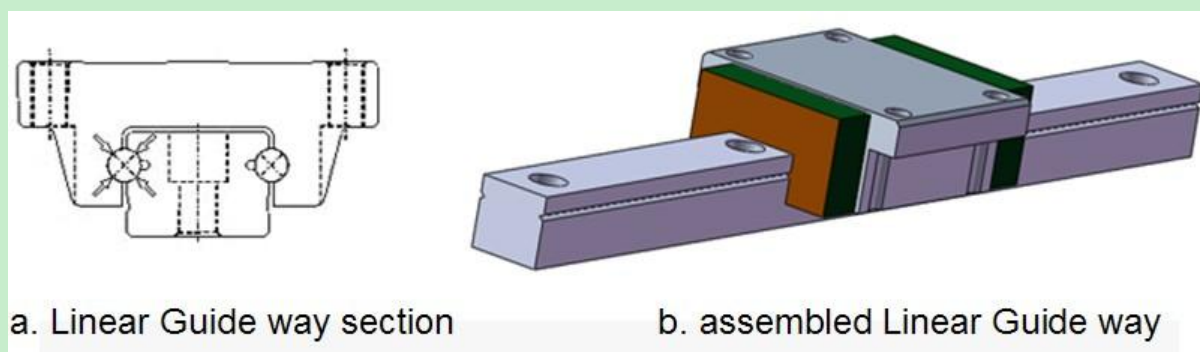


Diagram 3-4

Linear guideway installation should pay attention: ① be careful take down and up light, avoid the bump in order to influence a straight line accuracy of guideway. ② Do not split away from the slide onto the rail or exceed the trip back.

Processing unit mobile material sliding mechanism consists of two linear guide rail, pay attention to adjust two parallel linear guides. The installation method of mobile feeding platform and the sliding mechanism will be discussed in the task of "processing unit installation skills training" .

3.2.2 Pneumatic components

Processing unit use pneumatic actuators including standard linear cylinder, thin cylinder and pneumatic fingers, the following describes the thin cylinder and pneumatic fingers.

1. Thin cylinder

Thin cylinder belong to a space-saving cylinder category, namely, the cylinder axial or radial dimensions smaller than the standard cylinder. Has a compact structure, light weight, small footprint etc. Diagram 3-5 is the thin cylinder of case graph.



The characteristic of thin cylinder: rodless cylinder with die-casting into one side of cover, rod cover spring collar fixed for the square cylinder. This cylinder is commonly used fixture in the fixed and moving parts and so on. YL-335B in the processing unit, thin cylinder used for stamping, it is mainly to consider the characteristics of a short stroke cylinders .

2. Pneumatic fingers (Gripper)

Gripper is used to grasp and clamp the workpiece. Gripper usually have sliding rail type, pivot open-close-and rotary-driven and other working methods. YL-335B Processing unit use a sliding rail type pneumatic fingers, show in diagram 3-6 (a). The working principle can seen from profile (b) and (c).

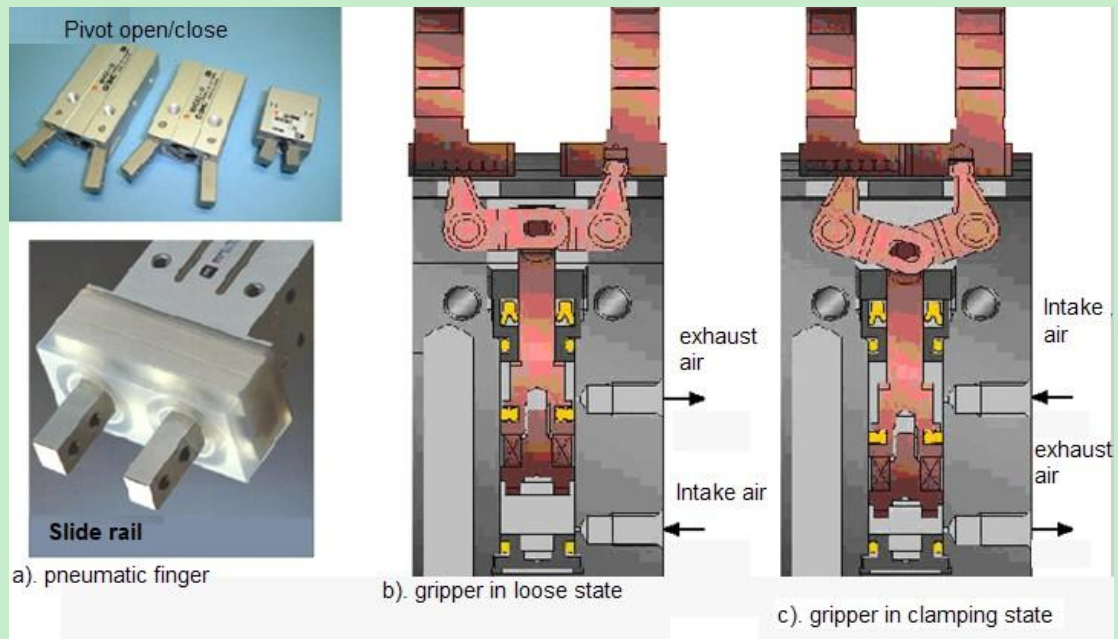


Diagram 3-6

3. Pneumatic control circuit

Pneumatic control components of processing unit use two position five way single-electric control solenoid valve, which with manual direction and lock button. They are mostly installed as a valve fixed to the press supporting frames behind.

The working principle of pneumatic control circuit as shown in Diagram 3-7. 1B1 and 1B2 stamping cylinder is installed in the two extreme positions of the magnetic induction near the switches, 2B1 and 2B2 is installed in the processing limits work location of the two magnetic proximity switches, 3B1 gripper cylinder installed in the work of the location of magnetic proximity switch. 1Y1, 2Y1 and 3Y1, respectively control the press cylinder, processing, telescopic cylinders and the gripper cylinder solenoid valve solenoid control terminal .

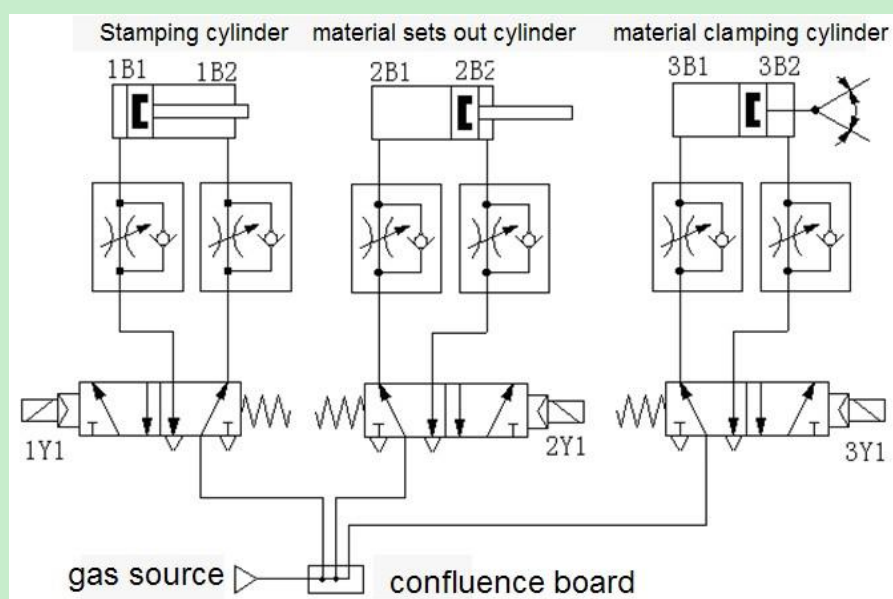


Diagram 3-7

3.3 Skills training for the installation of processing unit

1.The training objectives

The mechanical part of the processing unit assemble into components and spare parts, and then assembled into intact. Focus on control of mechanical equipment installation, adjustment methods and techniques.

2.Installation procedures and methods

The attention on the mechanical and spare parts we have related in the processing unit in the form, and then we focus on control of mechanical equipment installation, adjustment methods and techniques.

Processing module assembly process consists of two parts, one processed sector component assembly; second sliding component assembly processing units. And then final assembly, diagram 3-8 is a processing agency component assembly diagram, diagram 3-9 is a slide processing station component assembly diagram, diagram 3-10 is the final assembly of the entire processing unit.

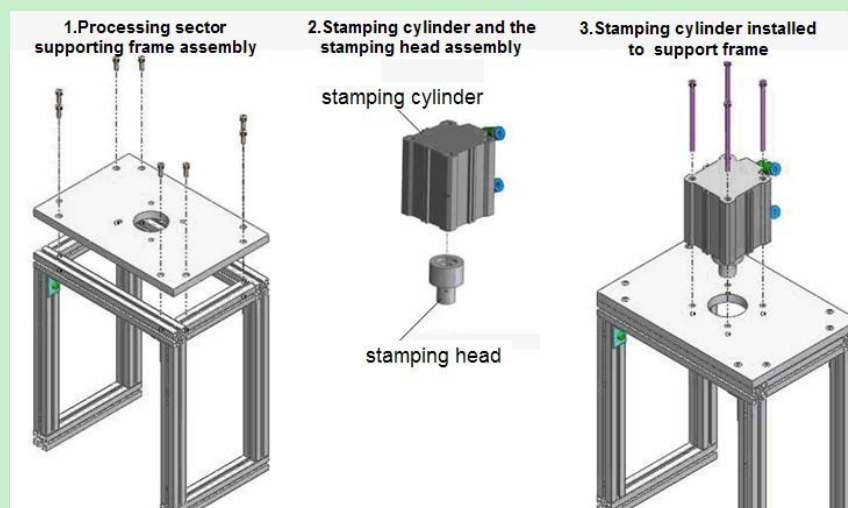


Diagram 3-8

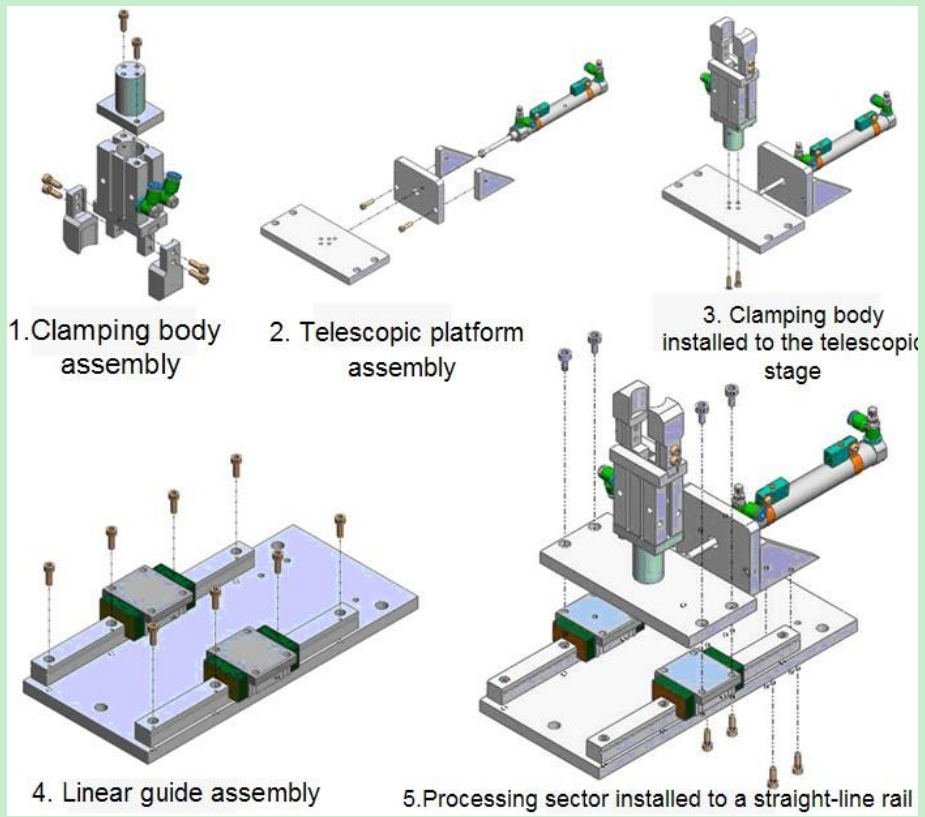


Diagram 3-9

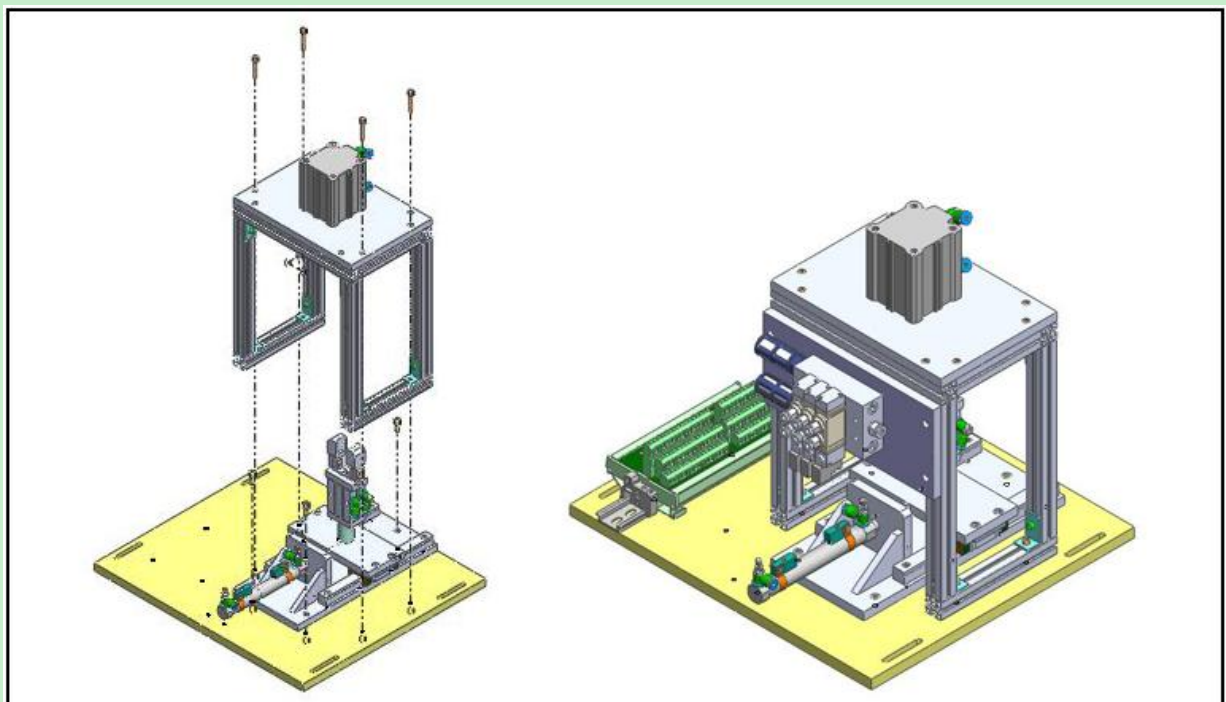


Diagram 3-10

Upon completion of the above components assembly, firstly clamping materials ,movement materials and the installed feed part of the backplane connecting to fixed,then installed the large aluminum supporting frames on the floor, finally put processing components in the supporting frames fixed to the aluminum alloy on the completion of the module assembly.

Precautions during installation:

- (1) Adjust the parallel of two linear guide rail, moving the mounting plate. while tightening the bolts fixed rail.
- (2) If the press head and processing center of the workpieces is not right now, you can adjust the depth of the pusher cylinder screwed two rail connecting plate to carry out correction.

3. Questions and thinking :

- (1) according to the above method after the completion of the assembly, the linear guide movement is still not particularly smooth, how should adjust the material clamping?
- (2) After the installation is complete, but the running time is not long will cause the material clamping and the linear cylinder seal damage of motion feeding part, imagine what causes?

3.4 Processing Unit PLC Control System Design

3.4.1 Tasks

Only considering the processing unit as an independent device running situation, the button / indicator work module selector switch should be placed in the "single-station approach" position. The specific control requirements as follows:

1.The initial state: power and gas source connected to the device, the sliding telescopic cylinder is extended processing location, processing units pneumatic gripper release the state of press cylinders in retracted position, emergency stop button is not pressed.

If the equipment in the initial state, then the "normal work" light **HL1** light, that equipment is ready. Otherwise, the indicator light flashes 1Hz frequency.

2. If the equipment is ready, press the start button, equipment start, "equipment running" indicator **HL2** light. When the workpiece sent to the processing stage and was detected,the implementation of the workpiece clamping device, sent to the processing region for stamping, punching action completed and returned to be material manufacturing processes. If there is no stop signal input,when the another workpiece sent to the processing stage, the processing unit started to work the next cycle.

3.In the course of their work, if press the stop button, processing unit in action after complete this cycle, stop working. **HL2** indicator light turns off.

Required to complete the following tasks:

- (1) Planning PLC, I / O allocation and distribution terminals.
- (2) A systematic way to install wiring and gas connections.
- (3) Development of PLC programs.
- (4) For debugging and running

3.4.2 I / O distribution and installation of wiring systems of PLC

1. Device side wiring port signal distribution as shown in Table 3-1

Table 3-1

Middle layer input port			Middle layer output port		
Terminal No.	Equipment Symbols	Signal line	Terminal No.	Equipment Symbols	Signal wire
2	SC1	materials testing	2	3Y	Clamping solenoid valve
3	3B2	Clamping Detection	3		Clamping solenoid valve
4	2B2	Processing sets out in place	4	2Y	Telescopic solenoid
5	2B1	Processing units in place retractable	5	1Y	Stamping solenoid valve
6	1B1	Processing pressure head cap			
7	1B2	Processing pressure head lower			
8 # ~ 17 # terminal not connected			6 # ~ 14 # terminal not connected		

2. Processing unit adopts **FX3U -32MR** main station, total 16 point input and 16 point relay output. I/O signal table of PLC is in table 3-2, and wiring schematic as shown in diagram 3-11.

Table 3-2 I / O signal table of PLC processing unit

input signal				Output signal			
No.	PLC input point	Signal Name	Signal Source	No.	PLC output point	Signal Name	Signal Source
1	X000	materials testing	Device Side	1	Y000	Clamping solenoid valve	Device Side
2	X001	Clamping Detection		2	Y001		
3	X002	Processing sets out in place		3	Y002	Material telescopic solenoid valve	
4	X003	Processing units in place retractable		4	Y003	Processing pressure head solenoid valve	
5	X004	Processing pressure head cap		5	Y004		
6	X005	Processing		6	Y005		

		pressure head lower					
7	X006			7	Y006		
8	X007			8	Y007		
9	X010			9	Y010	Work instructions	Button / LED Module
10	X011			10	Y011	Running instructions	
11	X012	Stop button	Button / LED Module				
12	X013	Start Button					
13	X014	E-STOP button					
14	X015	Single station / whole line					

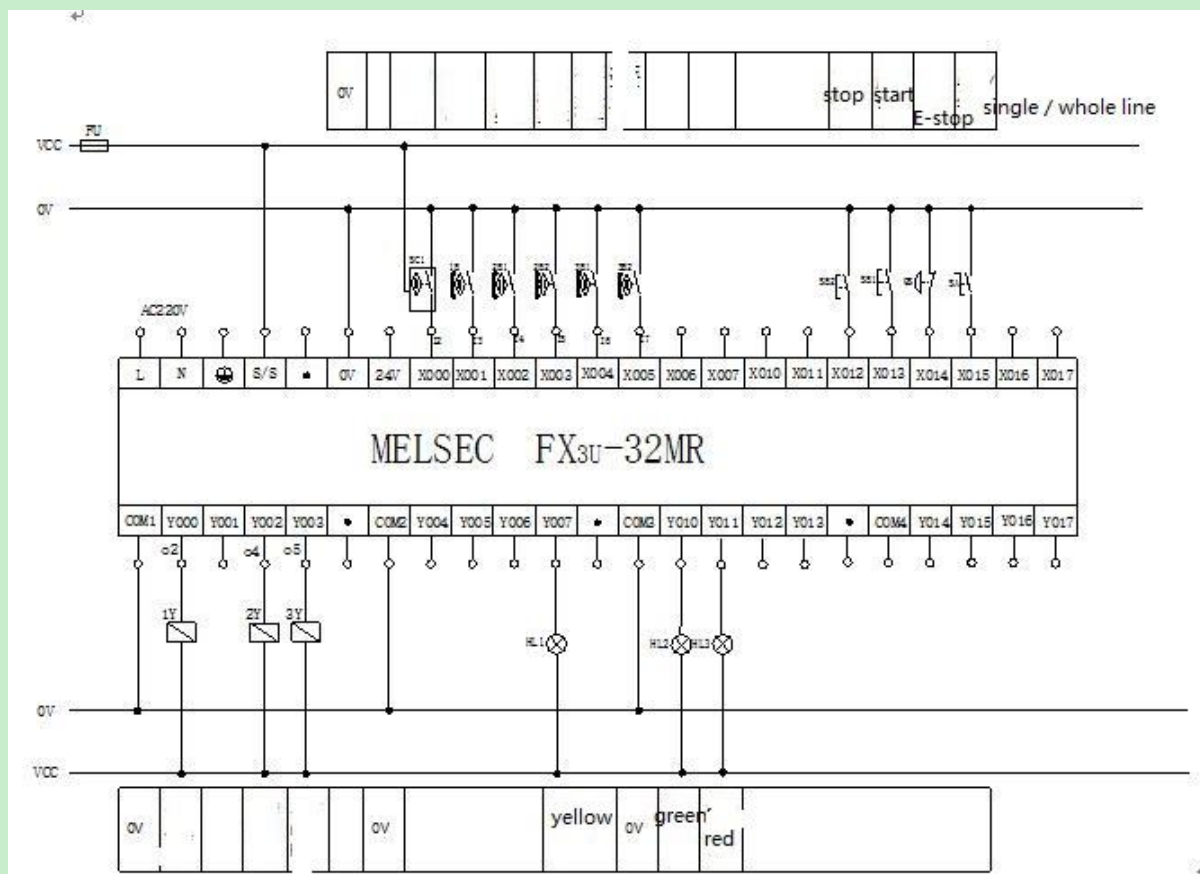


Diagram 3-11

3.4.3 Programming and debugging PLC control program

1. Programming ideas

The processing unit working flow is similar to feeding unit. PLC firstly reach the initial testing stage after connecting to power. For emergency stop, it will restart after that and reset, from the emergency stop point, continue, in two ways: condition jump (**CJ**)command, or by master command. Here we talk about jump (**CJ**)command.

By condition jump command, for emergency stop signal program, it is shown in diagram 3-12. in the diagram, when emergency stop button is pushed, X014 OFF, jump command condition is satisfied, program will jump to the command P0 and start. The program after the jump command will not be run.

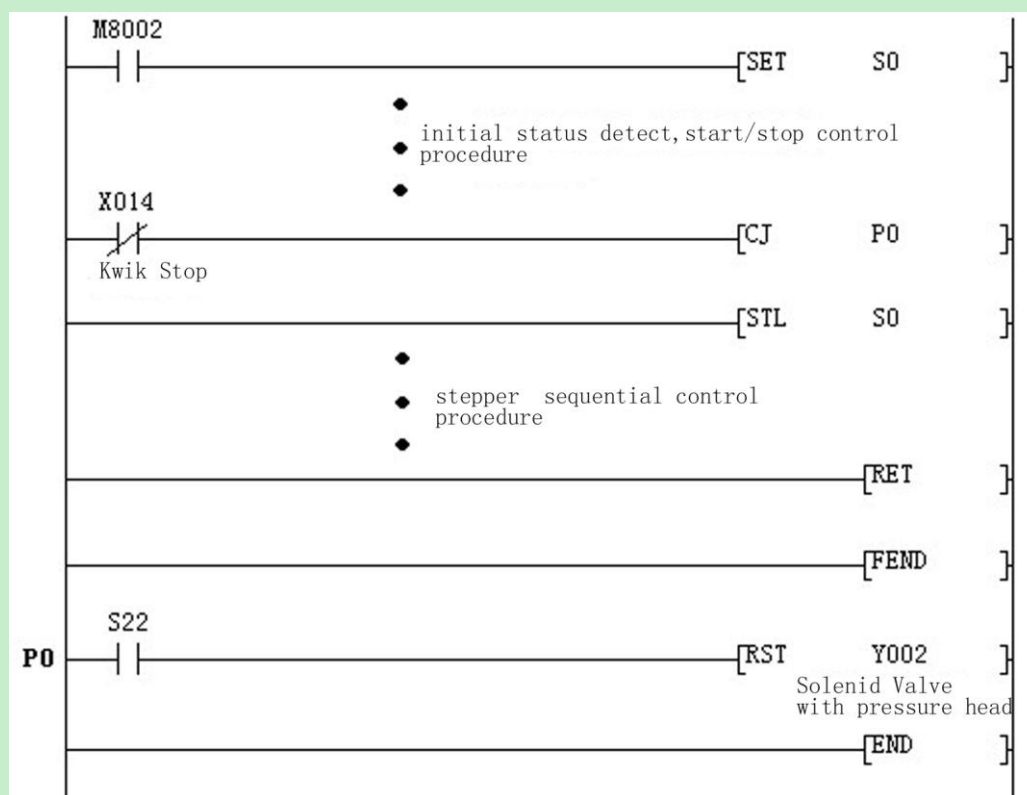


Diagram 3-12

CJ commands, the program after jump will not be scanned, the output status before will be kept, step sequence program status will be kept, till the emergency stop button is reset and restarted. If the emergency stop is on the S22 step, the positive punch will done, after the program jump, the status will be kept, so you need reset the punch solenoid program between **FEND** command and **END** command.

When the emergency stop button is not pushed, **X007** is **ON**, the program will run in the sequence, till the master program command **FEND** is over.

2.Commissioning and operation

(1) Adjusting pneumatic parts, check the gas line wether correct, pressure wether reasonable, appropriate action cylinder speed wether appropriate.

- (2) Check the magnetic switch installation location whether in place, magnetic switches working whether correct.
- (3) Check the I / O wiring whether correct.
- (4) Check whether it is reasonable to install photoelectric sensor, sensitivity is appropriate to ensure the reliability of detection.
- (5) Put into the workpiece, run the program, observe the processing unit action whether meets the mission requirements.
- (6) Debug a variety of possible scenarios, such as in any case are likely to join the workpiece, the system must be able to work reliably.
- (7) Optimization process.

3. Questions and thinking

- (1) Summary and learn to check the pneumatic connection, sensor wiring, I / O testing and troubleshooting methods.
- (2) How to deal with unexpected situations in the process
- (3) If we adopt the network control how to achieve?
- (4) Various problems in the processing unit.

Item 4 Assembly Unit Control System Training

4.1 Assembly unit structure and working process

Assembly unit function is to complete the assembly process of the black or white silos small cylindrical workpiece placed in the assembly hopper into the parts to be assembled.

Assembly unit includes: tube-shaped hopper, feeding mechanism, transfer material sets, mechanical hand, to be assembled workpiece positioning mechanism, pneumatic systems and valves, signal collect and automatic control system, as well as the terminal row components for electrical connections, indicator lights for the entire production line status and the aluminum bracket and the floor for installation of others and the sensor and other accessories. Mechanical assembly as shown in diagram 4-1.

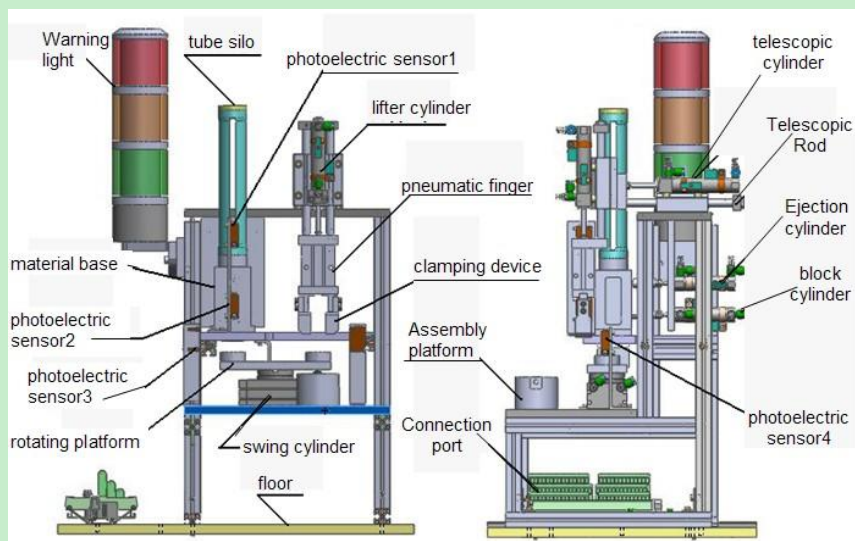


Diagram 4-1

1. Tube-shaped hopper

Tube-shaped silo is used to store the assembly metal, black and white small cylindrical parts. It consists of hollow plastic pipe and the base. On top of plastic tube is strengthening metal ring in order to prevent breakage. Workpiece vertically into the hollow silo, there is a certain gap between the two so that it can free-fall with gravity.

In order to alarm about feed shortages and out of workpiece, we installed two diffuse photoelectric sensor (E3Z-L type) in the plastic pipe and the bottom, and the silo plastic cylinder on a vertical milling tank so that the infrared photoelectric sensor reliably spot to the material to be detected. As shown in diagram 4-2. Photoelectric sensor sensitivity adjustment should be able to detect the black material.

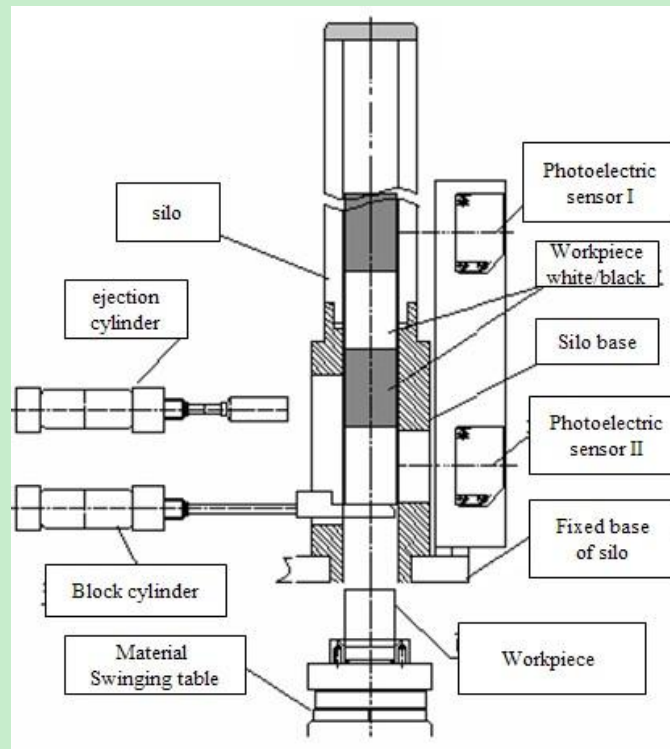


Diagram 4-2

2. Blanking mechanism

Diagram 4-2 shows a sectional view of blanking mechanism. We install two linear cylinders in the back of the silo base. The above cylinder named lift materials cylinder, the bottom cylinder named block material cylinder.

After gas source connected to the system, the initial position of the lift materials cylinder in retracted state, the initial position of the block material cylinder in the stick out state. Thus, when the piece down from the hopper above, the workpiece will be blocked by the block cylinder piston rod terminal so that can't fall.

When need blanking operation, first of all to stick out the lift materials cylinder, secondly clamping the lower workpiece, then retracted the block material cylinder, workpiece will fall into the transfer station material disk. After that the block material cylinder reset and stick out, lift materials cylinder retracted, the lower parts fall to block materials cylinder terminal block, prepare for the next feeding.

3. Transfer materials platform

The mechanism is composed of pneumatic swing set and two feed tray, pneumatic swing sets can drive feeding tray rotate 180 degrees, in order to achieve the function that to move the workpiece falls from the feeding tray to the under assembly. As shown in diagram 4-3, the photoelectric sensors 1 and photoelectric sensors 2 are used to detect the left and right trays whether there are spare parts. Two photoelectric sensors are selected CX-441-type .

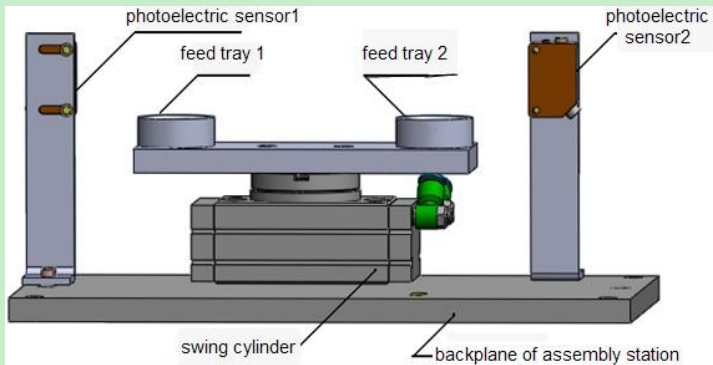


Diagram 4-3

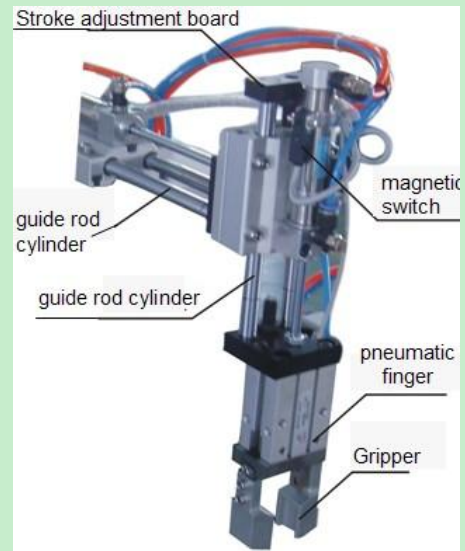


Diagram 4-4

4. The assembly mechanical arm

The assembly mechanical arm is the entire core. When there are small cylindrical parts on the material feed station, and the fiber optic sensors in the side station detect there are parts to be assembled, the mechanical arm started from the initial state to the process of assembly operations. The whole outlook shown in diagram 4-4.

The assembly mechanical arm is a three-dimensional body, which consists of two fingers form-oriented and pneumatic cylinders of horizontal movement and vertical movement.

The operation process of mechanical arm assembly is as follows:

Electromagnetic valve between PLC-driven and connected with vertical movement of cylinder action, after it in place, pneumatic finger-driven clamping materials, and send the clamping signal with the magnetic switch to the PLC, under the PLC control, the vertical moving cylinder reset, the material being clamped together with the pneumatic fingers raised, leave transfer materials stage, workpiece elevated to the highest level, the horizontal cylinder driven in the corresponding reverding valve, the piston rod stick out, moving to the cylinder front-end location, the vertical moving cylinder will again be driven down, move to the bottom position, pneumatic finger release, after a brief delay, the vertical movement and horizontal movement cylinder retracted, mechanical arm reset the initial state.

in the whole process of mechanical arm, in addition that the pneumatic fingers loosened in place without the sensor detects, the remaining actions in place are signal detection used with the magnetic switch that the cylinder matching, the signal will be collected back to the PLC as the input signal, the signal output by the PLC driven reversing solenoid valve, so that

the fingers formed by the pneumatic cylinder and a mechanical hand according to the procedure to run automatically.

5. Assembly platform hopper

The workpiece from transport unit directly put into the positioning hole and a smaller gap between the workpiece and the position hole position, thus completing the assembly precise movement and positioning accuracy. As shown in diagram 4-5.

In order to determine the assembly unit whether placed within the hopper assembly parts to be used in fiber optic sensor for testing. Hopper side open an M6 screw, the fiber optic sensor fixed in the screw holes.

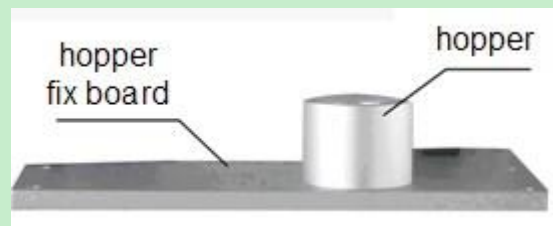


Diagram 4-5

6. Alarm Light

The working unit install red, orange and green alarm lights, which is used as a warning for the whole system. There are five pin-out wire in warning lights, including yellow-green color line named "ground line"; red line: the red light control line; yellow line: the orange light control line, green line: the green light control line; black line: signal of public control lines. Wiring as shown in diagram 4-6.

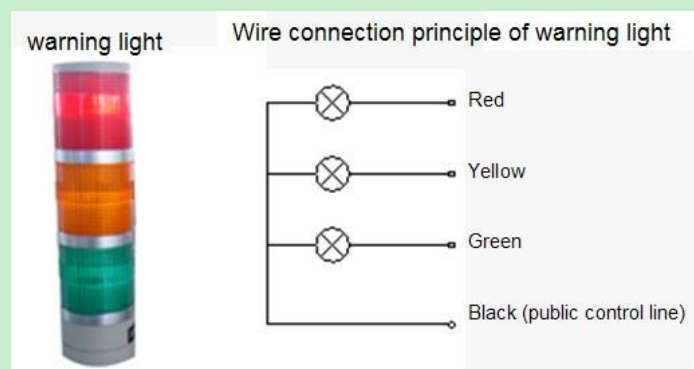


Diagram 4-6

4.2 Related knowledge points

4.2.1 Pneumatic components of assembly unit

Assembly unit used pneumatic actuators including standard linear cylinders, pneumatic fingers, pneumatic swing sets and a guide cylinder, the pneumatic swing set and guide cylinder will be described as follow;

1. Pneumatic swing sets

The main device materials set is pneumatic swing sets, which is driven by a linear cylinder rack and pinion to achieve rotary motion, rotation angle can be any adjustable between 0-90 degrees, 0-180 degrees, and the magnetic switches can be installed to detect rotation in place signal, multi-direction and position of the need for institutional change. As shown in diagram 4-7.

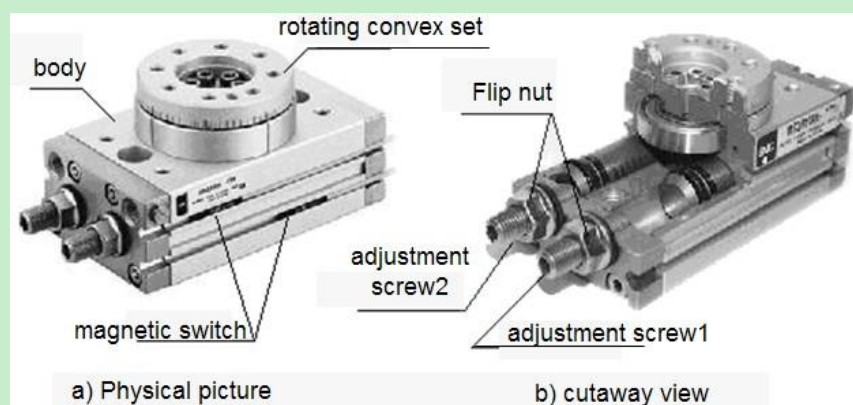


Diagram 4-7

Pneumatic rotary pendulum swing angle of units in the 0-180 degree range can be adjusted arbitrarily. When you need adjust the rotation angle of swing position accuracy, the first adjustment screw should be loosened Flip on the nut, through the spin in and spin out of adjustment screw, there by changing the rotation of the rotation angle of convex sets, regulate and adjust the screw 1 screw 2 for L-and D respectively, after angle adjustment, it shall Flip lock nut and the substrate to prevent the adjustment screw loosening. Caused by lower rotary precision .

Rotary in place signal is by adjusting pneumatic within the two magnetic slide switch position to achieve, diagram 4-8 is adjust the location of magnetic switch schematic. Magnetic switch installed inside the cylinder body slide, release the positioning screw fastening magnetic switch, magnetic switch can be along the slide and move around. Determine the switch position, the rotating screw firmly fixed position to complete the position adjustment.

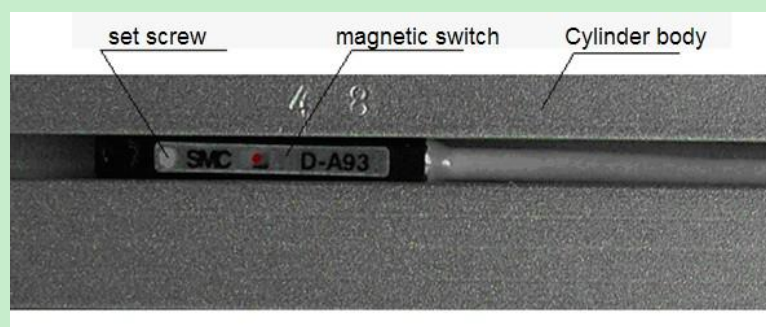


Diagram 4-8

2. Guide cylinder

The guide cylinder is guide functional cylinder. Generally with the standard and guide cylinder features, with high precision, great torque resistance, highly strong and stable in working etc.

The assembly unit is used to drive assembly mechanical arm horizontal movement, guide cylinder is shown in diagram 4-9. The cylinder is composed of linear cylinder, rail and accessories.

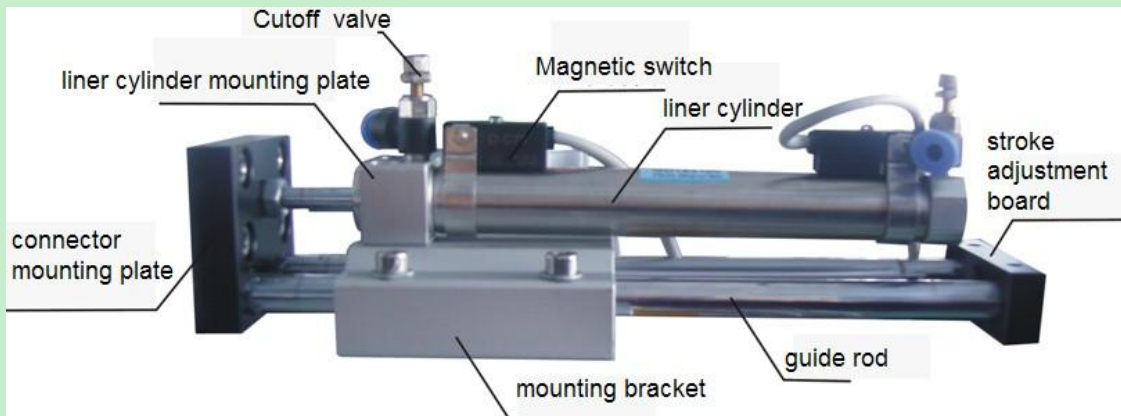


Diagram 4-9

The rack is used to install and fix guide cylinder, connector plates are for fixing and installation others need to be connected to the guide cylinder, and fix the relative position of the straight cylinder piston rod, guide rod, after one side of the straight line cylinder connect to the compressed air cylinder, the piston has been driven to make a straight-line movement, the piston shaft will move together, are connected together pieces of installation plate fixed to the two guide rod with the piston rod also extends or withdraw, in order to achieve guide cylinder as a whole function. installation at the end of the travel guide rod adjustment Plates to adjust the guide rod in cylinder extendtravel. adjustment: loosen travel panel screw, to make it move on the rail, when it arrives to an ideal position, then completely fix the crew,the stroke adjustment finished.

3. Solenoid valve group and penumatic control circuit

The assembly unit is consist of 6 pcs two position five way single control magnetic direction valve ,as shown in diagram 4-10, which to control supply, position change and assembly circuits. The penumatic control circuit diagram is shown in diagram 4-11.



Diagram 4-10

When you connect the pneumatic circuit, each cylinder shall be in initial position, baffle cylinder extends, claw cylinder rises.

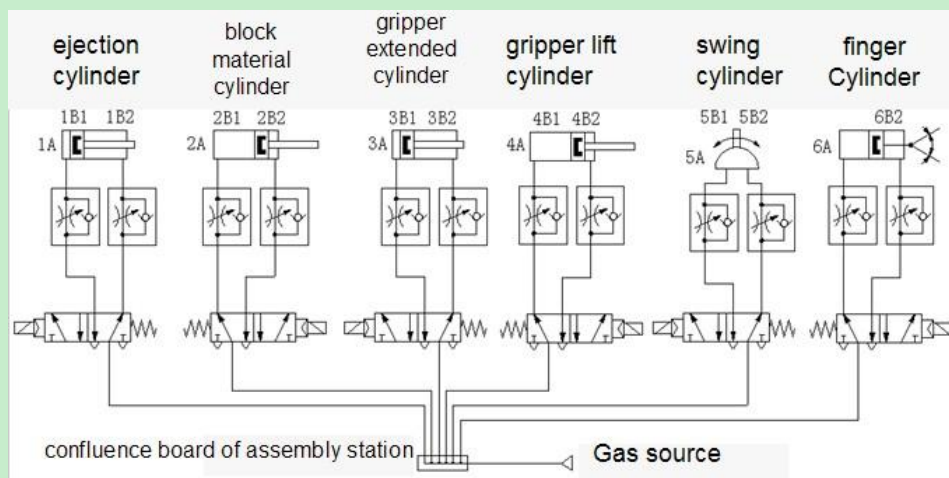


Diagram 4-11

4.2.2 Optical fiber sensor

The optical fiber sensor is composed of optical fiber, optical fiber amplifier, optical fiber is composed of two parts, in use, it can used for the two ends of the amplifier. The optical fiber sensor is shown in diagram 4-12. In diagram 4-13 is the amplifier.

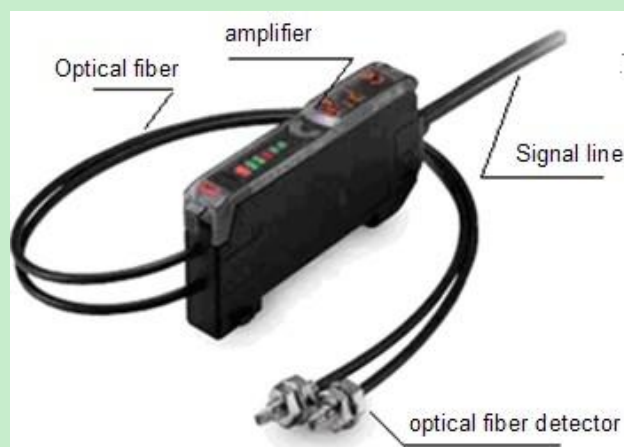


Diagram 4-12

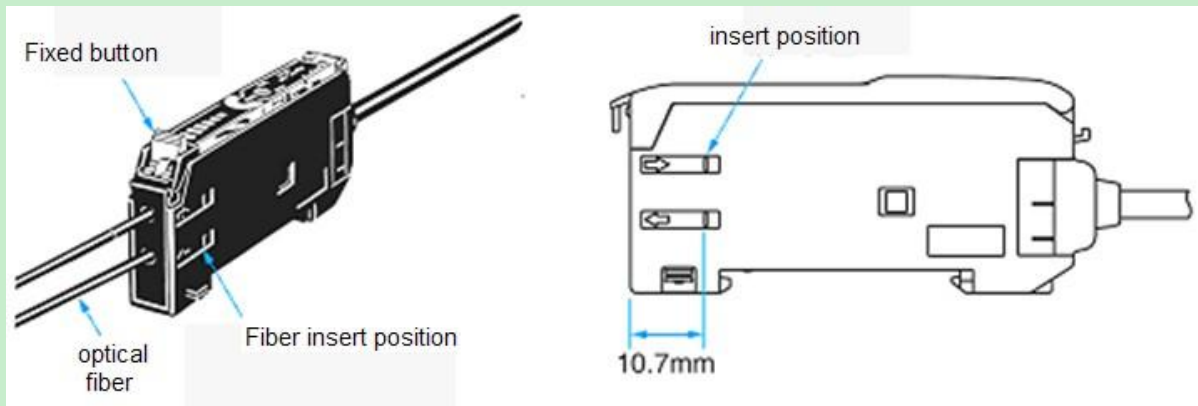


Diagram 4-13

The optical fiber sensor advantages: work in unfavorable environments, long travel distance, long life, and it can be installed in small places.

The optical fiber proximity switch amplifier has a large adjustment range. When the optical fiber sensor sensitivity has little adjustment range, the sensor cannot be used widely. If the optical fiber sensor sensitivity is improved, it can be used more widely.

Diagram 4-14 is amplifier station, you can rotate the knob to adjust amplifier sensitivity. Then you can see “light” indicator light changes. When workpiece is detected, the light can tell that workpiece is detected.

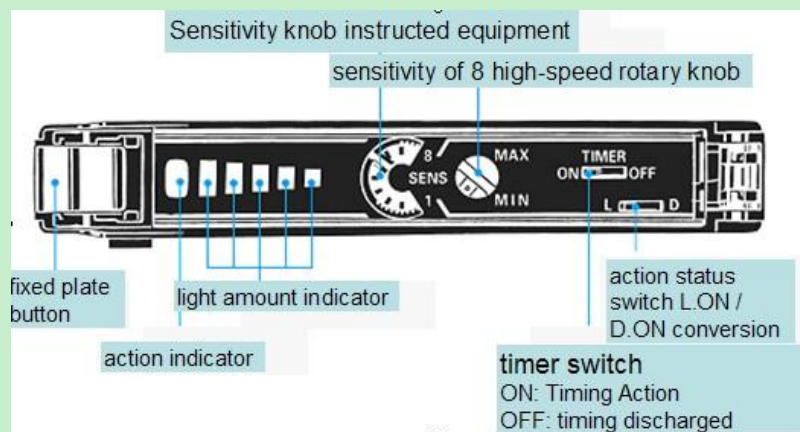


Diagram 4-14

E3Z-NA11 optical fiber sensor circuit diagram is diagram 4-15. in wiring, pay attention to color and polarity, signal output, do not connect signal output to power +24V.

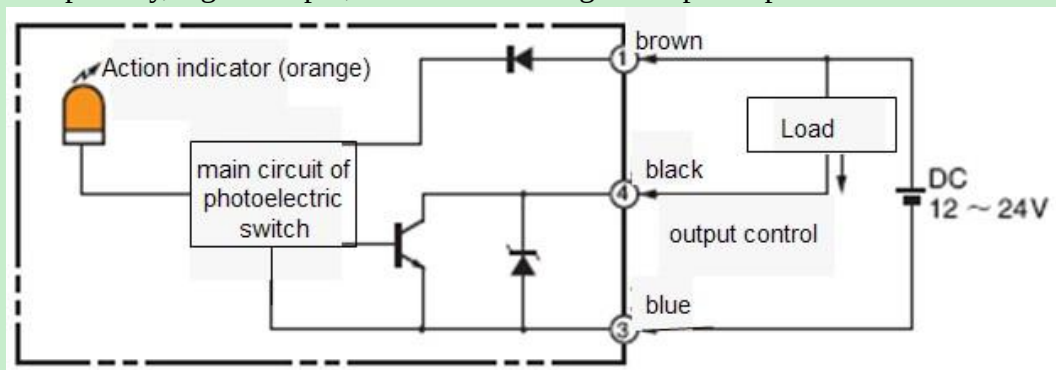


Diagram 4-15

4.3 Assembly station installation skills training

1. Training target

The mechanical part of the assembly unit is disassembled into components and parts, and then assembled into the original. Focus on the installation of mechanical equipment, adjustment methods and techniques.

2. Steps and methods

The assembly unit is a complicate unit in **YL-335B system**, before assembly,analyze structure and composition. The assembled components as shown in diagram 4-16:

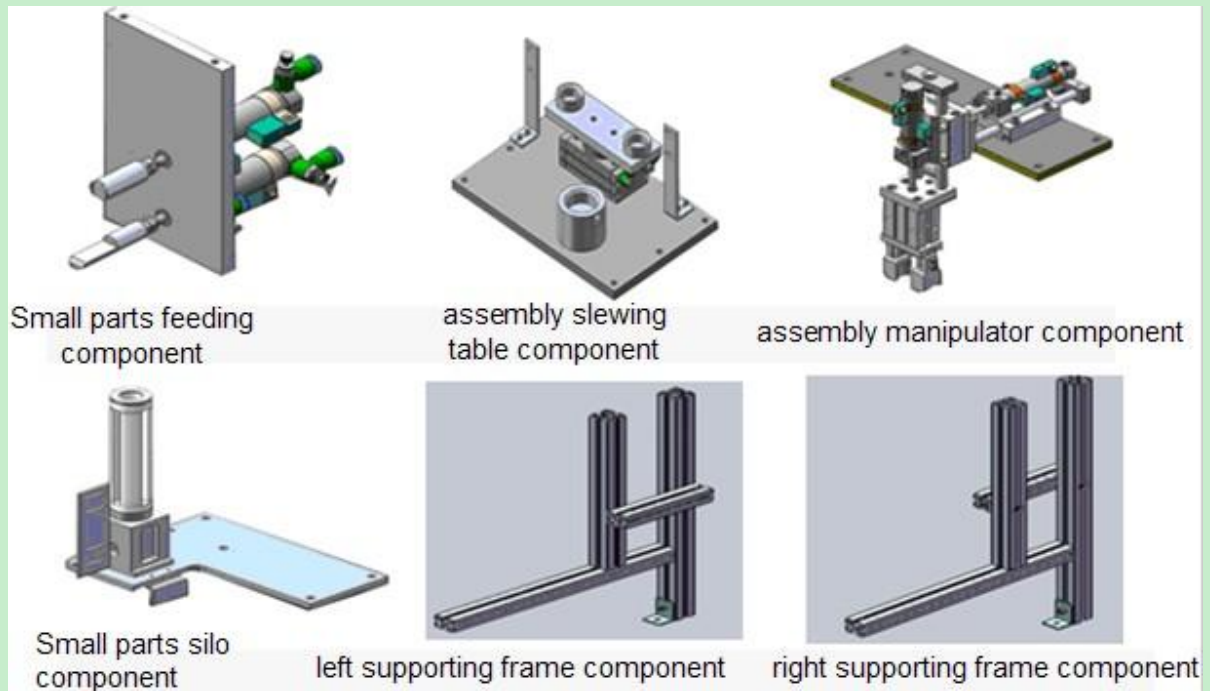


Diagram 4-16

When assembly is done, use “L” connectors to fix bolts, the assembly station rack is shown in diagram 4-17 .

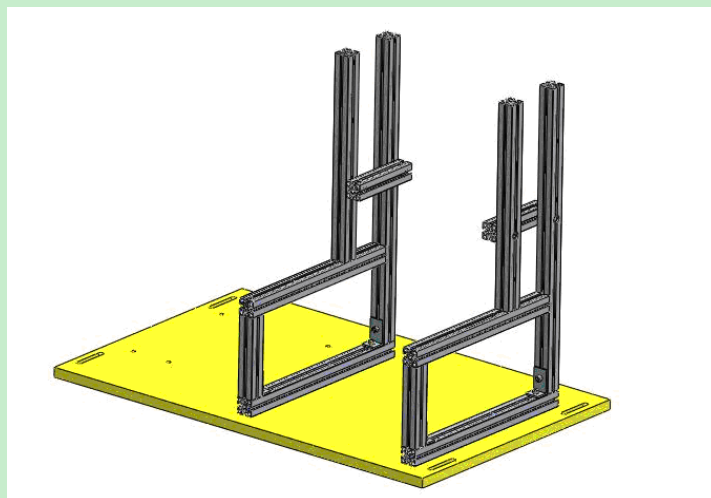


Diagram 4-17

Install components gradually as diagram 4-16, sequence: assemble rotary platform assembly → small workpiece bin assembly → small workpiece supply assembly → manipulator assembly.

Finally, install alarm lights and sensors, then complete the mechanical assembly.

Precautions:

- (1) Pay attention to initial position of sway bench.
- (2) The place reserved for bolts shall be enough.
- (3) Assemble, confirm position and fix.

4.4 Assembly station PLC control system design

4.4.1 Tasks

1. Each cylinder of assembly unit is in initial position: the block cylinder extends; lift material cylinder is retract status, there are already small enough silos column components; assembly mechanical arm lifting cylinder in the upgrade status,; telescopic cylinder is in retract status, the paw is in loosen status.

When power is on, if each cylinder is in initial position, and with enough column components; workpiece assembly bench does not have to be assembly workpiece. Then "normal" indicator light is HL1. If then, the indicator light to 1Hz frequency of blinking.

2. Push start button, assembly station starts, "run" indicator light is HL2. If rotary bench on the left disc there is no small column components, on the implementation of the next feed operation; if left disc has components, but right disc has no components, the rotary bench will start.

3. If rotary bench right disc has column components, assembly bench has workpiece for assembly, assembly mechanical arm catches column components, put workpiece for assembly.

4. When assembly is finished, assembly mechanical arm shall return to its initial position, wait for the next-time assembly.

5. Push stop button, the supply stops, when assembly is ready, assembly station will stop when the assembly is finished.

6. In running, in case of "components shortage" alarm, indicator light HL3 flashes in 1Hz, HL1 and HL2 are on; in running, in case of "no components" alarm, indicator light HL3 is on for 1s, off for 0.5s, HL2 is off, HL1 is on.

4.4.2 I/O allocation and system wiring of PLC

The assembly unit wiring port signals are in table 4-1.

Table 4-1

Input port			Output port		
End	Code	signal line	End	Code	signal line
2	SC1	components shortage	2	1Y	baffle solenoid
3	SC2	no components	3	2Y	push solenoid
4	SC3	left disc components	4	3Y	rotary solenoid

5	SC4	right disc components	5	4Y	claw tighten solenoid
6	SC5	assembly bench workpiece	6	5Y	claw drop solenoid
7	1B1	push arrival	7	6Y	arm extends solenoid
8	1B2	pushreset	8	AL1	red alarm light
9	2B1	baffle status	9	AL2	orange alarm light
10	2B2	drop status	10	AL3	green alarm light
11	5B1	sway cylinder left limit	11		
12	5B2	sway cylinder right limit	12		
13	6B2	claw tighten	13		
14	4B2	claw drop arrival	14		
15	4B1	claw rise arrival			
16	3B1	arm withdraw arrival			
17	3B2	arm extends arrival			

In the assembly unit, there are many I/O ports, and it adopts Mitsubishi FX3U-48MR main unit, total 24 point input, 24 point relay output. I/O table is shown in the table 4-2. the diagram 4-19 is PLC wiring schematic .

Table 4-2

input signal				output signal			
No.	PLC input	signal	signal source	No.	PLC output	signal	signal source
1	X000	Components shortage	device side	1	Y000	baffle solenoid	device side
2	X001	No components		2	Y001	push solenoid	
3	X002	Left disc components		3	Y002	rotary solenoid	
4	X003	Right disc components		4	Y003	claw tighten solenoid	
5	X004	assembly bench workpiece		5	Y003	claw drop solenoid	
6	X005	push arrival		6	Y004	arm extends solenoid	
7	X006	pushreset		7	Y005	red alarm light	
8	X007	baffle status		8	Y007	orange alarm light	
9	X010	drop status		9	Y010	green alarm light	

10	X011	sway cylinder left limit		10	Y011		button/indicator light module
11	X012	sway cylinder right limit		11	Y012		
12	X013	claw tighten		12	Y013		
13	X014	claw drop arrival		13	Y014		
14	X015	claw rise arrival		14	Y015	HL1	
15	X016	arm withdraw arrival		15	Y016	HL2	
16	X017	Arm extends arrival		16	Y017	HL3	
17	X020						
18	X021						
19	X022						
20	X023						
21	X024	stop button	button/indicator light module				
22	X025	start button					
23	X026	emergency stop button					
24	X027	Off-line/on-line					

Note: alarm lights used to show **YL-335B** running status, tasks are done by assembly unit, and alarm lights may not be connected with the PLC.

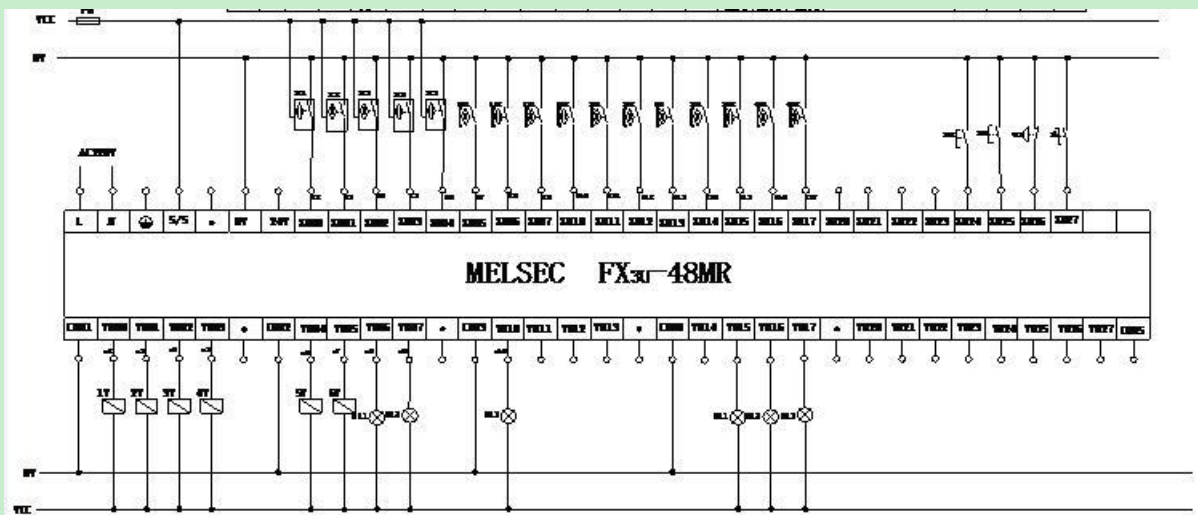


Diagram 4-18

4.4.3 PLC control programming and debugging

1. Programming

(1) In running process, work process of assembly unit include two independent subprocess, one is supply process and the other one is assembly process.

The supply process: drop column components to sway bench left disc; then the sway bench rotates to make disc move to the rightside, for the assembly mechanical arm to catch the components.

The assembly process: when there is workpiece on the assembly bench, and the column components are on the downside of the mechanical arm, it will start assembly.

In the master program, when the initial status check is done, the station is ready to start, pushstart button, you shall call supply control and assembly control programs in a same time.

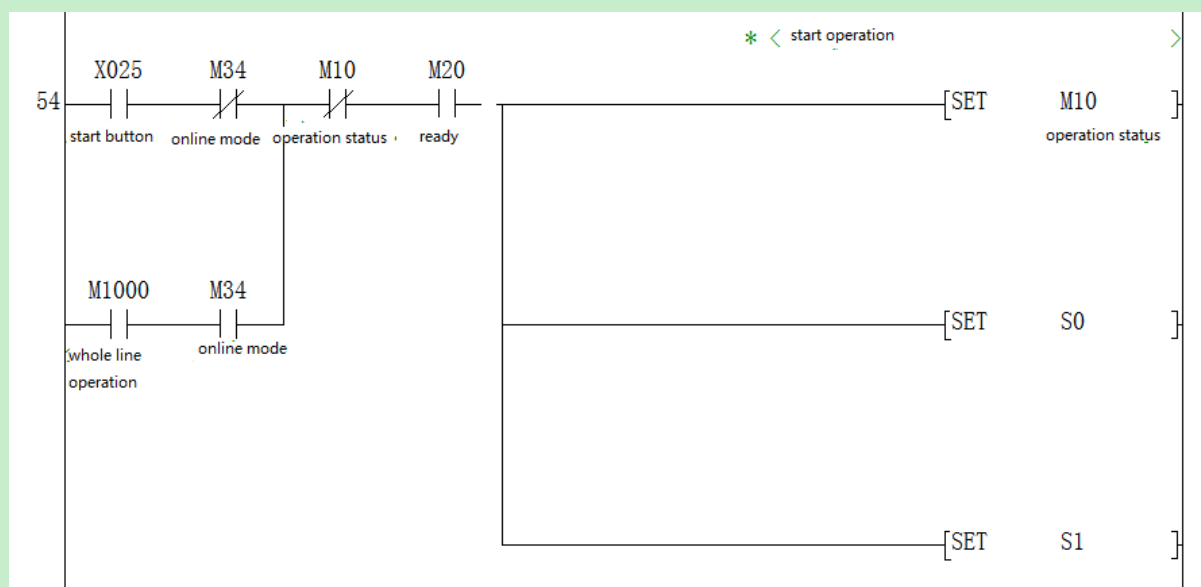


Diagram 4-19

(2) Supply control process includes two interlocked process: drop process and sway bench rotation, disc transfer process. When column components drop to left disc, it is forbidden to rotate sway bench; otherwise, in sway bench rotation, it is forbidden to open warehouse.

Inter-lock: ① When sway bench left limit or right limit magnetic switch action, and left disc is not expected, confirmed by the timing, it begins to drop process; ② when baffle cylinder extends arrival so that hopper off, left disc with workpiece and the right disc space, by the timing confirmed, we began to sway bench rotation, until it reaches the limit position.

(3) Supply process drop control and assembly control process is single sequence stepping sequence control, programming steps are mentioned before.

(4) Stop operation, there are two situation. One: push stop button, stop command is set; the second: when the last component drops, no sensor works (**X001 OFF**), the material lacking alarm will be sent.

For supply process drop control, shall be done at the moment when push cylinder arrives and returns. But for sway bench rotation control, once stop command is sent, then stop bench rotation.

For assembly control, it shall be done in one-time assembly, assembly mechanical arm then returns to the origin and stops.

When drop mechanism and assembly mechanical arm return to the origins, they can be reset to run status and stop. The stop can be made in the program, the ladder is shown in diagram 4-20.

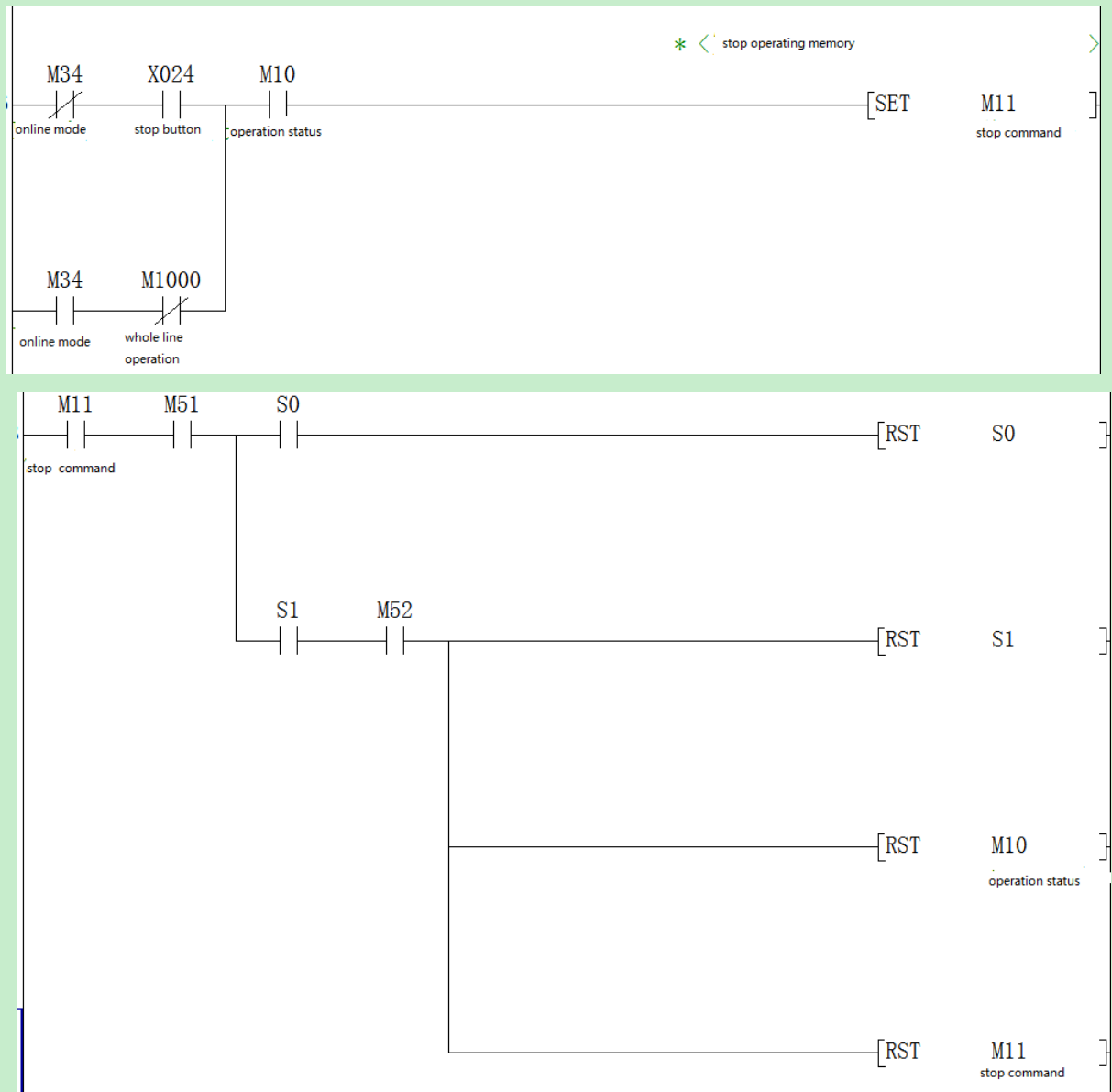


Diagram 4-20

2. Debugging and running

- (1) Adjust penumatic part, check the penumatic circuit and pressure, cylinder work speed.
- (2) Check magnetic switch positions and magnetic switch work status.
- (3) Check I/O wiring.
- (4) Check sensor installation and sensitivity.
- (5) Place workpiece, run program, check assembly station work.

ITEM5 Sorting unit control system training

5.1 Sorting unit structure and work processing

The sorting unit is the last unit of YL-335B. The workpieces in various colors are pushed to slots. When transfer station sends workpiece to convey belt, detected by the optical sensor, it starts inverter to start sorting.

The sorting unit structure: transfer and sorting mechanism, convey belt drive mechanism, inverter module, solenoid valve group, wiring ports, PLC module, button/indicator light module ,backplate and etc. The mechanical assembly part is shown in diagram 5-1.

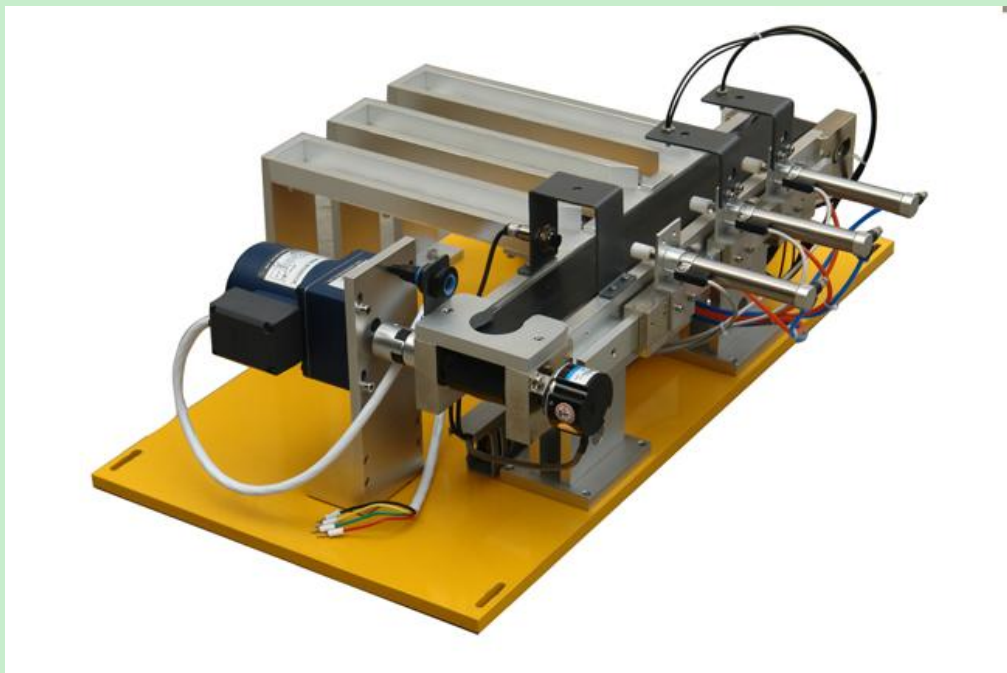


Diagram 5-1

1. Transfer and sorting mechanism

The transfer and sorting mechanism is mainly composed of conveyor, discharge slide slot, pushing (sorting) cylinder, diffuse photoelectric sensor, optical fiber sensor, magnetic proximity sensor, to transfer the processed and assembled workpiece and detected by optical fiber sensor, then to sort .

2. Convey belt drive mechanism

The convey belt drive mechanism is shown in diagram 5-2. The 3-phase reduction motor is used to drive the conveyor to transfer material ,which consist of motor bracket, motor, coupling etc .

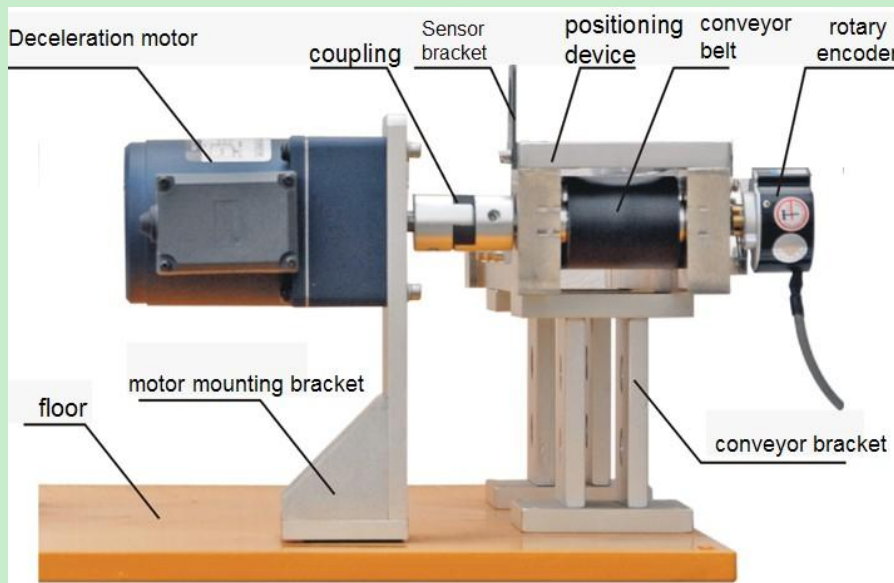


Diagram 5-2

Three-phase motor is the main part of transmission mechanism , motor speed is controlled by the inverter, to transfer workpiece. Motor bracket is used for fixing motor. Coupling due to the motor shaft and transfer with a master link the dynamic axle to group into a transmission mechanism.

3. Magnetic valve group and penumatic control circuit

The sorting unit magnetic valve group use 3 pcs two position five way single electrical control magnetic valve with manual switch, which are installed in bus board and control the gas path of metal, white and black workpiece which pushing cylinder, to change respective work status.

The unit penumatic control circuit schematic is shown in diagram 5-3. 1A, 2A and 3A are No. 1 cylinder, No. 2 cylinder and No. 3 cylinder. 1B1, 2B1 and 3B1 are proximity switches installed before each allocation cylinder. 1Y1, 2Y1 and 3Y1 are the electromagnetic control terminals for controlling the three cylinder valves of the sorting cylinder

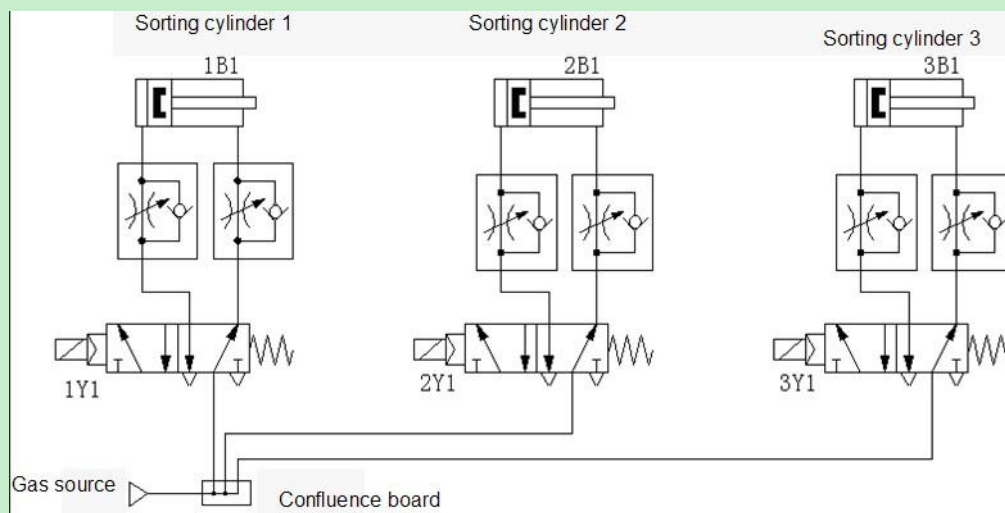


Diagram 5-3

5.2 Sorting unit installation skills training

1. Target

On the basis of understanding the composition of the sorting unit, the mechanical parts of the sorting unit are disassembled into components and parts, and then assembled into the original. Require to master the installation of mechanical equipment, adjustment methods and techniques.

2. Steps and methods

The sorting unit mechanical assembly can be carried out in the following four stages:

- (1) Complete the assembly of the conveyor, assemble the conveyor belt mechanism and its bracket, and then install it on the baseplate. As shown in diagram 5-4.

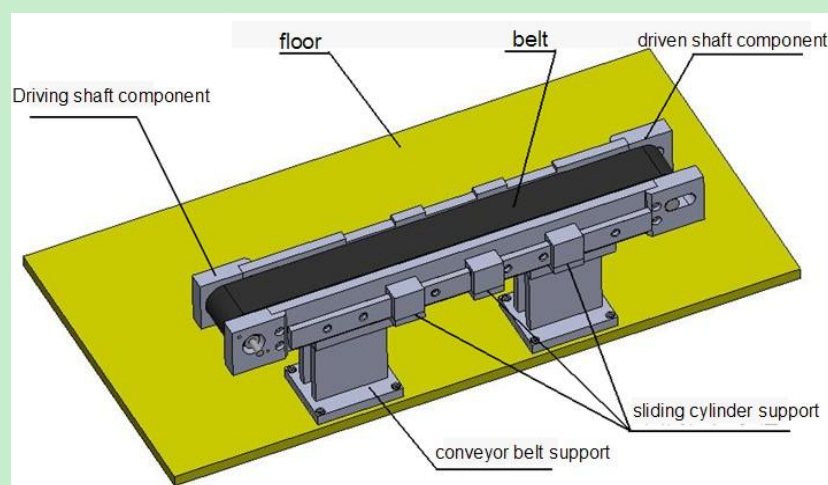


Diagram 5-4

- (2) Complete the drive motor components assembly, and further assembly of coupling, the drive motor components connected with the transfer mechanism and fixed on the baseplate, see diagram 5-5.

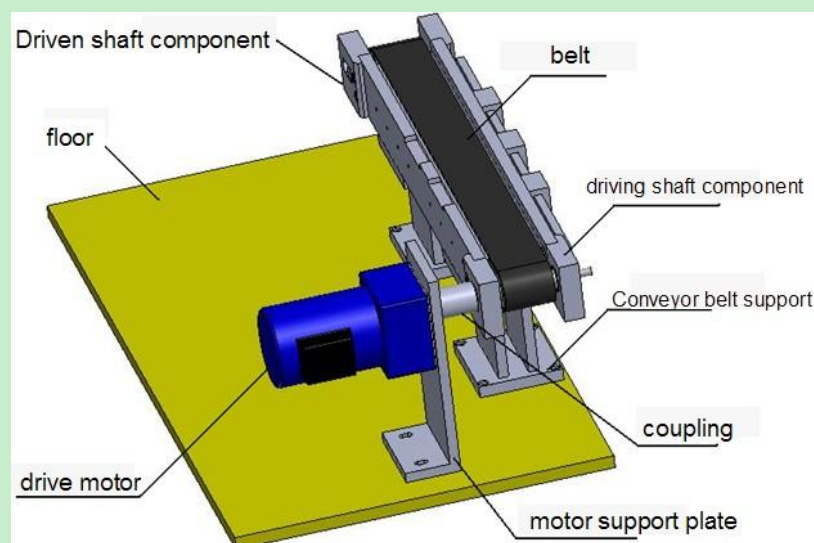


Diagram 5-5

(3) Assemble push cylinder rack, push cylinder, sensor rack, output slot, see diagram 5-6.
The process, refer to allocation station assembly slide film.

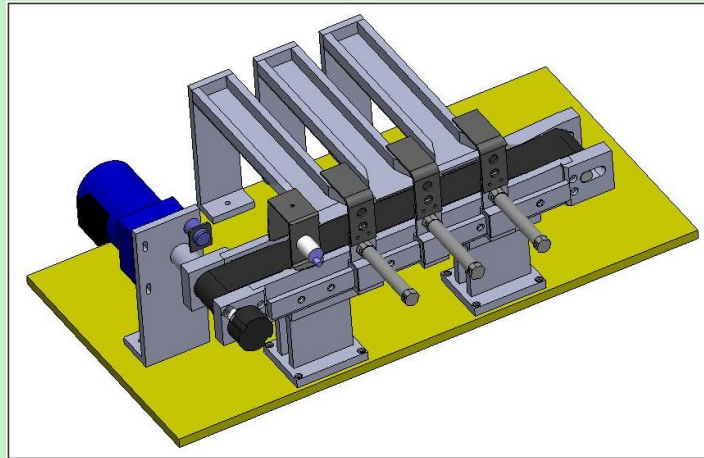


Diagram 5-6

(4) Assemble all sensors, solenoid group, wiring.

(5) Precautions:

Notes: ① adjust transfer belt positions. ② ensure the driving shaft and the driven shaft positions correct, and the driving shaft and the driven shaft panel positions shall not be exchanged. ③ the transfer belt tightness shall be proper. ④ ensure the driving shaft and the driven shaft parallel. ⑤ rolling bearing rotary part is used, but it is a precise part, special skills and tools are needed, do not do randomly.

5.3 Relative points

5.3.1 Rotation encoder

The rotation encoder transforms output mechanical, geometry displacement into pulse or number signal sensors, for speed or position. The fundamental diagram is diagram 5-7; the rotation encoder output pulse quantity each second will reflect current motor rotation speed.

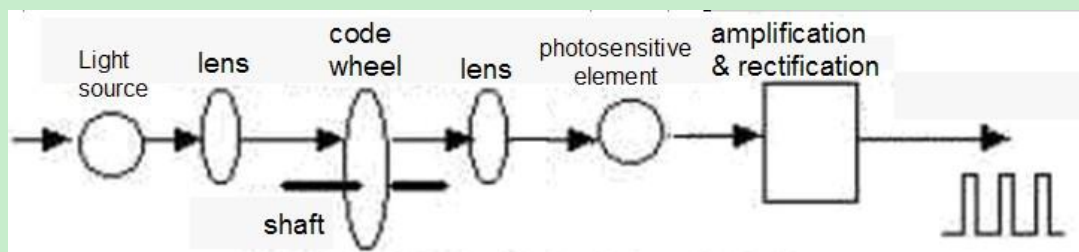


Diagram 5-7

Generally, rotation encoder generates pulse in various modes. The automatic production line often uses increment rotation encoder.

The increment rotation encoder transform output 3 groups of square pulses A, B and Z; A, B group pulse phase difference 90°: when A phase is in advance of B phase, and phase B pulse is in advance of A phase, it is reversed. Z phase is used for base point location, shown in diagram 5-8.

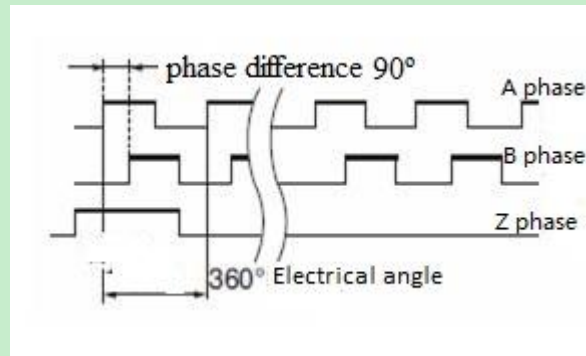


Diagram 5-8

YL-335B sorting unit uses the A, B phase 90° for the general rotation encoder. Used to calculate the location of the workpiece on the conveyor belt. Encoder directly connected to the shaft of conveyor belt. The 3-phase pulse of this rotary encoder use NPN open-collector (OC) output with a resolution of 500 lines, working power supply is DC 12 ~ 24V.

This work unit does not use Z-phase pulse, A, B 2-phase output terminal directly connected to the high-speed counter input terminal of PLC (**FX3U -32MR**) .

When calculate the location of the workpiece on the conveyor belt, the distance between every two pulses or pulse equivalent should be known. diameter of driving shaft in sorting unit is $d = 43 \text{ mm}$, so every time the reducing motor rotate one circle, the workpiece on the belt move a distance of $L = \pi \cdot d = 3.14 \times 43 = 136.35 \text{ mm}$. Therefore,when pulse equivalent μ is $\mu = L/500 \approx 0.273 \text{ mm}$. As the installation size shown in diagram 5-8, when the workpiece move from the feed downpipe to the center of sensor data, the rotary encoder issue about 430 pulses; when move to the center of the first push rod, issues about 614 pulses ; when move to the center of the 2nd push rod, about to issue 963 pulses; moved to the 3rd putt center, about 1284 pulses issued.

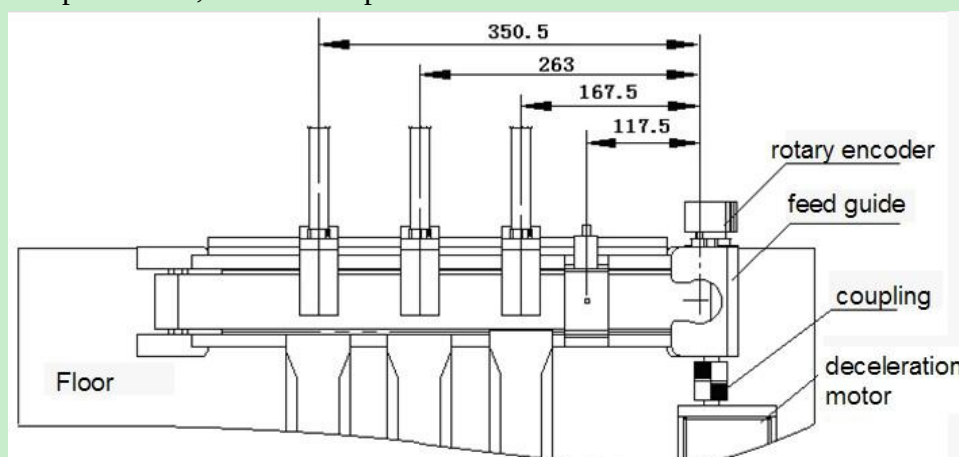


Diagram 5-9

It should be noted that the above is only a theoretical calculation of pulse equivalent. In fact a variety of error factors unavoidable, such as the measurement error of belt (including diameter and thickness), the installation deviation of belt, tension degree, sorting unit as a whole in the work on the stage positioning deviation, etc. will affect the theoretical calculation value. Therefore, the calculated values only serve as estimates. cumulative error caused by Pulse equivalent error will rapidly increased with the increasing distance the workpiece covered on the conveyor belt, even to an intolerable degree. Thus in the installation and commissioning of sorting unit, in addition to fine-tune to minimize the deviation, you'd better test pulse equivalent value.

Method of test pulse equivalent in field and how to counting pulses input to the PLC in high-speed to calculate the location of the workpiece on the conveyor belt,in this guide book we will combine the tasks of project in the PLC programming to introduce it.

5.3.2 Mitsubishi FR-E740 inverter

5.3.2.1 FR-E740 inverter installation and wiring

The inverter adopts Mitsubishi **FR-E700** inverter, **FR-E740-0.75K-CHT** inverter, the inverter rated voltage: 3-phase 400V, motor capacity **0.75kW** and under. **FR-E700** series of inverters' apperance and type definition are shown in the diagram 5-10.

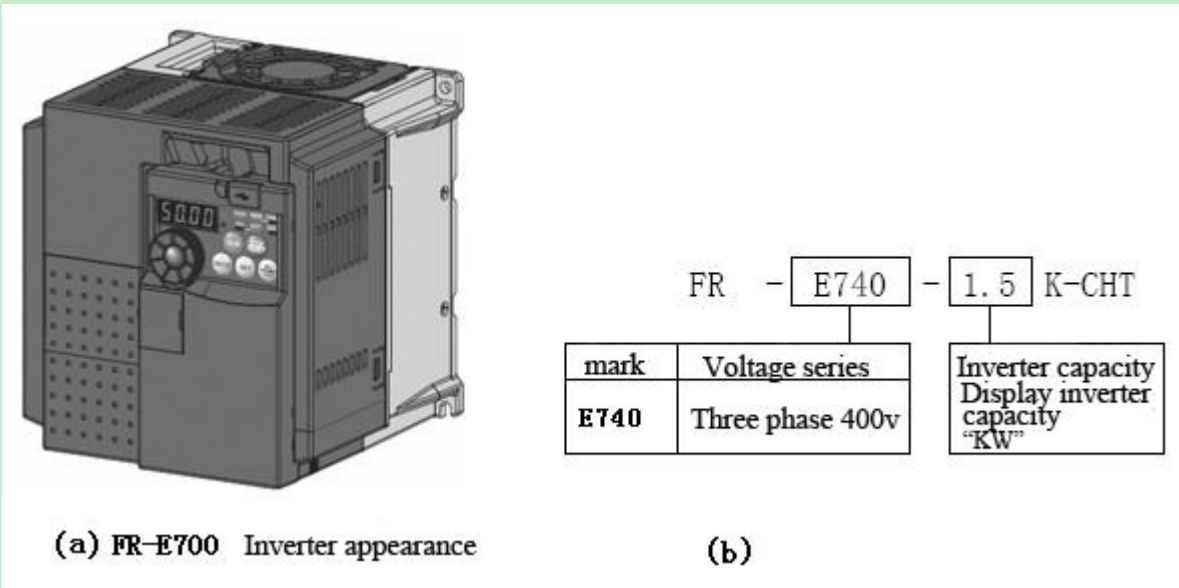


Diagram 5-10

FR-E740 series of inverter main circuit wiring is shown in the diagram 5-11.

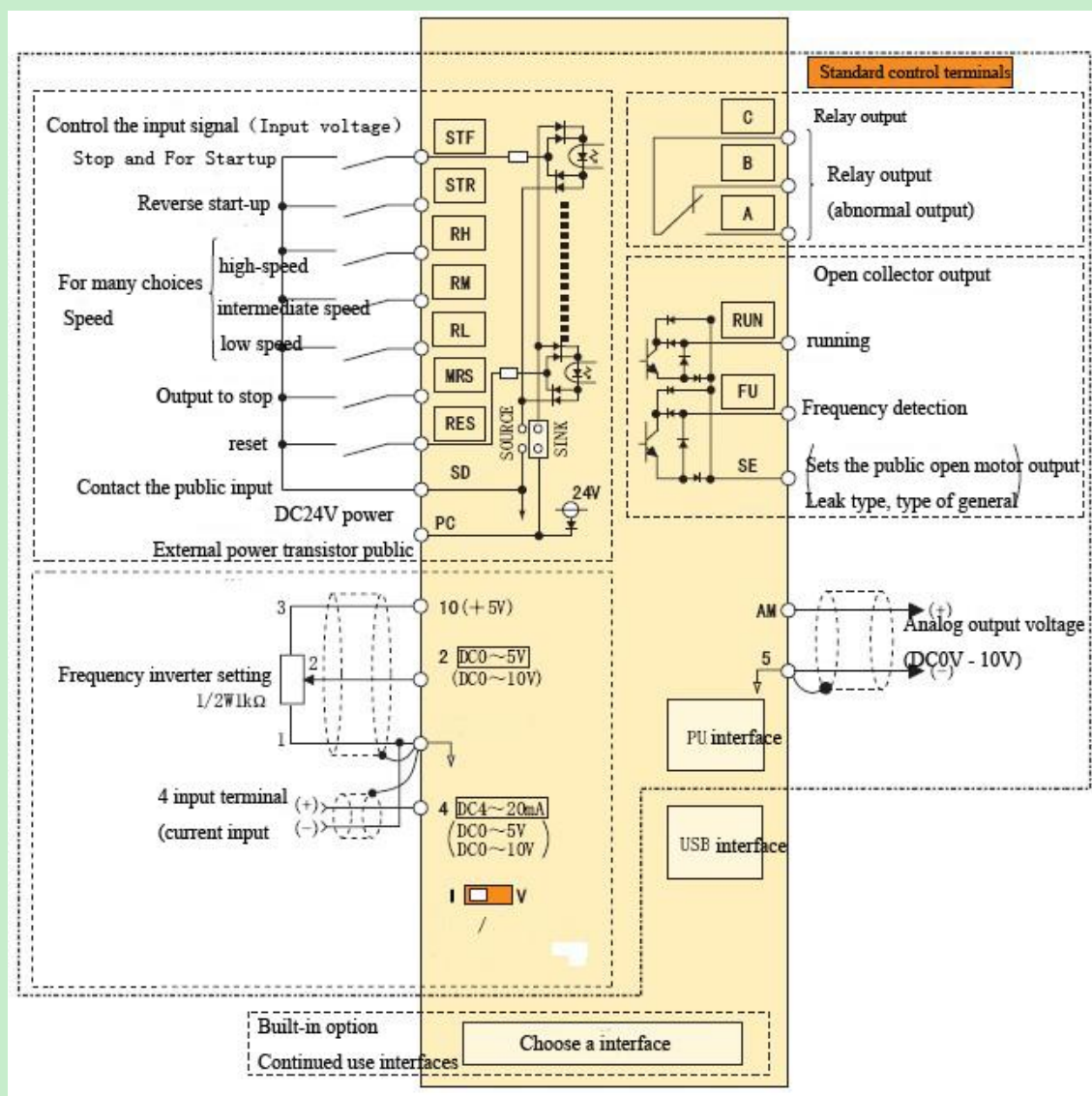


Diagram 5-12

The control circuit ends are control input, frequency setting (analogy input), relay output (abnormal output), collector open-circuit output(status detection) and analogy voltage output, each end functions can be changed by parameters change, with initial values, the control circuit end functions are shown in table 5-1, 5-2 and table 5-3.

Table 5-1 Control circuit input end functions

Type	Code	end name	end functions	
Input	STF	forward start	STF signal ON, forward, OFF stop	STF, STR signals are ON, it is stop command.
	STR	back start	STR signal ON, back, OFF, stop.	
	RH RM RL	Multi-speed selection	RH, RM and RL signals can select multi-speed	

	MRS	Output stop	MRS signal is ON (20ms or above), inverter output stops. Use electromagnetic brake to stop motor, cut off inverter output.
	RES	Reset	Cancel protection circuit alarm output. set RES signal ON status 0.1s or above, then cut off. The initial setting can be reset. But after Pr.75 setting, only when inverter alarm starts, it can be reset. The reset time is 1s.
	SD	input common (initial setting)	input end and common end .
		external transistor common (active)	Active logic, when it is connected to transistor output, such as programming controller (PLC), transistor output is used to connect external power supply common.
		DC24V power supply common	DC24V 0.1A power supply(end PC) common output end. and end 5 and end SE insulated.
	PC	external transistor common(sink) (initial setting)	Sink logic, when transistor output is connected, such as programming controller (PLC), transistor output will be connected to the external power supply common, to avoid faults by leak.
		input common (active)	To input end, common end .
		DC24V power supply	Can be DC24V, 0.1A power supply.
	10	frequency setting power supply	As external frequency setting (speed setting), power supply.
	2	frequency setting (voltage)	if input DC0~5V (or 0~10V), 5V (10V), is maximum output frequency, input/output are proportional. By Pr.73 DC0~5V (initial setting) and DC0~10V input switch.
Frequency setting	4	frequency setting (current)	If input DC4~20mA (or 0~5V,0~10V), for 20mA, maximum output frequency, input/output are proportional. Only when AU signal is ON, end 4 input signal is valid. By Pr.267, 4~20mA (initial setting) and DC0~5V, DC0~10V input switch. Voltage input (0~5V/0~10V), turn voltage/current input switch to “V” .
	5	frequency setting common	frequency setting signal(end 2or 4)and AM common. Do not make it grounded.

Table 5-2 Control circuit output end functions

Type	Code	name	functions	
Relay	A, B, C	relay output (abnormal output)	Indicate inverter, by protection function, works, output stops. When it is abnormal: B-C is cut off, normal: B-C is through.	
Collector open circuit	RUN	inverter runs	inverter output frequency is above or equal to start frequency (initial value 0.5Hz) low level, stopped or DC brake, high level.	
	FU	frequency detection	Output frequency is above or equal to any set frequency, low level, or high level.	
	SE	collector open-circuit output common	end RUN, FU common.	
Analogy	AM	analogy voltage output	can select one item to output. The inverter is reset, no output. The output signal and monitor item are proportional to each other.	output: Output frequency (initial setting)

Table 5-3 Control circuit network interface functions

type	code	name	functions
RS-485	--	PU interface	by PU interface, for RS-485 communication. •standard specification: EIA-485 (RS-485) •transition mode: multi-station communication •communication rate: 4800~38400bps •total distance: 500m
USB	--	USB interface	When PC and USB are connected, you can do FR Configurator operation. • interface: USB1.1standard •transition speed: 12Mbps •connector: USB Mini-B connector(socket: Mini-B)

If mechanical part of sorting unit is assembled, after main circuit wiring, you can use the inverter to drive the motor. if inverter mode parameters **Pr.79** is initial value, you connect ①, ②, ③ to inverterend ⑩, ②, ⑤, rotate potentiometer rightward; turn on power supply, toggle **STF** end knob switch, rotate potentiometer rightward, you will see motor forward rotation, the inverter output frequency will increase, motor rotation speed will increase.

When mechanical part of sorting unit is assembled, it is necessary to try motor running, this can be used to check assembly quality for adjustment.

5.3.2.2 Inverter operation panel training

1. FR-E700 series of panel

Before you use the inverter, you shall know the panel, and keyboard well, as well as the station (control station). **FR-E700** series of inverter parameters setting is done on panel, and it also can be connected to inverter **PU** interface parameters station (**FR-PU07**). The panel can be used in running, frequency setting, command monitor, parameters setting. The panel is shown in the diagram 5-13, the functions of all parts and buttons are shown in the table 5-4 and table 5-5.

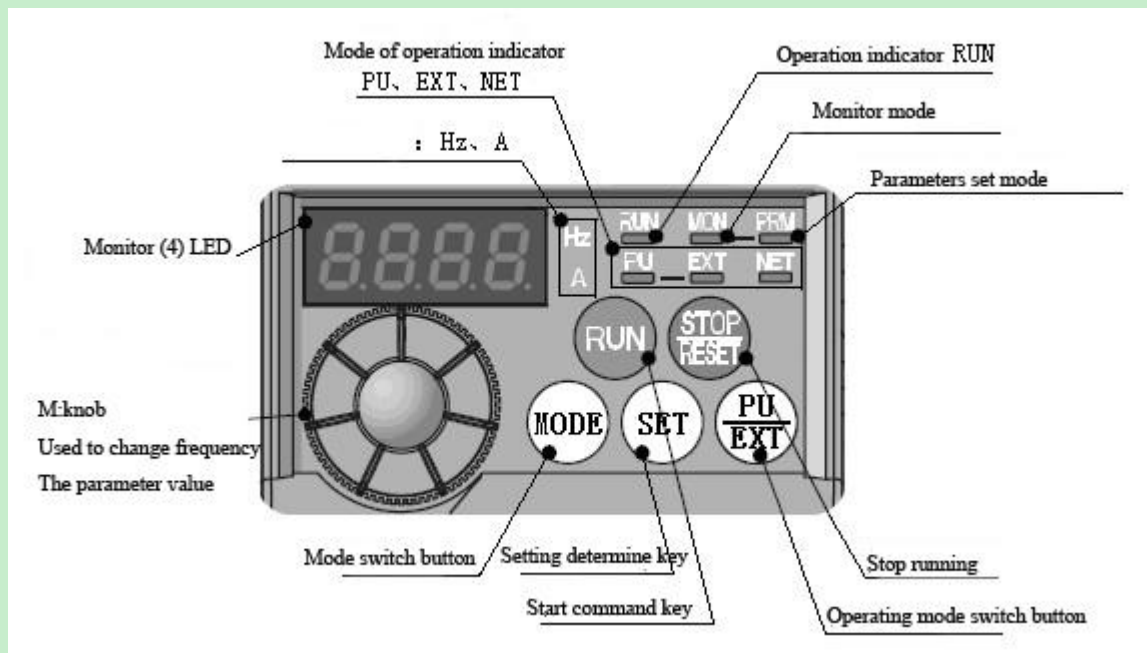


Diagram 5-13

Table 5-4 Knob, button functions

Knob and button	Functions
M knob (Mitsubishi inverter knob)	rotate the knob to change frequency setting, parameters setting value. monitor mode frequency setting correct current setting value alarm history mode sequence
Mode switch key MODE	Used for switch setting mode and run mode switch. push the key(2s)can lock.
Conform key SET	To confirm.
run mode key PU/EXT	switch PU /external run mode. external run mode, push the key, run mode EXT ON light status . switch to the mode, push MODE key 0.5s, or change parameters Pr.79.
Start command key RUN	In PU mode, push the key to start. by Pr.40 setting ,can selectionrotation direction .
Stop run key STOP/RESET	In PU mode, push the key to stop. protection functions.

Table 5-5 Run status display

Display	functions
Run mode display	PU : PU run mode, ON; EXT : external run mode, ON; NET : network run mode, ON.
Monitor (LED)	Display frequency, parameters
Monitor data display	Hz : display frequency, ON; A : display current, ON. (display voltage, OFF, display setting frequency monitor.)
run status display RUN	when inverter, ON: ON light--forward run; Slow flash (1.4s cycle)--back run; Quick flash (0.2s cycle): • button or input start command and cannot run • start command, frequency command, start frequency • input MRS signal
Parameters setting mode PRM	parameters setting mode, ON
Monitor MON	monitor mode, ON

2. Inverter run mode

From table 5-4 and table 5-5, in various inverter run modes. The run mode is input to inverter start command and setting frequency command source.

The control circuit, external set potentiometer and switch is “external run mode”, panel or parameters station input start command, setting frequency is “PU run mode”, by PU interface for RS-485 communication or is “network run mode (NET run mode)”. In inverter operation, you shall know each run mode.

FR-E700 series of inverter by parameters **Pr.79** value is used to appoint inverter run mode, setting value range is 0, 1, 2, 3, 4, 6, 7; the 7 run modes are shown in **LED** indication light status, table 5-6.

Table 5-6 Run mode selection (Pr.79)

setting value	Content	LED display status ( : OFF  : ON)
0	external/PU switch mode, by PU/EXT key switch PU and external run mode. note: turn on power supply, external run mode	external run mode:  PU run mode: 
1	PU run mode	
2	external run mode in external, network run mode.	external run mode:  network run mode: 

3	external /PU run mode1		
	frequency command	Start command	
	Panel setting or parameters station setting, or external signal input	external signal input (end STF, STR)	
4	external/PU run mode 2		
	Frequency command	Start command	
	external signal input (end 2, 4, JOG selection)	by panel RUN key, or by parameters station FWD, REV key input	
6	switch mode can be in run status, PU run, external run , network run switch		PU run mode:  external run mode:  network run mode: 
7	external run mode (PU run interlock) X12 signal ON, switch to PU run mode (external run output stop) X12 signal OFF, forbid switch to PU run mode		PU run mode:  external run mode: 

The initial parameters **Pr.79** setting value is 0. when it stops, user can set value.
 change **Pr.79** setting value: push **MODE** keymake inverter parameters setting mode;rotate **M** knob , selectionparameters **Pr.79**, **SET**key; then rotate **M** knob selection setting value , **SET**key; push **MODE** key, inverter run mode changes setting mode.
 diagram 5-14 is setting parameters **Pr.79** an example. inverter from external run mode changed to run mode 1.

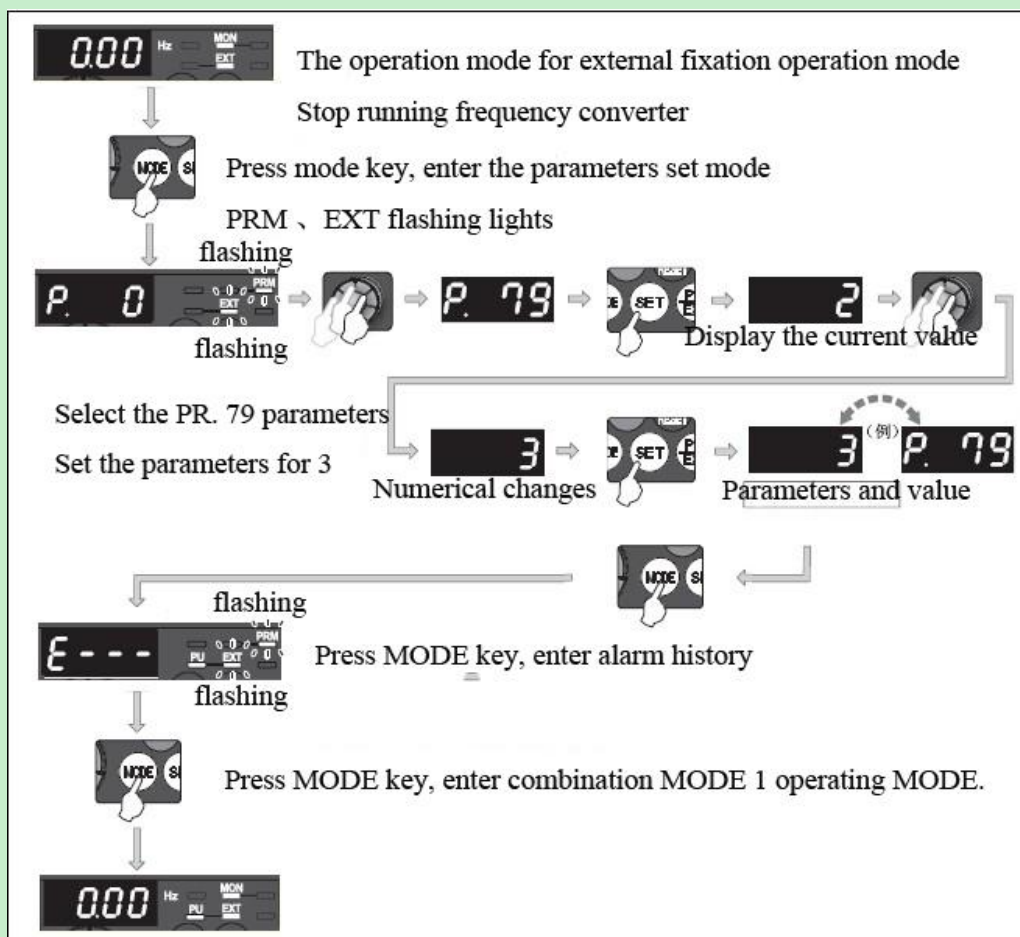


Diagram 5-14

3. Parameters setting

The inverter parameter setting value is simple. Load and operation setting parameters, enter parameters setting mode, parameters number, then set parameters value. setting parameters, **STOP** mode setting parameters, set all parameters; run setting, allow setting parameters, but all parameters number and parameters. In diagram 5-15 is parameters setting process, finished operation is parameters **Pr.1** (up limit frequency) from setting value 120.0Hz change is 50.0Hz, current run mode is external /PU switch mode (**Pr.79**=0).

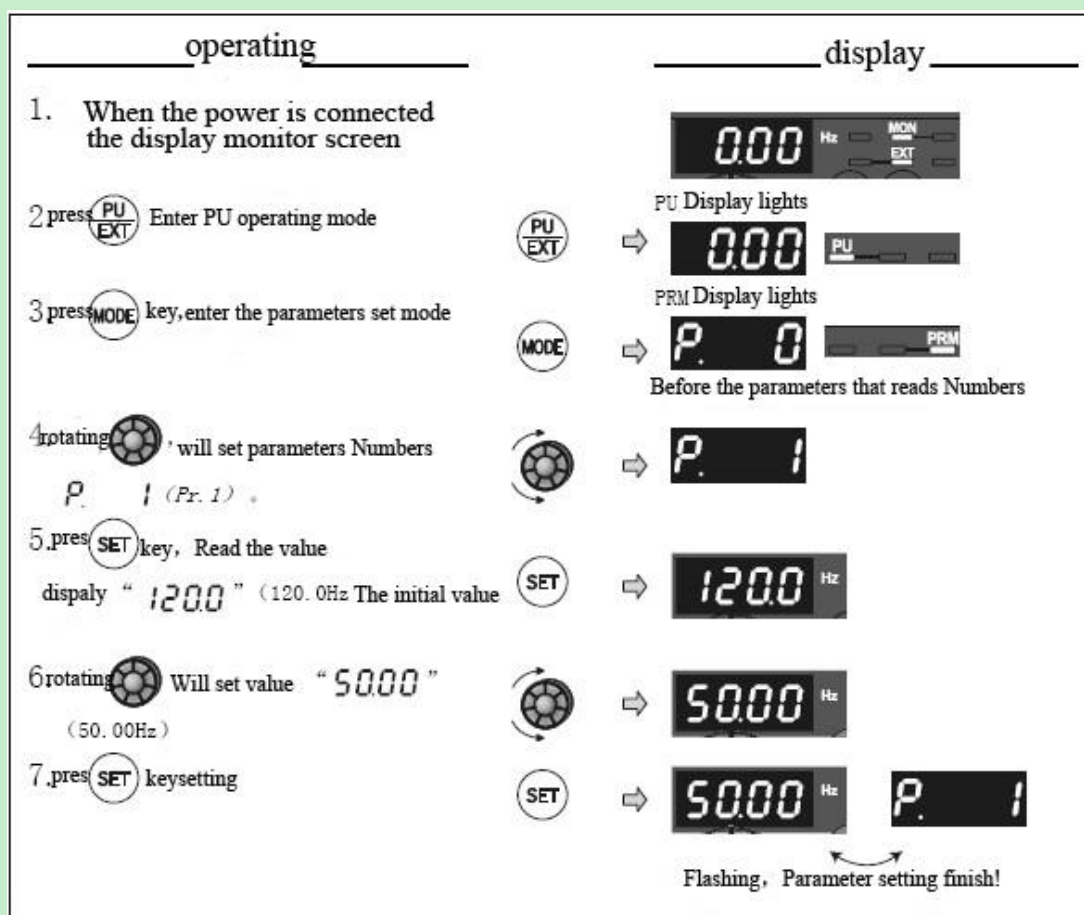


Diagram 5-15

The parameters setting process, switch to **PU** mode, enter parameters setting mode, and diagram 5-15 methods. In a run mode, Push **MODE** key, to enter parameters setting, as shown in diagram 5-15.

5.3.2.3 Normal parameters setting training

FR-E700 inverter has many parameters, here are some common parameters mostly used.

The distribution station technical process, inverter, parameters setting. inverter: drive motor specification, run limit; parameters initialization; motor start, run speed, brake command source, frequency set.

1. Output frequency limit (Pr.1, Pr.2, Pr.18)

To limit motor speed, inverter output frequency limit. **Pr.1** "upper limit frequency" and **Pr.2** "lower limit frequency" setting.

When the frequency is above 120Hz, parameters **Pr.18** "high-speed upper limit frequency" setting high-speed output frequency upper limit.

Pr.1 and **Pr.2** setting range is 0~120Hz, setting value is 120Hz and 0Hz. **Pr.18** setting range is 120~400Hz. The output frequency and setting value relation is shown in diagram 5-16.

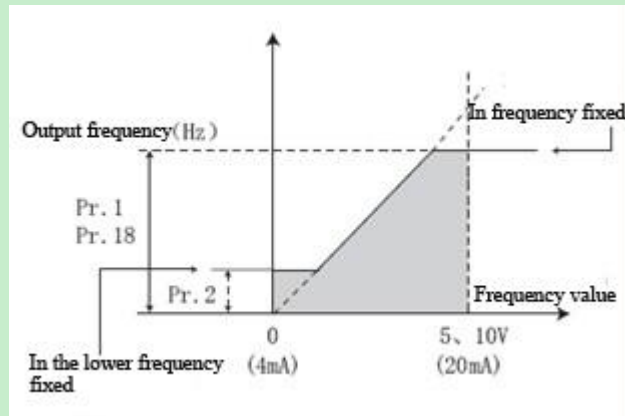


Diagram 5-16

2. Speed up/down time(Pr.7, Pr.8, Pr.20, Pr.21)

Each parameter meaning and setting range is shown in table 5-7.

Table 5-7 Speed up/down time parameters meaning and setting range

parameters number	parameters meaning	setting	setting range	note
Pr.7	speed up time	5s	0~3600/360s	Pr.21 speed up/down time singlebit setting value setting . initial value setting range is “0~3600s”, setting singlebit is “0.1s”
Pr.8	decrease speed time	5s	0~3600/360s	
Pr.20	increase/decrease speed frequency	50Hz	1~400Hz	
Pr.21	increase/decrease speed time bit	0	0/1	0: 0~3600s; singlebit : 0.1s 1: 0~360s; singlebit : 0.01s

Setting :

- ① **Pr.20** is increase/decrease speed frequency, 50Hz.
- ② **Pr.7** speed up time setting from stop to Pr.20 speed up/down frequency speed up time.
- ③ **Pr.8** decrease speed time setting from Pr.20 speed up/down frequency to stop decrease speed time.

3. Multi-speed run mode operation

Inverter external operation mode or operation mode 2, inverter can change input end status. The control frequency mode is multi-speed control function .

FR-E740 inverter speed control ends are **RH**, **RM** and **RL**.

Switch: rotation speed, binary (0 status) rotation speed. 3 speeds: **RH**, **RM**, **RL**. 7 speeds: **RH**, **RM**, **RL**. 7 speeds, each run frequency parameters **Pr.4 - Pr.6** (set 3 speed frequency), **Pr.24- Pr.27**. The control status and parameters relation is shown in diagram 5-17.

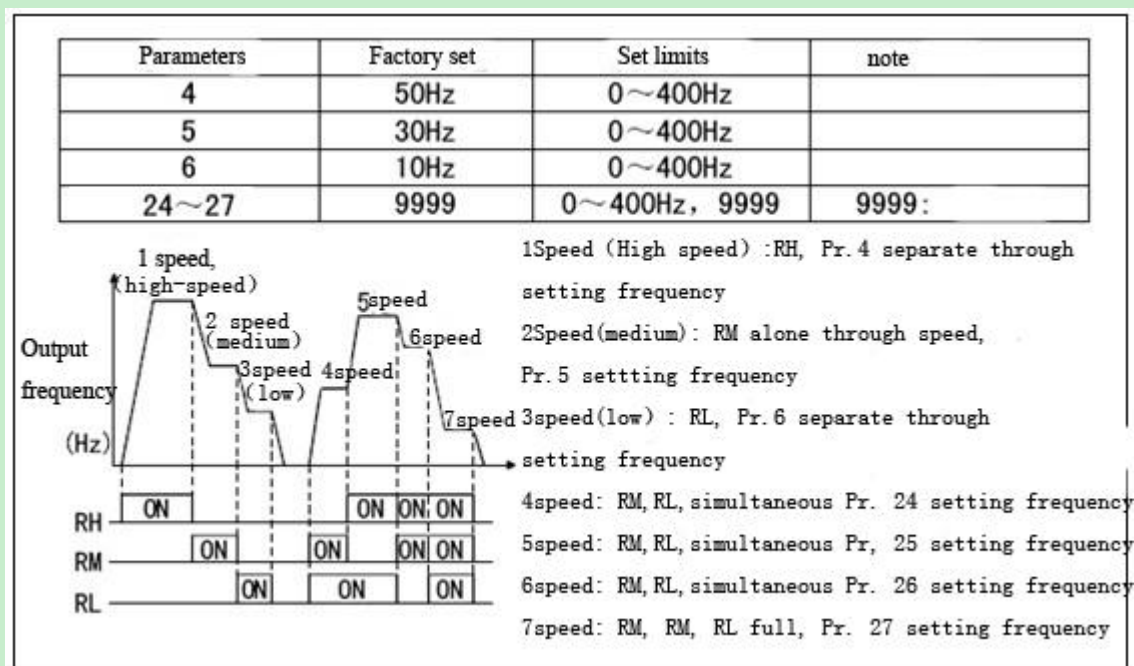


Diagram 5-17

The speed setting can be set on **PU**.

In 3 speed setting, 2 speeds above are selected, Low speed signal setting frequency is in the first place.

Then we need to talk that, if parameters **Pr.183** is 8, **RMS** end functions speedcontrol **REX**, are **RH, RM, RL** and **REX**, 15 speeds.

4. By analogy input (end 2, 4) setting frequency

The sorting station inverter frequency setting, **PLC** output end control speed setting, setting frequency. Such as inverter installation and wiring run test, speed potentiometer is connected to inverter analogy input signal, speed test. On the touch screen, inverter frequency. Note: if analogy input (end 2, 4) setting frequency, **RH, RM, RL shall be disconnected**.

(1). Analogy input signal end selection

FR-E700 series of inverter 2 analogy input signal ends (end 2, 4) change frequency setting. In the setting, end 2, end 4 invalid.

Make end 4, input end **STF, STR, ...RES**, functions definition is **AU** signal input. When end and **SD is connected**, **AU** signal is **ON**, end 4 change is valid, end 2 change is invalid.

The example: select **RES** end **AU** signal input, set parameters **Pr.184** = "4", **RES** end and **SD** connection, **AU** signal is **OFF**, end 2 is valid; when it is **ON**, **AU** signal is **ON**, end 4 is valid.

(2). Analogy signal input specification

If you make end 2, analogy signal is 0~5V or 0~10V voltage signal, parameters **Pr.73**, setting value is 1, is 0~5V input specification. The parameters **Pr.73** value range is 0, 1, 10, 11, shown in table 5-8.

If end 4, analogy signal is voltage input (0~5V, 0~10V) or current input (4~20mA initial value), parameters **Pr.267** and voltage/current input switch setting, and input and setting analogy signal. **Pr.267** value range is 0, 1, 2, shown in table 5-8.

Note: if switch and input signal error(setting is current input I, but end input is voltage signal;), external input equipment or inverter faults .

For frequency setting signal (DC0~5V, 0~10V or 4~20mA) output frequency parameters **Pr.125** (end 2) or **Pr.126** (end 4) setting,input gain (maximum) frequency. The initial setting value is 50Hz, setting range is 0~400Hz.

Table 5-8 Analogy input selection (Pr.73, Pr.267)

parameterser	name	initial value	setting range	content	
73	analogy input selection	1	0	end 2 input 0~10V	run
			1	end 2 input 0~5V	
			10	end 2 input 0~10V	run
			11	end 2 input 0~5V	
267	end 4 input selection	0	0	voltage/current input switch	content
					end 4input 4~20mA
			1		end 4input 0~5V
			2		end 4input 0~10V

Note: voltage input : input resistor $10k\Omega \pm 1k\Omega$, maximum voltage DC20V
current input : input resistor $233\Omega \pm 5\Omega$, maximum current 30mA

5. Parameters clear

If user parameters debugging process, wants to restart debugging, parameters clear operation methods. In **PU** run mode, setting **Pr.CL** parameters clear, **ALLC** parameters cleared to “1”, make parameters recover to initial values.

Parameters clear operation, in parameters setting mode, M knob select parameters **Pr.CL** and **ALLC**, value is 1, steps are shown in diagram 5-18.

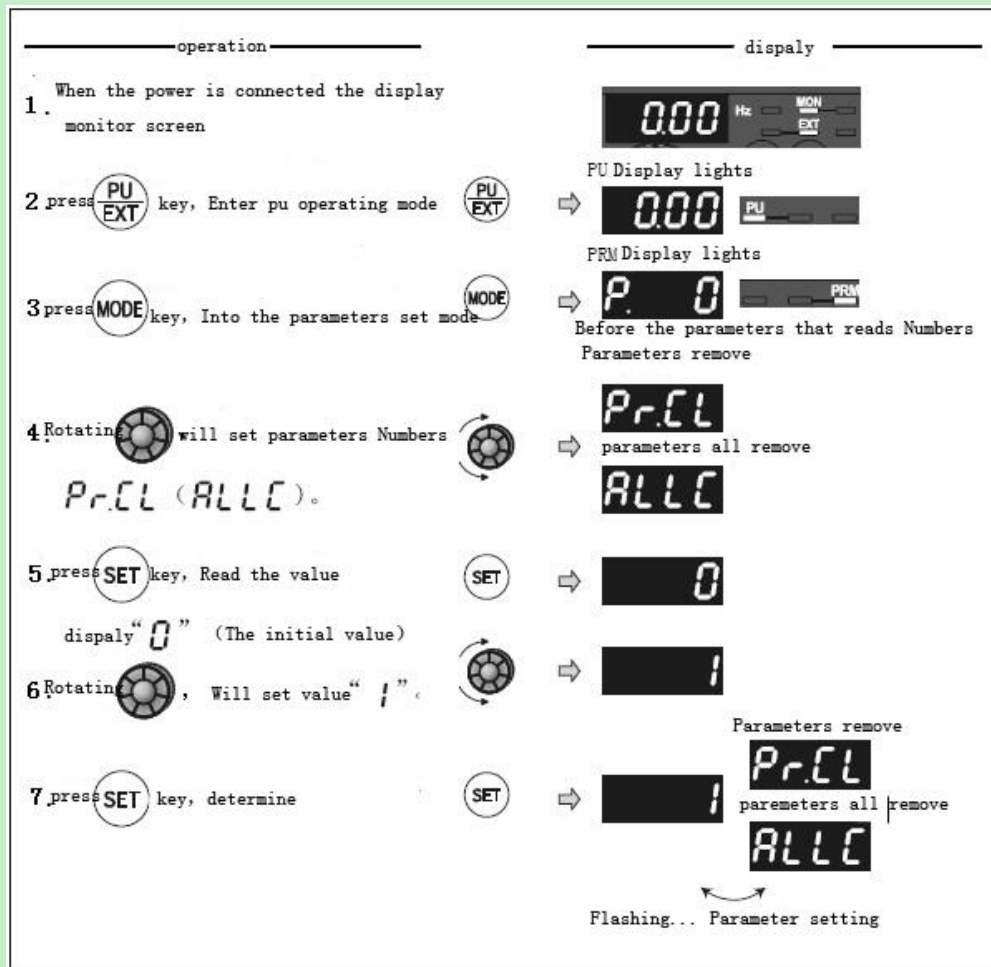


Diagram 5-18

5.4 Sorting Unit PLC control and programming

5.4.1 Tasks

1. Device target is to sort the white-core metal parts, plastic parts white core and black core of metal or plastic parts. In order to launch the accuracy and sorting artifacts, use rotary encoders for position detection. And the workpiece material and the core body color properties should be detected before the pusher cylinder positions .

2. After connection of power and gas source, if the three-cylinder at retracted position, then the "normal work" light HL1 lighting for long, that means equipment is ready. Otherwise, the light blinking as frequency of 1Hz.

3. If the equipment is ready, press the start button, the system starts, "equipment running" light HL2 lighting for long. When the conveyor belt feeding port has workpiece, the inverter start immediately, drive motor to drive at a fixed frequency of 30Hz to take the workpiece to sorting area.

If the workpiece is white core of metal parts, then the workpiece to reach No. 1 slide on the middle, conveyors stops, workpiece was pushed the No. 1 slide; If the workpiece is white core of plastic parts, then the workpiece to reach No. 2 slide on the middle, conveyors stops, workpiece was pushed the No. 2 slide; If the workpiece is black core parts, then the workpiece to reach No. 3 slide on the middle, conveyors stops, workpiece was pushed the No.3 slide. After workpiece been launched to the slide, one working cycle end. Only when

the workpiece has been launched to the slide can feed conveyor belt material.
If you press the stop button during operation, the unit of work stop running after ending this working cycle.

5.4.2 I/O connection of PLC

According to the work task requirements, equipment assembly and sensors installation as shown in diagram 5-19.

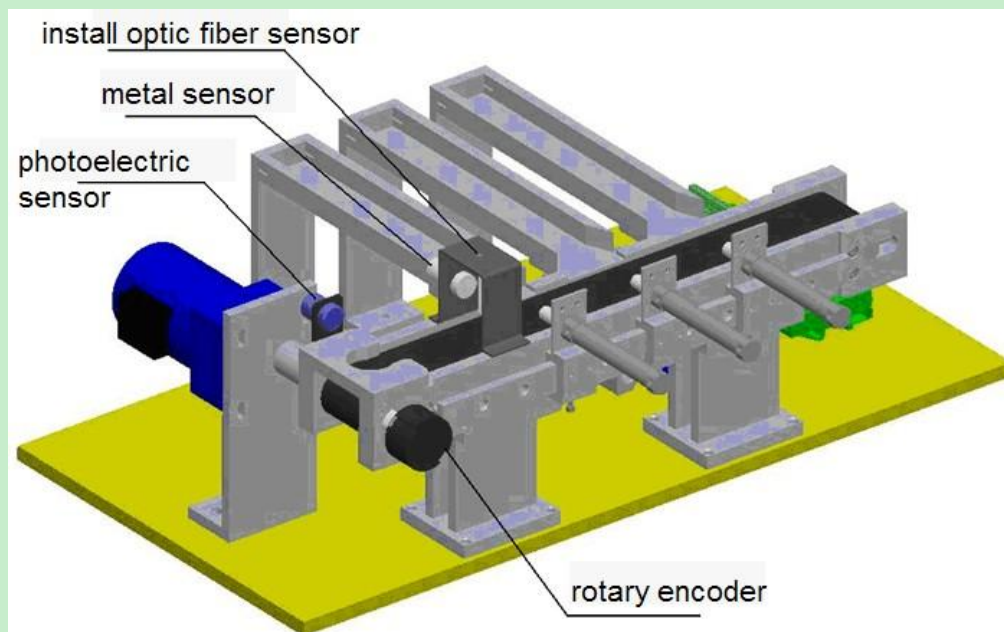


Diagram 5-19

Sorting unit device side of the terminal port terminal distribution of the signal as shown in Table 5-9. As the sensors for distinguish the color of the workpiece material and the core attributes only need to install on the inductive sensor and a fiber optic sensor, so the optical fiber sensor 2 donot need.

Table 5-9 sorting station side port signals

input port			output port		
end	equipment code	signal cable	end number	equipment code	signal cable
2	DECODE	rotary encoder B phase	2	1Y	pusher 1 solenoid
3		rotary encoder A phase	3	2Y	pusher 2 solenoid
4	SC1	optical fiber sensor 1	4	3Y	pusher 3 solenoid
5	SC2	optical fiber sensor 2			
6	SC3	inlet port workpiece detection			
7	SC4	inductive sensor			
8					
9	1B	pusher 1 arrival			

10	2B	pusher 2 arrival			
11	3B	pusher 3 arrival			
12#~17#end no connection			5#~14#end no connection		

Adopt Mitsubishi **FX3U-32MR** as main unit, 16 points input and 16 points relay output. The work motor run frequency is 30Hz, connected to a inverter speed control end, such as “**RH**”, setting parameters **Pr.79** =2 (external run mode), setting **Pr.4** =30Hz. When **FR-E740** end “**STF**” and “**RH**” **ON**, motor start and frequency is 30Hz.

PLC I/O signals shown in table 5-13, **I/O** wiring schematic is shown in diagram 5-20.

Table 5-13 Sorting stationPLC I/O signal table

input signal				output signal			
No.	PLC input	signal name	signal source	No.	PLC output	signal name	signal output target
1	X000	rotary encoder B phase	same side	1	Y000	STF	inverter
2	X001	rotary encoder A phase		2	Y001	RH	inverter
3	X002	rotary encoder Z phase		3			
4	X003	inlet port workpiece detection		4			
5	X004	inductive sensor		5			
6	X005	optical fiber sensor		6	Y004	pusher 1 solenoid	
7	X006			7	Y005	pusher 2 solenoid	
8	X007	pusher 1 arrival		8	Y006	pusher 3 solenoid	
9	X010	pusher 2 arrival		9	Y007	HL1	button/indicatio n light module
10	X011	pusher 3 arrival		10	Y010	HL2	
11	X012	start button	button/indic ation light module	11	Y011	HL3	
12	X013	stop button					
13	X014	emergency stop button					
14	X015	single station /full-line					

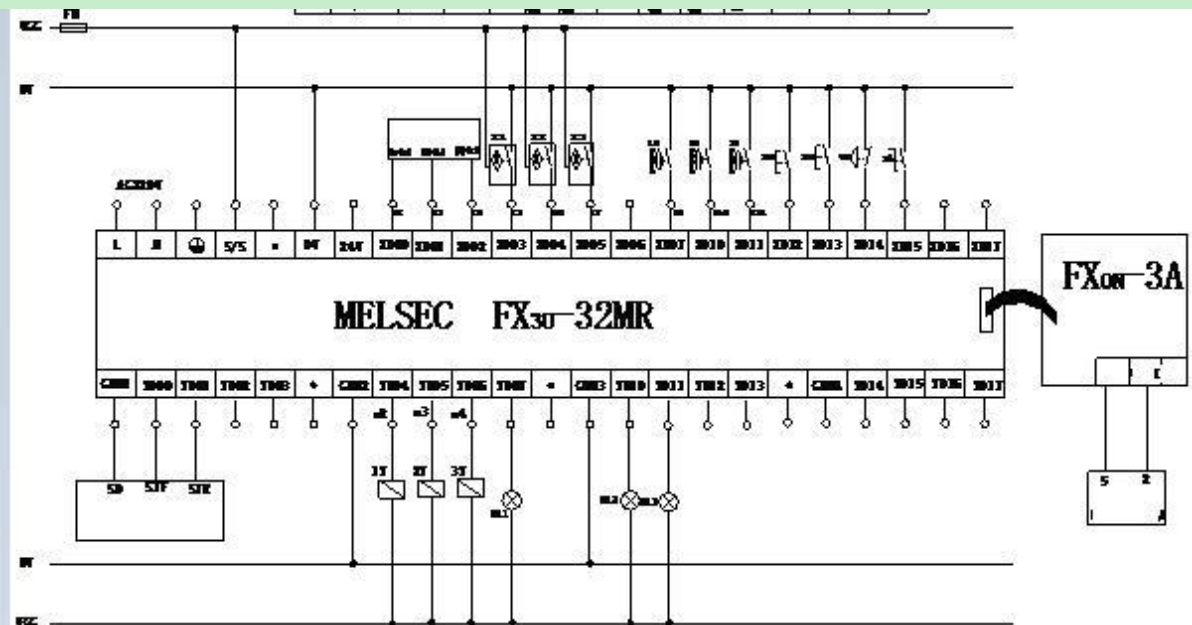


Diagram 5-20

5.4.3 Sorting Unit Programming point

5.4.3.1 Programming of high-speed counter

1. FX2N PLC high-speed counter

The high-speed counter is **PLC** programming software element, for common counters, high-speed counter count signal frequency is high.

FX2N PLC 21 high-speed counter C235~C255, each high-speed counter functions and input.

(1) High-speed counter functions:

- C235~C245 totally 11 high-speed counter a count input high-speed count, each counter 1 high-speed count input, count direction can be increase sequence or decrease sequence count, special assistant relay M8□□□ status. C245 X002 is high-speed count input, when special assistant relay M8245 is set, increase sequence count. C245, X003 and X007 are counter external reset and set input.
- C246~C250 totally 5 high-speed counter count input high-speed count, each counter 2 high-speed count input, 1 is increase count input, the other is decrease count input. C250 X003 is increase count input, X004 is decrease count input, X005 is external reset input, X007 is external set input. The counter count direction can be made by programming special assistant relay M8□□□ status.
- C251~C255 totally 5 high-speed counter, input high-speed count, each counter 2 high-speed count input, one is A phase count input, the other is A phase phasebit difference 90° B phase count input. C251~C255 functions and input are shown in table 5-14.

table 5-14 high-speed counter C251~C255 functions and input

	X000	X001	X002	X003	X004	X005	X006	X007
C251	A	B						
C252	A	B	R					
C253				A	B	R		

C254	A	B	R				S	
C255				A	B	R		S

The sorting station A, B 90 °phase bit difference general rotary encoder, and Z phase pulse signals. in table 5-14, adopts high-speed counter C251. rotary encoder A, B pulse output is connected to X000 and X001 .

(2).Each high-speed counter has inputs, but all high-speed counter input is within X000~X007 range. For example, C251, X000, X001 are connected high-speed counter, C252, C254 cannot be used.

2. High-speed counter programming

if external high-speed count source (rotary encoder output) is connected to PLC input, in the program, you can make high-speed counter count. For example, in diagram 5-21, C255 value is 100, when C255 current value is 100, counter output works. The control output Y010 ON.

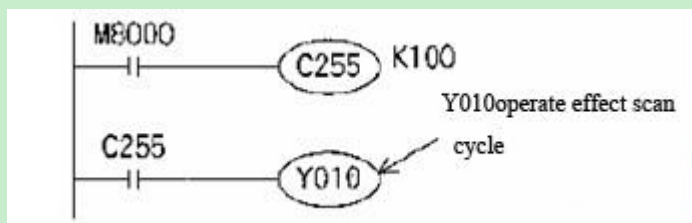


Diagram 5-21

The current value = preset value, counter will work, but output signal is from the scan cycle . if counter outputs signal, you shall make high-speed counter commands, FX2N PLC high-speed process commands for high-speed counter, as 32bit commands. The methods are in FX2N programming manual.

The test rotary encoder pulse is example of high-speed counter.

Example: equivalent field test of rotary encoder pulse

As already noted before, calculate rotary encoder pulse equivalent according to the belt driving shaft diameter, the result is an approximation. In the installation and commissioning, in addition to fine-tune the installation to minimize the deviation, yet to test pulse value.

Testing steps as follow:

(1) When installation, motor must be carefully adjusted in conjunction with the driving shaft and transmission shaft concentricity degree of belt tensioning. Tension adjustable degree of balance between the two adjustment bolts should be adjusted to avoid the belt run-time deviation. Belt tensioning degree is appropriate of electrical input frequency of 1Hz when a smooth start, lower than 1Hz it difficult to start. The test inverter is **Pr.79** =1, **Pr.3** =0 Hz, **Pr.161** =1; then you can do start/stop operation on panel, and M knob is potentiometer for frequency adjustment.

(2) Installed, inverter parameters set is :

Pr.79 = 2(external run mode), **Pr.4**=25Hz(high-speed run frequency setting value)

(3).Write the program as diagram 5-22, transferred to **PLC**.

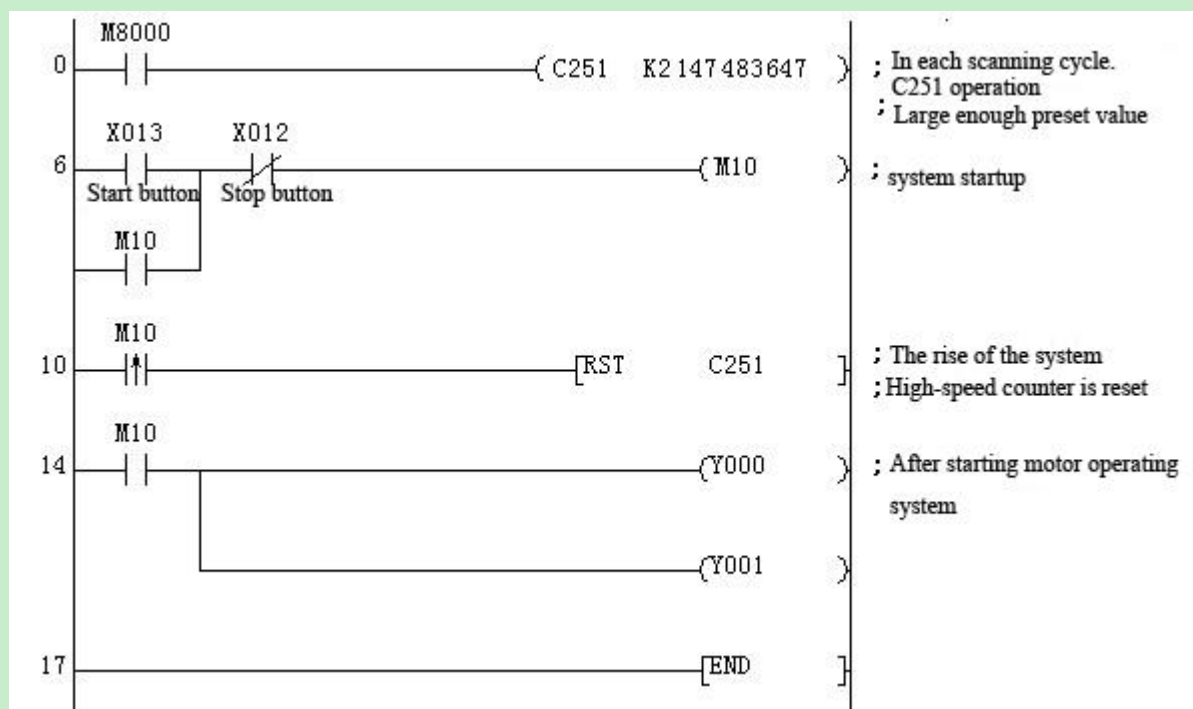


Diagram 5-22

(4). Run **PLC** program, monito rmode. Belt inlet port drop workpiece, push start button. workpiece transfer to a distance, push stop button. See C251 number, write to table 5-12 “high-speed count pulse quantity”. Measure workpiece movement distance; see high-speed count pulse value, calculate value = workpiece movement distance / high-speed count pulse quantity.

Table 5-12 Pulse test data

content No.	workpiece movement distance (measured value)	high-speed count pulse quantity (test value)	Pulse μ (calculated value)
1.	357.8	1391	0.2571
2.	358	1392	0.2571
2.	360.5	1394	0.2586

(5)Feeding again, press the start button to the second test. Conducted three tests, find the average pulse equivalent μ : $\mu = (\mu_1 + \mu_2 + \mu_3) / 3 = 0.2576$.

As shown in diagram 5-8 re-calculation the pulse number of rotary encoder to various locations by the installation size, When the workpiece material from the bottom centerline move to sensors centre, rotary encoders send 456 pulses, Go to putt a focal point for the first time, send 650 pulse; Go to the second putt center, send 1021 pulse; moved to third center putter, send 1361 pulses.The task, Programming high-speed counter, is C251 current value workpiece position, and save to variable position data memory. The positions:

- Discrimination position of the workpiece property should be behind the centre position from the import to sensors, so they chose the pulses No. 470, stored in the D110 cell (double integer)

- From position 1, push out workpiece , the stop position inlet port to pusher 1 position, pulse quantity is 600, saved in D114 station
- From position 2, push out workpiece , the stop position inlet port to pusher 2 position, pulse quantity is 970, saved in D118 station
- From position 3, push out workpiece , the stop position inlet port to pusher 3 position, pulse quantity is 1325, saved in D122 station.

Note: position data are from inlet port, when distribution workpiece is dropped to inlet port, motor starts, C251 current value shall be reset (clear).

5.4.3.2 Programming structure

1. Sorting station main work process is sorting control. After initial status check, confirm system is ready, push start button, to enter run status, and it starts distribution process control. The initial status check program flow is almost as samer as previous supply, processing station. The previous position data, in 1 scan cycle write to data memory.

When system is in run status, check whether stop button is pushed. if stop command is sent, then when a system cycle is over, it returns to the initial step, reset status and initial step, system stops.

2. Distribution process is a step sequence program, programming:

- ① When distribution workpiece drops to inlet, reset high-speed counter C251, inverter drives the motor.
- ② When workpiece passes optical fiber head and inductive sensor, 2 sensors will detect workpiece property.

C251current value and sensor position value command. The functions ladder is shown in diagram 5-23and diagram 5-24.

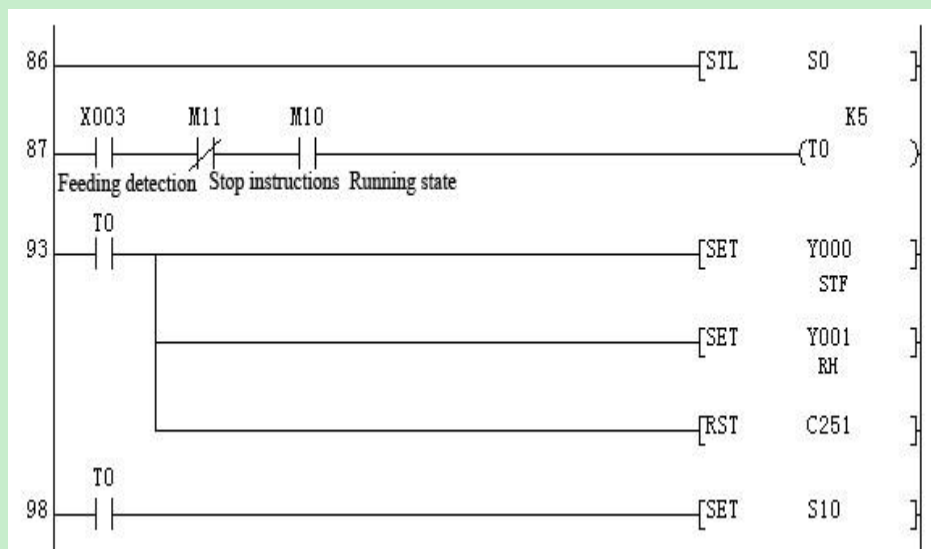


Diagram 5-23

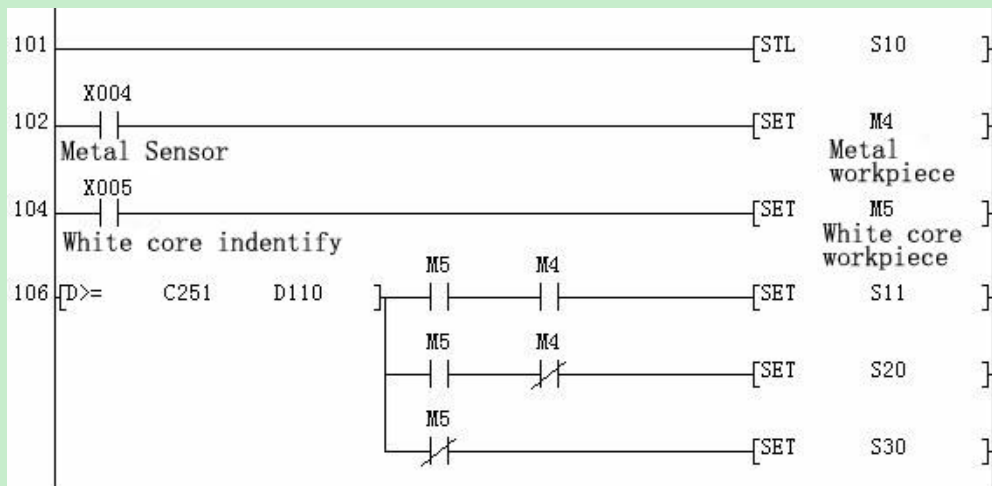


Diagram 5-24

③ According to mission requirements of workpiece properties and sorting, push cylinder in the corresponding position of the workpiece. After pusher cylinder return, step sequential control subroutine to return the initial step. This part of the preparation process, please complete by the reader.

ITEM6 Control the Sorting Unit Operation with Human Machine Interface

6.1 TPC7062KS human-machine interface

YL-335B uses the man-machine interface TPC7062KS. It is a real-time multi-tasking embedded operating system running WindowsCE environment, MCGS embedded configuration software configuration.

The product is designed with a 7-inch high-brightness TFT LCD screen (resolution 800 × 480), four-wire resistive touch screen (resolution 4096 × 4096), Color up to 64K color.

CPU board: ARM architecture low-power embedded CPU core, clocked at 400MHz, 64M memory space.

6.1.1 TPC7062KS human-machine interface hardware connection

TPC7062KS man-machine interface of the power into the lines, a variety of communication interfaces were carried out in back, as shown in diagram 6-1.

In which USB1 port used to connect the mouse and U disk, etc., USB2 port for the project download, COM (RS232) to connect PLC. Download cable and communication lines as shown in diagram 6-2

TPC7062KS back view

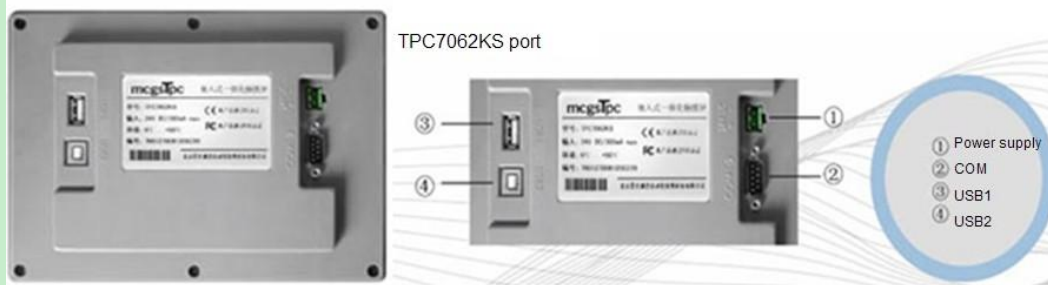


Diagram6-1



Diagram 6-2

1. TPC7062KS touch screen with personal computers connection

The TPC7062KS touch screen through a USB2 port and connected to a personal computer, before connection the personal computer should install the MCGS configuration software. When you need to download the information to HMI from MCGS configuration software, select "Connect to run", click "Project Download" button to download. As shown in diagram 6-3. If the project to be in the computer simulation tests, select the "mock run", and then download the project.



Diagram 6-3

2. TPC7062KS touch-screen with the FX Series PLC connection

In YL-335B, the touch screen directly with the transfer station PLC (FX3U-48MT) programming port through the COM port to connect. The used communication cable with RS232/RS422 converter as diagram 6-2 .

In order to achieve normal communication, in addition to the proper hardware connection, also would need to set the serial port 0 attributes on the touch screen, which is implemented in the device configuration window, setting method will later tasks in detail.

6.1.2 Touch-screen configuration

In order to operate the machine or systems through the touch-screen, must give the touch-screen device configuration user interface, the process is called "the configuration stage." System configuration is approach to operating units and mechanical equipment or communication between the processes through the PLC "variable". PLC variable value is written on a storage area (address), by the operating unit reads from the region.

Run MCGS embedded version of the configuration environment software, in the interface appears, click the menu "File" → "New Project", pop-up interface, as shown in diagram 6-4. MCGS embedded version of "workbench" window to manage the composition of the five parts of the user applications, 5 Tags on working platform: master window, equipment windows, user window, real-time database and operating strategies, corresponding to five different window pages, each page is responsible for managing a part of the user applications, using the mouse to click different window can select different window page, on the application of the corresponding parts of the system configuration operations.

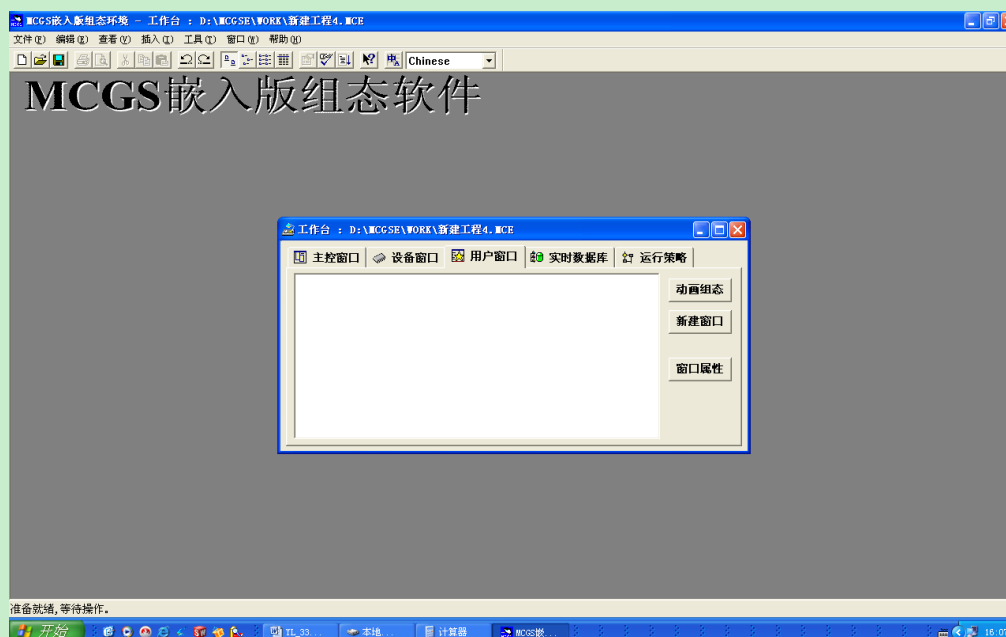


Diagram 6-4

(1) .Master window

MCGS embedded version of the master configuration of the project window is the main window, and is parent window of all windows and user window, that is equivalent to a large container which can be placed a device window and multiple users windows, undertake the

management of these windows and scheduling, and scheduling user policy from running. Meanwhile, the main control window is the main frame structure, the configuration works, you can set the host window system is running processes and characteristic parameters, user-friendly operation.

(2). Equipment window

Equipment window is the background operating environment of MCGS embedded version of the object system and a monitoring and control of external devices, responsible for driving the external device, to control external devices working condition. System through the channel between the device and data to come in external devices running data collection, into the real-time database for other parts of the system call, and put real-time database in the data output to external devices, to realize the operation and control of external devices .

(3). Users window

The user window is a "container" used to place a variety of graphical objects (primitives, icons and animation components), different graphical objects corresponding to different functions. Through the user window configuration of multiple graphic objects to generate a beautiful graphical interface, in order to achieve the animation effects of preparation.

(4). Real-Time Database

In the MCGS embedded version, with the data object to describe the system real-time data, using the object variable to replace the traditional sense of the variable value, the database technology management of the collection of all data objects known as real-time database.

Real-time database is the core of the system MCGS embedded version, the application system's data processing center. The various parts of the system are real-time database for the common areas and data exchange among the various parts to achieve the coordination of movements.

Equipment window drive external devices through the device components, collected the data into the real-time database; the graphic objects composed by the user window, real-time data in the database to establish a connection between an object in order to animate data visualization; Operation strategy through the strategic components of the data operations and processing. As shown in diagram 6-5:

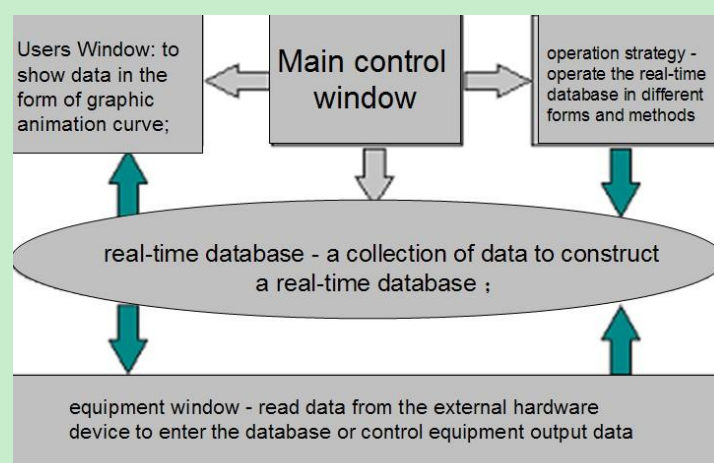


Diagram 6-5

(5) Operation strategy

For complex projects, monitoring system must be designed multi-branch, multi-nested loop structure, in accordance with predetermined conditions, operation of the system processes and equipment operating status of a targeted selection and precise control. To this end, MCGS embedded version of the introduction of the concept of running strategy to address the issue.

The so-called "operation strategy" is a general term for a series of functional blocks which help uses to realize the freedom of controlling the system running processes. MCGS embedded version provides users with the configuration window and toolbox of conduct policy-specific. The establishment of operating strategies, enabling the system operate real-time database, control the user a window open and close, and equipment components of the working conditions accordance with setting the order and conditions, in order to achieve precise the working process of the system control and the orderly scheduling management purposes.

6.2 Working tasks using human-machine interface

To further illustrate the man-machine interface configuration of specific methods and steps, the following is given a practical training task slightly modified based on the project's fifth task, and from the man-machine interface provides the main signal and display system state.

1. Target, power and gas source connected to the initial position of Equipment, the specific sorting requirements are the same as the original tasks, start and stop operations and work status indicator, not through the button indicator light box instructions, but in touch-screen. At this time, sorting station I / O wiring principle as shown in diagram 6-6.

2. When the conveyor belt fed artificial with has been assembled workpiece, the drive immediately start, drive transmission with touch-screen for a given motor speed, the workpiece taken to sorting area. Frequency can be adjusted in 40-50HZ.

The Trough workpiece accumulated data give displayed on the touch screen and touch screen data can be cleared.

Based on the above man-machine interface configuration required to complete the preparation and sorting procedures.

6.2.1 HMI Configuration

Sorting station in the picture effect diagram as shown in diagram 6-7.

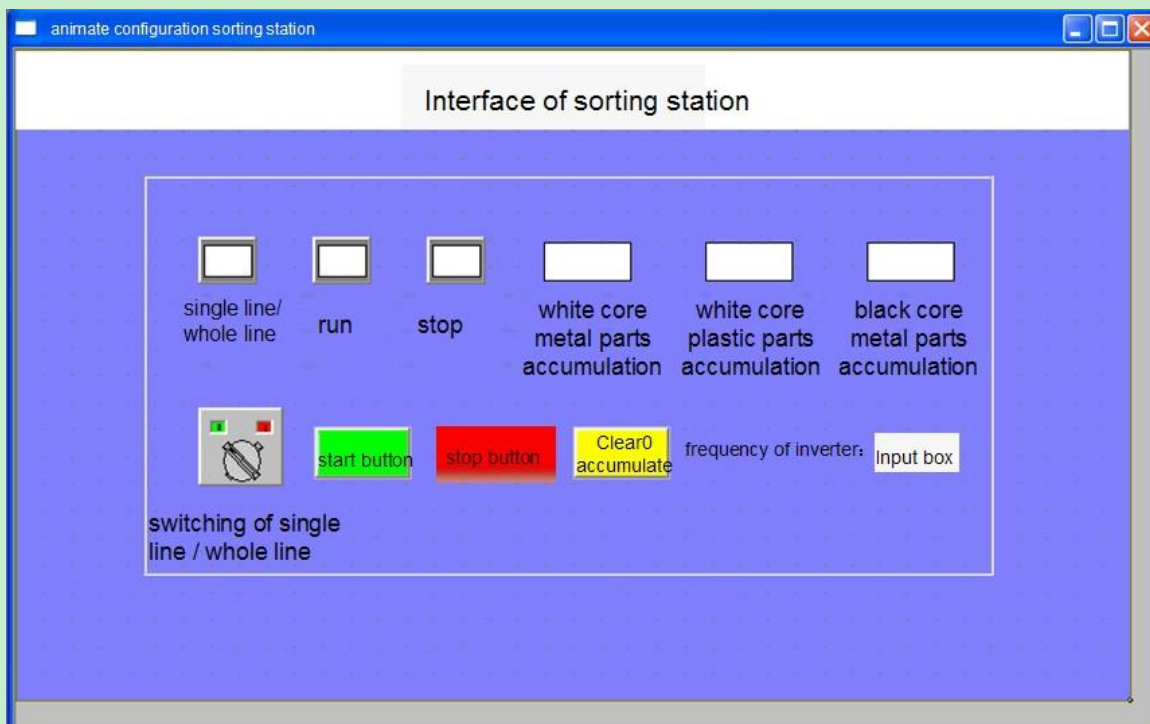


Diagram 6-7.

Screen contains the following aspects:

- Status Indication: Single / across the board, run, stop;
- Switch Knob: single line of switching;
- Buttons: start, stop, clear the accumulated button;
- Data Input: inverter input frequency settings;
- Data output shown: white-core metal parts, total white-core plastic parts, black core body parts;
- Rectangular box.

The following lists the configuration touch screen PLC addresses each component corresponds to the following table

Table 6-1 PLC addresses each touch screen component configuration corresponds to

Component Category	name	Input address	Output Address	Remark
Bit state switch	Single / full range switch	M0001	M0001	
Bit state switch	Start Button		M0002	
	Stop button		M0003	
	Clear Accumulated button		M0004	
Bit Status Indicator	Single / full range of light	M0001	M0001	
	Running lights		M0000	

	Stop lights		M0000	
Numerical Input devices	The inverter frequency given	D0000	M0000	min 40,max50
Numerical output devices	White-core metal parts aggregate	D0130		
	White-core plastic parts aggregate	D0131		
	Black Core workpiece cumulative	D0132		

Following is the configuration step and method of man-machine interface.

1. Create Project

If you can not find" TPC7062KS "in TPC type, then select" TPC7062K "Named "335B-sorting station."

2. Definition data objects

According to the previous given Table 1, define the data objects, all of the data objects are listed in the following table 6-2.

Table 6-2 Configuration screen touch screen PLC addresses each component

Data Name	Data type	Remark
running	Switching	
Single line of switching	Switching	
Start button	Switching	
Stop button	Switching	
Clear Accumulated button	Switching	
the inverter given frequency	Numeric	
White-core metal parts aggregate	Numeric	
White-core plastic parts aggregate	Numeric	
Black Core workpiece cumulative	Numeric	

The following data-object "running" is as an example to introduce the definition of data objects steps:

- (1) Click the "real-time database" window tab, enter the real-time database window page.
- (2) Click the "Add new object" button on the window list of data objects, add new data objects, the Department of unified definition of the default named "Data1", "Data2", "Data3", etc. (click on the button several times, you can increase numbers of data objects).
- (3) In the table, double click "general serial port father equipment", then double click "Mitsubishi _FX series of programming", there will pop up "general serial port father equipment", "Mitsubishi _FX series of programming", as shown in diagram 6-8:
- (4) Change the name of the object to: Run state; Object Type options: switch type; click "Confirm."

In accordance with this step, according to the above list, set the other data object.

3. Devices connection

In order to make touch screen and PLC communications connection is required to defined

the data object and PLC internal variables connection, concrete steps are as follows:

- (1) In the "device window", double-click the "device window" icon to enter.
- (2) Click the toolbar in the "tool box" , open the "Device toolbox."
- (3) In the optional equipment list, double-click the "universal serial parent device", then double-click "Siemens _S7200PPI" in the next issued a "universal serial parent device", "Siemens _S7200PPI", the window interface device as shown in diagram 6-8.



Diagram 6-8

- (4) Double-click "Universal Serial Father equipment", into the general-purpose serial port of the basic properties of the parent device settings, see below as shown in diagram 6-9, Universal serial device interface, the following settings:

- Serial port number (1 ~ 255) is set to: 0 - COM1;
- Communication baud rate is set as follows: 8-19200;
- Data validation mode is set to: 2 - even parity;
- Other settings as the default.



Diagram 6-9

- (5) Double click “Mitsubishi _FX series of programming”, to enter edit window, shown in diagram 6-10. the left menu is CPU 2-FX2NC CPU. The right menu “channel name” default is X000~X007, you can single click “delete all channels” button to delete.

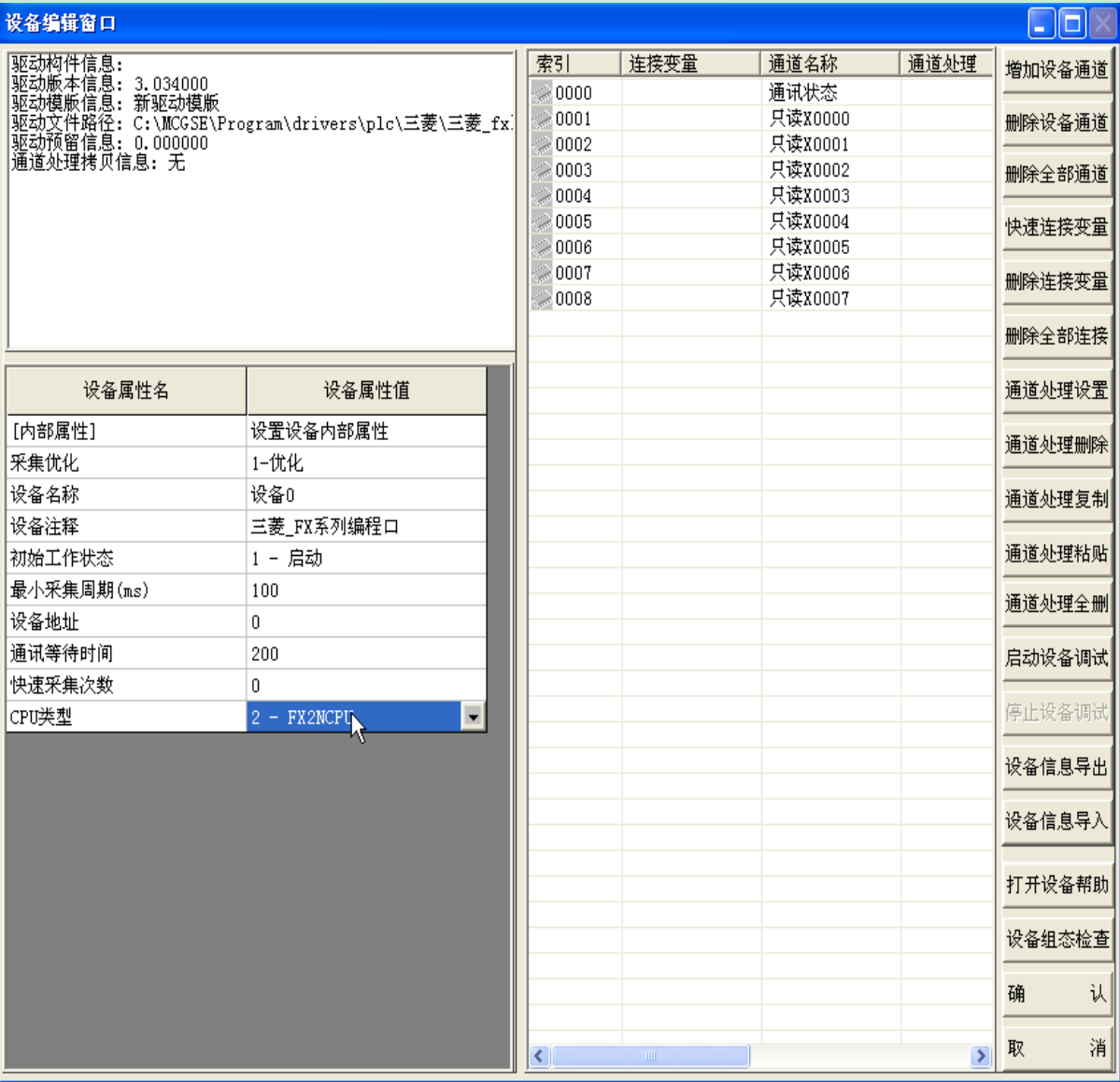


Diagram 6-10

(6)Connection of variable, the "running status" variable as an explanation with an example.

- ① Click the "Add Device Channel" button, the window appears as shown in diagram 6-11.

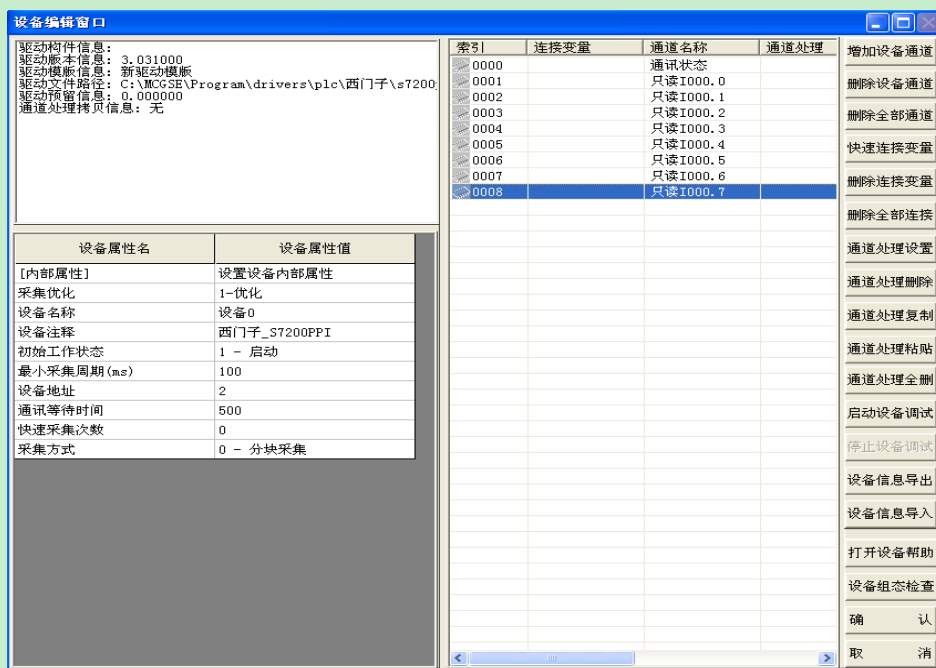


Diagram 6-11

Parameter settings are as follows:

- Channel type: M register ;
- Data types: channel No. 00;
- Channel Address: 0;
- Channel number: 1;
- Read-write mode: read-only.

② Click "Confirm" button to complete the basic attribute setting.

③ Double-click variables the "Read-only M000.0" channel corresponds to, select the variable from the data center: "running."

With the same method, add other channels, link variables, shown in diagram 6-12 to complete click "Confirm" button.

索引	连接变量	通道名称	通道处理
0000		通讯状态	
0001	运行状态	只读M000.0	
0002	单机全线切换	读写M000.1	
0003	启动按钮	只写M000.2	
0004	停止按钮	只写M000.3	
0005	清零累计按钮	只写M000.4	
0006	变频器频率给定	只写VWUB072	
0007	白芯金属工件累计	只写VWUB074	
0008	白芯塑料工件累计	只写VWUB076	
0009	黑色芯体工件累计	读写VWUB1002	

Diagram 6-12

4. The screen and components production

(1). New images, and property

- ◆ In the "User windows", click "New Window" button to create a "window 0." Select the "window 0", click "Window Properties" and enter to the user to set the window properties.
- ◆ Change the name of the window to: Sorting picture; the window should read: Sorting pictures
- ◆ Click the "window background" in the "Other Color" and select the desired color, as shown in diagram 6-13.

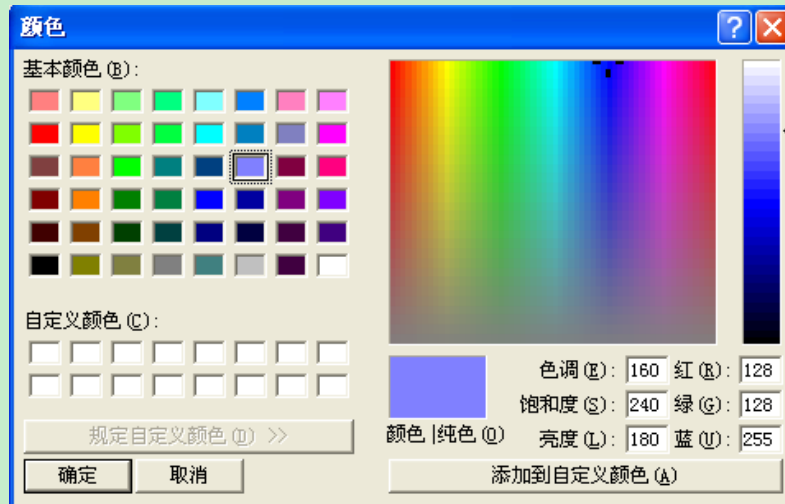


Diagram 6-13

(2). Create Text Block: The title of the production of the text as examples.

- ◆ Click the Tools section in the "tool box"  button, open the drawing toolbox.
- ◆ Select "tool box" in the "tab" button , the mouse cursor was "cross"-shaped, drag the mouse on top of the center of the window, pulled out a rectangle the size of fit according to the need.
- ◆ To enter text at the cursor blinking position, "sorting station interface", press the Enter key, or anywhere in the window with the mouse click, complete text input.
- ◆ Select the text box, make settings as following:
 - Click on the  (fill color) button of toolbar, set the background color: white;
 - Click on the  (line color) button of toolbar, set the text box edges color: No foul line;
 - Click on the  (and character font) button of toolbar, set the text fonts:Chinese thin black; font as follows: bold; size: II;
 - Click on the  (and character color) button of toolbar to set the text color: navy blue.
- ◆ Other attributes of the text box set as follows:
 - Background color: background color same as the screen
 - Sideline Color: No foul line;

Text fonts: Chinese thin black; fonts as follows: normal; font size as follows: II;

(3) Status Indicator. The "single / full range " Indicators as examples:

- ① Click the  (insert component) icon in the Drawing toolbox, pop-up object Element Management dialog box, select the indicator 6, press "Confirm" button. Double-click the indicators, pop-up dialog box as shown in diagram 6-14:

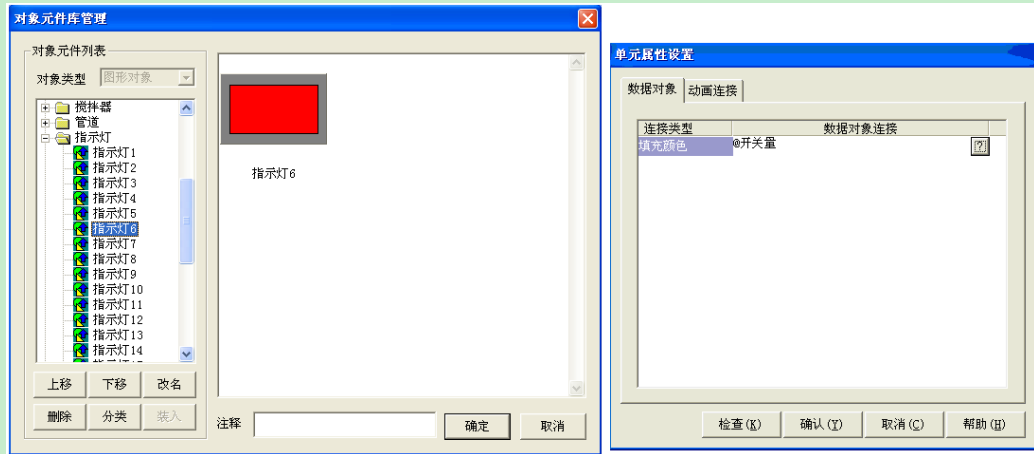


Diagram 6-14

- ② Data object, click the right corner of the "?" Button, select "Single line of switch" variable from the data center.
- ③ Animation connection, click the "Fill Color"  button on the right, as shown in diagram 6-15.

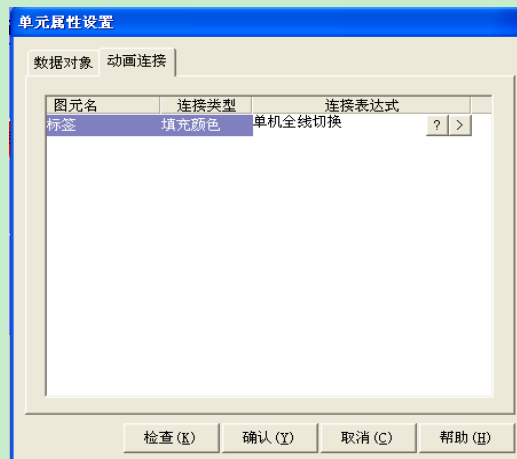


Diagram 6-15

- ④ Click the ">" button, appears the following dialog box. As shown in diagram 6-16.

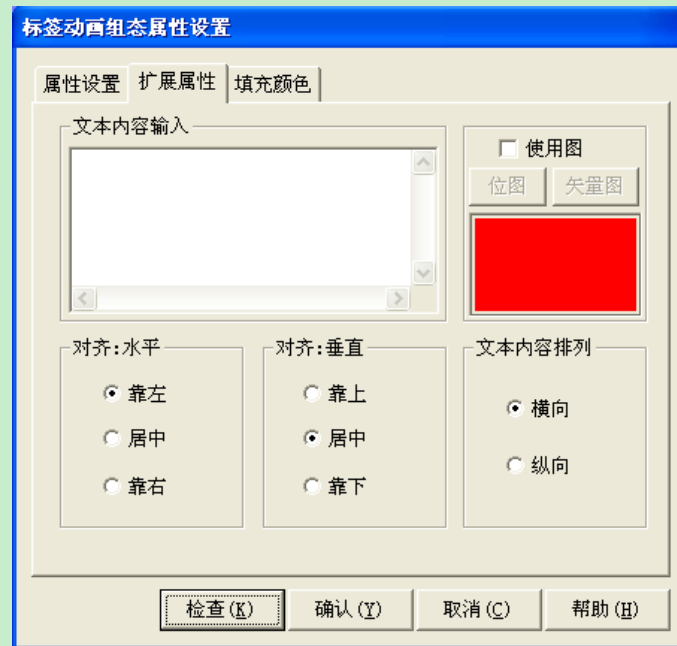


Diagram 6-16

- ⑤ "Properties Setting" page, fill color: white;
- ⑥ "Fill Color" page, sub-point 0 corresponds to the color: white; sub-point 1 corresponds to the color: light green. As shown in diagram 6-17, click the "Confirm" button to finish.

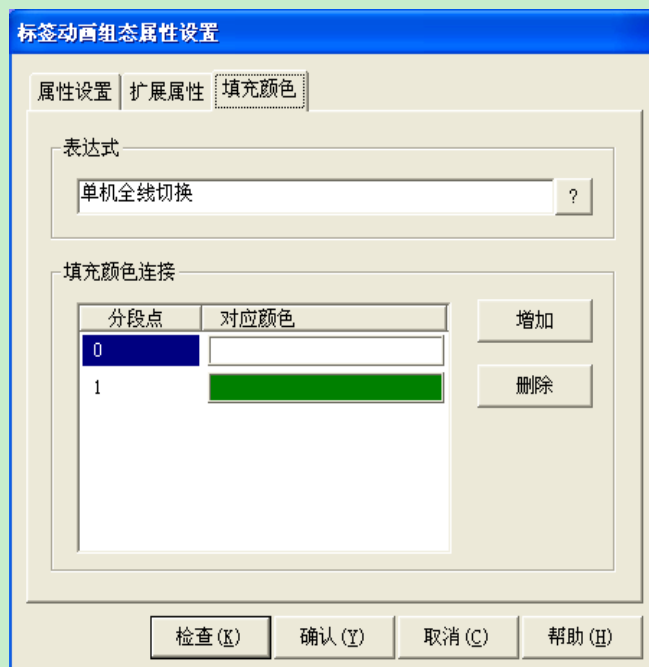


Diagram 6-17

(4). Production switch knob

Click the Drawing icon  (insert component) in toolbox, pop-up object Element Management dialog box, select switch 6, press "Confirm" button. Double-click the knob, the pop-up dialog box in diagram 6-18. Button on the page in the data object type and visibility of connection data object "stand-alone switch across the board."

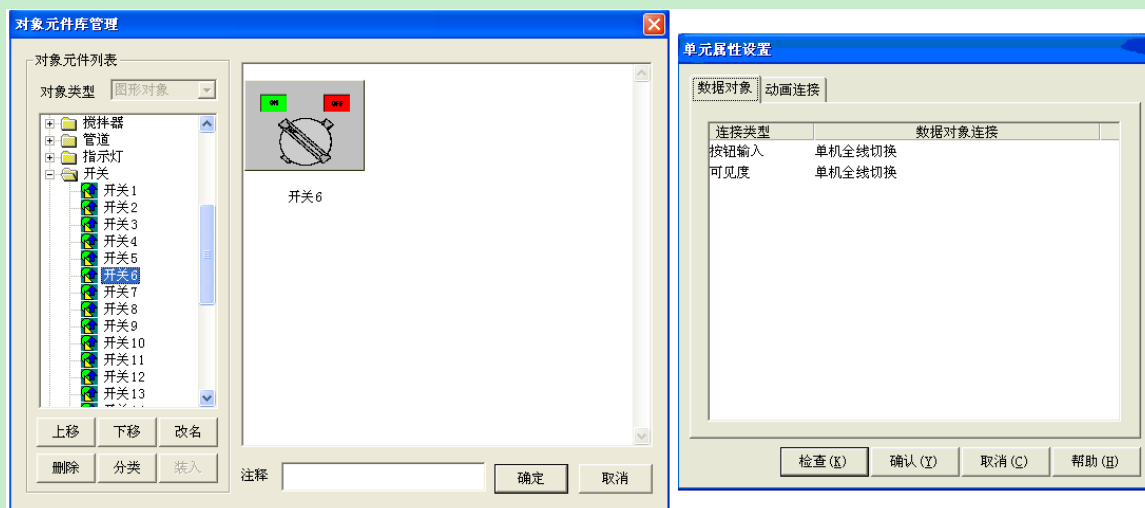


Diagram 6-18

(5).Production Button,e.g.the start button

- ① Click, " icon in the the drawing toolbox, drag a button to the right size in the window, double-click the button, the following figure window, and property are as follows:

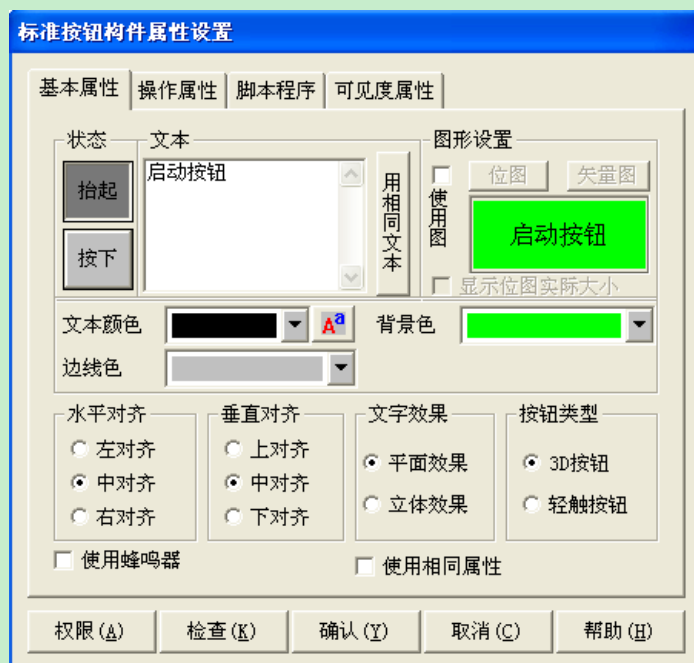


Diagram 6-19

- ② in "Basic Properties" page, whether pop-up or pressed, the text are set to start button; "Pop-up function" attribute to set to the font Arial, font size is set to the 5th, the background color is set to light green; "Press function" as follows: the font size set to small

5th, the other with pop-up feature.

- ③ In "Action Properties" page, pop-up function: data objects operating-ching 0, start button; Press features: data object operation set 1, the start button.
- ④ Other default. Click "Confirm" button to finish.

(6). Value input text

- ① Select the "input box"  icon in "tool box", drag the mouse, draw an input text.
- ② Double-click the icon "input box" to set property. Only need to set the action attribute:
 - Data Object Name: the highest frequency setting;
 - The units : Hz
 - min : 40 ;
 - max : 50 ;
 - Decimal places : 0。

Setting result as diagram 6-20 :

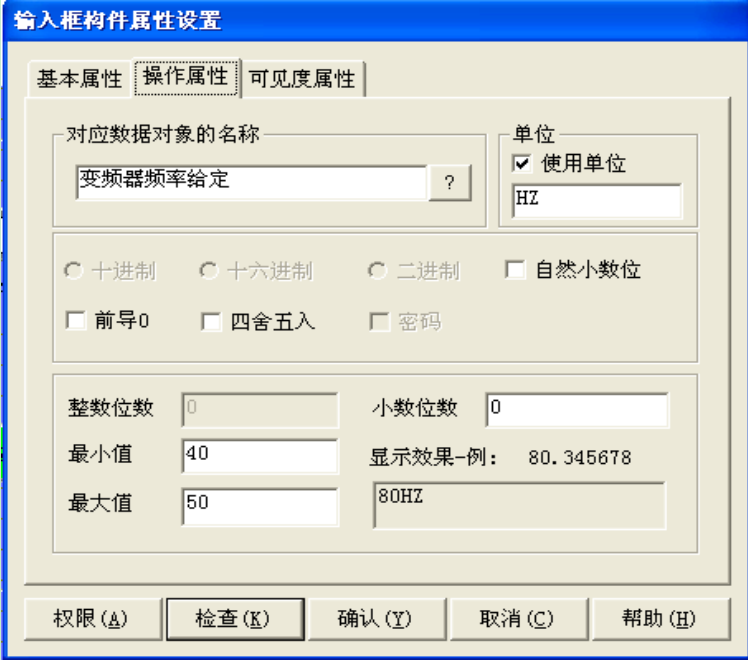


Diagram 6-20

(7). Data show that take the white metallic material accumulated data as an example:

- ① Select the icon  in the "tool box", drag the mouse, draw a display box.
- ② Double-click the display text, appears a dialog text, in the input and output connections in the domain, select the "Show Output" option, then in the Configuration Properties settings window will appear "Display Output" tab, shown in diagram:



Diagram 6-21

③ Click the "Show Output" tab, set the display output properties. Parameter settings as follows:

- Expression: white metallic material accumulated;
- unit: pcs;
- Output Value type: numeric output;
- Output format: Decimal;
- Integer digits: 0;
- Number of decimal places: 0.

④ Click "Confirm" to produce finished.

(8). makerectangular text

Click the icon  in the Toolbox, on the top left of the window pulled out a rectangle, double-click the rectangle, appears as shown in diagram 6-22, the property set as follows:

- Click on the  toolbar of the (fill color) button, set the background color of a rectangular text: no filling;
- Click on the  toolbar of the (line color) button, set the rectangular text sideline color: white;
- Other default. Click "Confirm" button to finish.



Diagram 6-22

6. Download see in 6.1.1.

6.2.2 Inverter output analogy control

To achieve inverter output frequency continuous adjustment, the distribution station **PLC** is connected to special functions analogy module **FX0N-3A**. The start/stop is controlled by external ends. Then inverter parameters are adjusted, as shown in the table 6-1:

table 6-1 inverter parameter set

Parameters	Parameter name	Default value	set value	Remark
Pr.73	analogy input selection	1	0	0--10V
Pr.79	Run mode selection	0	2	external run mode

FX0N-3A functions and wiring

1. Special functions module FX0N-3A main performance

FX0N-3A has 2 input channels and 1 output channel, the maximum resolution is 8-bit analogy I/O module, analogy input and output mode can be voltage or current mode, or user wiring mode.

FX0N-3A input channel main performance is shown in table 6-2, output channel main performance is shown in table 6-3.

Table 6-2 FX0N-3A input channel main performance table

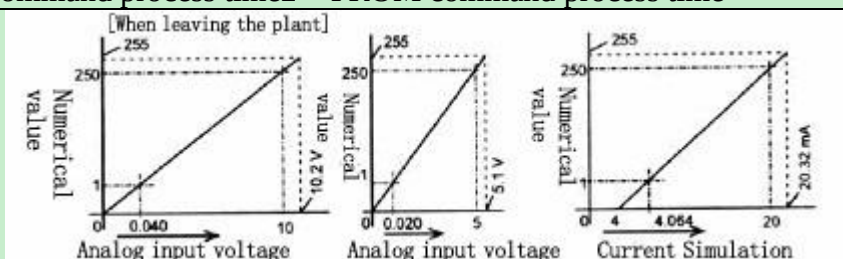
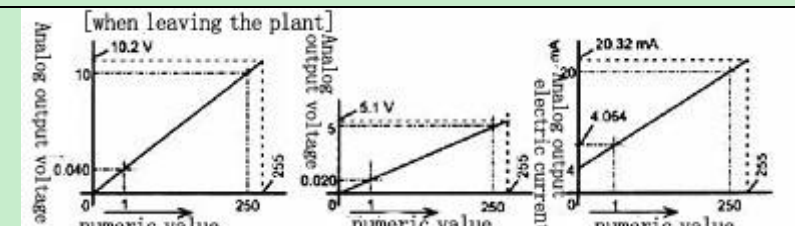
	Voltage input	Current input
Analogy input range	Initial, 0 to 10VDC input 0 to 250range. if FX0N-3A current input or 0 to 10V voltage input , you shall adjust voltage and gain.	
	0~10V,0~5V DC,input resistor is 200K Ω note: input voltage exceeds -0.5V, +15V	4~20mA,input resistor 250 Ω . note: input current exceeds -2 mA, +60 mA
Digital resolution	8bit	
Minimum input signal resolution	40mV: 0~10V/0~250	64 μ A: 4~20mA/0~250
Resolution	$\pm 0.1V$	± 0.16 mA
Process time	TO command process time2 + FROM command process time	
Input		

Table 6-3 FX0N-3A output channel main performance table

	voltage output	current output
analogy output range	0 to 10VDC outputselection 了 0 to 250range. if FX0N-3A current output or 0 to 10V voltage output, you shall adjust bias and gain.	
	0~10V, 0~5V DC, external load is : 1K Ω to 1M Ω	4~20mA, external load: 500 Ω or less
digital resolution	8bit	
minimum output signal resolution	40mV: 0~10V/0~250	64 μ A: 4~20mA/0~250
Resolution	$\pm 0.1V$	± 0.16 mA
process time	TO command process time $\times 3$	
output		

Note:

- ① Power supply module is from **PLC** master station circuit, analogy circuit power supply is 24VDC $\pm 10\%$, 90mA, Digital circuit power supply is 5VDC 30mA.
- ② analogy and digital circuit are insulated, but analogy channel is not insulated.
- ③ 8 I/O (input or output)

2. Wiring

The analogy input and output wiring schematics are shown in diagram 6-23, 6-24. in wiring, make current input, end [Vin] and [Iin] shall be connected; make current output, you shall not connect [VOUT] and [IOUT] end.

If voltage input/output voltage changes or noise are great, you shall connect 25V, 0.1 to 0.47 μ F capacitor.

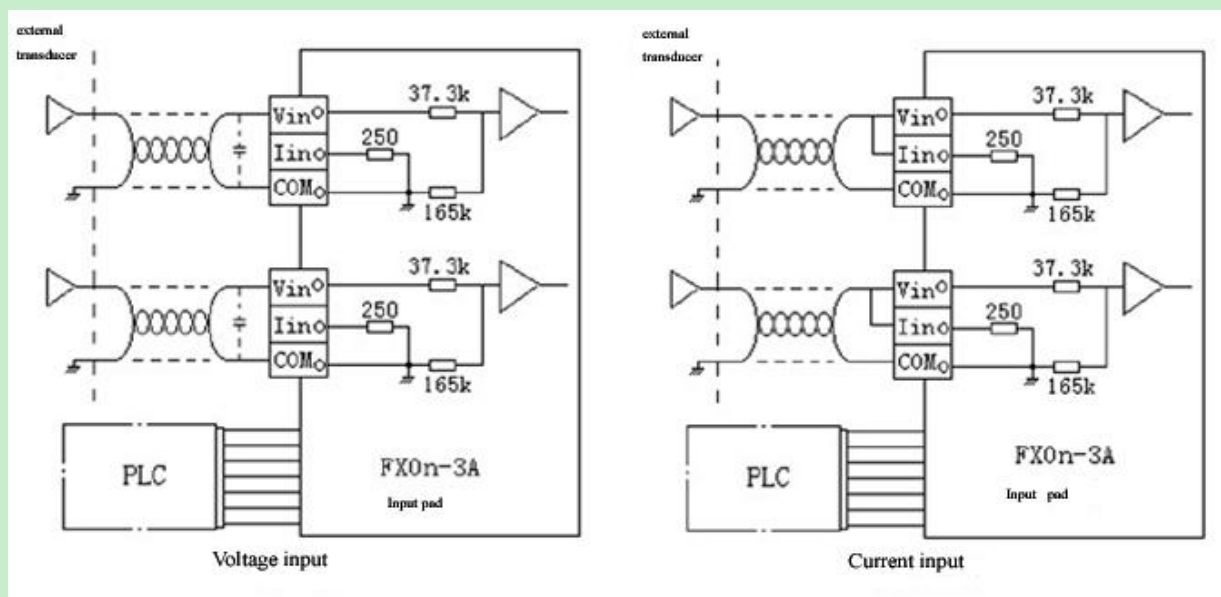


Diagram 6-23

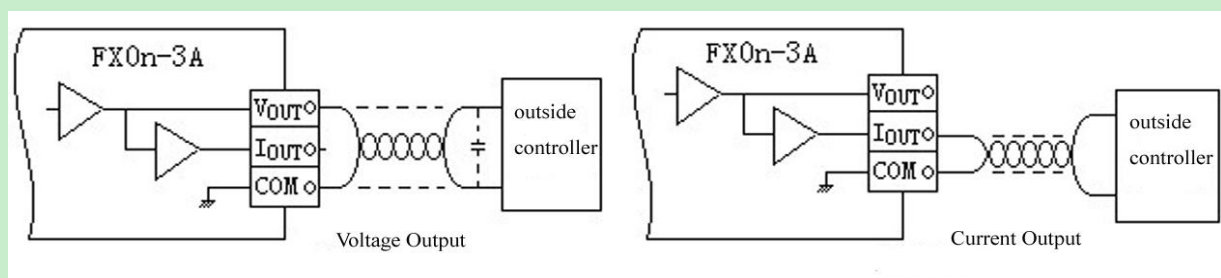


Diagram 6-24

3. Programming and control

Make special functions module read command **FROM(FNC78)** and write command **TO(FNC79)** read/write **FX0N-3A** module analogy input and output.

FROM command from special functions module BFM, read data, as shown in diagram 6-25(a). The module number is m1 special functions module, from BFM number is m2, data is read into PLC, saved from **D** data register.

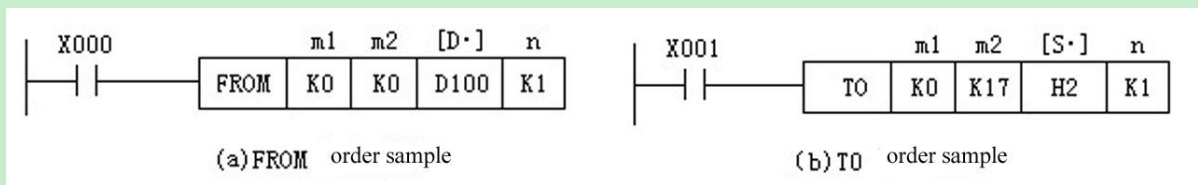


Diagram 6-25

TO command from PLC special functions module BFM write data, shown in diagram 6-25(b).

The module number is from PLC No.0→No.1→No.2..... sequence connection, module number FROM/TO command module work.

Special functions module is by BFM and PLC, **FX0N-3A** totally 32 channel 16bit **BFM**, as shown in table 6-14.

Table 6-14 FX0N-3A BFM

Channel number	b15-b8	b7	b6	b5	b4	b3	b2	b1	b0
#0	reserved	currentinput channelA/D value							
#16		Current D/A output channelset value							
#17							D/A start	A/D start	A/D channel selection
#1~#15 #18~#31	Reserved								

#17channel bit meaning:

b0=0, selectionanalog input channel1;b0=1, selectionanalog input channel2.

b1 from 0 to 1, A/D start

b2 from 1 to 0, D/A start

Diagram 6-26 is D/A example, Diagram 6-27 is A/D example

Example 1: write module number is 0FX0N-3A module, D2 is D/A value.

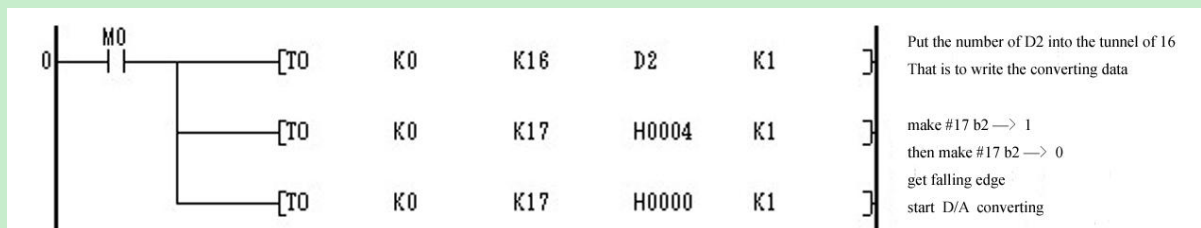


Diagram 6-26

Example 2 : read modulenumber is 0 **FX0N-3A** module, channel 1 A/D value is saved to D0, channel 2 A/D value is saved to D1.



Diagram 6-27

The sorting station inverter speed ajustment program as follow:

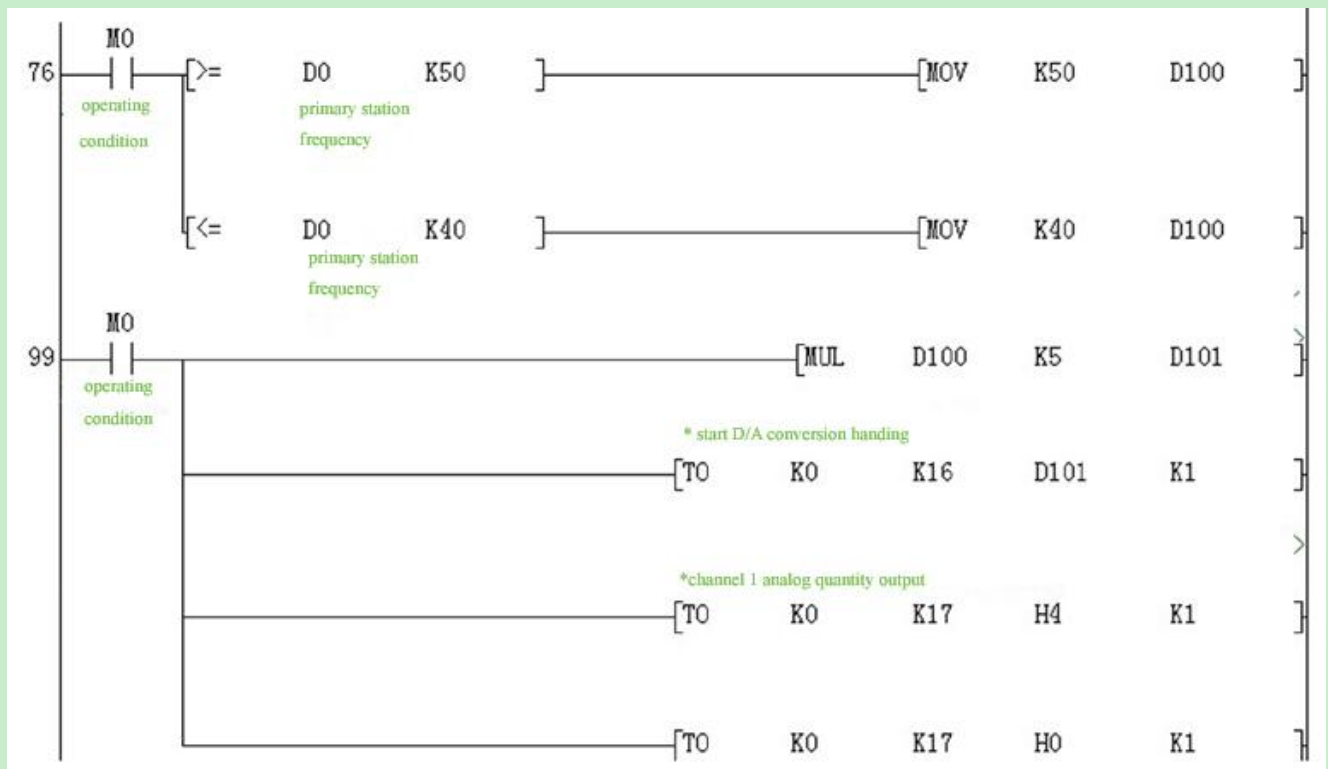


Diagram 6-28

ITEM7.Transmission unit control system training

7.1 Transmission unit structure and working process

Transmission unit process functions are: to drive its grasping robot precise positioning device to a designated cell material sets the stage grasping in workpiece material, the captured workpiece transported to the designated place and then to lay down their functions. YL-335B factory configuration, the transport unit in the network system is a main role, it receives system main order signal from the touch-screen, reading the various network status information of the station, to be a comprehensive treatment, then to all subsidiary stations to send control requirements, coordinating the the entire system work.

Transmission unit consists of grasping robot devices, linear motion transmission components, tow equipment, PLC module and wiring ports, as well as the button / indicator module parts. As shown in diagram 7-1 is delivery unit device-side part installed on a countertop.

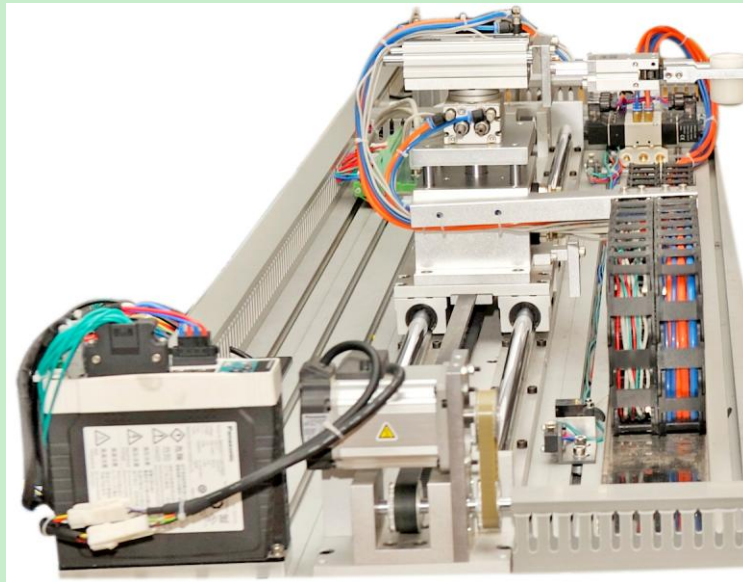


Diagram 7-1

1. Grasping robot device

Grasping robot device is the work unit to achieve a 3-DOF motion (lift, telescopic, pneumatic finger clamping / release and along the vertical axis of the four-dimensional movement), the unit is installed in the whole transmission components, linear motion slide board, led by the drive components for linear reciprocating motion as a whole, navigate to the other units of work materials units, and then complete the crawl and drop function of the workpiece. Diagram 7-2 is the device physical diagram.

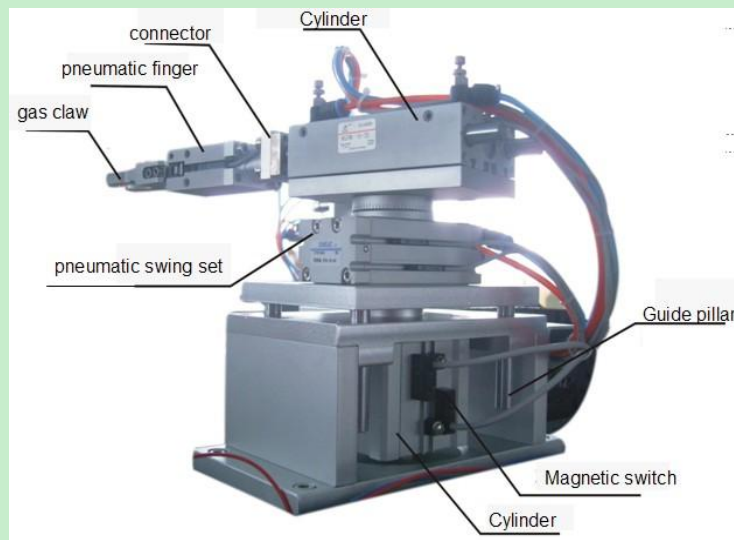


Diagram 7-2

The specific composition is as follows:

- (1) Pneumatic gripper: used to grab / drop down workpiece on various workstation material platforms. control by a 2-way 5 position bi-direction electric control valve.
- (2) extension cylinder: used to drive arm move, controlled by an electric solenoid.
- (3) rotary cylinder: used to drive arm rotate in 90 degrees forward and back, controlled by an electric solenoid.
- (4) lift cylinder: used to control mechanical arm lift and drop, controlled by an electric solenoid.

2. Linear movement drive assembly

The linear movement drive assembly is used to drive catch mechanical arm in linear movement, to finish precise positioning function. The diagram 7-3 is the assembly.

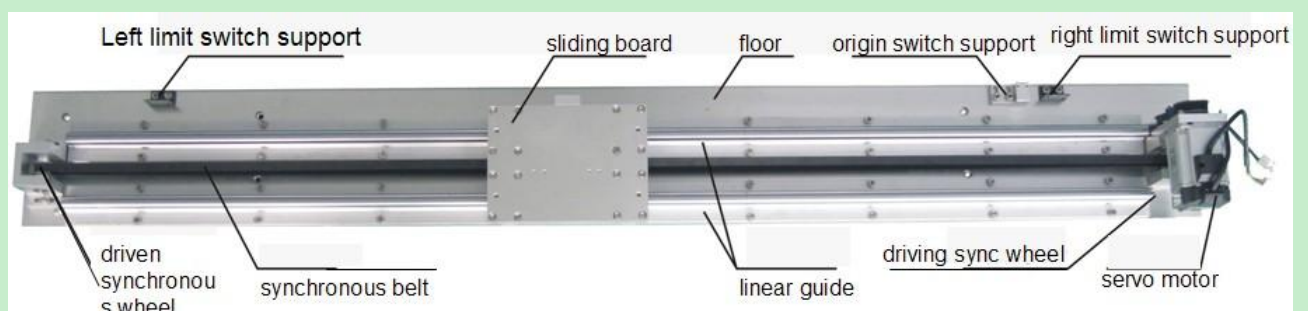


Diagram 7-3

The diagram 7-4 is linear movement drive assembly and catch mechanical arm composition diagram .

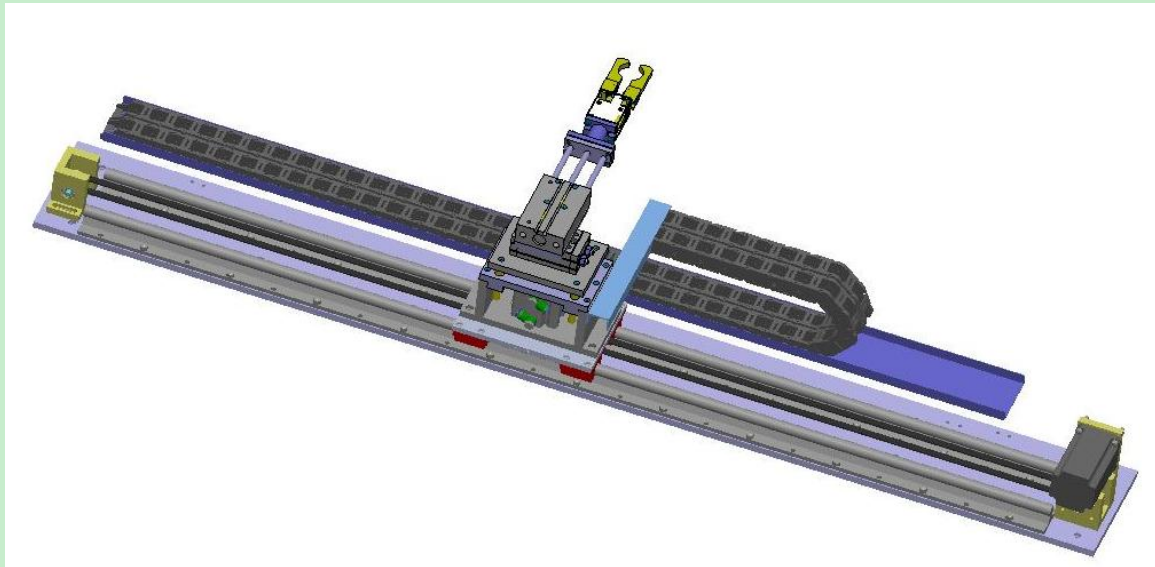


Diagram 7-4

The drive assembly is composed of linear rail base, servomotor, servo amplifier, synchronous wheel, synchronous belt, linear rail, sliding base, link and origin proximity switch , left, right proximity switches.

The servo motor is driven by servomotor amplifier, through synchronous wheel and synchronous belt, sliding base, linear rail, in forward-back linear movement. Fixed on the sliding base, catch mechanical arm will start forward-back linear movement. The synchronous wheel tooth clearance is 5mm, so totally 12 teeth displacement is needed to rotate for a cycle to transfer the mechanical arm at displacement 60mm.

The catch mechanical arm's tubes and wires are connected to solenoid wiring ends underside.

The origin proximity switch and left, right proximity switches are installed on the linear rail base, as shown in diagram 7-5.



Diagram 7-5

The origin proximity switch is a contactless inductive proximity sensor, to provide linear

movement initial origin signal. For the inductive proximity sensor fundamental, the precautions are mentioned before in project one.

The left, right proximity switches are contact proximity switches, as the over-travel fault protection signals: when sliding base travels longer than left or right proximity positions in movement, the proximity switch will work and send overtravel fault signal to the system.

4. Pneumatic control circuit

The transfer station catch mechanical arm, all cylinder tubes are connected to the solenoids, the pneumatic control circuit is shown in the diagram 7-6.

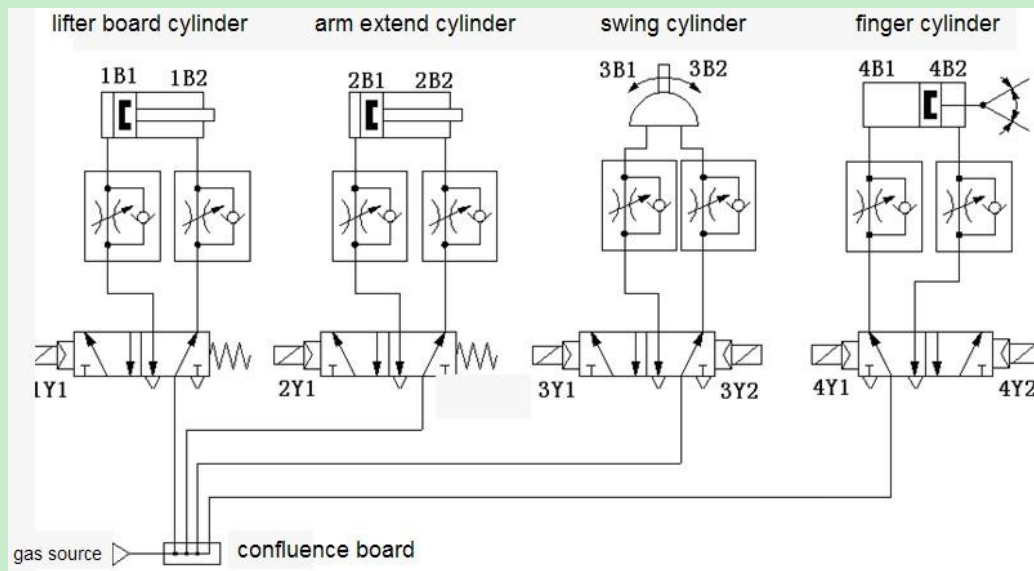


Diagram 7-6

In the pneumatic control circuit, the cylinder solenoid drives the sway cylinder and pneumatic arm, and the solenoid is shown in diagram 7-7.

The double electric control solenoid and single electric control solenoid difference: for the single electric control solenoid, when no power signal, the single electric control solenoid will be reset, but for the double electric control solenoid, when there are no electric control signals on the double sides, the double electric control solenoid position is determined by the first electric control signal.

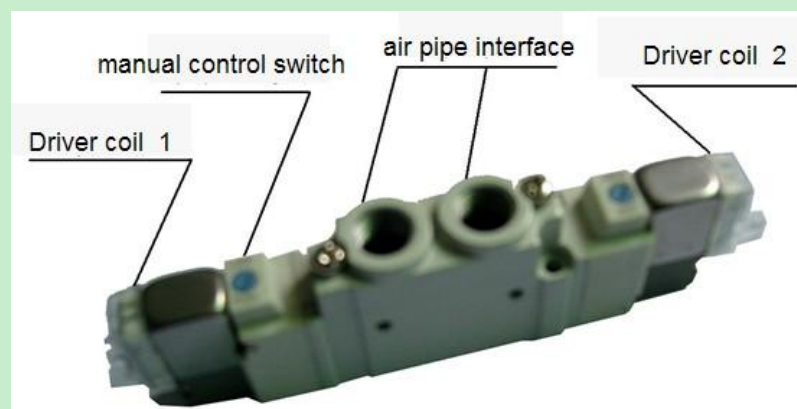


Diagram 7-7

Note: the double electric control solenoid signals cannot be “1” in one time, in control, it is not allowed to make the two coils powered in one time, otherwise, the the electric coils maybe burnt, and in this case, the solenoid position is not determined.

7.2 Transmission station installation skills training

1. Target

Decompose the transfer station mechanical parts, then assemble again. You shall learn the skills and methods in mechanical equipment installation, adjustment and electrical wiring.

2. Mechanical part installation steps and methods

To improve installation speed and accuracy, you shall install stations one by one.

(1). Compose linear movement assembly:

① The linear rail is a precise mechanical movement part, the installation and adjustment shall be done in correct methods and steps, the rail is long, you shall adjust the relative position of the two rails, for stable movement and little noise.

② To assemble base, and sliders: fix properly with bolts, till the sliding is smooth.

③ To connect synchronous belt: take 4-slider base out from the rail. Install the two synchronous belts on the back of the base.

Then you shall adjust the synchronous wheel, install rack assembly, the synchronous wheel shall be installed on the rack assembly, the synchronous wheel, shall be inserted into the synchronous belt, you shall pay attention that the synchronous wheel rack assembly installation direction, the relative positions, then fix the synchronous belt onto its own synchronous belt seat, in a same time, you shall keep the installed synchronous belt uniform and smooth. Then, on the rail, insert, do not damage the sliding balls and support in the sliders.

④ Synchronous wheel rack assembly: fix synchronous wheel and rack assembly with bolts properly on the rail base, then adjust connection of synchronous wheel rack assembly and base, and adjust synchronous belt tightness.

⑤ Servomotor installation: fix panel on motor rack position, connect motor, then connect synchronous wheel, install synchronous belt, adjust motor position, and fix bolts. Then install left, right proximity switches and origin sensor rack.

Note: among the parts, bearing and bearing seat are precise mechanical parts, so disassembly and assembly need good skills and tools, then, do not randomly disassemble or assemble. In the diagram 7-2 is the linear movement assembly.

(2). assemble mechanical arm. The steps:

① lift mechanism assembly is shown in diagram 7-8.

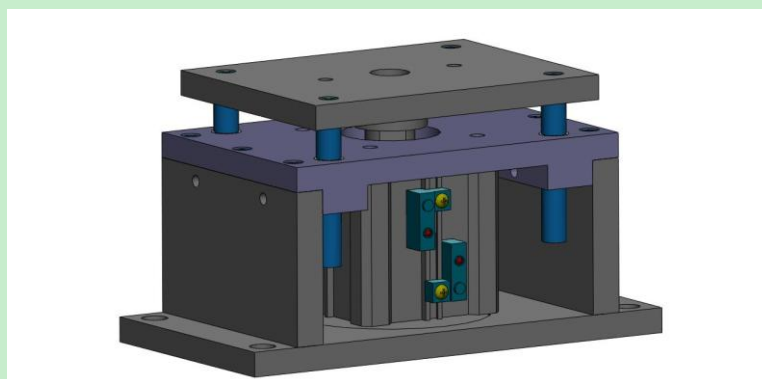


Diagram 7-8

② Fix the pneumatic sway bench on the lift mechanism, then on the pneumatic sway bench, fix lead cylinder base, find the origin positions for connection properly in advance for the lead cylinder base and the pneumatic sway bench.

③ Connect to pneumatic arm and lead cylinder, then fix the lead cylinder onto the lead cylinder base. And finish catch mechanical arm assembly

(3). Fix the catch mechanical arm to the linear movement assembly base, as shown in diagram 7-9. Then check sway bench, lead cylinder, pneumatic arm positions, and adjust properly.

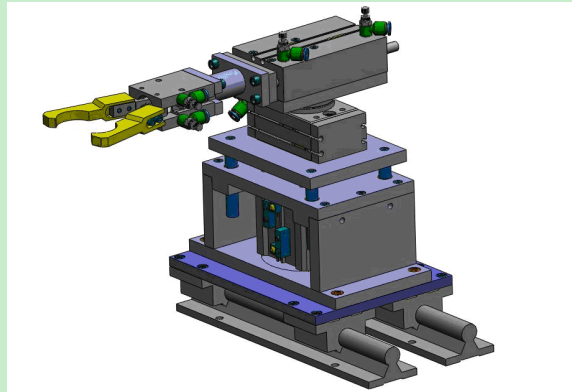


Diagram 7-9

3. Pneumatic circuit and electrical wiring

When catch mechanical arm is in forward-back movement, the tubes and wires connected also move. To ensure the tubes and wires connected move smoothly is important.

The tubes and wires connected to the mechanical arm are be long enough, and they shall not be intercrossed.

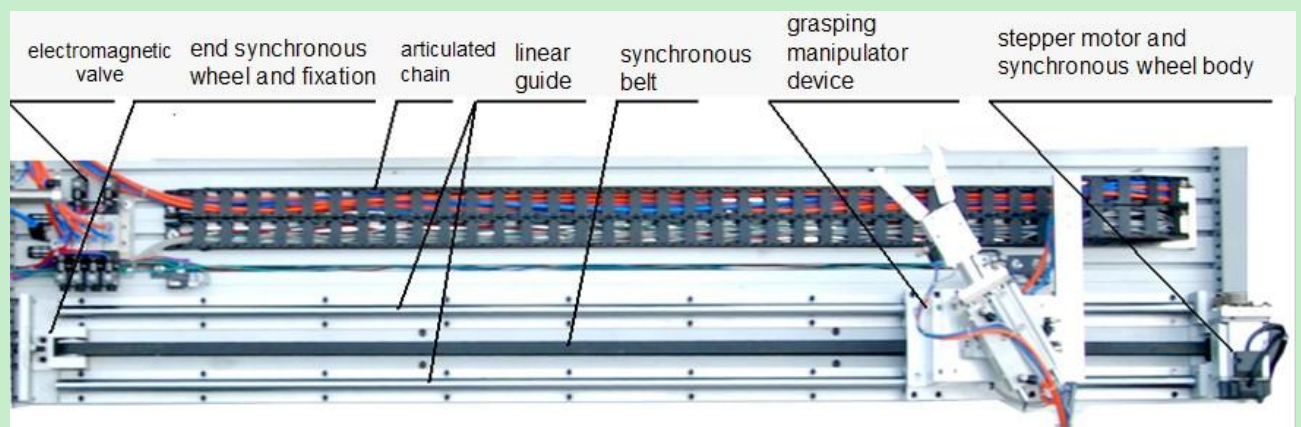


Diagram 7-10

7.3 Related points

In the transmission station, to drive catch mechanical arm in linear rail forward-back movement, the driver can be stepper motor, or servomotor. And for different projects, stepper motor and servomotor are used, and it is easy to replace.

The stepper motor and servo motor introduction as follow;

7.3.1 stepper motor and driver

1. Stepper motor

Stepper motor is a motor with electric pulse signal transformed into angle displacement or linear displacement. Each input electric pulse signal, motor will rotate in an angle, and its movement form is stepping, as the stepper motor.

(1) Stepper motor fundamental

We take a simple 3-phase reaction stepper motor as the example, to specify stepper motor fundamental.

In diagram 7-11 is a 3-phase reaction stepper motor schematic. There are 3 pairs of magnetic polarities (6pcs), each two correspond to a control winding. The rotor is made of soft magnetic material.

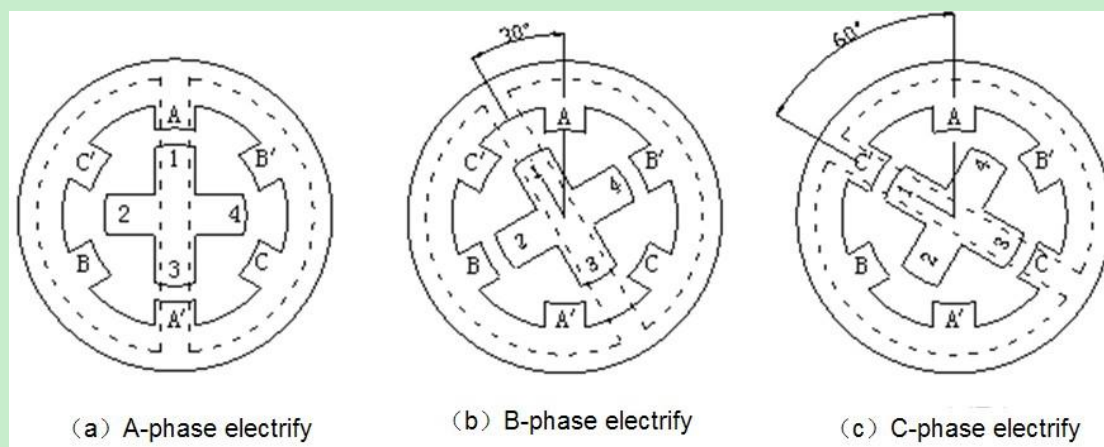


Diagram 7-11

When A-phase control winding is powered, other two phases are depowered, stator A-phase magnetic field is created. The rotor 1, 3 axes shall be directed with the stator A-phase polar axes, as shown in diagram 7-11(a). if A-phase control winding is depowered, B-phase control winding is powered, rotor, under the reaction torque, will be in the counterclockwise rotation by 30° ; in order to make the rotor tooth 2, 4 axes directed with the stator B-phase polar axes, a rotor step, as shown in diagram 7-11(b). if B-phase is off, C-phase control winding is on, rotor in counterclockwise direction 30° , to make the rotor tooth 1, 3 axes directed with the stator C-phase polar axes, as shown in diagram 7-11(c). in this way, A-B-C-A sequence, wheel is powered, rotor will rotate counterclockwise step by step. The rotation speed is determined by each control winding ON and OFF frequency, the rotation direction is determined by the control winding wheel powered ON sequence. If in A-C-B-A sequence powered, then the motor will be in clockwise direction rotation.

The electric powered mode is 3-phase mode. “3-phase” is 3-phase stepper motor; “single step” is in one time, only one phase control winding is powered ON; the control winding changes once, it is one step, “three steps” is changing power status for three times in a cycle. In each step, rotor rotation angle is the step angle. The 3-phase, single step, in this mode, step angle is 30° . Then the angle is too great to carry out.

If control winding is A→AB→B→BC→C→CA→A, for a cycle, it will change electric

power status for 6 times, 3-phase, and double 2-step power mode. When A-phase and B-phase windings are powered in a same time, only when A-phase polarity and B-phase polarity are in the magnetic pull force balance middle position, the rotor can be in the balanced position. Thus, in the 6-step power mode, the rotor balance positions will be more for one time, and the step angle is 15 °.

To reduce step angle, you shall reduce the quantity of the teeth in the stator magnetic polarity, more rotor teeth, more teeth structure, this structure stepper motor can has smaller step angle.

Besides step angle, there are other technical parameters of the stepper motor such as hold torque, damping torque and so on, these parameters meaning, please see more technical documents for stepper motor. In **3S57Q-04056**, is the **technical** parameter, as shown in table 7-1.

Table 7-1 3S57Q-04056 technical parameter

Parameter name	Step angle	Phase current(A)	hold torque	Damping torque	motor inertia
Parameter value	1.8 °	5.8A	1.0Nm	0.04Nm	0.3kg.cm2

(2) Stepper motor usage, correct installation and correct wiring are important.

To install stepper motor, you shall do according to specification. The stepper motor is precise, do not tap its shaft in installation, do not disassemble the motor.

For various stepper motors, wirings are different; **3S57Q-04056** wiring diagram is shown in diagram 7-12, the 3 phase winding, 6 leads, shall be connected. To change winding power-on sequence, you can change stepper motor rotation direction.

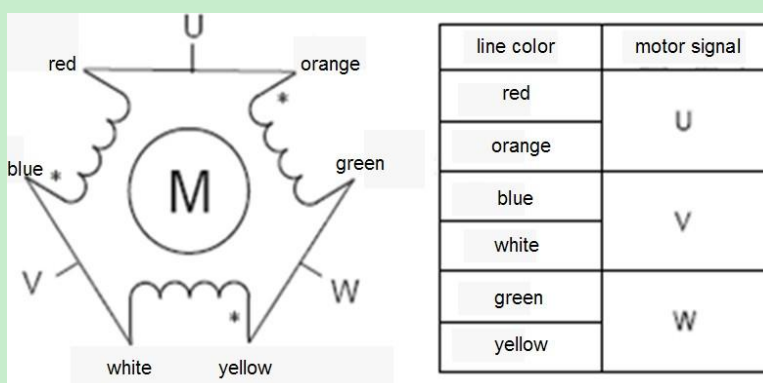


Diagram 7-12

2. Stepper motor driver

The stepper motor needs a driver power, the driver and stepper motor is an integration, the stepper motor performance is the composition of motor and driver performances.

Generally, a stepper motor needs a driver, **Kinco** 3-phase stepper motor **3S57Q-04056**, its

driver is **Kinco 3M458** 3-phase stepper motor driver. In diagram 7-13 and diagram 7-14 are the typical wiring diagrams. In the diagram, driver can adopt DC 24V~40V power supply. The power supply is transfer station switching DC power supply (DC24V 8A). The output current and input signal specification:

- ① Output phase current 3.0A~5.8A, output phase current, set by toggle switches; driver is of natural wind cooling;
- ② Control signal input current 6~20mA, control signal input circuit is optical coupledly insulated. The transfer station **PLC** output common end **Vcc** is for **DC24V** voltage, current limiting resistor: **R1** is 2K Ω .



Diagram 7-13

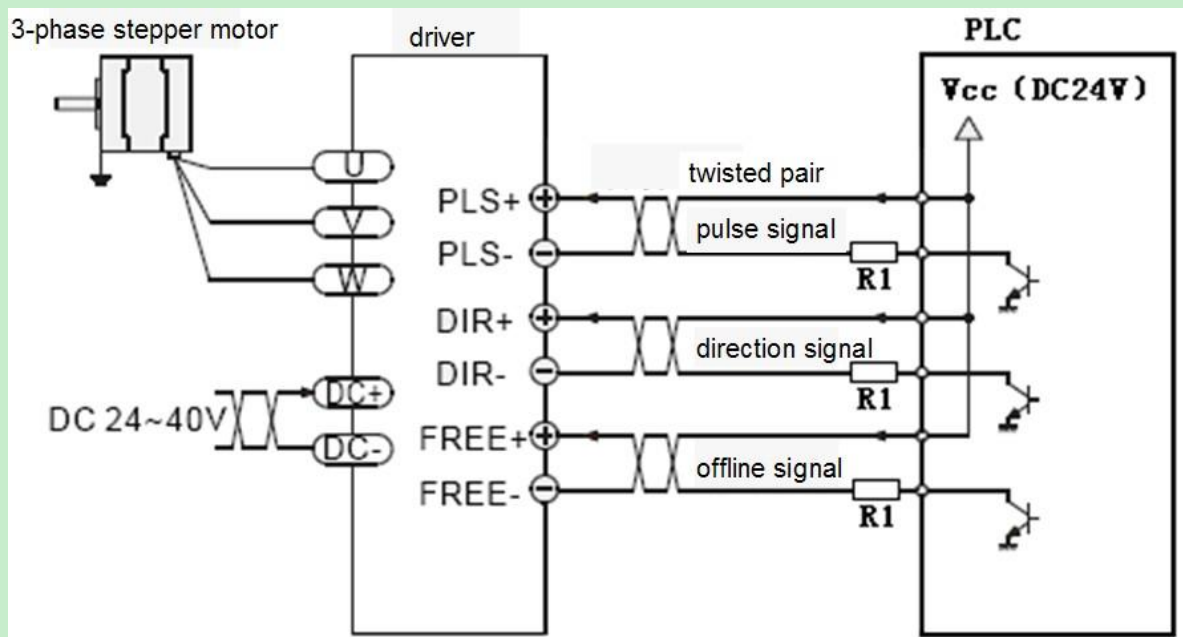


Diagram 7-14

See from the diagram, stepper motor driver function is to receive controller (PLC) frequency pulse signals, and motor rotation direction signal, stepper motor output 3-phase power pulse signals.

The stepper motor driver is composed of pulse distributor and pulse amplifier, for stepper motor each phase winding output pulse distribution and power amplification.

The pulse distributor is a logic station, it receives controller pulse signals and rotation signals, pulse signals are sent to each phase pulse amplifier, to make the stepper motor work in the specific mode. As the stepper motor each phase winding will cycle in a specific power sequence for stepping function, so the pulse distributor also can be called as the loop distributor.

The pulse amplifier is to amplify pulse power. So the pulse distributor can output small current, and stepper motor need great current in work, so power shall be amplified. Then, the output pulse waveform, amplitude, waveform, also affect stepper motor performance.

3M458 driver takes following actions to improve stepper motor performance:

- Internal drive DC voltage reach 40V.
- Motor static locked status: automatic half-current function, to reduce motor heating.

For easy debugging, the driver has a pair of off-line signal input **FREE+** and **FREE-** (see diagram 7-14), when the signal is **ON**, driver will turn off stepper motor power supply circuit.

- **3M458** driver is of AC servodrive fundamental, by PWM technology, to transform DC voltage into Sino waveform current, as shown in the diagram 7-15.

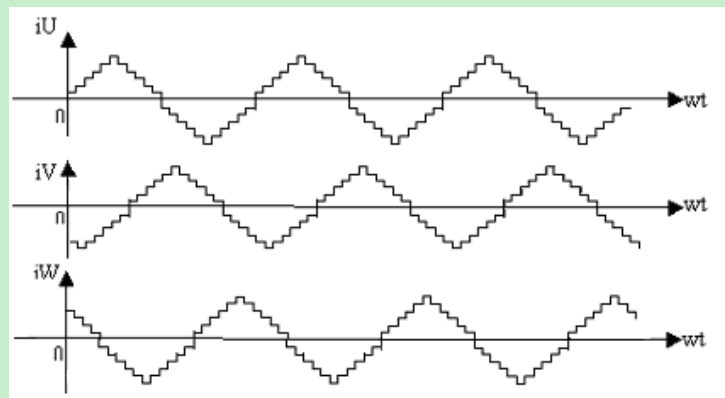


Diagram 7-15

The step Sino waveform current will flow through the three windings in a specific sequence. By changing driver output Sino current frequency, to change motor rotation speed, and the output step quantity will determine each step rotation angle, when angle is smaller, then the output step quantity will be more. **3M458** can achieve 10000steps/r in maximum for drive subdivision function, subdivision can be set by the toggle switches.

The subdivision drive can reduce stepper motor step angle, improve resolution, reduce low-frequency vibration, make motor running more stable and uniform.

On **3M458** driver side is a red 8-bit DIP function setting switch, can be used to set driver work mode and working parameters, including subdivision setting, static current setting and running current setting. In the diagram 7-16 is the DIP switch function specification, table 7-2 (a) and (b) are current setting table.

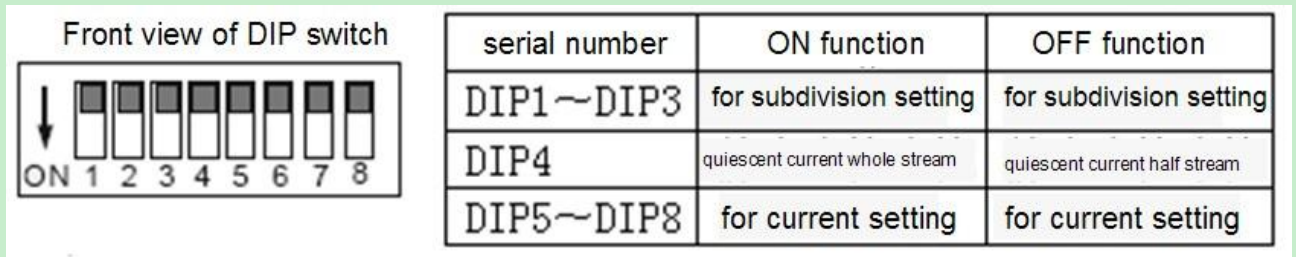


Diagram 7-16

Table 7-2(a) setting

DIP1	DIP2	DIP3	
ON	ON	ON	400steps/r
ON	ON	OFF	500steps/r
ON	OFF	ON	600steps/r
ON	OFF	OFF	1000steps/r
OFF	ON	ON	2000steps/r
OFF	ON	OFF	4000steps/r
OFF	OFF	ON	5000steps/r
OFF	OFF	OFF	10000steps/r

Table 7-2(b) output current setting

DIP5	DIP6	DIP7	DIP8	output current
OFF	OFF	OFF	OFF	3.0A
OFF	OFF	OFF	ON	4.0A
OFF	OFF	ON	ON	4.6A
OFF	ON	ON	ON	5.2A
ON	ON	ON	ON	5.8A

The stepper motor drive assembly basic technical data:

3S57Q-04056 stepper motor step angle 1.8 degrees, without subdivision, there will be 200 pulse motor rotations.

For **YL335-B** system stepper motor driver, the initial driver setting: 10000steps/r. The linear movement assembly synchronous wheel tooth clearance is 5mm, totally 12 teeth, a rotation cycle will drive mechanical arm displacement 60mm. And each step will achieve mechanical arm displacement 0.006mm; the motor drive current is set to 5.2A.

3. Problems of stepper motor

To control the stepper motor running, pay attention to problems of stepper motor step loss or over step in running.

The stepper motor step loss or over step. In case of step loss, the rotor step quantity will be less than the quantity of the pulses, in case of over step; the rotor step quantity will be more than the quantity of the pulses. In case of serious step loss, the rotor will stop will vibrate around one position; In case of serious over step, the equipment will be shocked.

When the mechanical arm returns to origin, over step often occurs. When the mechanical arm returns to origin, origin switch works, to turn command input **OFF**. If before it arrives the origin, the speed is too high, inertia torque will be greater than stepper motor to maintain torque Ershi stepper motor more steps. So back to origin should ensure that adequate

low-speed operation should; when stepper motor drive mechanical arm assembly of high-speed operation when the emergency stop, the more steps there the situation is inevitable, it should be taken after the first emergency stop reset, and it will be in low-speed to return to origin for adjustment.

As the motor winding itself is inductive load, input frequency higher excitation current smaller. Frequency high magnetic flux change intensified, eddy current loss increase. Therefore, input frequency increased, output torque reduction. Maximum operating frequency output torque can only achieve the low-frequency torque 40-50%. for high-speed positioning control, if you specify the frequency is too high, there will be the problem of step loss.

Then, if mechanical part adjustment is incorrect, the mechanical load will be greater. The stepper motor cannot run in overload condition, if it is serious, it will vibrate in one position quite irregularly and repeatedly.

7.3.2 Servo motor and servo amplifier

7.3.2.1 Permanent magnetic AC servo system

The modern servo systems, mostly are permanent magnetic AC servo systems, including permanent magnetic synchronous AC servomotor and digital AC permanent magnetic synchronous servodriver.

1. AC servomotor fundamental: servomotor internal rotor is a permanent magnet, the driver control U, V, W 3-phase power forms the electric magnetic field, rotor, in this magnetic field, rotates, then the encoder feedback signals will be sent to the driver, the driver, comparing feedback data and target data, will adjust rotor rotation angle. The servomotor precision is determined by the encoder precision.

AC permanent magnetic synchronous servodriver mainly is composed of servo control station, power drive station, communication station, servomotor and feedback unit, the servo control station is composed of position controller, speed controller, torque and current controller. The structure is in diagram 7-17.

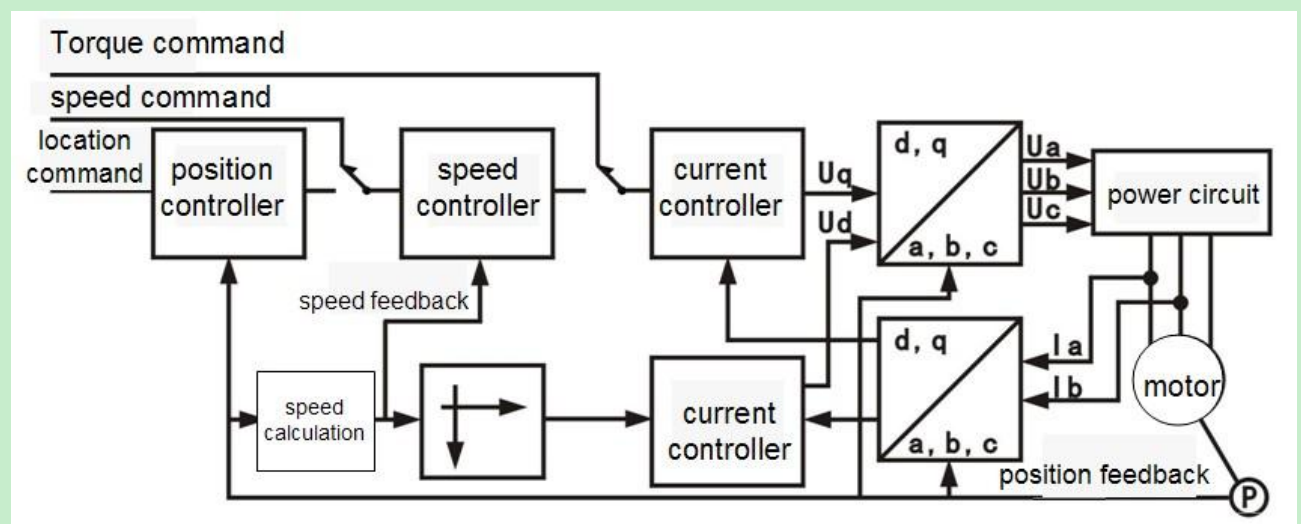


Diagram 7-17

The servo driver has digital signal processor (**DSP**) as controller, for complicate control, to be digital and intelligent. The power devices generally are in intelligent power module (**IPM**) designed drive circuit, IPM internal IC drive circuit, with over-voltage, over-current, over-heat, under-voltage faults protection circuits, in the main circuit, soft start circuit is made to reduce shocking in start.

The power drive station, with rectification circuit, for input 3-phase power, to obtain DC power. Then by 3-phase Sino PWM voltage inverter, to drive 3-phase permanent magnetic synchronous AC servomotor .

The inversion part (**DC-AC**) adopts power IC drive circuit, to protect circuit and power switch intelligent power module (**IPM**), the main structure is of 3-phase bridge circuit, the schematic is in diagram 7-18. By **PWM** (Pulse Width Modulation), to change power transistor transition On time and to change inverter output waveform frequency, to change each half cycle transistor duty ratio, that is to say, to change pulse width, is to change inverter output voltage, in order to adjust power.

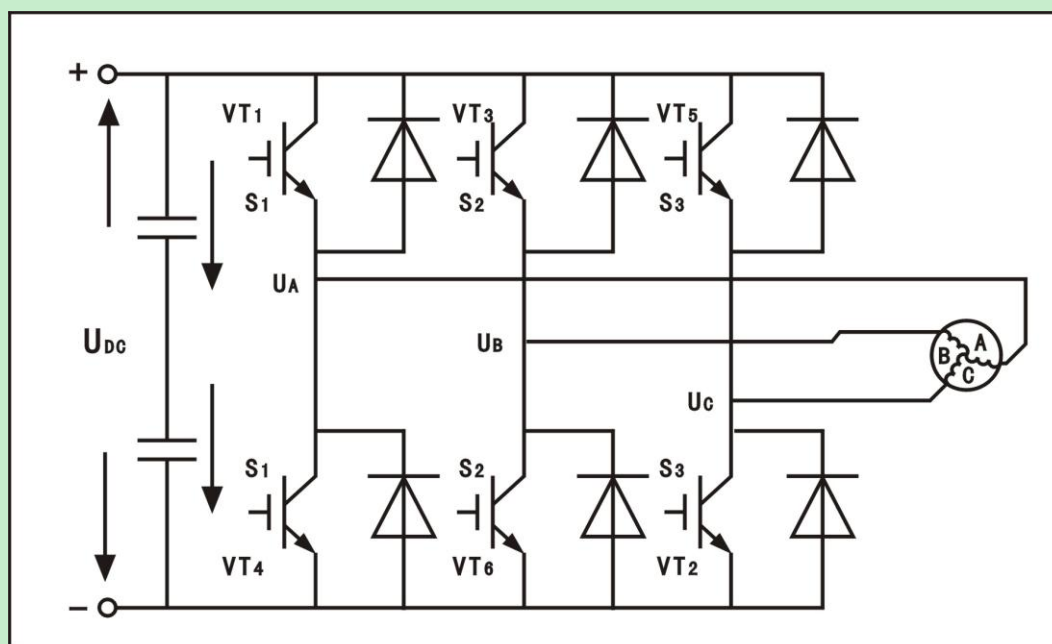


Diagram 7-18

2. AC servo system position control mode

In diagram 7-17 and diagram 7-18:

- (1) servo driver output to servomotor: 3-phase voltage waveforms mostly are Sino waveforms, not stepper motor 3-phase pulse sequence.
- (2) When the servo system is used for position control, the position command is input to position controller, and the speed controller input electric switch will be switched to the position controller output, in a same way, current controller input electric switch will be switched to the speed controller output. Therefore, in position control mode, the servo system is a close-loop control system, the two internal loops are current loop and speed loop. From automatic control theory, we can know that the system structure can improve system performance greatly. With an enough open-loop gain, stable system error is almost zero. That is to say, the servomotor in order to command pulse and feedback pulse approximation-phase, etc. speed run. On the contrary, before steady state is reached, the system with error signal will be under the action of drive motor acceleration or deceleration. If the command pulse suddenly disappeared, servomotor will run to the feedback pulse number is equal to the

number of command pulse disappeared before the pulse stops.

3. position control mode, electronic wheel tooth

In position control mode, the close-loop system is shown in diagram 7-19.

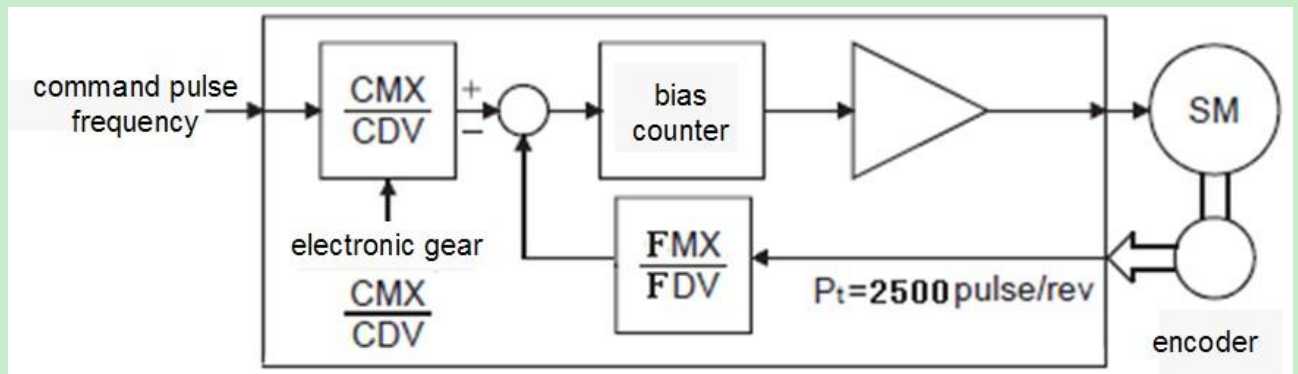


Diagram 7-19

In the diagram, command pulse signal and motor encoder feedback pulse signal enter the driver, and then through the electronic tooth wheel conversion offset calculation. The electronic tooth wheel actually is a frequency multiplier, a proper arrangement can make the command pulse travel setting flexible.

7.3.2.2 Panasonic MINAS A4 series AC servo motor driver

In the transmission unit, Panasonic **MHMD022P1U** permanent magnetic synchronous AC servo motor, and **MADDT1507E** digital AC permanent magnetic synchronous servo driver is used as the mechanical arm motion control device.

MHMD022P1U: **MHMD** is motor inertia, **02** is motor rated power 200W, **2** is voltage 200V, **G** is encoder increment encoder, pulse is 20 bit, resolution 1048576, output signal 5 cables.

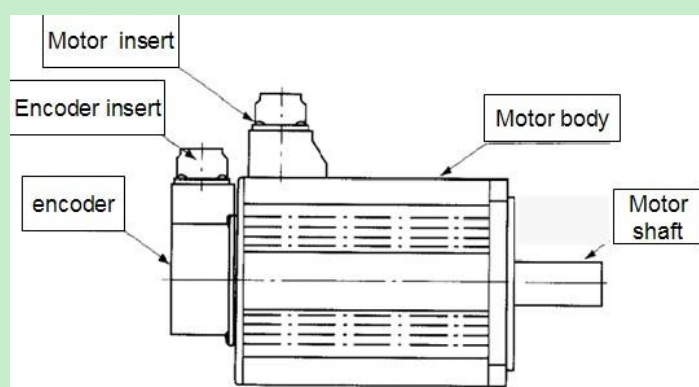


Diagram 7-20

MADDT1507E: MADDT Panasonic A driver, T1 is the transient output current 10A, 5 is power supply voltage single-phase 200V, 07 is current monitor rated current 7.5A. Driver and panel are shown in diagram 7-21.

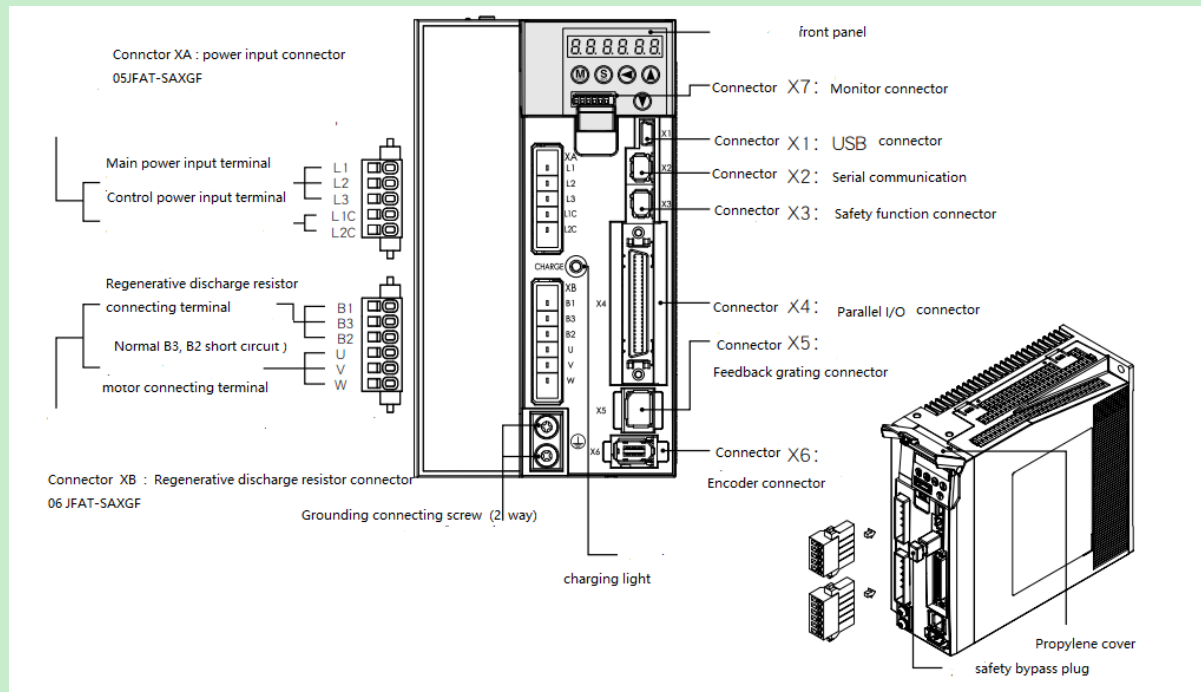


Diagram 7-21

2. Wiring:

MADHT1507E servo driver panel has some wiring terminals:

XA: power supply input, AC220V power supply is connected to L1, L3 power supply ends, as well to control power supply L1C, L2C ends.

XB: motor port and an external regenerative discharge electric resistor port. U, V, W terminals used to connect motor. must be noted,: power supply voltage must be in accordance with the instructions on the driver plate, motor wiring terminals (U, V, W) can not be grounded or short-circuit, AC servomotor rotation direction, unlike induction motor can be through the exchange of 3-phase-phase sequence to change, we must ensure that driver on the U, V, W, E wiring terminals and the motor main circuit wiring terminal according to the provisions of the order-one correspondence, otherwise it might driver causing damage. The motor wiring ends and driver earth ends in order and filter ground terminal to ensure reliable connection to the same an access locations. fuselage must also be grounded. YL-335B release did not use an external discharge resistance.

X6: connected to motor encoder signal port, the belt should be used to connect electric cables have shielding twisted electric cables, the shield should be receiving the ground terminal on the motor side, and should ensure encoder electric cable shield connected to the plug shell (FG).

X4: I/O control signal port, The pin signal definitions and selection of control mode related to the different wiring refer to the next mode, "Panasonic A series Servomotor Manual". In the YL-335B transfer station, servo motor used to position control, choice of position control mode. is to simplify the wiring methods used, such as shown in diagram 7-22.

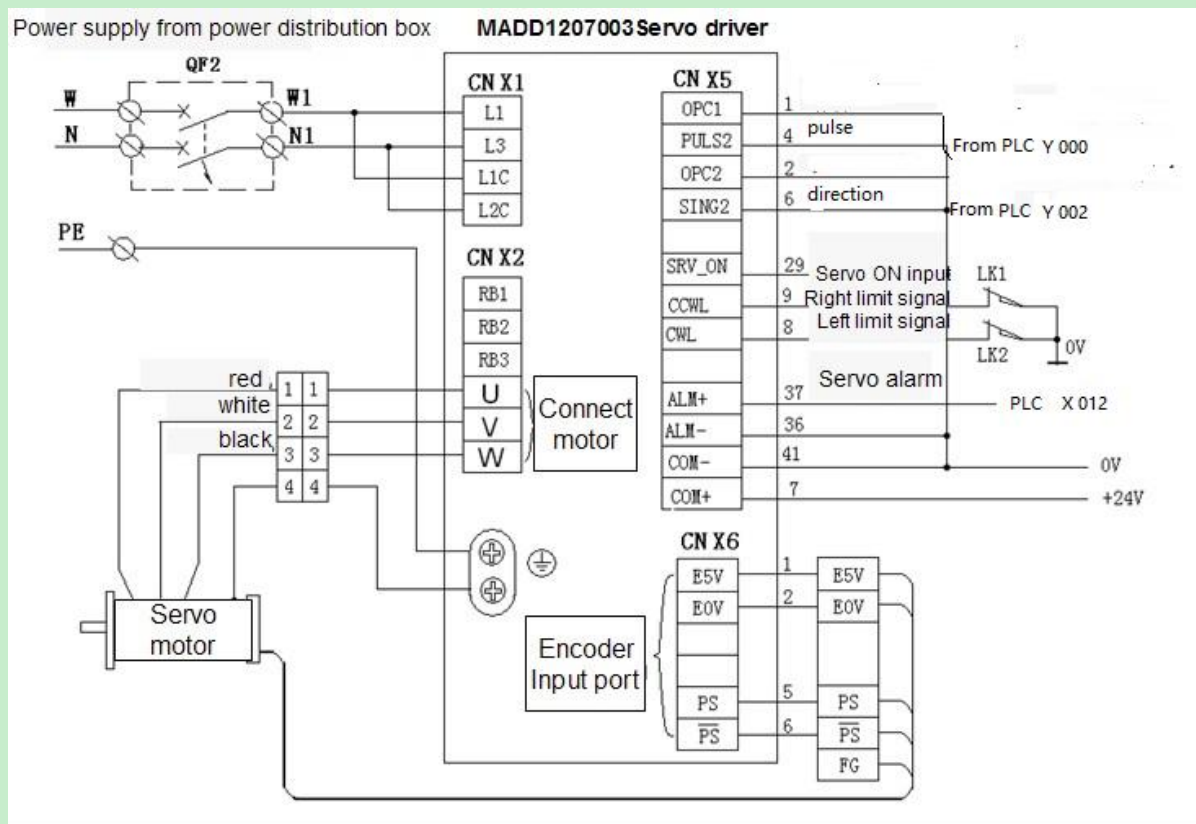


Diagram 7-22

3. Servo driver parameter setting

Panasonic servo driver has 7 control modes, position control, speed control, torque control, position/speed control, position/torque, speed/torque, close-loop control. The position mode is to make the input pulse string running motor position, motor rotation speed and the pulse string frequency-phase closed, motor rotation angle and the pulse number-phase closed; speed in two ways, one through the input DC-10V to - +10 V Speed command voltage, and second, optional driver within the speed within a setting to speed; torque is through input DC-10V to - +10 V command voltage regulator motor output torque, in this way run speed limits must be carried out, there are two ways: 1). setting driver within the parameter to limit; 2) input analog voltage speed limit.

3. Parameter setting

MADHT1507E servo driver 218 parameters, Pr00-Pr6.39, can be connected with the PC software for commissioning in the specialized setting, the driver can also be carried out above the board setting.

In the PC, installation, set up communications with the servodriver, servodriver parameter can be read or written to the state, see diagram 7-23. When when field conditions do not allow, or modify a small parameter, they can operate through the driver panel to complete. Operation panel is shown in diagram 7-24. Each button is shown in table 7-3.

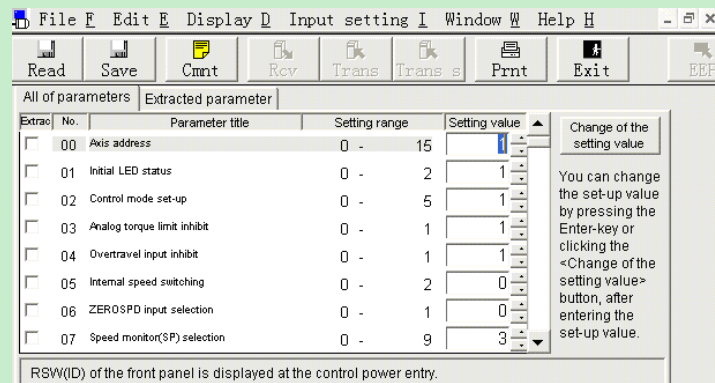


Diagram 7-23

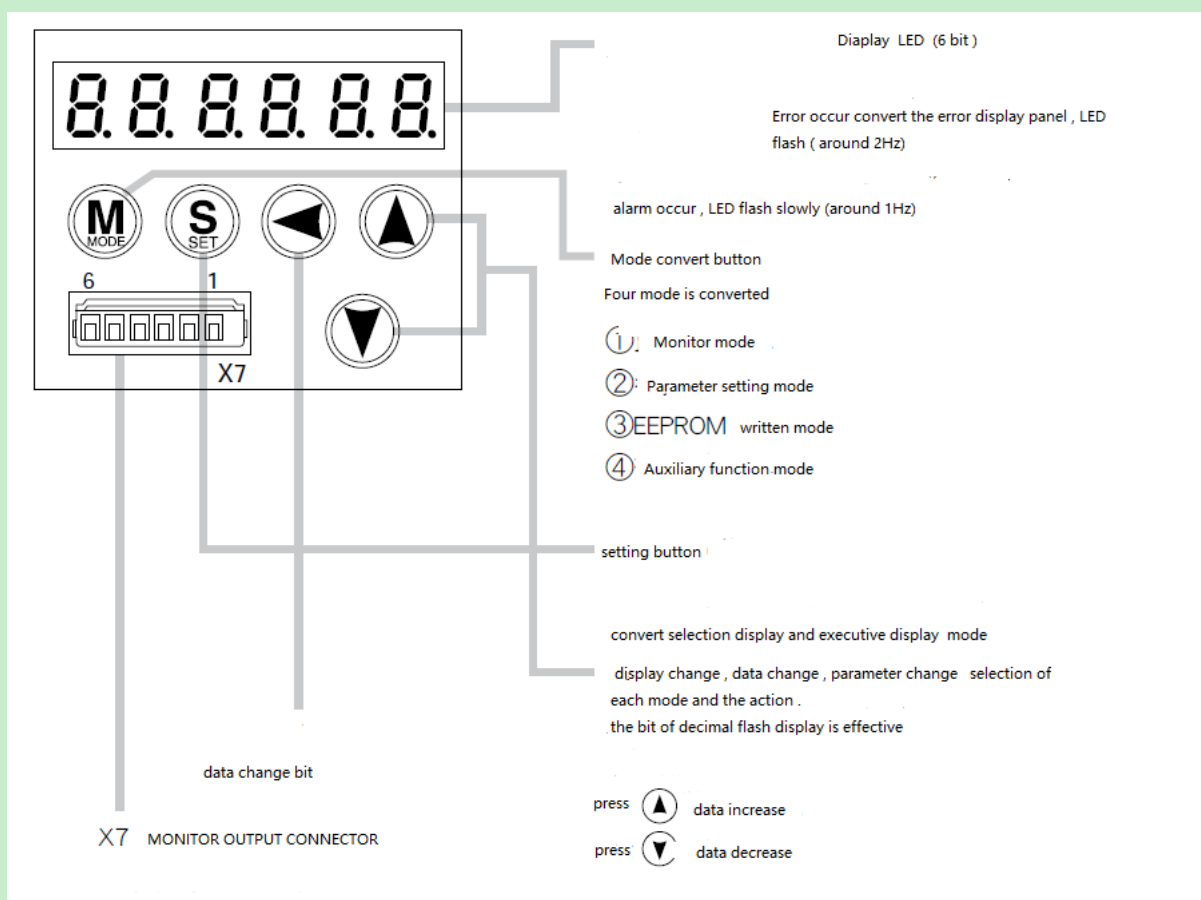


Diagram 7-24

Table 7-3 servo driver buttons

Button	Activation	function
MODE	Valid when mode displayed	Switch among 5 modes: 1)monitor mode; 2)parameter setting mode;

		3)EEPROM write mode; 4)automatic adjustment mode; 5) Auxiliary function mode.
SET	Always valid	Switch between display and run
▲ ▼	Flash and valid	Change display or parameters, select parameter to operate
◀		Move to the highest bit

Panel operation:

1. Parameter setting, push “Set”, and “Mode” to select “Pr00”, then select general parameters menu, push “Set” to enter. Push to adjust parameters, then push “S” to return.
2. Parameter saving, push “M”, then “EE-SET”, and “Set” to confirm, to show “EEP -”, push upward 3s, to show “FINISH” or “reset”, then save.
3. JOG, push “Mode”, to “AF-ACL”, then select “AF-JOG”, push “Set”, to show “JOG -”, then push upward 3s to show “ready”, then push leftward 3s to show “sur-on”, the **S-ON** shall be off.

4. Part parameters

In **YL-335B**, servo drive works in position control mode, Y000 of **FX3U-48MT** output pulse is servodriver position command, pulse quantity decides servomotor rotation displacement, mechanical arm linear displacement, the pulse frequency decides servomotor rotation speed, that is, the mechanical arm movement speed, Y002 of **FX3U-48MT** output pulse is servodriver direction command. The servodriver can adopt automatic gain adjustment mode. According to the requirements above, servodriver parameter setting is in table 7-4.

Table 7-4 Servo parameter setting

NO.	Parameter		Setting value	Function and meaning
	Parameter No.	Parameter name		
1	Pr5.28	LED initial status	1	Display motor rotation speed
2	Pr0.01	Control mode	0	Position control (code P)
3	Pr5.04	Driver forbid input setting	2	When left or right (POT or NOT) limit works, then Err 38 travel limit forbid input signal ,error alarm. Setting parameter value shall be changed when control power is reset.
4	Pr0.04	Inertia ratio	250	
5	Pr0.02	Real-time automatic gain setting	1	Stanard model will be adjusted automatically real time , little change of the load inertia when operation
6	Pr0.03	Real-time automatic gain mechanical selection	13	The more value of parameter setting , the quick response .

7	Pr0.06	Command pulse rotation direction	0	
8	Pr0.07	Command pulse input mode	3	
9	Pr0.08	The number of pulses per every rotation of the motor	6000	

7.3.3 FX3U pulse output function and programming

The transistor output **FX3U** series of **PLC CPU** unit support high-speed pulse output function, but only Y000 and Y001. The output pulse frequency highest is 100KHZ. For transfer unit servo motor control is return to origin and position control.

The pulse output command FNC157 (PLSY), with acceleration /deceleration pulse output command FNC59 (PLSR) , variable speed pulse output command FNC157 (**PLSV**), the origin return command FNC156 (ZRN), position control FNC158(**DRVI**) , absolute position control FNC158(**DRVA**) of FX 3U can be achieved . The behind three command will be introduced here.

1. Origin return command FNC156(ZRN)

When programming controller is turned off, it will return to origin.

The mechanical part origin position data will be written. The origin return command format is shown in diagram 7-25.

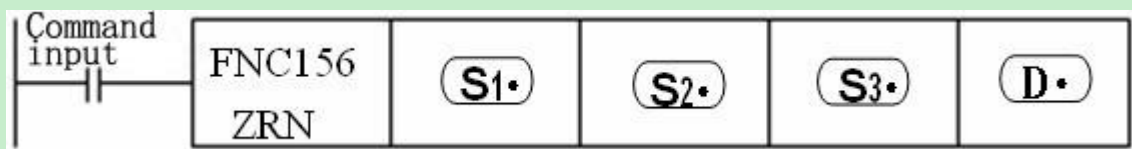


Diagram 7-25

(1). Origin return command format

① **(S1)**: origin return speed. [16bit command]: 10~32,767(HZ) ; [32bit command] : 10~100(KHZ).

① **(S2)**: speed
Signal (DOG) is ON, low speed.

② **(S3)**: signal input.

③ **(D)** pulse output **Y** number(only Y000 or Y001).

(2) Origin return sequence is shown in diagram 7-26:

① Drive command, origin return speed **(S1)**.

- When the command drive contact turns OFF during the origin return process, it will not decelerate and stop.

- When the command drive contact turns OFF, monitoring (Y000: M8340, Y001:

M8350) turns on in the pulse output, and the command will not be driven again.

- ② when the signal (DOG) OFF to ON, speed (S_2) .
 when the signal (DOG) ON to OFF, current value register (Y000: [D8141, D8140], Y001: [D8143, D8142]) write 0. M8140 (clear signal output functions)ON, output clear signal. When M8029 is same, Pulse output monitor is OFF

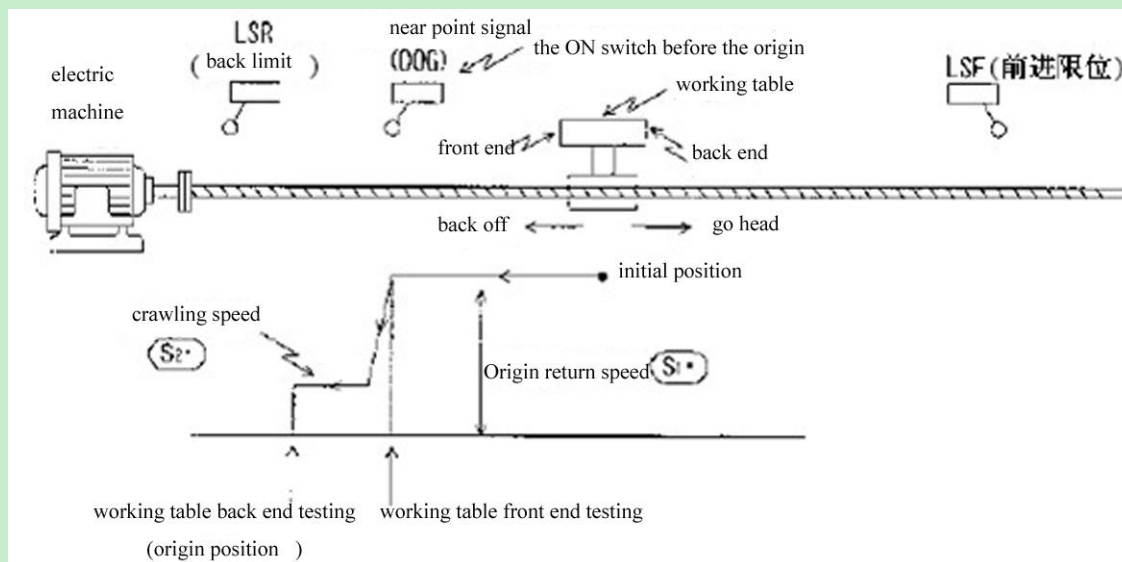


Diagram 7-26

2. Relative position control command FNC158(DRVI)

The relative drive mode execute single speed position control command, command format is shown in diagram 7-27.



Diagram 7-27

The command format:

- ① (S_1) : output pulse quantity (relative)
 [16bit command]: -32,768~+32,767
 [32bit command]: -999,999~+999,999
- ② (S_2) : output pulse quantity
 [16bit command]: 10~32,767(HZ)
 [32bit command]: 10~100(KHZ)
- ③ $(D1)$: pulse output address, command is only for Y000, Y001.

- ④ **(D2-)**: rotation direction signal output address, according the positive and negative of **(S1-)**: [+ (positive)]→ ON [- (negative)]→ OFF

● Output pulse quantity **(S1-)**, current value register position .

[Y000] output→[D8341(high bit), D8340(low bit)](use 32bit)

[Y001] output→[D8351(high bit), D8350(low bit)](use 32bit)

back,current value register number value decreases.

● Rotation direction through output pulse quantity **(S1-)** position and negative symbol command .

● During the execution of the command, even if the contents are changed, it can not be displayed in the current operation. It is valid only when the next command is executed.

● During the execution of the command, command driver contact becomes OFF, it will decelerate. At this time the execution completion symbol M8029 does not operate.

● When the command drive contact becomes OFF, pulse output symbol (Y000:[M8340],Y001:[M8350] is ON, will not re-driven by command .

In addition , programming DRVI command should also pay attention to the interoperability of the operation:

① Speed up/down speed number is 10, the speed is highest frequency 1/10. in drive step motor, setting highest frequency is within step motor step range.

② speed up/down time is no less than **PLC** scan time maximum value (**D8012** value)10 times, otherwise speed up/down each time is not uniform

3. absolute position control command FNC158(**DRVA**)

The absolute drive mode execute single speed position control command, command format is shown in diagram 7-28



Diagram 7-28

Command format:

① **(S1-)**: output pulse (absolute)

[16 bit command]: -32,768~+32,767 ;

[32 bit command]: -999,999~+999,999

② **(S2-)**: output pulse quantity

[16 bit command]: 10~32,767 (HZ) ;

[32 bit command]: 10~100 (KHZ)

③ **(D1-)**: pulse output address, command only for Y000, Y001.

④ **(D2-)**: rotation direction signal output address, **(S1-)** and current position difference value:

[+ (positive)]→ ON [- (negative)]→ OFF

● Target position **(S1-)**, current value register is absolute position .

[Y000] output→[D8341(high bit), D8340(low bit)](32 bit)

[Y001] output→[D8351(high bit), D8350(low bit)](32 bit)

back,current value register number value decrease.

-
- Rotation direction through output pulse quantity  positive and negative symbol command .
 - During the execution of the command, even if the contents are changed, it can not be displayed in the current operation. It is valid only when the next command is executed.
 - During the execution of the command, command driver contact becomes OFF, it will decelerate. At this time the execution completion symbol M8029 does not operate.
 - When the command drive contact becomes OFF, pulse output symbol (Y000:[M8340],Y001:[M8350] is ON, will not re-driven by command .

4. Pulse output functions main special memory

[D8341, D8340] output to **Y000** pulse quantity

[D8351, D8350] output to **Y001** pulse quantity

[M8349] **Y000** pulse output stop command (stop)

[M8359] **Y001** pulse output stop command (stop)

[M8340] **Y000** pulse output monitor

[M8350] **Y001** pulse output monitor

Each data register, you can use “**(D)MOV K0 D81□□**” to clear.

7.4 PLC control and programming of Conveyor unit

7.4.1 working tasks

The function of transmission unit is to test the function of equipment conveying the workpieces. Require other working units in place, as shown in diagram 7-29. And placed workpieces on discharging platform of feeding unit. Specific testing requirements are as follows:

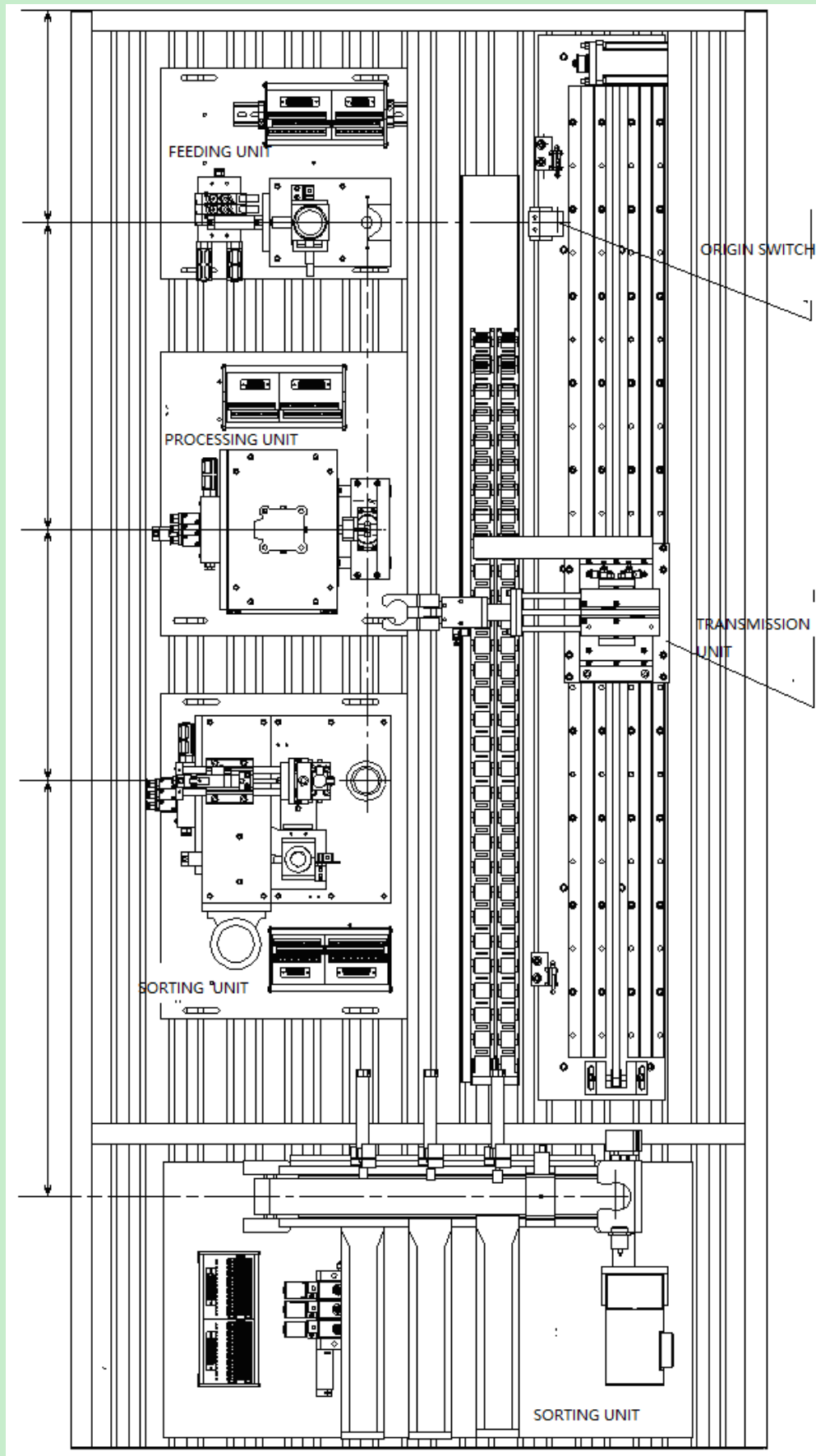


Diagram 7-29

1. Transmission unit in power and press the reset button SB1, execute the reset operation so that grasping manipulator device returned to its original position. In the reset process, the "normal work" light HL1 blinking at the frequency of 1Hz.

When the grasping manipulator device returned to its original position, and the cylinders of transport units should meet all the requirements of the initial position, then the reset completed, the "normal work" HL1 constantly lights on. Press the start button SB2 to start up equipment, light HL2 "equipment running" constantly lights on, and began the function testing process.

2.The normal function test

(1)Manipulator device grab workpiece from the feeding station, sequence is: grab arm stretched out → gripper clamp the workpiece → up → arm retract.

(2)After crawl action finished, servo motor drive manipulator move to the processing station ,speed is not less than 300mm / s.

(3)Manipulator moved to the front of processing station, drop the workpiece onto the material platform of processing station in this order : the arm stretched out → down → gripper release to drop workpiece Taiwan to lay down their arms retracted → workpiece.

(4) Down 2 seconds after the completion of workpiece movement, grasping grasping robot installations in the operation of the workpiece processing station. The order of grasping grasping with feeding stations the same as the order of the workpiece.

(5) Crawl action is completed, robot servo motor drive unit to move to assembly points in front of materials to Taiwan. And then the workpiece material into the assembly station platform. Their actions in order to lay down the workpiece and the processing stations in the same sequence.

(6) Down 2 seconds after the completion of workpiece movement, grasping grasping robot installations in the operation of the workpiece assembly station. The order of grasping grasping with feeding stations the same as the order of the workpiece.

(7) Robot arm retracted, the swing sets rotated counterclockwise 90 °, servo motor drive unit from the assembly station robot picking stations to transport the workpiece, to reach the top of the feed conveyor belt sorting station, after the mouth piece down, move down the order and processing station the same as the order of the workpiece.

(8) Down moves the workpiece is completed, robot arm retracted, and then perform the origin return operation. Servo motor drive robot device to 400mm / s rate of return, return to 900mm, the swing sets clockwise rotation 90 °, then to 100mm / s rate of return to the origin of low-speed stop.

When the crawl robot device to return the origin after the end of a test cycle. When the feeding unit of a material placed on the workpiece table, and then once the start button SB2, start a new round of tests.

3. Non-normal operation of the functional test

If the process of pressing the emergency stop button in the work of QS, then the system immediately stops running. Reset the emergency stop after the emergency stop should be started before the break to continue running. However, if the emergency stop button is pressed, the transmission station manipulator devices are moving to a target point, then the emergency stop reset after the transfer station manipulator should first return to the origin location, and then back to the original target point movement.

In the E-STOP state, the green indicator light flashes HL2 to the frequency of 1Hz, until the emergency stop reset to resume normal operation, HL2 recovery .

7.4.2 Selection and I/O wiring of PLC

The transmission unit I/O, input signals: button/indicator light module button, switch; master signals, each sensor signal; output signal: mechanical arm each solenoid control signals, servo motor driver pulse signals and drive direction signals; you shall consider output signals to button/indicator light module, indicator lights, to display station or system work status.

As output drive servo motor high-speed pulse is needed, **PLC uses** transistor output .

Mitsubishi FX3U-48MT PLC, 24 point input, 24 point transistor output. In table 7-7 is **PLC** I/O signal table, I/O wiring schematic is shown in diagram 7-30.

Table 7-7 Transfermission PLC I/O signal table

input signal				output signal			
No.	PLC input	signal name	signal source	No.	PLC output	signal name	signal source
1	X000	Origin sensor	device side	1	Y000	pulse	device side
2	X001	Right limit guard		2	Y001		
3	X002	Left limit guard		3	Y002	direction	
4	X003	mechanical arm lift lower limit		4	Y003	lift bench rise solenoid	
5	X004	mechanical arm lift upper limit	device side	5	Y004	rotary cylinder leftward solenoid	
6	X005	mechanical arm rotation leftlimit		6	Y005	rotary cylinder rightward solenoid	
7	X006	mechanical arm rotation right limit		7	Y006	claw extends solenoid	
8	X007	mechanical arm extends		8	Y007	claw tighten solenoid	
9	X010	mechanical arm withdraw		9	Y010	claw loosen solenoid	
10	X011	mechanical arm tighten		10	Y011		
11	X012	servo alarm		11	Y012		
12	X013 ~ X013			12	Y013		button/indicator light module
13				13	Y014		
14				14	Y015	alarm	
15				15	Y016	run	
16				16	Y017	stop	
17							
21	X024	start button	button/indicator light module				
22	X025	reset button					
23	X026	Emergency stop button					
24	X027	Mode selection					

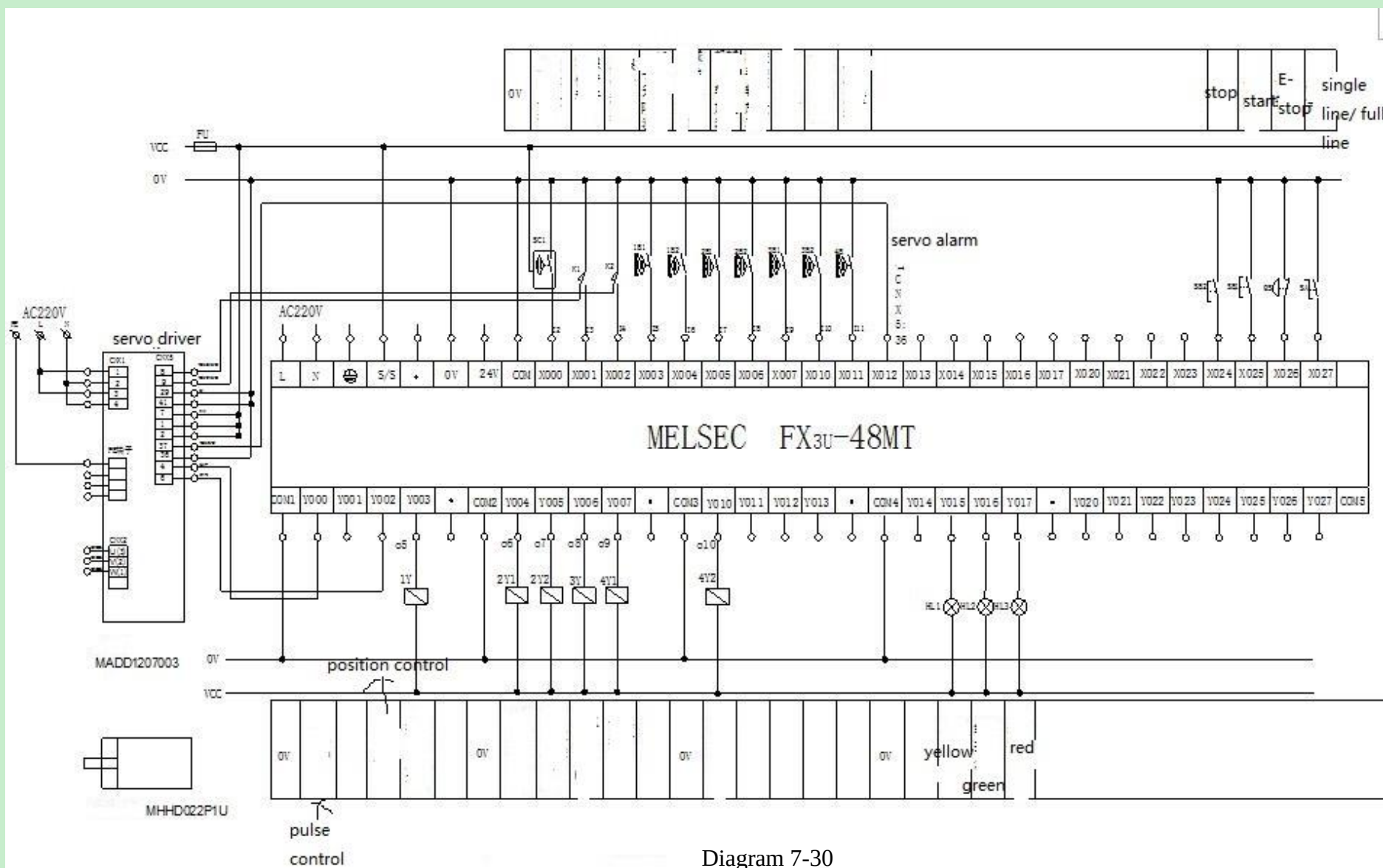


Diagram 7-30

In the diagram, moving contact of the left and right limit switches **K2** and **K1** are connected to **PLC input point X002 and X001**.

It is important to note that K2, K1 provide a pair of switching contacts whose static contacts should be connected to the common point COM and the dynamic contact must be connected to the servo driver's control port CNX5 CCWL (pin 8) and CWL (pin 9) as a hard interlock protection (see diagram 7-30)

Transistor output **FX3U** series **PLC**, power supply is **AC220** power supply, and same as previous each unit relay output

After system electrical wiring, you shall make servo motor driver parameter setting, shown in table 7-8.

Table 7-8 Servomotor driver parameters setting table

No.	Parameter		value	Function and meaning
	No.	Name		
1	Pr5.28	LED initial status	1	display motor rotation
2	Pr0.01	Control mode	0	position control (code P)
3	Pr5.04	driver forbid input device	2	when left or right (POT/NOT) limit works, then Err 38 travel limit input signal error alarm. Setting parameter value shall be changed when control power is reset.
4	Pr0.04	Inertia ratio	250	
5	Pr0.02	Real-time automatic gain setting	1	Standard mode will be adjusted automatically, load inertia changes little during operation.
6	Pr0.03	real-time automatic gain mechanical selection	13	The greater the parameter value, the faster the response .
7	Pr0.06	command pulse rotation direction setting	0	
8	Pr0.07	command pulse input mode	3	
9	Pr0.08	The number of pulses per rotation of the motor	6000	

7.4.3 PLC control programming and debugging

1. Master program

Table 7-9 Servo motor operation motion position

NO	Station	Pulse quantity	Movement direction
0	Low-speed zero return (ZRN)		
1	ZRN (zero) →supply station 22mm	2200	
2	Feeding unit →processing unit 430mm	43000	DIR
3	Feeding unit→assembly unit 780mm	78000	DIR
4	Feeding unit→sorting unit 1040 mm	10400	DIR

The master program includes initialization, reset process (subprogram), the main program list as diagram 7-31,7-32,7-33,7-34 .



Diagram 7-31

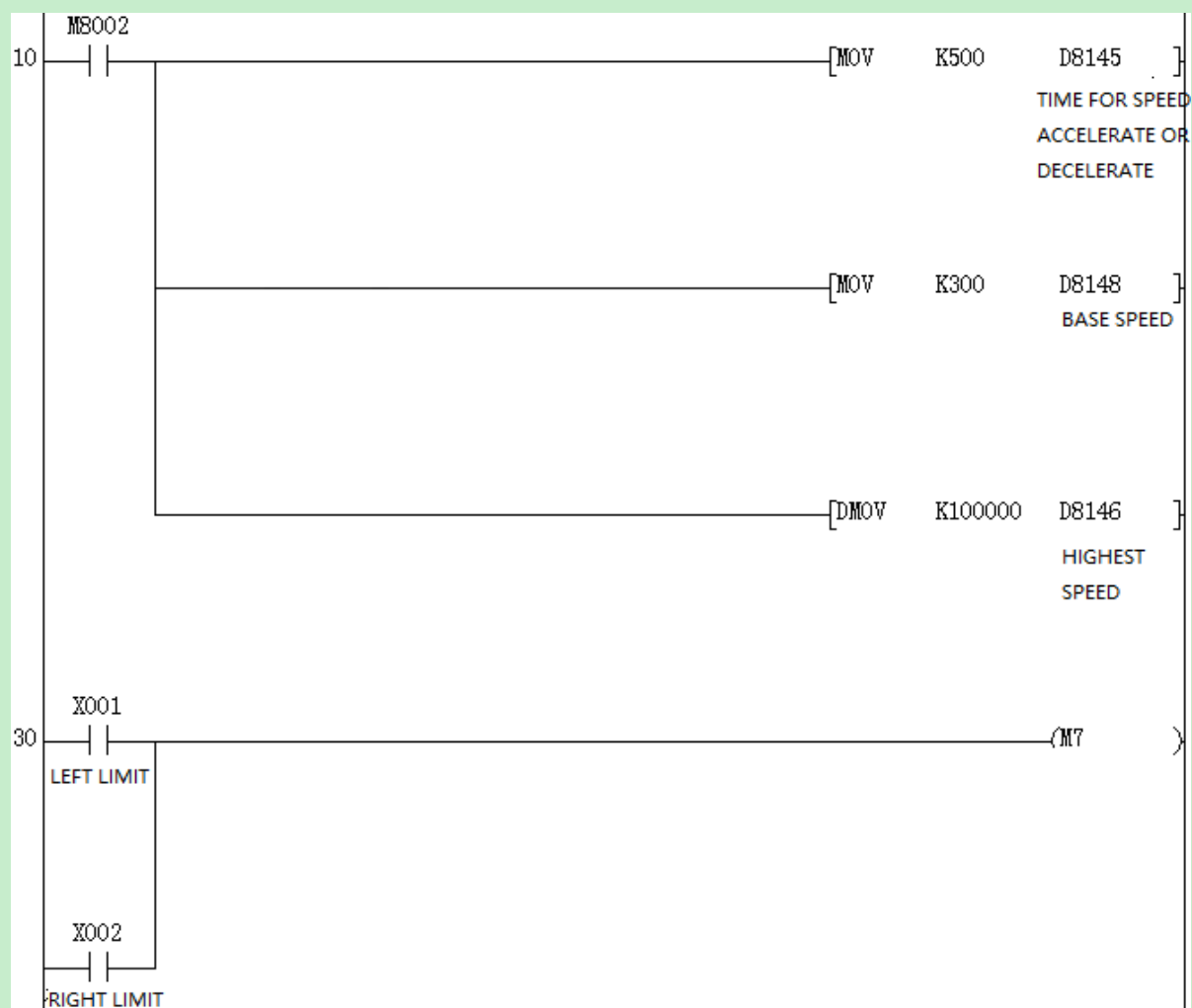


Diagram 7-32

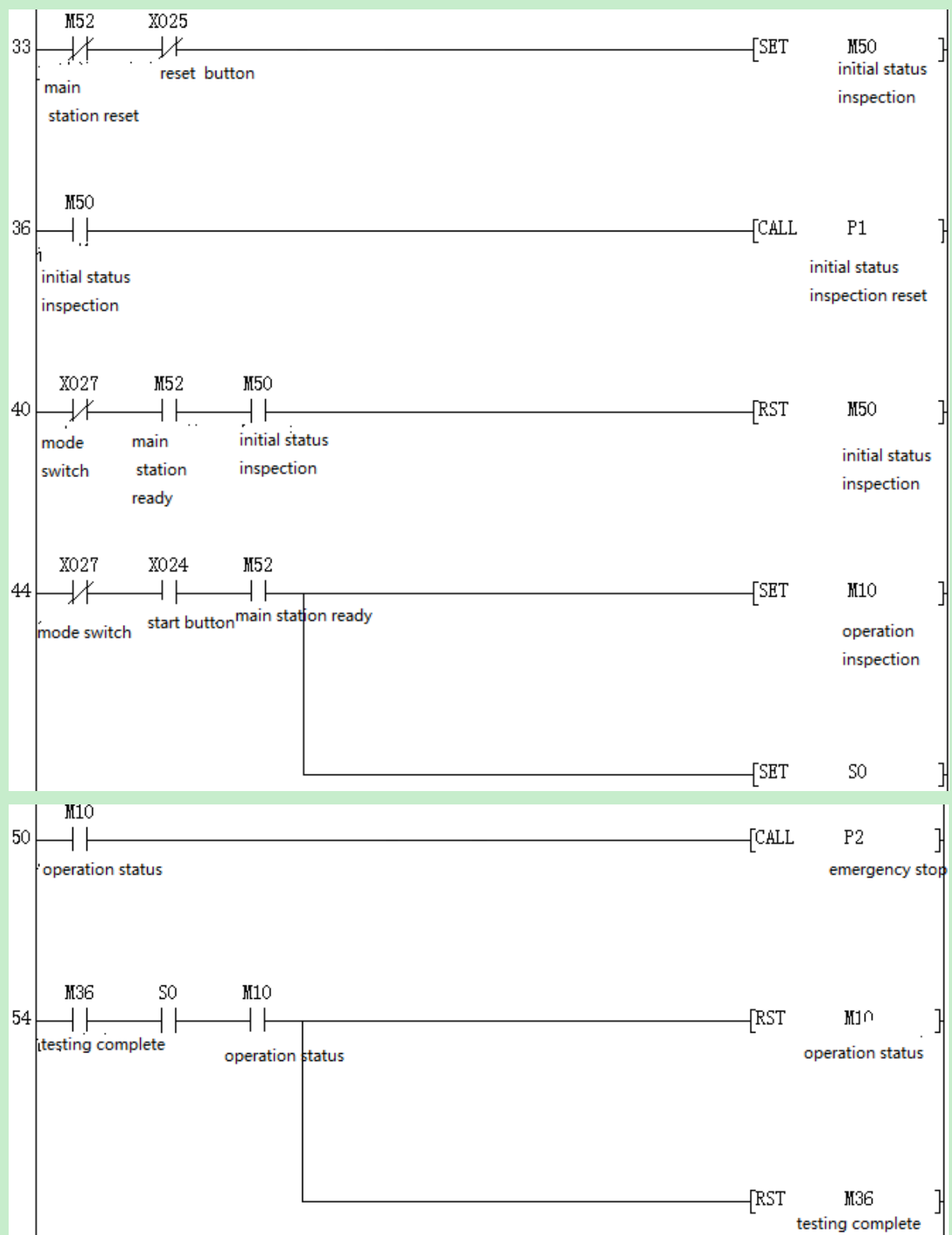


Diagram 7-33

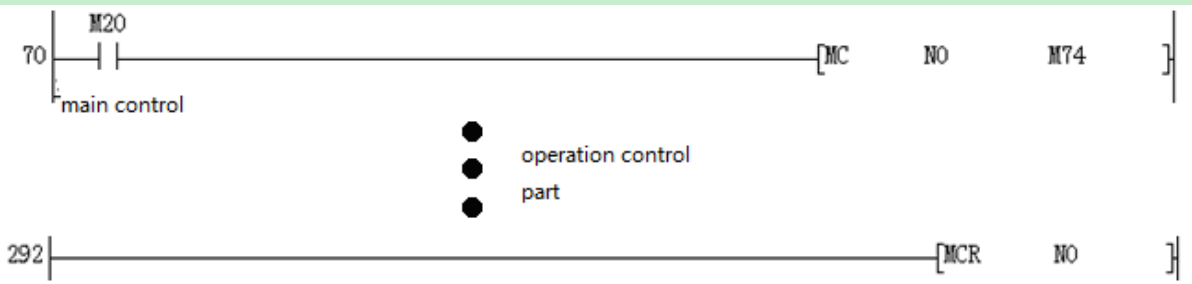


Diagram 7-34

2. Initial status check reset subprogram and origin program

The zero subprogram as diagram 7-33, and the zero program part as diagram 7-35,7-36,7-37,7-38.



Diagram 7-35

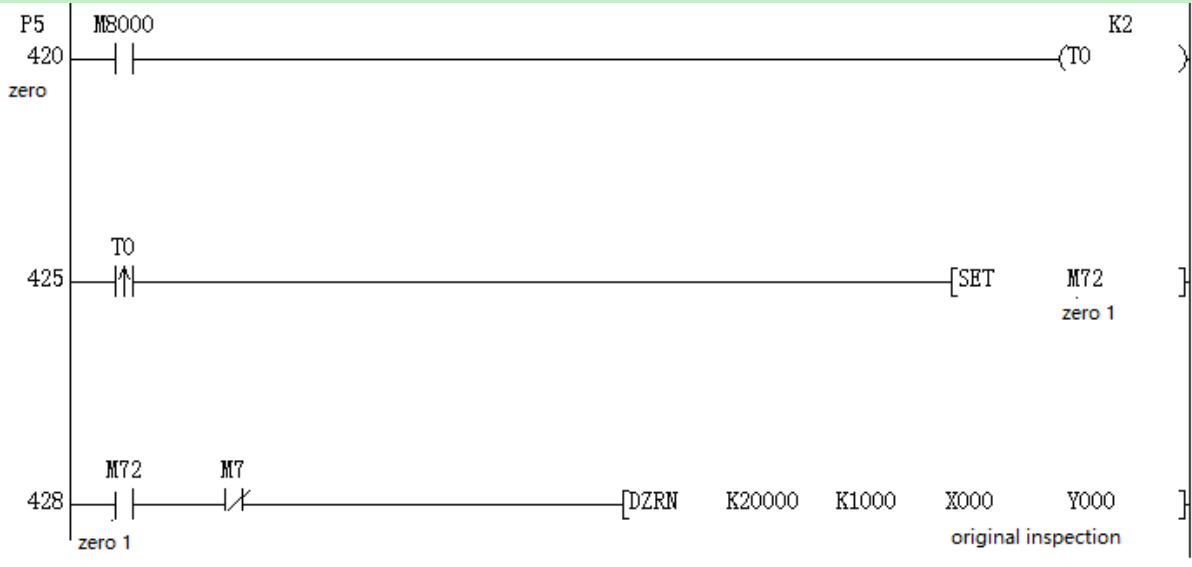


Diagram 7-36

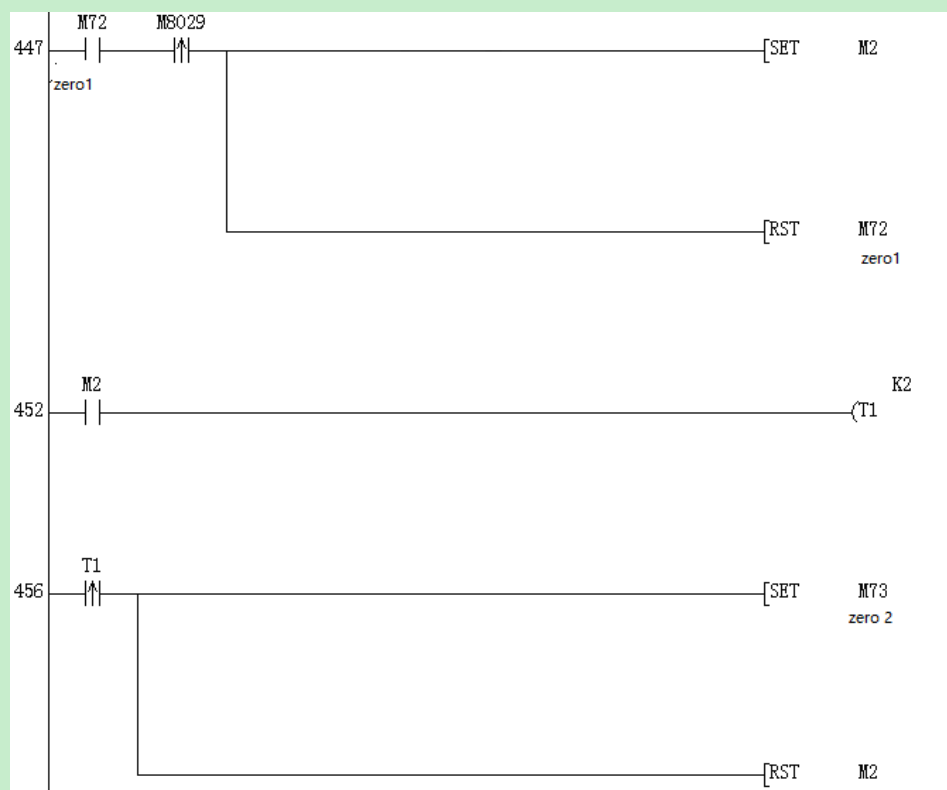


Diagram 7-37

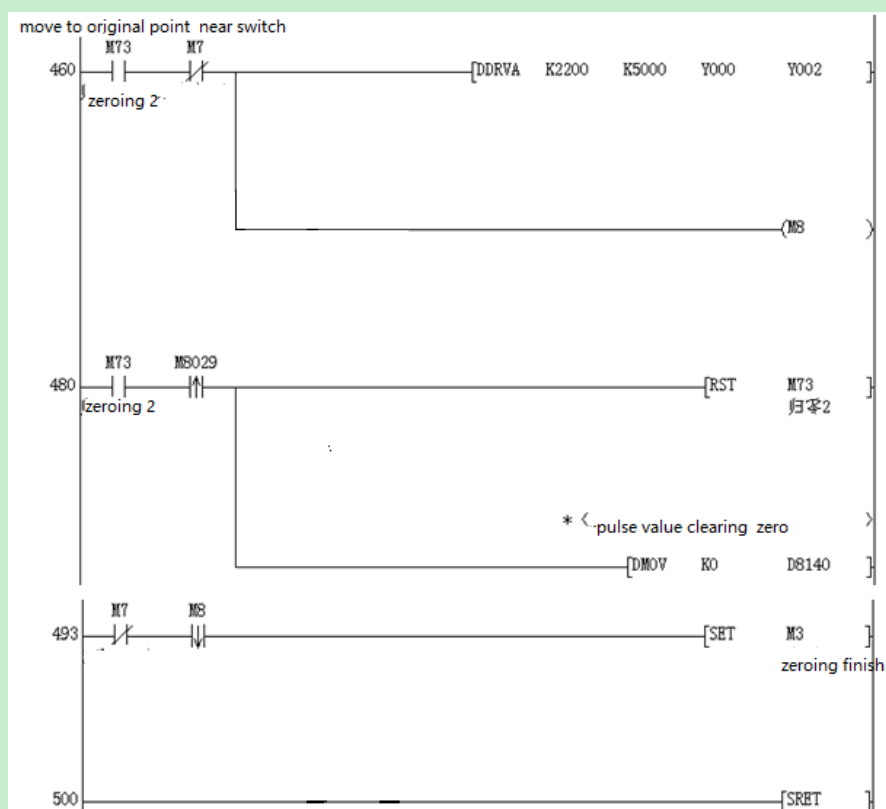


Diagram 7-38

3. Emergency stop process subprogram

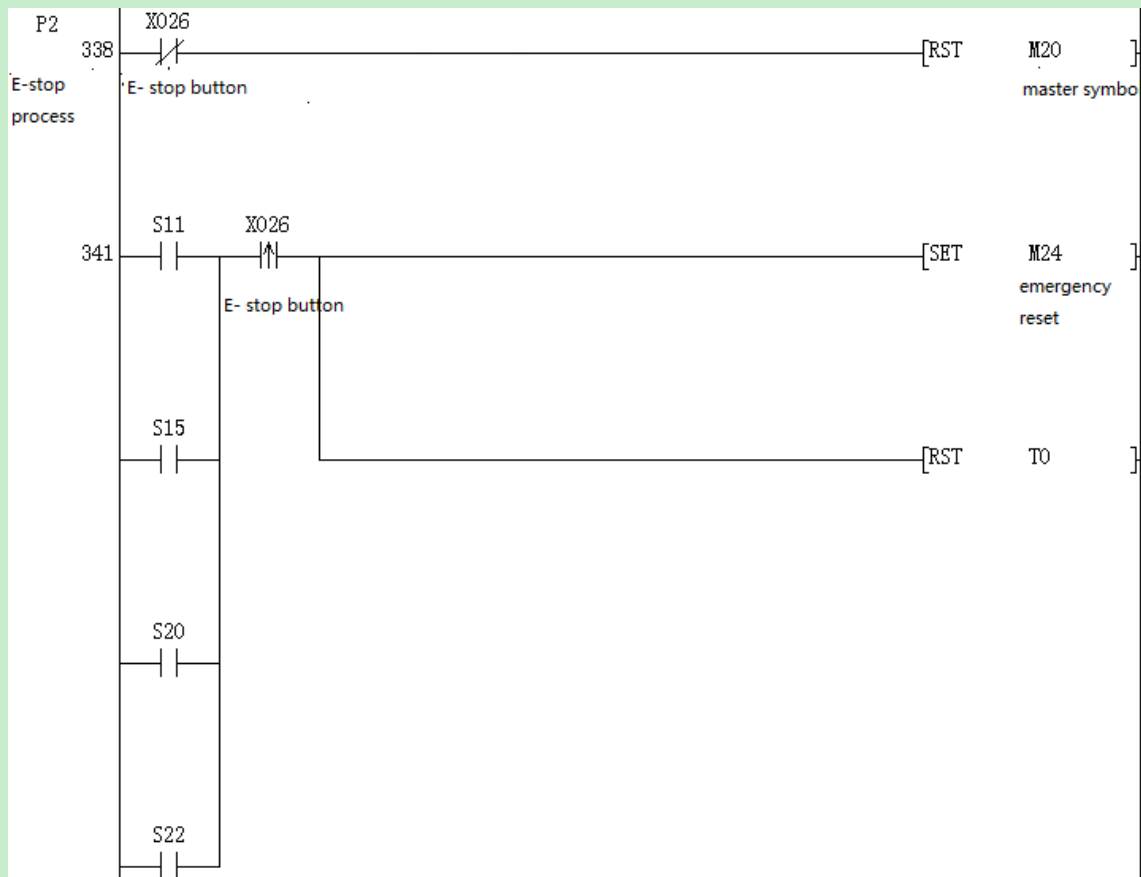


Diagram 7-39

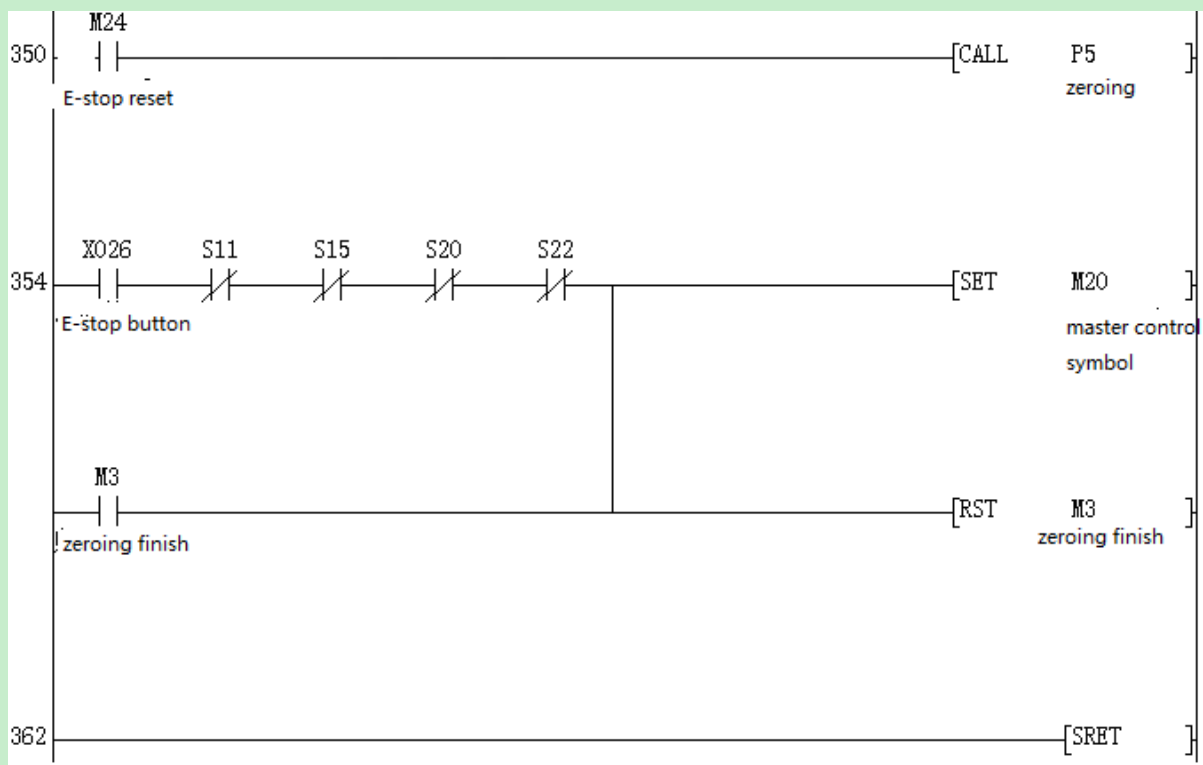


Diagram 7-40

ITEM 8. Whole Control of YL-335B

Here we will take **YL-335B** initial routine as an example to specify how PLC realize control function with several stations combined.

YL-335B system control method, each station bench **PLC** is in charge of its control tasks, each **PLC** communicate with each other by **RS485** serial communication control method. When a network is composed, in the system, each station is a work station.

8.1 Mitsubishi FX series of PLC N:N communication

8.1.1 Mitsubishi FX series of PLC N:N communication network

FX series of PLC supports the 5 types of communication:

- (1) **N: N** network: **FX2N, FX2NC, FX3U, FX0N** PLC data transition, **N: N**. The network, can link small system data. In maximum 8 **PLCs** (**FX2N, FX2NC, FX3U, FX0N**) network.
- (2) Parallel link: the network adopts 100 auxiliary relay and 10 data registers based on 1: 1 data transition.
- (3) Computer link: **RS485(422)** station data transition 1: n(16).
- (4) No protocol communication: each **RS232** station, with computer, barcode reader and printer, for data communication, by no-protocol communication, the communication mode needs **RS** commands or **FX2N-232IF** special functions module.
- (5) Programming port: for **FX2N, FX2NC, FX3U, FX1S** series of **PLC**, when the port is connected to **FX3U-232BD, FX0N-232ADP, FX2N-422BD**, it can be connected with other external equipments.

Mitsubishi **FX** series of **PLC** adopts **N: N** network each station data communication, here we only talk about **N:N** communication network basic features and network methods.

N: N network is based on **RS485** transimission standard, network has a **PLC** as master station, other **PLC** is slave station, network station quantity shall not exceed 8 as diagram 8-1

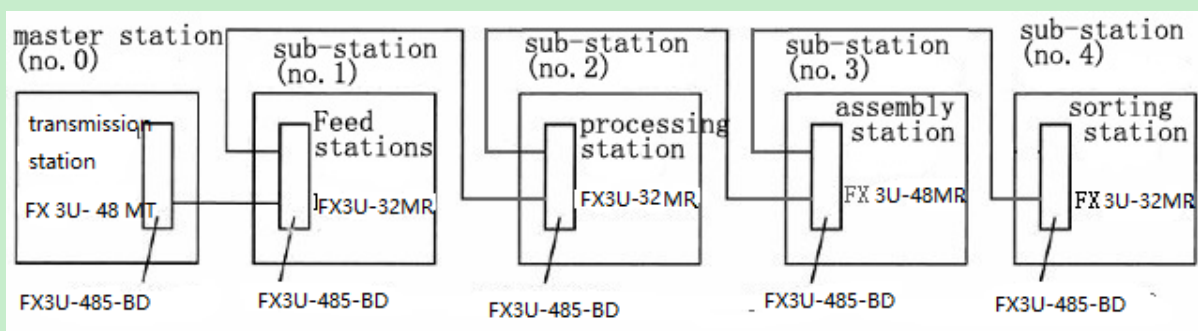


Diagram 8-1

The system **RS485** communication interface panel FX3U-485-BD, maximum distance 50m, network station number is 5.

N:N network communication protocol is: half duplex communication, baud rate (**BPS**) is 38400 BPS.

N:N network adopts broadcast mode in communication: in the network, each station has a special auxiliary relay and special data register to compose the link zone, each station links zone address is same. Each station link zone data will be sent. In the network, any **PLC** zone status will be reflected to network, then data from **PLC** link can be shared by all **PLCs**, and all station data can be updated in a same time.

8.1.2 Installation and connection N:N communication network

Before network installation, you shall open power supply. Each station **PLC** shall be connected to **485-BD** communication panel. The **LED** display/ends are shown in diagram 8-2.

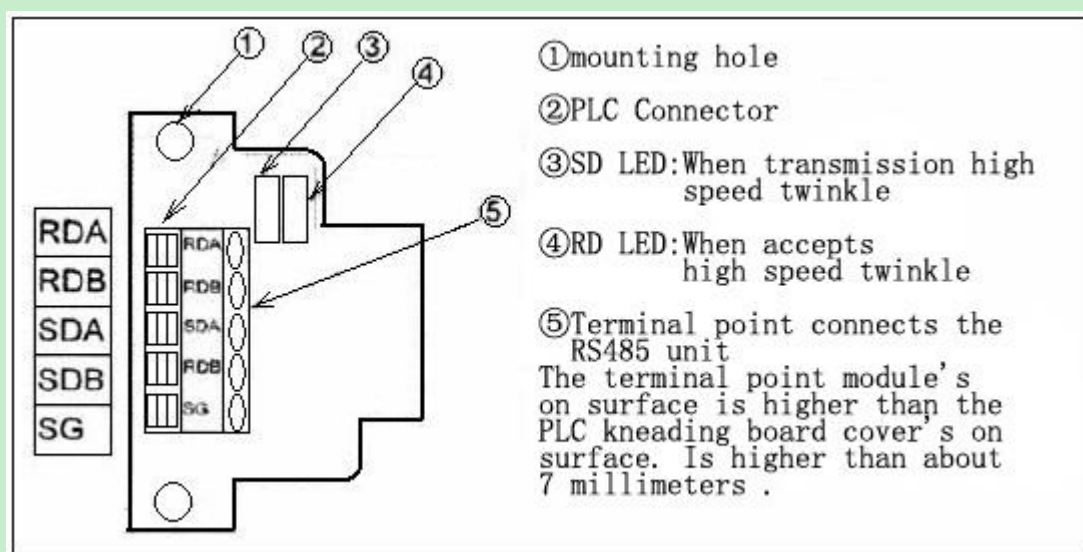


Diagram 8-2

YL-335B system **N: N** link network, each station, as shown in diagram 8-3, in wiring, 110ohm resistor is needed (485BD panel).

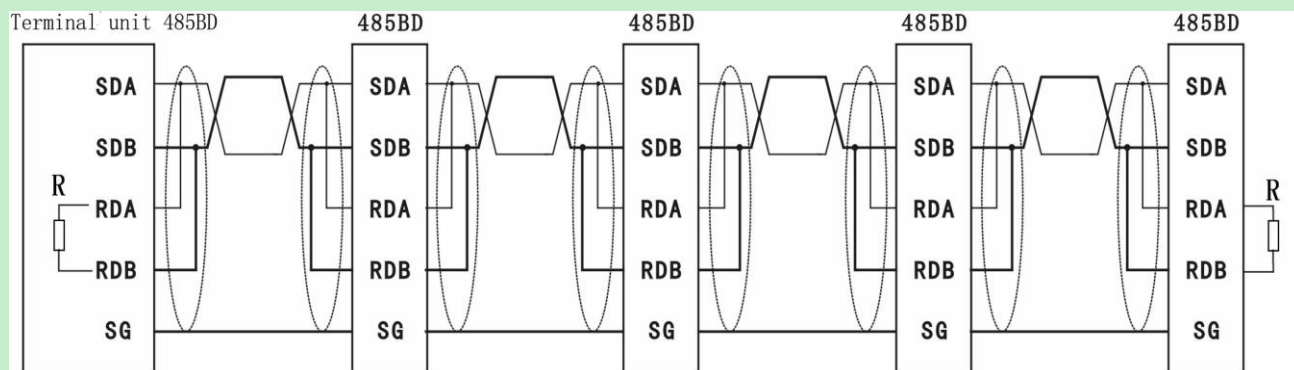


Diagram 8-3

Network connection:

1. Diagram 8-3, **R** is resistor. Between end **RDA** and **RDB** is resistor (110ohms)
2. Connect end **SG** to programming controller end, and 100ohms resistor is grounded.
3. If the cable diameter shall be within AWG26~16.if each station **PLC** network parameters are set, then in finished network connection, turn on each **PLC** work power supply, you can see that, on each station communication panel, **SD LED** and **RD LED** indication light are ON/OFF status, **N: N** network is composed.

If **RD LED** indication light is ON/OFF status, and **SD LED** no flash, you shall check station number, transition rate and slave station quantity.

8.1.3 Build N:N communication network

1. Network and process

FX series of **PLC N:N** communication network each station **PLC** programming mode network parameters.

FX series of **PLC N: N** network sign bit and save network parameters and network status specialdata register.when **PLC** is **FX3U** or **FX2N(C)**, **N: N** network is shown in table 8-1, table 8-2.

Table 8-1 Special auxiliary relay

code	assistant relay	name	description	Response
R	M8038	N: N network parameters setting	set N: N network parameters	M, L
R	M8183	master station communication error	when master station communication error ON	L
R	M8184~M8190	slave station communication error	when slave station communication error ON	M, L
R	M8191	data communication	when and other stations communication ON	M, L

Note: R: read only ; W: write only; M: master station ; L: slave station
CPU error, program error or stop status, for each station, it will produce communication error quantity count.

M8184~M8190 is slave station communication error sign, No. 1 slave station M8184,...No. 7 slave station M8190.

Table 8-2 Special data register

Code	Data register	Name	Description	Response
R	D8173	station number	save station number	M, L
R	D8174	slave station quantity	save slave station quantity	M, L
R	D8175	refresh range	save refresh range	M, L
W	D8176	station number set	set station number	M, L

W	D8177	slave station quantity set	set slave station quantity	M
W	D8178	refresh range set	set refresh range modenummer	M
W/R	D8179	retry times set	set times	M
W/R	D8180	communication set	set communication	M
R	D8201	current network scan time	save current network scan time	M, L
R	D8202	maximum network scan time	save maximum network scan time	M, L
R	D8203	master station communication error quantity	save master station communication error quantity	L
R	D8204~D8210	slave station communication error quantity	save slave station communication error quantity	M, L
R	D8211	master station communication error code	save master station communication error code	L
R	D8201~D8218	slave station communication error code	save slave station communication error code	M, L

Note: R: read only; W: write only; M: master station; L: slave station

CPU error, program error or stop status, station communication error quantity cannot be counted.

D8204~D8210 is slave station communication error quantity, No.1 slave station D8204,...No.7 slave station D8210.

In table 8-1, special auxiliary relay **M8038**(N: N network parameters setting relay, read only) is used to set N: N network parameters .

For the master station, programming methods network parameters, program starts from step 0 (**LD M8038**), special data register **D8176~D8180** with parameters. For slave station, in step 0 (**LD M8038**)**D8176**.

For example, in the diagram 8-4 are transmission station (master station) network parameters setting program.

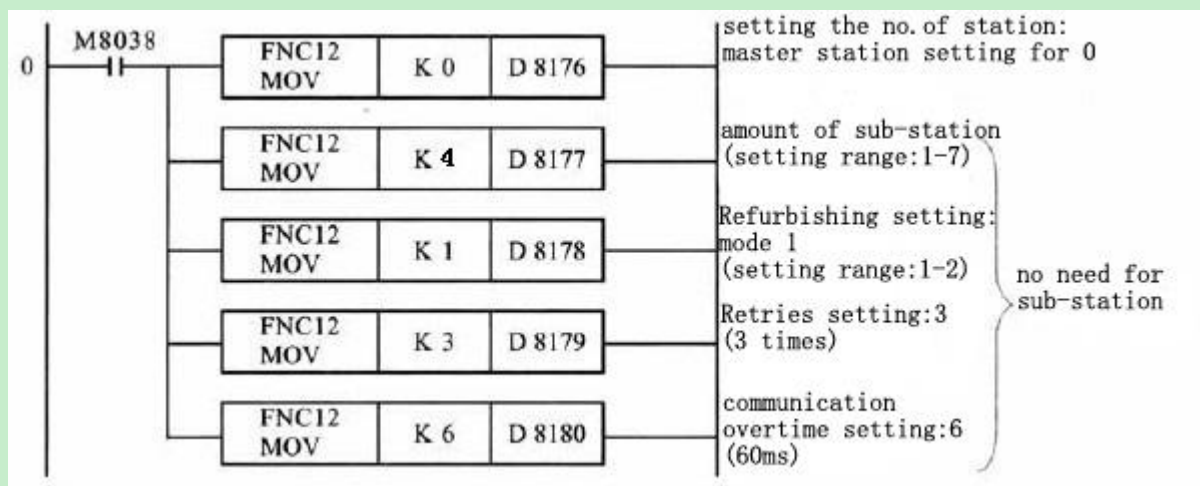


Diagram 8-4

For the programs:

- (1) In the programming: **N: N** network parameters setting program is from step 0.
- (2) Special data register **D8178** set refresh range, refresh range is each station link save zone. For slave station, the setting is not needed. The network information exchange data, and the 8-3(mode0), table 8-4(mode1) and table 8-5(mode2), 3 refresh modes. In this mode, element **N: N** network all stations.

Table 8-3

station	Element	
	bit soft element (M)	word soft element (D)
	0	4
No. 0	-	D0~D3
No. 1	-	D10~D13
No. 2	-	D20~D23
No. 3	-	D30~D33
No. 4	-	D40~D43
No. 5	-	D50~D53
No. 6	-	D60~D63
No. 7	-	D70~D73

Table 8-4

station	Element	
	bit soft element (M)	word soft element (D)
	32	4
No. 0	M1000~M1031	D0~D3
No. 1	M1064~M1095	D10~D13
No. 2	M1128~M1159	D20~D23
No. 3	M1192~M1223	D30~D33
No. 4	M1256~M1287	D40~D43
No. 5	M1320~M1351	D50~D53
No. 6	M1384~M1415	D60~D63
No. 7	M1448~M1479	D70~D73

Table 8-5 Mode 2 station number and bit, word element table

station number	Element	
	bit soft element(M)	word soft element(D)
	64	4
No. 0	M1000~M1063	D0~D3
No. 1	M1064~M1127	D10~D13

No. 2	M1128~M1191	D20~D23
No. 3	M1192~M1255	D30~D33
No. 4	M1256~M1319	D40~D43
No. 5	M1320~M1383	D50~D53
No. 6	M1384~M1447	D60~D63
No. 7	M1448~M1511	D70~D73

In the diagram 8-4, refresh range setting is mode 1. Then each station 32×8bit soft element, 4×8 word soft element is linksave zone. In run, for No. 0 number station (master station), network switch data write bit soft element **M1000~M1063**, network digital data write word soft element **D0~D3**,....., for other stations.

(3) Special data register **D8179** setting retry times, setting range is 0~10(default=3), for slave station, the setting is not needed. if a master station retries the diagram in this times(or higher) and slave station communication,.

(4) Special data register **D8180** setting communication value, setting range is 5~255 (default=5).

(5) For slave station, network parameters set needs setting station number, such as that supply station (1number station) setting is shown in diagram 8-5.

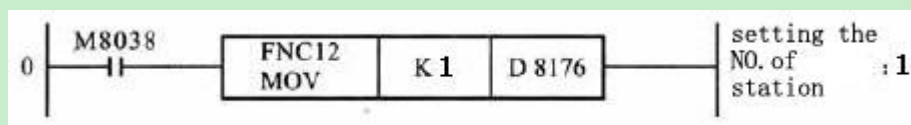


Diagram 8-5

For master station and each slave station programming above, after network connection, you shall turn on each **PLC** work power supply, in **STOP** status, communication will be on.

2.N: N network debugging and operation

(1). Tasks

The feeding station , processing station , assembly station , sorting station , transfer station PLC are connected by FX3U-485-BD communication panel, the transmission station is the master station, the number of the station is 0, supply station, processing station, assembly station, sorting station's numbers: feeding station No. 1, processing station No. 2, assembly station No. 3, sorting station No. 4. The functions:

1)No. 0 station X1~X4 are No. 1 station ~ No. 4 station Y0 (note: when network is normal, push 0 number station X1, No. 1 number station Y0 output).

2)No.1 station ~4number station D200 value is 50, station Y1, Y2, Y3, Y4 output.

3).From No.1 station read No.4 station D220 values, save to 1 number station D220.

(2). Connection network and write, debug program

Link communication port, write master station program and slave station program, in programming software, you shall monitor, change input and data register status, observe station changes, if it does not work, you shall correct the hardware and software, and do debugging.

In the diagram 8-6 and diagram 8-7 are transmission station and feeding station programs. In the programs, station communication error sign bit (special assistant relay **M8183~M8187**,

in table 8-1). For example, when a slave station has communication faults, the master station is not allowed to get data from the slave station in the network. make station communication error sign bit programming, it is good for communication data reliability, but the station cannot recognize the self error, is each station error program is not needed.

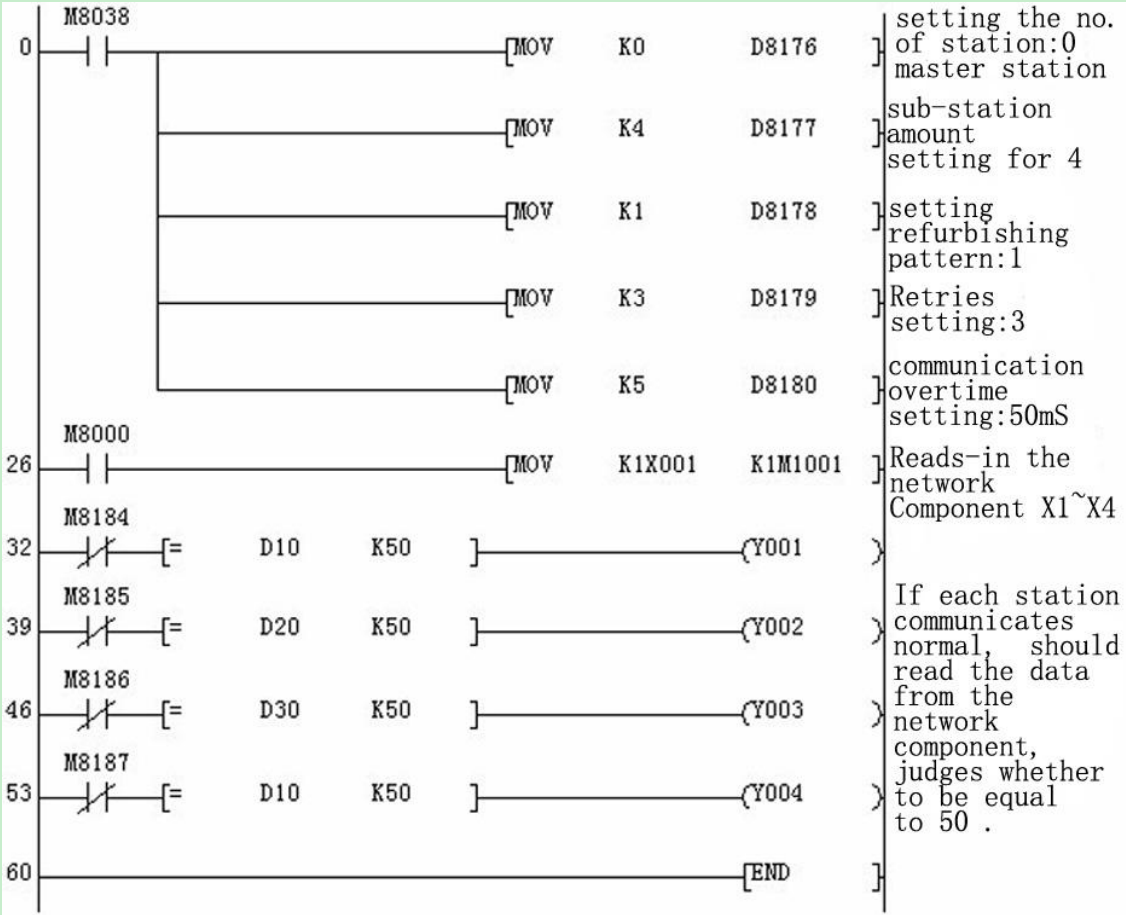


Diagram 8-6

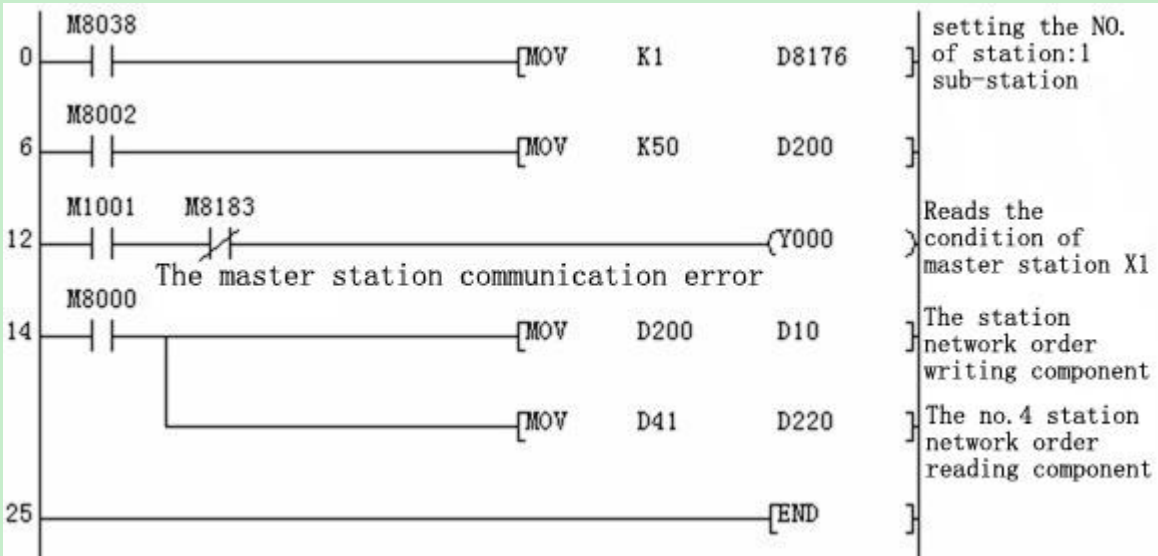


Diagram 8-7

3.Communication time

The time needed for data transferred in the network, **N:N** network adopts broadcast mode in communication, the refresh time is communication time(ms), in the network, the more stations, the larger data refresh range, and the longer communication time. The relation among communication time and network station quantity, and communication refresh range are shown in table 8-6.

Table 8-6

		communicationequipment		
		Mode 0 bit soft element: 0 word soft element: 4	Mode 1 bit soft element: 32 word soft element: 4	Mode 2 bit soft element: 64 word soft element: 8
station quantity	2	18	22	34
	3	26	32	50
	4	33	42	66
	5	41	52	83
	6	49	62	99
	7	57	72	115
	8	65	82	131

For **N: N** network, for communication mode, each station **PLC** scan time will increase by 10%.

In order to ensure network communication time features, when you make the network programs, you shall select the proper refresh mode according to the network communication rate. Then, you shall consider communication time in network programming.

8.3 System whole control training work task

YL-335B automatic production line training work is a comprehensive work, for 3 students to finish in 5 hours.

■ **Automatic production working target:** transfer supply station workpiece to processing station bench, when processing is finished, transfer processed workpiece to assembly station, then insert white and black column component into workpiece, then transfer to allocation station to output. The finished processing and assembly work workpiece is shown in diagram 8-8.



Diagram 8-8

■ Tasks

(1) Automatic production line installation

To finish **YL-335B** automatic production line feeding, processing, assembly, sorting station and transmission station assembly work, then install on **YL-335B** bench.

YL-335B automatic production line each station installation is shown in diagram 7-38.

2. Assembly requirements:

- (1) supply, processing and assembly station assembly work has been finished.
- (2) Finish sorting station installation and adjustment. Assembly is shown in see diagram 5-36.
- (3) Transmission station linear rail and base are assembled, you shall install onto the bench, and do other parts assembly, to finish the whole station installation and adjustment.

(2) pneumatic circuit connection and adjustment

1. Sorting and transmission station pneumatic system diagram, finish pneumatic circuit connection.
2. Check each station cylinder initial position, then adjust.
3. Adjust pneumatic circuit, ensure each cylinder running well.

(3) Circuit design and circuit connection

According to running requirements, design allocation and transfer station circuit and connect circuit.

1. Design allocation station electrical control circuit, connect circuit according to circuit schematic. The circuit schematic includes **PLC I/O** and inverter Main circuit and control circuit. According to running requirements, set parameters, provide circuit schematic .

2. Design transfer station electrical control circuit, according to circuit schematic, connect circuit; circuit schematic includes **PLC I/O**, servomotor and driver control circuit. Connect circuit according to running requirements, set servomotor driver parameters.

(4) PLC network connection

The system control is of **PPI** protocol communication distribution network control,

transfer station is the Main station. The system Main signals are from touch screen, but the system emergency stop signal is from transfer station button/ indicator light module emergency stop button, installed on the bench, the alarm lights can show the system work status.

(5) Touch screen configuration user menu interface

The touch screen is connected to system Main station **PLC**.

TPC7062K menu requirements :user window includes Main menu and Welcome menu, Welcome menu is the start menu, touch screen text already moves rightward.

When touch welcome menu, it is switched to main window menu. Main window menu Group functions:

1. Provide system work mode (single station/full-line)select signal and system reset, start and stop signal.
2. Sorting station inverter input running frequency(40Hz~50Hz).
3. Dynamic display transfer station mechanical arm position.
4. Indicate network running status (normal, fault).
5. Indicate each station running, fault status, includes:
 - (1) Feeding station supply shortage status and no material status.
 - (2) Assembly station supply shortage status and no material status.
 - (3) Transmission station mechanical arm over-travel fault.
6. Indicate full-line running, system emergency stop status.

Welcome menu and main menu are shown in diagram 8-9 and diagram 8-10.

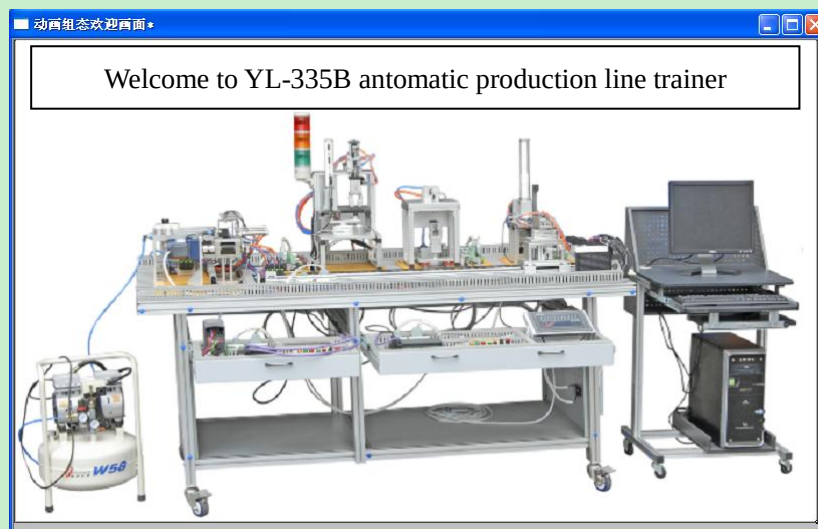


Diagram 8-9

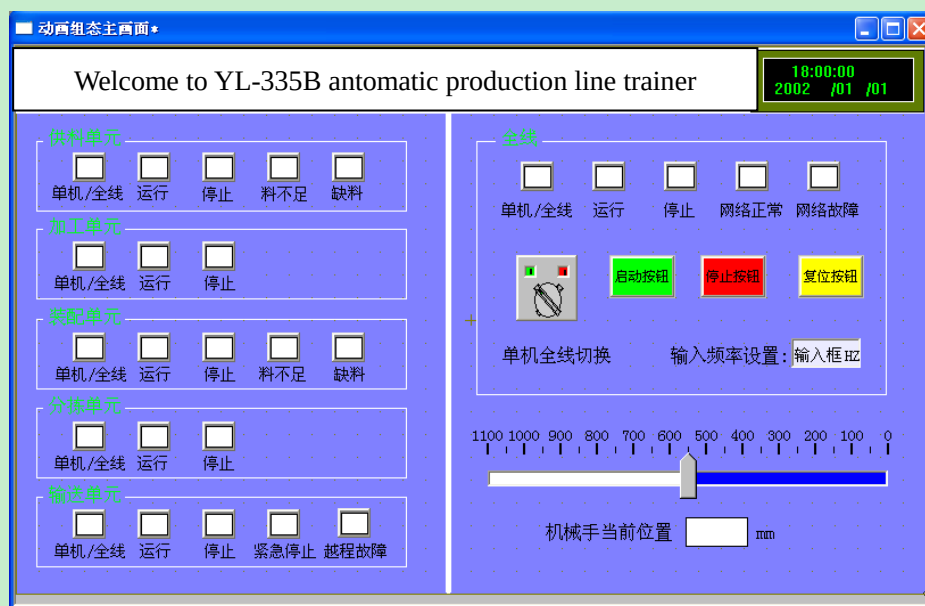


Diagram 8-10

(6) programming and debugging

The system work modes divided into single station work and full-line running mode.

From single station work mode to full-line running mode: each station is in stop status, each station button/ indicator light module work mode is in full-line mode, then if in the man-machine menu, you turn switch to full-line running mode, the system will enter full-line running status.

From full-line running mode to single station work mode, only when the current work cycle man-machine menu switch is turned to single station running mode, this can be valid.

In full-line running mode, each station only receives man-machine menu signals through network, except Main station emergency stop button, all other station signals are invalid.

1.Single station running mode test

In single station running mode, each station Main signal and work status signal are PLC button/ indicator light module. The button/ indicator light module work mode select switch SA is turned to “single station mode”. each station control requirements:

(1).Feeding station single station runningwork requirements

① if station two cylinders are in initial position, and there is no enough workpiece , then “ normal work” indicator light **HL1** is on, equipment is ready. Otherwise, the indicator light 1Hz frequency flashes.

② if equipment is ready, push start button,station start, “running” indicator light **HL2** is on.start, if output bench has no workpiece, then it pushes workpiece to output bench. The output bench workpiece is taken, if has no stop signal, then enter the next cycle workpiece push operation.

③ if in running, push stop button, then when the work cycle task is finished, each station stop work, **HL2** indicator light is off.

④ if in case of workpiece shortage, then station will continue, but “ normal work” indicator light **HL1 flashes in** 1Hz frequency, “running” indicator light **HL2** is on. if has no workpiece , then **HL1** indicator light and **HL2** indicator light will 2Hz frequency flashes.

When the station finishes the cycle task, it will stop. Then enough workpiece is needed to make the station restart.

(2).Processing station single station runningwork requirements

- ① if each cylinder is in initial position, then “normal work” indicator light **HL1** is on, means the equipment is ready. Otherwise , the indicator light 1Hz frequency flashes.
- ② if equipment is ready, push start button, equipment starts, “ running” indicator light **HL2** is on. When the workpiece arrives processing bench and is detected, equipment will tighten the workpiece, transfer to processing zone for punching, then the workpiece will return for next workpiece processing. If there is no stop signal input, when the workpiece arrives the processing bench, the processing station will start the next cycle work.
- ③ in work, if push stop button, when the processing station finishes a cycle, it stops work. **HL2** indicator light is off.
- ④when the workpiece is detected, and processing process starts, if you push emergency stop button, the station will stop running at once, **HL2** indicator light 1Hz frequency flashes. Emergency stop button reset, equipment from emergency stop point, will continue running.

(3).Assembly station single station runningwork requirements

- ① if each cylinder is in initial position, with enough column components; workpiece assembly station has no workpiece. Then “normal work” indicator light **HL1** is on, means that the equipment is ready. Otherwise , the indicator light 1Hz frequency flashes.
- ② if equipment is ready, push start button, assembly station start, “running” indicator light **HL2** is on. if no column component, it will drop; if left disc has component, right disc has no component, the rotary bench will rotate.
- ③ if rotary bench, right disc has column component and the assembly station has workpiece, assembly mechanical arm catches column component, and put the workpiece.
- ④ finish assembly tasks, assembly mechanical arm return initial position, wait for the next cycle assembly.
- ⑤ if in running process, you push stop button, then supply will stop supply, when assembly is ready, assembly station will stop when this cycle assembly is done.
- ⑥ in running, when “component shortage” alarm starts, indicator light **HL3** 1Hz frequency flashes, **HL1** and **HL2** light is on; in running, “no component” alarm, indicator light **HL3** is on for 1s, off for 0.5s, in this mode flashes, **HL2** is off, **HL1** is on.

(4).Sorting station single station runningwork requirements

- ① initial status: if station 3 cylinders are in initial positions, then “ normal work” indicator light **HL1** is on, equipment is ready. Otherwise the indicator light 1Hz frequency flashes.
- ② push start button, system starts, “running” indicator light **HL2** is on. When assembled workpiece is dropped, inverter starts, to drive motor frequency 30Hz, to transfer workpiece to allocation zone.
- ③if metal workpiece, column workpiece are white, then the workpiece is in No. 1 slot, the transfer belt stops, workpiece is pushed to No. 1 slot; if plastic workpiece column workpiece is white, then the workpiece will be in No. 2 slot, transfer belt stops, workpiece is pushed to No. 2 slot; if workpiece column workpiece is black, then theworkpiece will be in No. 3 slot,

transfer belt stops, workpiece will be pushed to No. 3 slot. The workpiece is pushed out of slot, the station a work cycle is finished. Only when workpiece is pushed out of slot, then it can drop workpiece to the transfer belt again.

if in running, push stop button, the station stops when a work cycle is finished.

(5). Transmission station single station running work requirements

The single station running is to transfer workpiece to test equipment. Each station shall be ready, and the drop workpiece onto the supply station output bench. The test process requirements:

① transfer station is turn on, push reset button SB1, for reset operation, to make mechanical arm returns origin position. In reset process, “normal work” indicator light **HL1** 1Hz frequency flashes.

When mechanical arm returns to origin, and the transfer station each cylinder is in initial position, then reset is finished, “normal work” indicator light **HL1** is on. Push start button **SB2**, equipment starts, “running” indicator light **HL2** is on, to start test process.

② mechanical arm from supply station output bench catch workpiece, catch sequence: arm extended→claw catches workpiece→bench rises→arm returns.

③ servomotor drives mechanical arm to move to the processing station, and the speed is no less than 300mm/s.

④ mechanical arm moves to processing station bench upside, drop workpiece onto processing station bench. mechanical arm dropping workpiece sequence: arm extended→bench drops→drop workpiece→arm returns.

⑤ 2s later, mechanical arm catch processing station workpiece operation. catch sequence is as same as supply station catch workpiece sequence.

⑥ servomotor drives mechanical arm moves to assembly station bench. Then put workpiece onto assembly station bench. The sequence is as same as the processing station sequence.

⑦ 2s later, mechanical arm catches assembly station workpiece. The catch sequence is as same as supply station sequence.

⑧ mechanical arm returns, sway bench rotates for 90° counterclockwise, servomotor drives mechanical arm from assembly station, to transfer workpiece to allocation station, when it arrives the allocation station transfer belt, it will drop the workpiece, the sequence is as same as processing station drop workpiece sequence.

⑨ When mechanical arm returns, then return origin operation. servomotor drives mechanical arm, in 400mm/s return, returns 900mm, sway bench rotate clockwise 90°; then 100mm/s returns to origin and stop.

When the mechanical arm returns to origin, 1 test cycle is end. When supply station output bench drops workpiece, push start button **SB2**, restart.

2. System normal full-line running mode test

In full-line running mode, each station work sequence and transfer station mechanical arm running speed requirements, and single station running mode shall be same. The full-line running steps:

(1). turn on power, when N: N network starts work. Touch menu reset button, to reset operation, in reset process, green alarm light flashes in 2Hz frequency. red and yellow light

are off.

Reset process: make transmission station mechanical arm return to origin, and check whether each station is in initial status.

Each station initial status:

- ① Each station parts are in initial positions.
- ② Feeding station has enough workpiece .
- ③ Assembly station has enough column component.
- ④ Transmission station emergency stop button is not pushed.

When transmission station mechanical arm returns origin, and each station is in initial status, then reset finished, green alarm light is on, to allow system start. Then if touch menu start button, system starts, green and yellow alarm light are on.

(2).Feedingstation running

When system starts, if supply station has no workpiece, then push workpiece to output bench, then sends output bench has workpiece signal to system. If supply station has no workpiece or workpiece shortage, then sends alarm or pre-alarm signal to the system. The output bench workpiece will be transferred to station mechanical arm, if system still needs to push the workpiece for processing, then it will start the next cycle workpiece operation.

(3) Transmission station running 1

When workpiece is pushed to supply station, transfer station mechanical arm catches supply station workpiece. Then servomotor drives mechanical arm moves to processing station processing bench, put workpiece on processing station processing bench .

(4).Processing station running

When the processing station processing bench workpiece is detected, it will start processing. When processed workpiece returns, it will send puching processing finished signal to the system.

(5)Transmission station running 2

When system receives processing finished signal, transfer station mechanical arm will catch processed workpiece. Then servomotor drives mechanical arm to assembly station. Then put workpiece onto assembly station bench.

(6).Assembly station running

When the assembly station bench sensor detected workpiece, it starts for assembly process. Then it sends signal to system.

if assembly station has no column workpiece or workpiece shortage, it sends alarm or pre-alarm signal to system.

(7). Transmission station running 3

When system receives assemblyfinishedsignal, transfer station mechanical arm catches assembled workpiece, then from assembly station to allocation station, move workpiece, put on allocation station, then returns to origin.

(8).Sorting running

The mechanical arm put off workpiece, returns, allocation station inverter will start, drives the motor 80% frequency, to transfer workpiece to allocation zone, the workpiece allocation is as same as the single station. When allocation cylinder pushes workpiece and returns, it will send allocation finished signal to the system.

(9)When allocation station allocation work is finished, and the transfer station mechanical arm returns to the origin, a system work cycle is finished. if you do not touch stop button,

system will delay for 1s to start the next cycle. If in the work cycle, you touch stop button, the system work is finished, yellow light is off, green light is on. Then system work is finished if you push start button again, then system restarts.

8. abnormal work status test

(1). Workpiece supply status signal alarm

If supply station or assembly station “workpiece shortage” pre-alarm signal or “no workpiece” alarm signal, then system actions:

① if “workpiece shortage” pre-alarm signal alarm light red light 1Hz frequency flashes, green and yellow light is on. System will continue work.

② if “no workpiece” alarm signal, alarm light red light is on for 1s, off for 0.5s, mode flashes; yellow light is off, green light is on.

if “no workpiece” alarm signal comes from supply station, and the supply station bench pushes workpiece, system will continue running, to the end. When the work is end, only when “no workpiece” alarm signal is off, system can start.

if “no workpiece” alarm signal comes from assembly station, and the assembly station rotary bench drops column workpiece ,system will continue running, till the work cycle is finished. when the cycle is finished, the system will stop, only when “no workpiece” alarm signal is off, system can start.

(2). Emergency stop and reset

In system work process, if you push transfer station emergency stop button, then transfer station stops. When emergency stop resets, from emergency stop point, it will continue running. But if emergency stop button is pushed, mechanical arm will move to a target, then emergency stop reset transfer station mechanical arm will return to the origin, then move to the target.

8.3 Work tasks

8.3.1 equipment installation and adjustment

YL-335B each station installation, electrical wiring, penumatic circuit connection and adjustment, the steps are mentioned above.

In system installation, we shall make a reference point. In diagram 7-38, the distances are shown (X), 310mm: ①origin position and feeding station output bench center on X direction. ②Feeding station output bench center to processing station processing bench center: 430mm. ③Processing station processing bench center to assembly station assembly station center: 350mm. ④assembly station assembly station center to sorting station input center: 560mm.

When origin position is decided, transmission station position is also decided.

On empty bench, start system installation:

(1) . Finish transmission station installation, includes moving parts, mechanical arm, link, electrical ports; and the mechanical arm each sensor is connected to each cylinder; connected electrical ports; as well as station penumatic circuit and so on.

(2). Feeding, processing and assembly station will be assembled and installed. On Y direction, transfer station mechanical arm extends, to catch and drop workpiece.

(3).Sorting station assembly. On Y direction, to ensure transfer belt inlet center line corresponds to the transfer station line rail center line, on X direction, to ensure transfer station mechanical arm moves workpiece to allocation station and can drop the workpiece

into the inlet center exactly.

In installation work, you shall check carefully.

8.3.2 Parameters setting and testing

The electrical wiring is finished, for inverter, servo driver parameters setting, rotary encoder.

8.3.3 Human-machine interface configuration

8.3.3.1 Project analysis and creation

According to work tasks, project plan:

(1) **Project frames:** 2 user windows, Welcome menu and Main menu, Welcome menu is start menu. Policy 1: cycling.

(2) **Data objects:** each station and full-line work status indicator light, single full-line knob, start, stop, reset button, inverter input frequency, mechanical arm and so on.

(3) **Diagrams:**

Welcome menu window: ①pictures: loading; ②text: labels; ③button: introduced by objects warehouse.

Main menu window: ①text: labels; ② each station and full-line work status indicator light, clock: introduced by objects warehouse; ③single full-line knob, start, stop, reset button: introduced by objects warehouse; ④ input frequency setting: by input block; ⑤ mechanical arm: labels sliding input.

(4)**Flow control:**

Then you can project, steps: in “user window”, click “creation window” button, create “window 0”, “window 1”, then set two window properties.

8.3.3.2 Welcome interface configuration

■ Create welcome menu

To select “window 0”, click “window properties”, to enter user window properties setting includes:

- ①Window name is changed to “Welcome menu”;
- ②Window label is changed to :Welcome menu.
- ③In “user window”, select “Welcome”, right click, select drawn menu “start window”.

2. “Welcome menu” configuration

■ Edit Welcome menu

Select “Welcome menu” window icon, click “Group”, to start window menu editing.

(1). Load diagram:

Select “diagram” button , use mouse to drag to make a rectangle to fill the window.

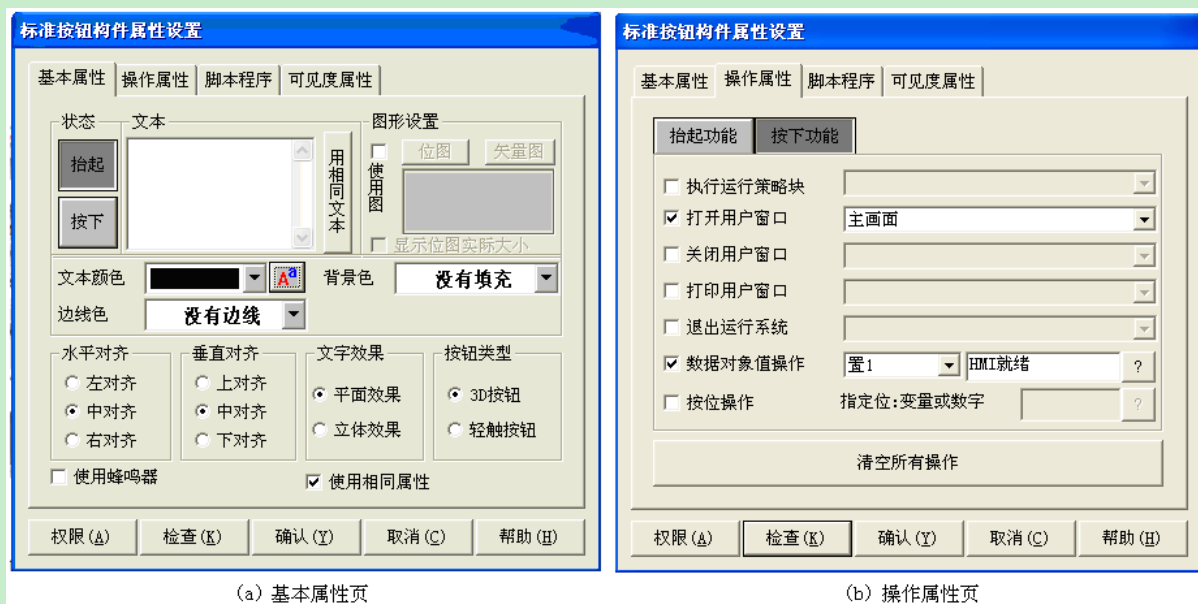
Right click, select “load diagram”, find the items, click to select, see diagram 8-11, then click “open” button.



Diagram 8-11

(2). Making button

Single click “” icon, in the menu, drag a button, double click the button, there will pop up the menu as shown in the diagram 8-12(a). you can select “button invisible”; then single click “push functions”: open user menu, select the main menu, and make data object “HMI ready” value is set to 1.



(a) 基本属性页

(b) 操作属性页

Diagram 8-12

(3). Making cycling text block

- ① select button , drag to window, then make for proper sizes for the rectangle. Input text “Welcome to use YL-335B Automatic Production Line Trainer”, push return to finish text input .
- ② static properties: text block background color: no; text block color: no; text color:

red.

③ to make text move, the properties setting is shown in diagram 8-13.

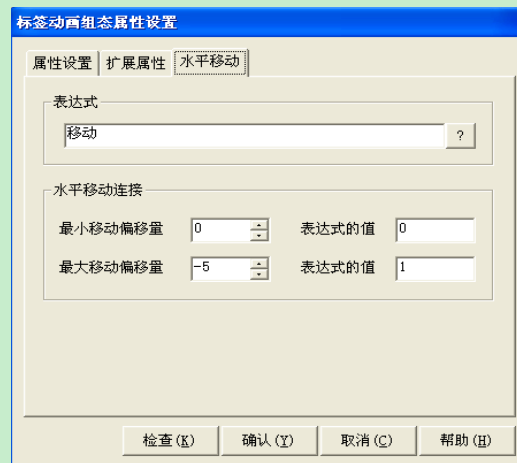


Diagram 8-13

Setting as follow:

- Firstly you shall confirm objects expression, then to definite expression offset. In the diagram 8-16, an internal data object “moves” is defined for the expression, and it is in the proportional to the text object offset, when the expression “moves” value is 0, the text object will be moved to point 0 (no movement), when the expression “moves” value is 1, the text object will be moved leftward for 5 points (-5), that is to say, “moves” variable and text object relation shall be -5 slope linear relation.

- the left corner coordinate origin, single is pixel, leftward is the negative direction, and rightward is positive direction. **TPC7062KS** resolution 800×480, “Welcome to use YL-335B Automatic Production Line Trainer!”, the text moves in a cycling policy, if text moves leftward and leaves, then return to initial position to restart.

- Configuration “cycling policy” operation:

- in “running policy”, double click “cycling policy” to enter cycling policy window .

- double click  icon to enter “policy properties setting”, to set the cycling time: 100ms, and push “confirm”.

- in cycling policy window, click “add a new policy”  icon, to add a policy line, as shown in the diagram 8-14

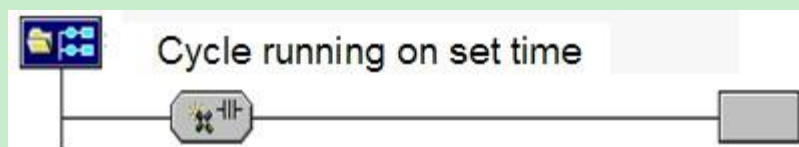


Diagram 8-14

- click “policy tools box”, “script program”, then move the policy icon  , left click, to add a script program, as shown in the diagram 8-15:

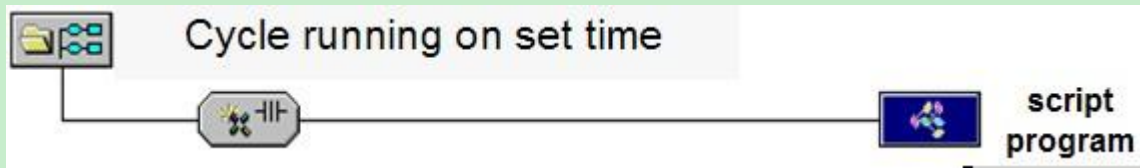


Diagram 8-15

- double click  to enter policy conditions setting, in the expression, input 1, for all conditions.
- double click  to enter script program editing, input the following program:


```
if moves<=140 then
  moves=moves+1
else
  moves=-140
endif
```
- click “confirm”, to finish script program creation.

8.3.3.3 Main interface configuration

1.Create Main menu

- ① select “window 1”, click “window properties”, to enter user window properties setting.
- ② change window name to: Main menu; window label is changed to: Main menu; “window background”, select the color needed.

2.Define data objects and device connection

■ to definite data objects

Each station and full-line work status indicator light, single full-line knob, start, stop, reset button, inverter input frequency, mechanical arm current value, by PLC, to exchange information data objects. To definite data objects:

- (1) click “real-time data warehouse” window icon, to enter the real-time window page.
- (2) click “add objects” button, in window data objects list, add new data objects.
- (3) select objects, push “objects properties” button, or double click select objects, to open “data objects properties setting” window . then edit properties. In table 8-4 are all PLC data objects.

Table 8-4

No.	objects	Type	No.	objects	Type
1	HMI ready	switch	15	single full-line _supply	switch
2	over-travel	switch	16	running_supply	switch

	fault_transfer				
3	running_transfer	switch	17	shortage_supply	switch
4	single full-line_transfer	switch	18	no material_supply	switch
5	single full-line_full-line	switch	19	single full-line_processing	switch
6	Reset button_full-line	switch	20	running_processing	switch
7	stop button_full-line	switch	21	single full-line_assembly	switch
8	start button_full-line	switch	22	running_assembly	switch
9	single full-line_full-line	switch	23	shortage_assembly	switch
10	network normal_full-line	switch	24	no material_assembly	switch
11	networkfault_full-line	switch	25	single full-line_allocation	switch
12	running_full-line	switch	26	running_allocation	switch
13	emergency stop_transfer	switch	27	claw_transfer	Data
14	inverter frequency_allocation	Data			

■Equipment connection

To definite data objects and **PLC** internal variables, the steps:

- (1) open “toolbox”, double click “general serial port father equipment ”, then double click “Siemens_S7200PPI”. There will pop up “general serial port father equipment”, “Mitsubishi _FX series of programming”.
- (2) set general serial port father equipment basic properties, as shown in the diagram 8-16.

通用串口设备属性编辑

基本属性

电话连接

设备属性名	设备属性值
设备名称	通用串口父设备0
设备注释	通用串口父设备
初始工作状态	1 - 启动
最小采集周期 (ms)	1000
串口端口号 (1~255)	0 - COM1
通讯波特率	6 - 9600
数据位位数	0 - 7位
停止位位数	0 - 1位
数据校验方式	2 - 偶校验

检查 (K)

确认 (Y)

取消 (C)

帮助 (H)

Diagram 8-16

(3)double click “ Mitsubishi _FX series of programming”, to enter the menu, then as shown in table 8-4, push “Add equipment channel”, as shown in diagram 8-17.



Diagram 8-17

3. Main menu making and configuration

Make main menu :

(1) To make main menu label text, insert clock, select lines, the text zone under the label will be divided into the two parts shown in the diagram 8-18. On the leftside of the zone, you can make each station menu, and on the rightside, you can make the main station menu and transmission station menu.

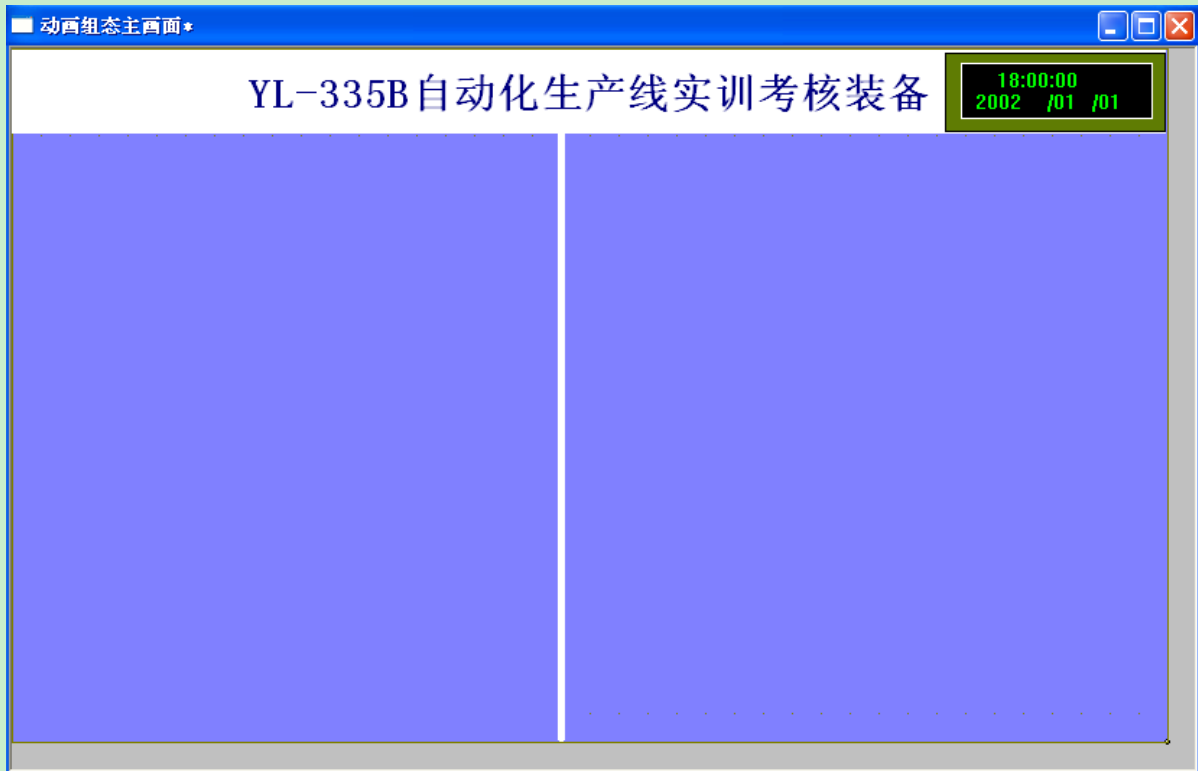


Diagram 8-18

(2). To make each from station menu. Take the feeding station as an example, as shown in the diagram 8-19, each parts are shown. These parts making and properties setting have been mentioned above, but “supply shortage” and “no material”, the two status indicator lights have alarming flash functions, below we will take the supply station no material alarm indicator light to introduce the properties setting methods.

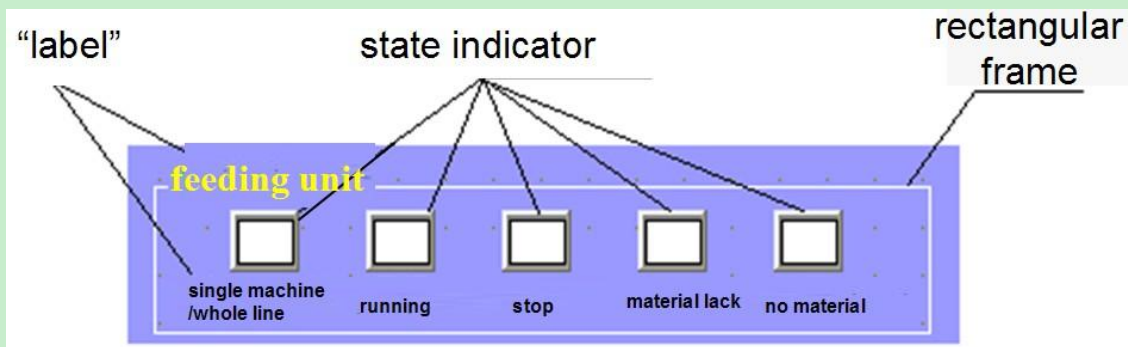


Diagram 8-19

Different from other indicator lights: no material alarm color is red, with flash function. Steps: in properties setting page, in special animation block, click “flash effect”, “fill color”, then “flash effect” page will pop up, as shown in diagram 8-20(a). click to select “flash effect” page, the expression will be selected to “no material_supply”; in flash mode block, click to select “Use properties change for flashing”; the filled color is yellow, as shown in (b).

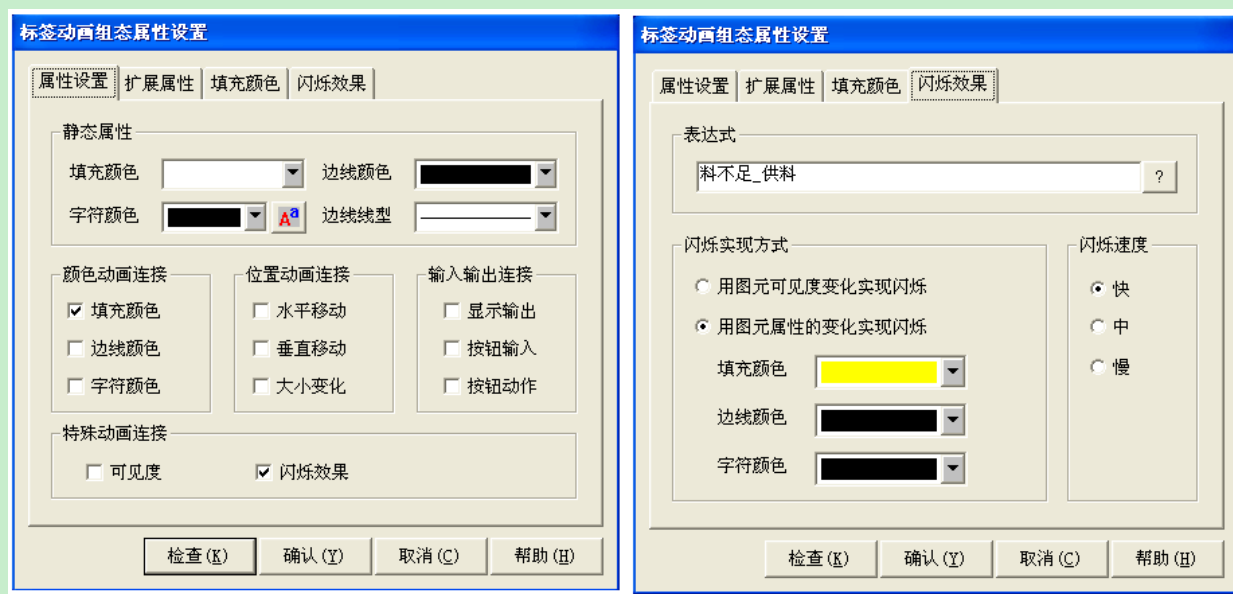


Diagram 8-20

(3). To make main station and transmission station menu. Here are the steps:

- ① select “tools box”,  icon, when mouse is shown as “+”, drag and adjust properly.
- ② double click the sliding part, to enter diagram 8-21 properties setting window. Then set each parameter:

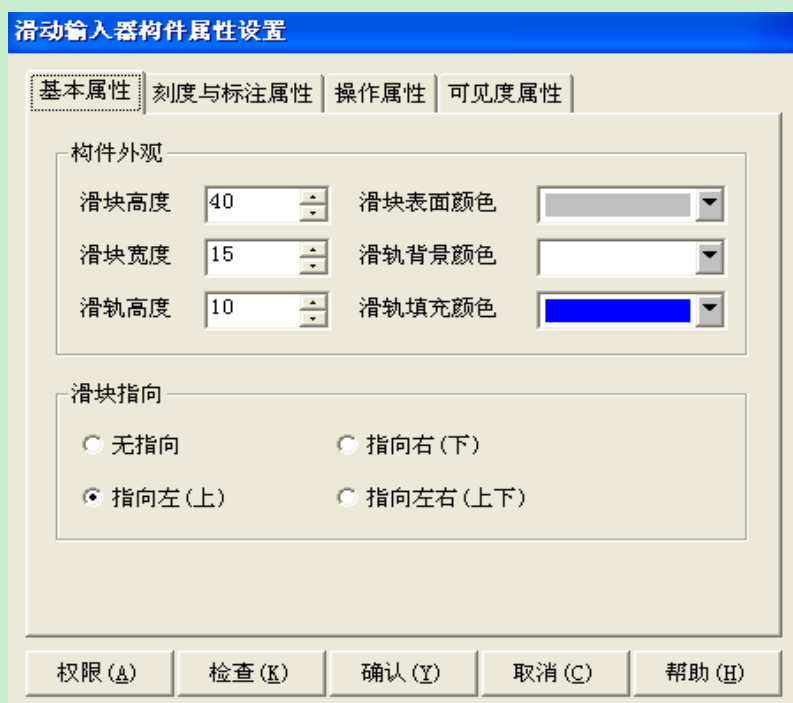


Diagram 8-21

In the “basic properties” page, pointed leftward (up);

In the “graduate and making properties”, “Main line number”: 11, “line number”: 2;

decimal fraction: 0;

In the “running properties”, for data object name: claw the current_transfer; the sliding block is in the leftside: 1100; and the sliding block is on the rightside: 0;

It is default.

④ click “right” button, to enter user right setting block, select manager group, push “confirm” button to finish. As shown in the diagram 8-22.

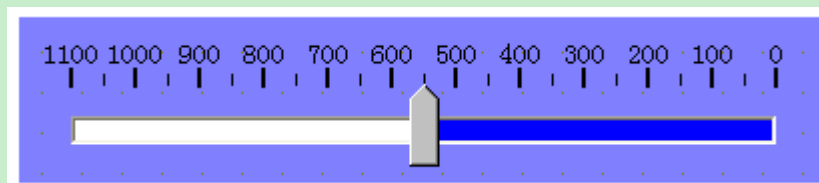


Diagram 8-22

8.3.4 Programming and debugging PLC control program

YL-335B distribution control automatic production line, in design, control program, from its system, by network connection, to plan communication data, and organize the system. Then according to each station technical tasks, make each station control programs.

8.3.4.1 Plan communication data

According to requirements, communication data

Table 8-5(a) Transmission station (0# station)data definition

transfer station address	data meaning	note
M1000	full-line run	
M1001		
M1002	allow processing	
M1003	full-line emergency stop	
M1004		
M1005		
M1006		
M1007	HMI on-line	
M1008		
M1009		
M1010		
M1011		
M1012	request supply	
M1013		
M1014		
M1015	allow sorting	
D0	highest frequency setting	

Table 8-5(b) Feeding station (1# station)data definition

supply station address	data meaning	note
M1064	initial status	
M1065	supplysignal	
M1066	on-line signal	
M1067	run signal	
M1068	shortage alarm	
M1069	workpiece loss alarm	

Table 8-5(c) Processing station (2# station)data definition

processing station address	data meaning	note
M1128	initial status	
M1129	processing finished	
M1130		
M1131	on-line signal	
M1132	run signal	

Table 8-5(d) Assembly station (3# station)data definition

supply station address	data meaning	note
M1192	Initial status	
M1193	on-line signal	
M1194	Run signal	
M1195	Parts shortage	
M1196	Parts loss	
M1197	Assembly finished	

Table 8-5(e) Sorting station (4# station)data definition

supply station address	data meaning	note
M1256	initial status	
M1257	distribution finished	
M1258	Distribution on-line	
M1259	Distribution	

8.3.4.2 slave station unit control programming

YL-335B workstation in the single-station operation of the programming ideas, in the previous items are made. In the case of online operation, the process of the slave station specified by the task book is basically fixed, and the process control program in the original single station program has little change. In the single-station program on the basis of the amendment, the preparation of online running procedures, the implementation is not too difficult. The following first to the feed station on-line programming as an example to illustrate the programming ideas.

The main changes in the case of online operation, one is different in the operating conditions, the master signal from the system through the network down the signal; Second, the workstations through the network constantly exchange of signals, thus determining the station flow And operating conditions.

For the former, first of all need to clear the current working mode of the workstation, in order to determine the current effective master signal. The task book clearly specifies the conditions for the work mode switch, the purpose is to avoid the occurrence of misuse, to ensure reliable operation of the system. The logical judgment of the operating mode switching condition is performed after power-on initialization (M8002 ON). Diagram 8-23 shows the implementation of the ladder.

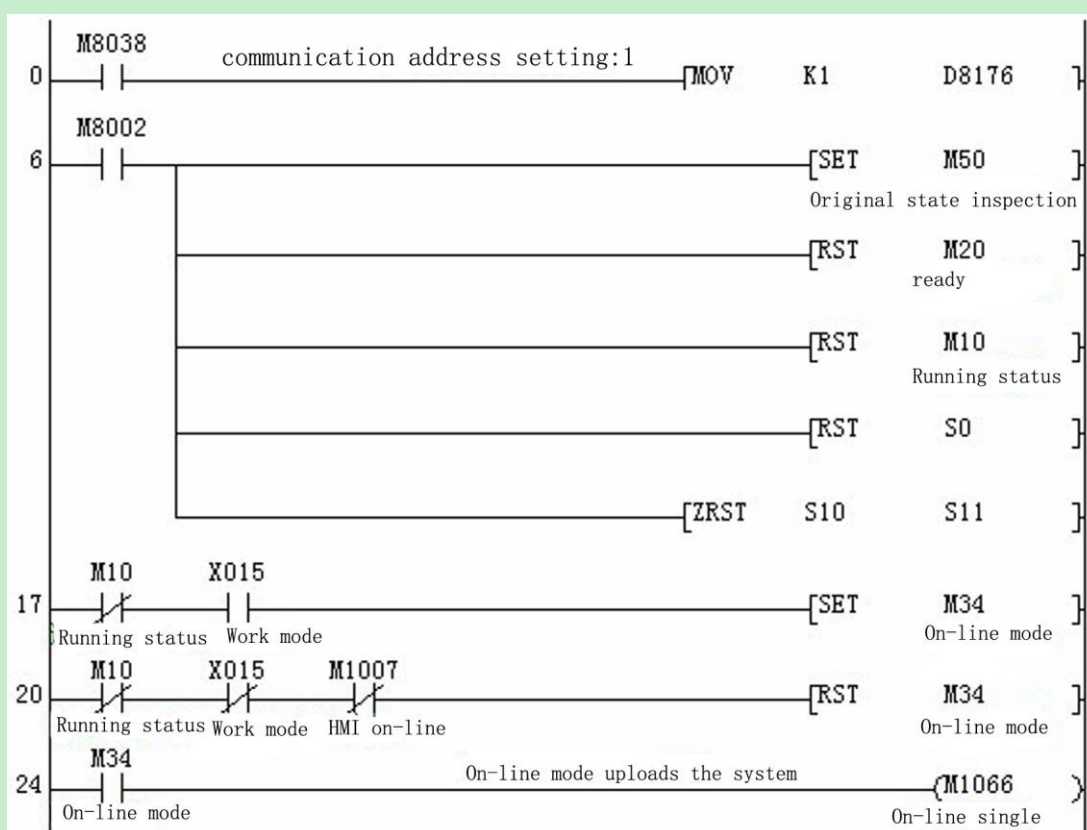


Diagram 8-23

The following work is similar as previous single station: ① initial status check, to judge the station is ready. ② if ready, it will receive full-line run signal or the station start signal, then starts. ③ in run status, it will monitor stop command, when it comes, stop command will be set, when the station finishes a technical process cycle, the station will stop. The ladder is shown in diagram 8-24.

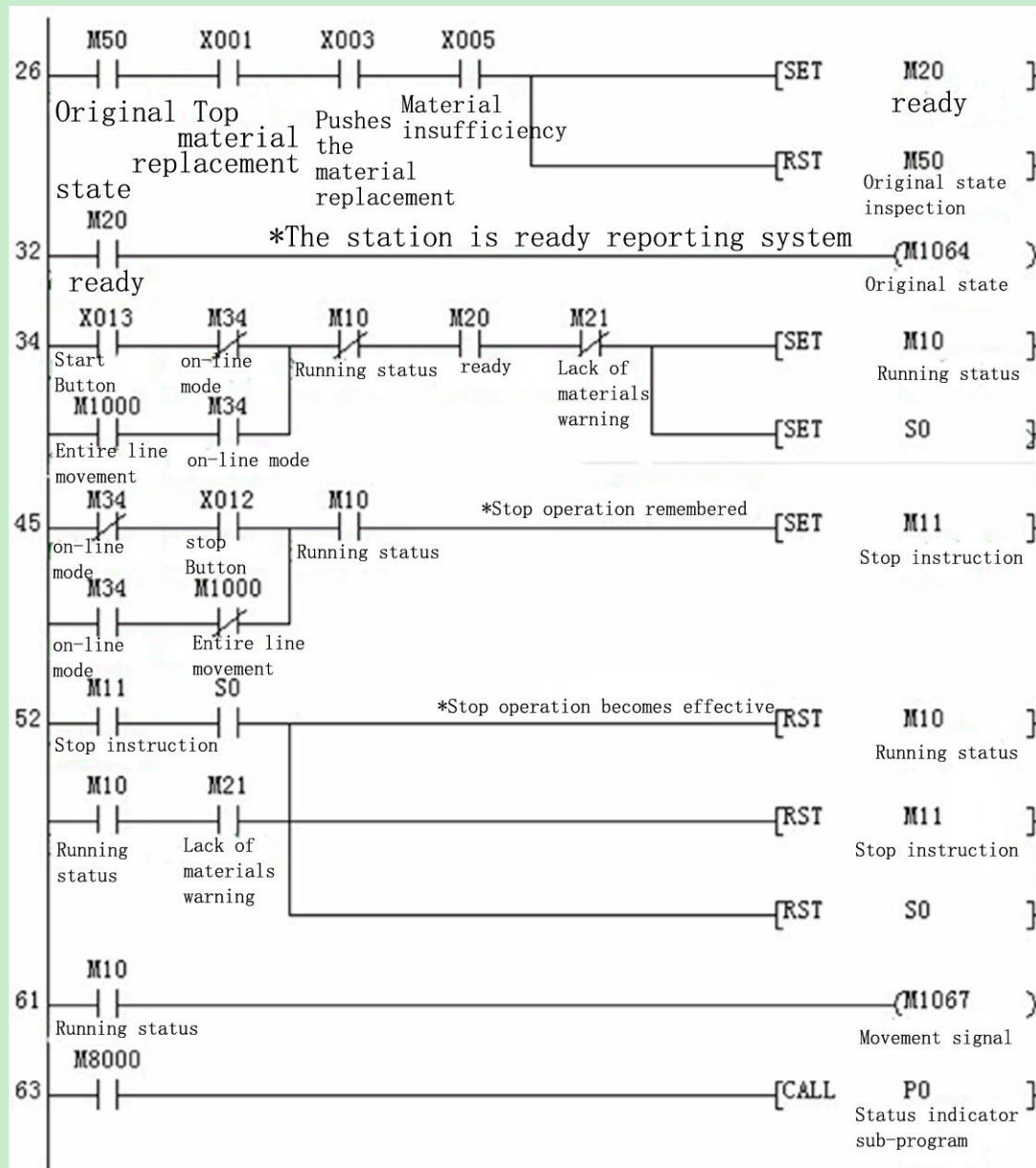


Diagram 8-24

The next step is technical control process, it is: from initial step S0, step sequence control process. The step program is almost as same as the previous single station. Other programming methods of other stations are as same as the feeding station. You can read each station single station program to compare and analyze.

8.3.4.3 Main station control programming

Transmission station is YL-335B system most important station: ① transmission station PLC and touch screen are connected, to receive touch screen signals, in the same time, feed back system status information to the touch screen .② The network main station, need a great quantity of network information to process. ③To finish the station, in the on-line mode, the technical production tasks are different from single station running. Therefore, you can change transfer station single station control program into on-line control mode, with more work. Below we will discuss on the problems and the way of programming.

1. Memory configuration

In order to make the program clear, you shall make good preparation for the program memory. Previously we have planed the network memory, and address range. In the menu, you also need to plan the PLC equipment channels, in the table format, as shown in table 8-6.

Table 8-6 Human-machine menu and PLC variables equipment channel

No.	variables	channel	No.	variables	channel
1	over-travel fault_transfer	M007(read)	14	single /full-line _supply	M1066(read)
2	running status_transfer	M0010(read)	15	running status_supply	M1067(read)
3	single /full-line _transfer	M0034(read)	16	workpiece shortage_supply	M1068(read)
4	single /full-line _full-line	M0035(read)	17	No workpiece_supply	M1068(read)
5	Reset button_full-line	M0060(write)	18	single /full-line _processing	M1131(read)
6	Stop button_full-line	M0061(write)	19	running status_processing	V1132(read)
7	Start button_full-line	M0062(write)	20	single /full-line _assembly	M1193(read)
8	Mode switch_full-line	M0063(write)	21	running status_assembly	M1194(read)
9	Network normal _full-line	M0070(write)	22	workpiece shortage_assembly	M1195(read)
10	Network fault_full-line	M0071(write)	23	No workpiece_assembly	M1196(read)
11	Running status_full-line	M1000(read)	24	single /full-line _allocation	M1258(read)
12	Emergency stop status_transfer	M1002.2(read)	25	running status_allocation	M1259(read)
13	Input	VM(read & write)	26	Claw_transfer	D2000(read)

frequency_full-line				
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Only after configuring the memory mentioned above can the other intermediate variables required for programming be considered. To avoid illegal access to internal memory, programming must pay attention to the problem.

2. Main program structure

As the transmission station bear more tasks, on-line running time, main program has great volatility.

(1). Every scan cycle, need to call the read and write subprogram and communication subprogram.

(2).Finish logic judgment of system working mode, in addition to its own transmission station should be in the on-line mode, the other slave station must be in the on-line mode.

(3).On-line mode, main order signal of system reset, issued by the HMI. In the initial status check, the condition of system ready, except transmission station itself in ready, the other slave stations are prepared ready. Therefore, reset subprogram in the the initial state check , in addition to finish transmission station initial statuscheck and reset operation, but also through the network to read each from station to prepare ready information.

(4).The whole running process is still : the initial state check → preparation ready, waiting to start → into running so gradually in several stages, but conditions of the end or the beginning will change.

The above is main program programming ideas, the following give main program list, as shown in diagram 8-25 each sub-diagram.

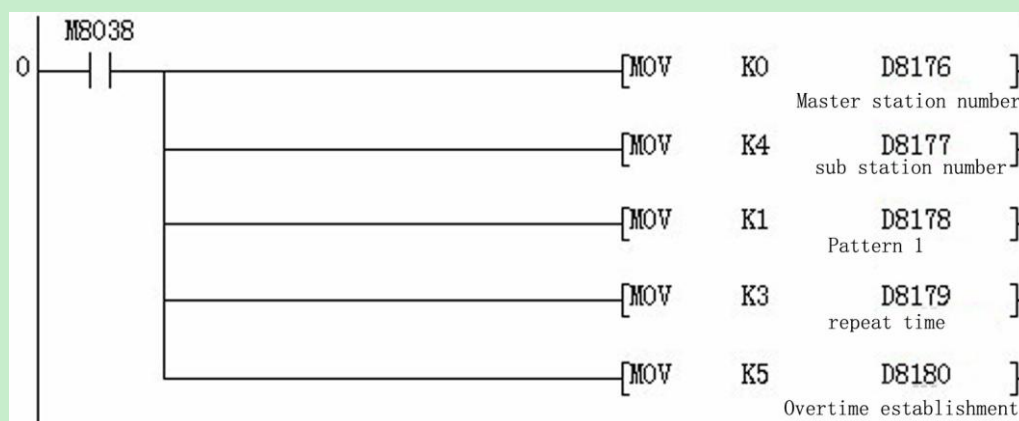


Diagram 8-25 (a)

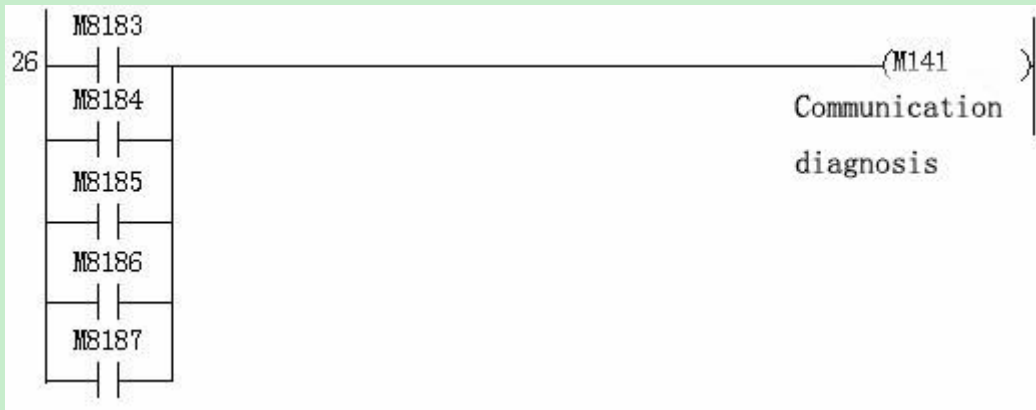


Diagram 8-25 (b)



Diagram 8-25 (c)

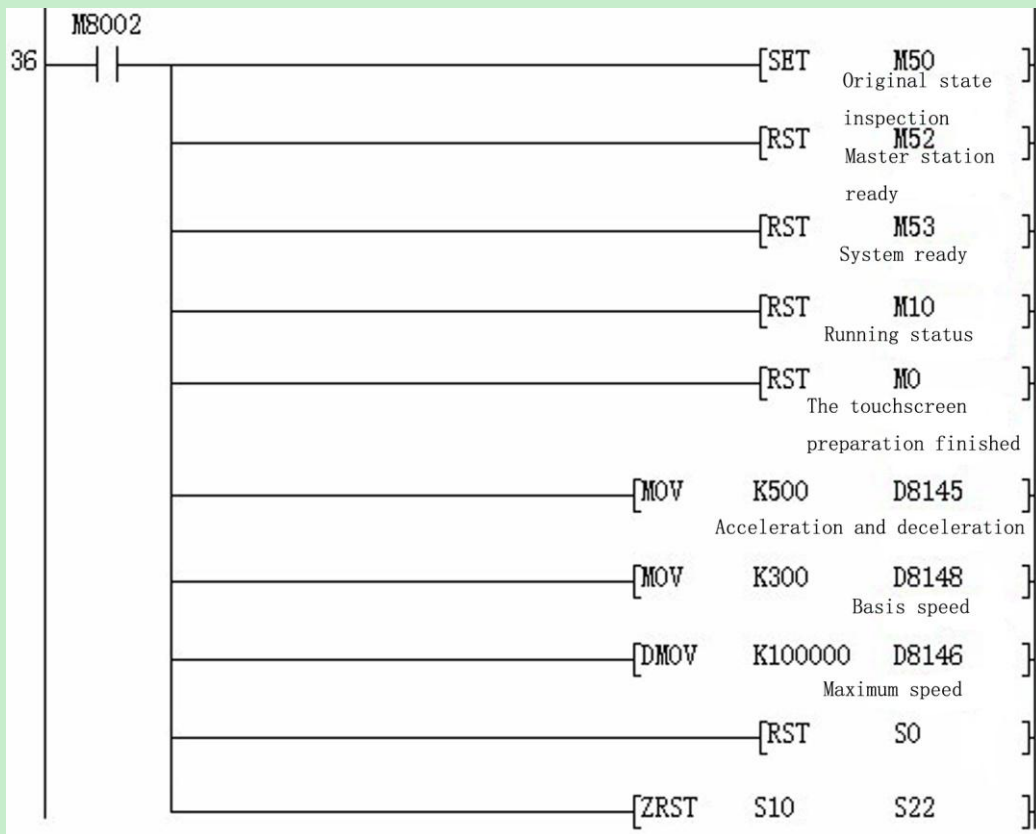


Diagram 8-25 (d)

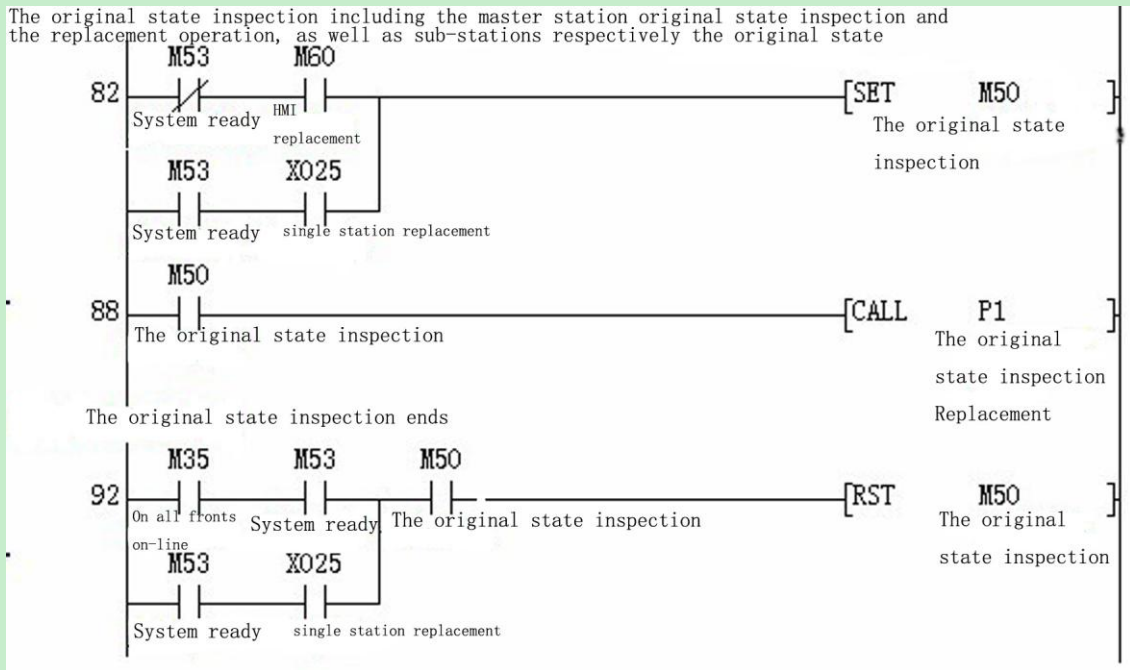


Diagram 8-25 (e)

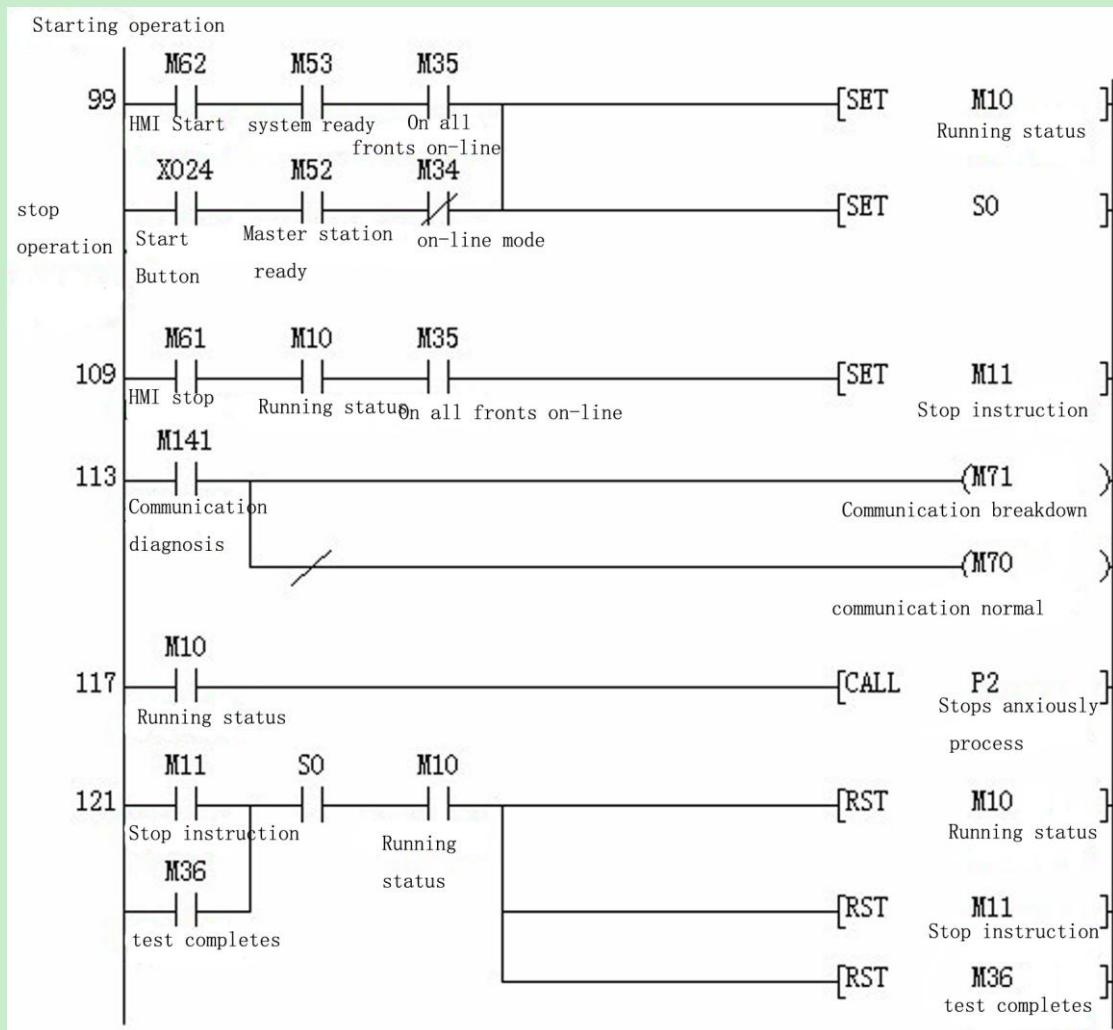


Diagram 8-25 (f)

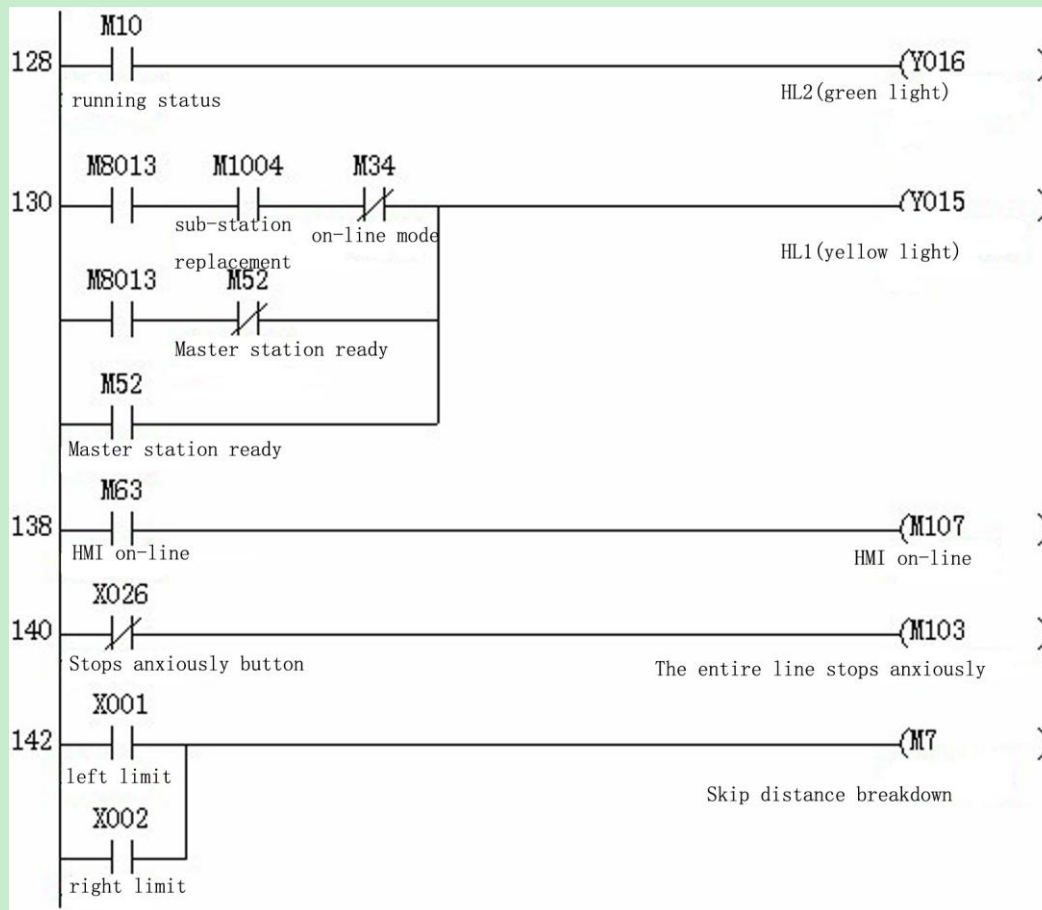


Diagram 8-25 (g)

3. “Running control ” subprogram structure

The transmission station on-line process process and single station process only has a little different, not much need to be modified. Main has the following:

- (1). Project 7 in work tasks, testing subprogram of transmission station started in the initial step on the mechanical arm to the supply station output bench catch workpiece, while the on-line mode the next, initial step operation as follows: supply station through the network to request supply, receipt of supply station signal, if has no stop command, then move a step that the implementation of the next catch workpiece.
- (2). Single station running time, mechanical arm to the processing station processing bench down workpiece, wait for 2 seconds to recover workpiece, while the on-line mode, the back workpiece conditions received from the network processing finished signal.assembly station conditions and the same.
- (3).Single station running time, test process ending that is out of running status.on-line mode the next, 1work cycle finished, return initial step, if has no stop command start the next 1 work cycle.

Thus, the operation control subprogram modified on the basis of the project 7 transmission function test subprogram is shown in diagram 8-26.

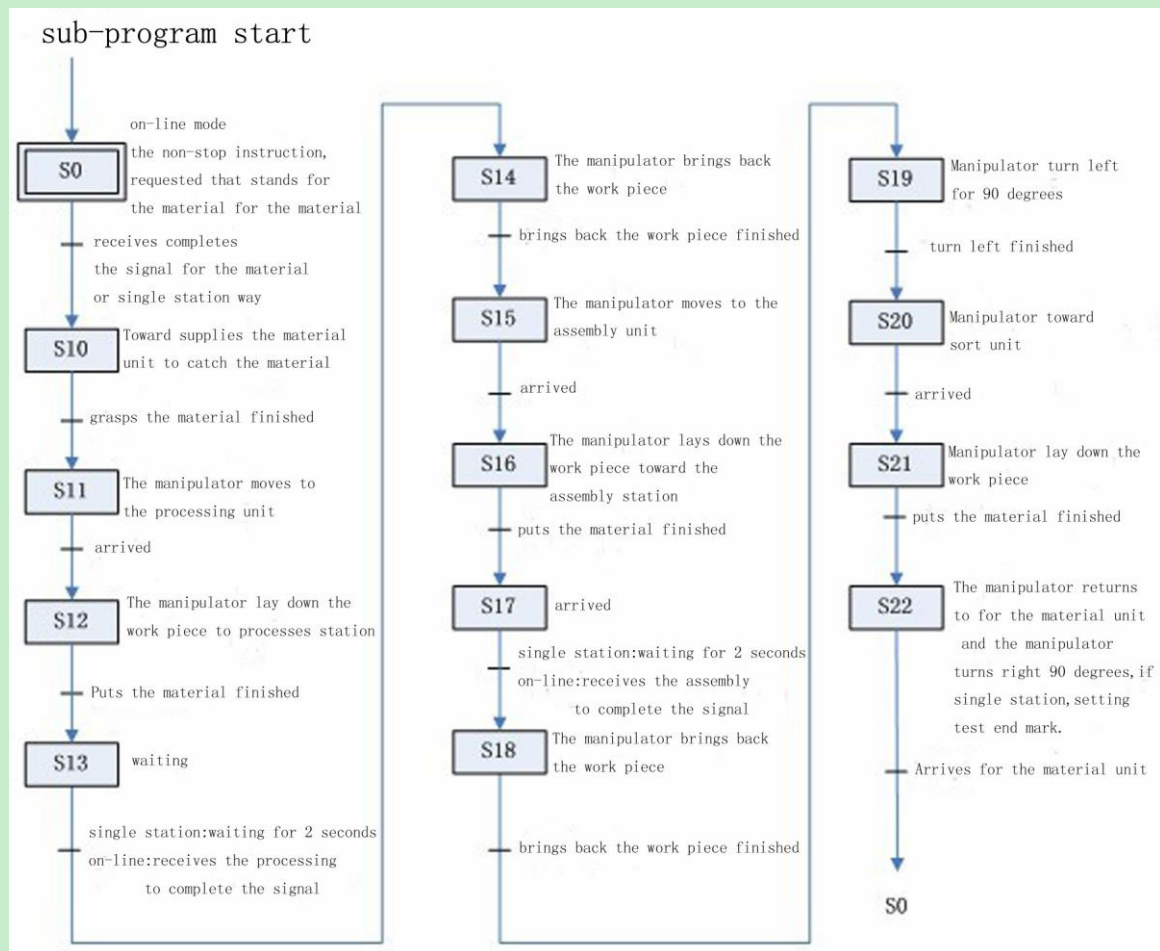


Diagram 8-26

4. “Communication” subprogram

“Communication ” subprogram includes from slave station alarm signal processing and transfer (between slave station, **HMI**), and to provide mechanical arm with current information of transmission station for **HMI**. The main program in each scanning cycle, always call this subprogram.

(1) Alarm signal processing and transfer include:

- Feeding station workpiece shortage and no workpiece alarm signal is transferred to the assembly station, provide information for alarm light.
- Handle the alarm signal of “no workpiece” in feeding station or “no component” in assembly station .
- Provide **HMI** network normal /fault information.

(2) Provide mechanical arm current position information of transmission station to **HMI**.

- In each scan cycle, pulse quantity indicates the current position converted length information (mm), then sends to **HMI** connection variable **D2000 (double words)**.
- When it returns to origin, pulse quantity will be cleared